

EXPLICIT BIOLOGY

A REVISION COURSE



GCE, SSCE, JAMB (UTME),
POST - UTME & PRE - DEGREE

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CHAPTER 1

LIVING ORGANISMS

Understand the following after reading this chapter.

Characteristics of living organisms
Prokaryotic and Eukaryotic cell
Cell structure and functions of cell components
Levels of organization.

CHARACTERISTICS OF LIVING ORGANISMS

1. **Movement:** an action by an organism or part of an organism causing a change of position, place, or aspect.
2. **Assimilation of food (Nutrition):** taking in of nutrients which are organic substances and mineral ions, containing raw materials or energy for growth and tissue repair, absorbing and assimilating them.
3. **Respiration:** the chemical reactions that break down nutrient molecules in living cells to release energy
4. **Reproduction:** the processes that make more of the same kind of organism
5. **Irritability or Sensitivity:** the ability to detect or sense changes in the environment (stimuli) and to make responses

6. **Adaptation:** the possession of factors (of structure or function) which enable organisms to live successfully and to survive in the respective environment

7. **Growth:** the ability to increase in size through acquisition of new protoplasm.

8. **Excretion:** removal from organisms of toxic materials, the waste products of metabolism (chemical reactions in cells including respiration) and substances in excess of requirements

Memory Tip: MARRIAGE

Viruses are category of extremely small microscopic parasites of plants, animals, and bacteria. Viruses are not cells but rather RNA or DNA molecules surrounded by a protein coating. Since viruses cannot reproduce without a host cell, they are not strictly speaking living organisms. Viruses are link between living and non-living things. The living properties of viruses are:

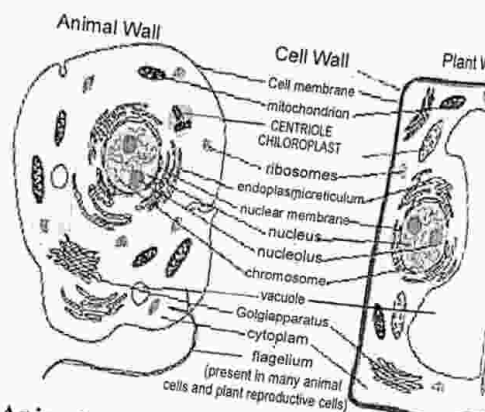
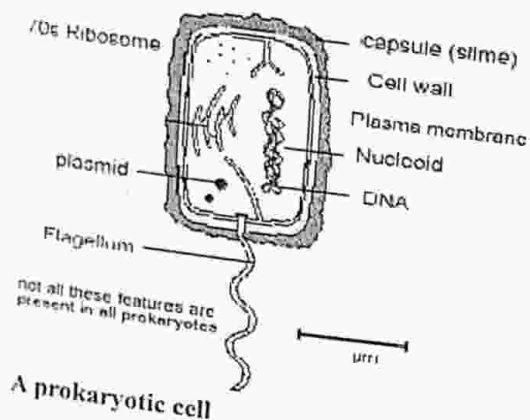
1. They reproduce by replicating in the living cell of a host
2. They possess characteristics which are transmitted from one generation to another, one generation to another.

Viruses and bacteria		
	Virus	Bacteria
Covered by	Protein coat	Cell wall
Cell membrane	No	Yes
Cytoplasm	No	Yes
Genetic material	DNA or RNA – only a few genes	DNA or RNA – enough for several hundred genes
Living or not?	Non-living unless in host	Living

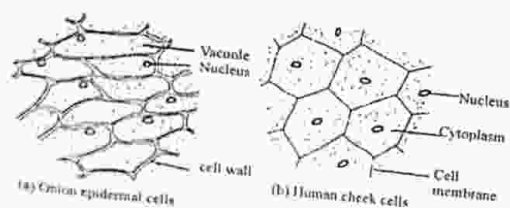
CELL

There are two major types of cell. **Eukaryotic cells**, which have membrane-bound organelles, like a nucleus, mitochondrion; and **prokaryotic cells** which do not have these organelles. The best example of a prokaryotic cell is bacteria cell.

	Prokaryotes	Eukaryotes
Typical organisms	bacteria	Protoctista, fungi, plants, animals
Typical size (smaller)	~ 1-10 μm	~ 10-100 μm (sperm cells) apart from the
Type of nucleus	Nuclear body No nucleus	real nucleus with nuclear envelope
DNA	circular (ccc DNA)	linear molecules (chromosomes) with histone
Ribosomes	70S	80S
Cytoplasmatic structure	very few structures	highly structured by membranes and a cytoskeleton
Cell movement	Flagellae/cilia made of flagellin	flagellae and cilia made of tubulin
Mitochondria	none	1 - 100 (though RBC's have none)
Chloroplasts	none	in algae and plants
Organization	usually single cells	single cells, colonies, higher multicellular
Cell division	Binary fission (simple division)	organisms with specialized cells Mitosis (normal cell replication) Meiosis (gamete production)



Animal and plant cell



Epidermal cells

CELL COMPONENTS AND THEIR FUNCTIONS

Nucleus
The nucleus is the membrane-bound region which contains the DNA in a Eukaryotic cell. In the Eukaryotic cell DNA is found in chromosomes which are linear molecules joined together by histones. The nucleolus is a dark area in the nucleus which is where ribosomal RNA is being made.

Nuclear Envelope
The nuclear envelope is a double membrane which surrounds the DNA in a Eukaryotic cell. There are pores in the nuclear envelope which allows mRNA to leave the nucleus and enter the cytoplasm.

Rough Endoplasmic Reticulum

Rough ER is made up of a large number of interconnected compartments known as cisternae. The ribosomes which are attached to the outside of the membrane. The role of the Rough ER is to make and transport proteins which are then transported to other parts of the cell.

Smooth Endoplasmic Reticulum

Smooth ER has a similar structure to the Rough ER but does not have the ribosomes. It is used to synthesize lipids, hormones and cholesterol.

Golgi Apparatus

This organelle is involved in the modification of proteins. As proteins arrive here in vesicles from the rough ER, they have carbohydrates or other molecules added to them to make them more complex and suited for their role. The modified proteins are then packaged into vesicles and transported out of the cell by the process of exocytosis.

Mitochondria

Mitochondria are sometimes known as the power house of the cell, it is here where aerobic respiration is used to release energy from organic molecules such as glucose. They make ATP. The mitochondria has a double membrane, the inner of which is folded to provide a large surface area.

Lysosomes

These are vesicles which contain hydrolytic enzymes. They can be exported out of the cell by the process of pinocytosis or used in the cytoplasm to break down organelles or proteins which are no longer needed.

Ribosomes

They have the same role that they have in prokaryotic cells. They read ribosomal RNA and make it into the protein needed.

Microtubules

These are found in both prokaryotic and eukaryotic cells. They are found throughout the cytoplasm and are made of a protein called tubulin. The role of microtubules is to assist in the transports of the cellular organelles and assist with cell division.

Chloroplasts

These are only found in plant cells where they contain chlorophyll which is used to transform light energy into carbohydrates. It is not unusual for chloroplasts to contain starch granules for energy.

Cell Wall

The roles of the cell wall in Plant cells are similar to the role of the cell wall in a bacterium. It prevents the cell rupturing due to osmotic pressures. In a plant cell the cell wall is made out of cellulose which is a very strong fibre but it is also flexible. The role of controlling what leaves and enters the cell is down to the cell membrane which is found in prokaryotic cells.

Rough Endoplasmic Reticulum

The Rough ER is made up of a large number of interconnected compartments known as **cisternae**. There are ribosomes which are attached to the outside of the membrane. The role of the Rough ER is to make and then break up proteins which are then transported to other parts of the cell.

Smooth Endoplasmic Reticulum

The Smooth ER has a similar structure to the Rough ER but does not have the ribosomes. It is used to synthesize steroid hormones and cholesterol.

Golgi Apparatus

This organelle is involved in the modification of proteins. The proteins arrive here in vesicles from the rough ER and have carbohydrates or other molecules added to them to make them more complex and suited for their role. The modified proteins are then packaged into vesicles and transported out of the cell by the process of exocytosis.

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Mitochondria are sometimes known as the power house of the cell, it is here where aerobic respiration is used to release energy from organic molecules such as glucose and make ATP. The mitochondria has a double membrane, the inner of which is folded to provide a large surface area.

Lysosomes

These are vesicles which contain hydrolytic enzymes. They can be exported out of the cell by the process of phagocytosis or used in the cytoplasm to break down organelles or proteins which are no longer needed.

Ribosomes

They have the same role that they have in prokaryotic cells they read ribosomal RNA and make it into the proteins needed.

Microtubules

These are found in both prokaryotic and eukaryotic cells. They are found throughout the cytoplasm and are made out of a protein called **tubulin**. The role of microtubules is to assist in the transports of the cellular organelles and assist with cell division.

Chloroplasts

These are only found in plant cells where they contain chlorophyll which is used to transform light energy into carbohydrates. It is not unusual for chloroplasts to contain starch granules for energy.

Cell Wall

The roles of the cell wall in Plant cells are similar to the role of the cell wall in a bacterium. It prevents the cell rupturing due to osmotic pressures. In a plant cell the cell wall is made out of cellulose which is a very strong fibre but is fully permeable the role of controlling what leaves and enters the cell is down to the cell membrane which is the same as the one found in prokaryotic cells.

Cell Membrane

The cell membrane forms the outside of the animal cell, which acts as a barrier and controls the transfer of materials into and out of the cell. It is composed of a double layer of phospholipids and proteins.

Organelles with DNA

The Mitochondria and Chloroplasts have their own DNA

LEVELS OF ORGANIZATION

Living organisms are composed of cells organized at different levels.

1. **Cell:** the basic structural and functional unit of life e.g. *Amoeba*, rods and cones (in the retina).
2. **Tissue:** a group of similar cells organized into a structural and functional unit e.g. *Hydra*, bone.
3. **Organ:** a collection of tissues which are similar in structure and perform similar functions e.g. Onion bulb, eye.
4. **System:** a group of specialized and co-ordinated organs performing a distinct major function e.g. Nervous system.

CRITICAL THINKING

If you read that a new organism had been discovered, what would you know about the organism without examining it in terms of cells?

HIGHLIGHT

IT'S ALIVE!? ...MAYBE.

An ongoing discussion in microbiology concerns the nature of viruses. Are they alive? Viruses are noncellular; that is, they lack cell membranes, cell walls, and most other cellular components. However, they do contain genetic material in the form of either DNA or RNA. (Few viruses have both DNA and RNA.) While many viruses use DNA molecules for their genes, others such as HIV and poliovirus use RNA for genes instead of DNA. No cells use RNA molecules for their genes. However, viruses have some characteristics of living cells. For instance, some demonstrate responsiveness to their environment, as when they inject their genetic material into susceptible host cells. Nevertheless, viruses lack most characteristics of life: they are unable to grow, reproduce, or metabolize outside of a host cell, although once they enter a cell they take control of the cell's metabolism and cause it to make more viruses. This takeover typically leads to the death of the cell and results in disease in an organism.

Revision Questions

1. CHARACTERISTICS

1. UME 79/1

All living organisms

A. Photosynthesize

C. Move

B. Breathe

D. Feed

E. Transpire

2. UME 94/2

The smallest living organisms which share the characteristics of both living and non living matter are

- A. Bacteria
- B. Fungi
- C. Virus
- D. Protozoa

3. PCE 2001/15 TYPE: 4

Which of this is a living thing?

- A. air
- B. water
- C. spirogyra
- D. star

4. PCE 2002/13 TYPE: 5

Which of this is a living thing?

- A. Water
- B. Sun
- C. Bread
- D. None of the above.

5. UME/2009/2

A characteristic exhibited by all living organisms is

- A. sexual reproduction
- B. aerobic respiration
- C. the ability to move from one place to another
- D. the ability to remove unwanted substances

6. UME/2008/2

A characteristic that can possibly be shared by both living and non-living organisms is

- A. locomotion
- B. irritability
- C. increase in biomass
- D. increase in size

1.2. CELL STRUCTURE AND FUNCTION OF CELL COMPONENTS

7. UME/2008/3

The cell of an onion bulb can be differentiated from a cheek cell by the presence of

- A. plasmalemma
- B. chloroplast
- C. cell wall
- D. nucleus

8. UME/2006/30

The role of the Golgi complex in a eukaryotic cell is to

- A. conduct ions in and out of the cell
- B. transport genetic materials out of the cell
- C. provide attachment for ribosome granules
- D. transport organic materials in and out of the cell

9. UME/2009/3

In a cell, the genes are carried by

- A. nuclear membrane
- B. lysosomes
- C. chromatin threads
- D. mitochondria

10. UME/2010/2/TYPE: D

Which of the following characterizes a mature plant cell?

- A. the nucleus is pushed to the centre of the cell
- B. the cell wall is made up of cellulose
- C. the nucleus is small and irregular in shape
- D. the cytoplasm fills up the entire cell space

11. UME/2010/3/TYPE: D

Which of the following is NOT a function of the nucleus of a cell?

- A. it translates genetic information for the making of proteins
- B. it stores and carries hereditary information
- C. it is a reservoir of energy for the cell
- D. it controls the life processes of the cell

12. UME 1982/2

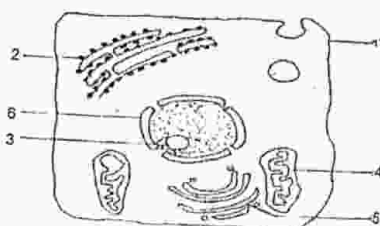
Which of these combinations is common to animal cells?

- A. Centriole, middle lamella, Golgi mitochondrion
- B. Cytoplasm, sap vacuole, starch grains, leucoplasts
- C. Plasma membranes, chromosomes, mitochondria, lysosomes
- D. Nucleus, nucleolus, cellulose cell wall, endoplasmic reticulum
- E. Cytoplasm, centriole, cellulose cell wall, nucleus

13. UME 1987/1

The function of endoplasmic reticulum is

- A. Protein synthesis
- B. Intracellular transport of materials
- C. Digestion and destruction of foreign bodies
- D. Production of energy from glucose



14. UME 89/1

The structure labeled 5 is the

- A. Nucleolus
- B. Golgi body
- C. Mitochondrion
- D. Vacuole

15. UME 89/2

Which of the labeled parts of the cell contains ribonucleic acid?

- A. 1
- B. 2
- C. 3
- D. 5

16. UME 89/3

Which structure is known as the power house of the cell?

- A. 6
- B. 5
- C. 4
- D. 5.

17. UME 93/1

On what structures are the units of inheritance situated?

- A. Golgi bodies
- B. Ribosomes
- C. Chromosomes
- D. Endoplasmic reticulum

18. UME 94/1

The membrane surrounding the vacuole in a plant cell is called the

- A. Plasmalemma
- B. Tonoplast
- C. Nuclear Membrane
- D. Endoplasmic reticulum

IME 2002/4 TYPE: Y

structures found only in plant cells are cell membrane and cytoplasm, chromatin and nucleus, cell wall and Chloroplast, cell membrane and lysosome

PCE 91/1

structure in a cell concerned with protein synthesis is the

- A. endoplasmic reticulum
- B. ribosome
- C. nucleus
- D. plasma membrane

PCE 91/2

mitochondrion is regarded as the power house of the cell because it

- A. Contains numerous enzymes involved in aerobic respiration
- B. Controls the synthesis of fatty acids
- C. Contains numerous enzymes involved in photosynthesis
- D. Contains numerous enzymes involved in protein synthesis

PCE 97/1

the cytoplasm in unicellular organisms is divided into

- A. Ectoplasm and Endoplasm
- B. Ectoplasm and Nucleus
- C. Endoplasm and Nucleus
- D. Protoplasm and Nucleus

3. PCE 99/4 TYPE: E

the organelle associated with detoxification of substances is

- A. Lysosome
 - B. Mitochondrion
 - C. Golgi body
 - D. Endoplasmic reticulum
4. The cell membrane consists of
- A. Carbohydrate and Lipids
 - B. Vitamins and Proteins
 - C. Lipids and Proteins
 - D. Water and Sugar
 - E. Starch and Cellulose

1.3. LEVELS OF ORGANIZATION

25. PCE 96/1

Which of the following is the correct sequence of biological organization

- A. Cell → Organ → System → Tissue → Organism
- B. Cell → Tissue → System → Organ → Organism
- C. Cell → Tissue → Organ → System → Organism
- D. Cell → Organ → Tissue → System → Organism

26. UME 85/12

A group of similar cells performing a common function is called

- A. an organ
- B. a system
- C. a tissue
- D. an organelle

27. UME 2002/1 TYPE: Y

An Amoeba and an Unlaid chicken egg are

- A. Animal tissues
- B. Organelles
- C. Single cells
- D. Organisms

19. UME 2002/4 TYPE: Y

The structures found only in plant cells are

- A. Cell membrane and cytoplasm
- B. Chromatin and nucleus
- C. Cell wall and Chloroplast
- D. Cell membrane and lysosome

20. PCE 91/1

The structure in a cell concerned with the synthesis of proteins is the

- A. Endoplasmic reticulum
- B. Ribosome
- C. Nucleus
- D. Plasma membrane

21. PCE 91/2

Mitochondrion is regarded as the "power house" of a cell because it

- A. Contains numerous enzymes involved in respiration
- B. Controls the synthesis of fatty acids
- C. Contains numerous enzymes involved in photosynthesis
- D. Contains numerous enzymes involved in metabolic activities

22. PCE 97/1

The cytoplasm in unicellular organisms is made up of

- A. Ectoplasm and Endoplasm
- B. Ectoplasm and Nucleus
- C. Endoplasm and Nucleus
- D. Protoplasm and Nucleus

23. PCE 99/4 TYPE: E

The organelle associated with detoxification of harmful substances is

- A. Lysosome B. Mitochondrion
- C. Golgi body D. Endoplasmic reticulum.

24. The cell membrane consists of

- A. Carbohydrate and Lipids
- B. Vitamins and Proteins
- C. Lipids and Proteins
- D. Water and Sugar
- E. Starch and Cellulose

1.3. LEVELS OF ORGANIZATION

25. PCE 96/1

Which of the following is the correct hierarchy of biological organization

- A. Cell → Organ → System → Tissue → Multicellular organism
- B. Cell → Tissue → System → Organ → Multicellular organism
- C. Cell → Tissue → Organ → System → Multicellular Organism
- D. Cell → Organ → Tissue → System → Multicellular Organism

26. UME 85/12

A group of similar cells performing the same function is

- A. an organ C. a tissue
- B. a system D. an organelle E. an enzyme.

27. UME 2002/1 TYPE: Y

An Amoeba and an Unlaid chicken egg are

- A. Animal tissues B. Organelles
- C. Single cells D. Organisms.

28. UME/2004/29 TYPE: Y

The rods in the retina of the eye are examples of

- A. Organs C. Systems
- B. Cells D. Tissues

29. UME/2009/1

What is the level of organization of an onion bulb?

- A. tissue B. organ
- C. systemic D. organism

30. PCE 91/3

Hydra is said to be at the tissue level of organization because

- A. it is unicellular
- B. it has many cells performing different functions
- C. it has many cells performing the same function
- D. it is cells are differentiated into tissues

31. PCE 2006/31/TTYPE: A

Cellular level of organization is found in the phylum

- A. nematode
- B. coelenterate
- C. protozoa
- D. platyhelminthes

MISCELLANEOUS QUESTIONS (1)

32. Which of the following does a virus have in common with animal cells?

- A. Nucleus B. DNA
- C. Glycogen D. Cytoplasm (UME 95/1)

33. Virus are considered to be living organisms because they

- A. possess transmittable characters
- B. move from one place to another
- C. respond to stimulation
- D. inject food materials (UME 90/1)

34. Which of these organelles are likely to be present in cells that are actively respiring and photosynthesizing?

- A. Nucleus and centrioles
- B. Mitochondria and chloroplasts
- C. Lysosomes and ribosomes
- D. Golgi apparatus and endoplasmic reticulum (UME 95/2)

35. The nucleus is considered the central organelle of a cell because it

- A. contains the genetic material
- B. contains the nuclear sap
- C. is bounded by the nuclear membrane
- D. is located at the centre of the cell (UME 98/1)

36. All living cells require water because it

- A. is a medium that neutralizes acids in cells
- B. is the main source of energy for the cells
- C. preventing the development of diseases in cells
- D. is a medium for all reaction (UME 2004/19)

37. Which of the following is likely to have a higher concentration of mitochondria?

- A. sperm cell
- B. white blood cell
- C. egg cell
- D. red blood cell (UME/2005/2)

38. The tiny organelles on the walls of a rough endoplasmic reticulum are the

- A. lysosomes
- B. microvilli
- C. microtubules
- D. ribosomes (PCE/2005/44)

39. The organelle common to both plant and animal cells is the

- A. centriole
- B. plasmalemma
- C. cell wall
- D. chloroplast (UME 2005/1)

40. The component of a bacterial cell that is not found in a nucleus as in other cells is the

- A. deoxyribonucleic acid
- B. ribonucleic acid
- C. internal membrane
- D. cell wall (PCE/99/8/TYPE: E)

41. Non-living things differ from living things in that

- A. are composed of one or more cells
- B. have a relatively low energy content
- C. obtain and use energy
- D. have a high degree of organization (PCE/06/29/TYPE: A)

CHAPTER 1

1. B	2. C	3. C	4. D	5. D
6. D	7. C	8. D	9. B	10. B
11. C	12. C	13. B	14. B	15. C
16. C	17. C	18. B	19. C	20. B
21. A	22. A	23. A	24. C	25. C
26. C	27. C	28. B	29. B	30. B
31. C	32. B	33. A	34. B	35. A
36. D	37. A	38. D	39. B	40. A
41. B				

Understand the following after reading this chapter:

Monera (prokaryote) e.g. bacteria and blue-green algae
Protista (protozoan and protophyta) e.g. Amoeba and Euglena
Fungi e.g. mushrooms and bread moulds
Plantae (plants)
Animalia (animals)

EVOLUTION AMONG LIVING ORGANISMS

Classification and diversity of living organisms

The system of classifying living organisms on a hierarchy is known as taxonomy. Below is the taxonomical hierarchy; Kingdom, Phylum, Class, Order, Family, Genus, Species.

Memory Tip: (King Play Card On Fine Green Sand)

Binomial system: a system of naming species in which the scientific name of an organism is made up of two parts showing the genus (starting with a capital letter) and species (starting with a lower case letter), written in italics when printed (therefore underlined when handwritten) e.g. *Homo sapiens*

The five kingdoms:

Animal: Multi-cellular ingestive heterotrophs (eat living organisms)

Plant: Multi-cellular photosynthetic autotrophic (make their own food) organism with a cellulose cell wall.

Fungi: Single celled or multi cellular heterotrophic organism with a cell wall not made of cellulose, saprotrophs (feed off dead organisms) or parasites

Monera: Single celled organism with no true nucleus

Protista: Single celled organism with a nucleus

CLASSIFICATION OF PLANTS

Plants are classified into division based on their structural differences, presence or absence of chlorophyll, presence or absence of flower, presence or absence of vascular bundle and life cycle.

1. **The Schizophytes:** These are unicellular and microscopic plants. They lack definite nucleus and possess cell wall without cellulose. Reproduction is asexual e.g. bacteria.

2. **The Thallophytes:** These have a simple body structure. They lack roots, stems and leaves. The two divisions are: the

- Algae e.g. *Spirogyra*, *Volvox* and *Chlamydomonas*
- Fungi e.g. *Rhizopus*, *Mushroom* and *Mucor*.

3. **The Bryophytes:** These are multicellular non-vascular and non-flowering plants. They possess root-like

structures called rhizoids. They are found in damp places, on the barks of trees, floor of grass-lands and forests e.g. mosses, hornworts and liver-worts.

4. **The Pteridophytes:** These are non-flowering vascular plant having structures resembling roots, stems and leaves. They reproduce by means of special gametes and spores e.g. ferns and selaginella.

5. **The Spermatophytes:** These are the seed producing vascular plants. They have true roots, stems and leaves. They are subdivided into two;

- Gymnosperms:** They bear naked seeds inside scale-like structures. They bear naked seeds inside seed boxes called cones e.g. conifers, cycads, whistling pine etc.
- Angiosperms:** These are flowering seed producing vascular plants. Their fruits are covered with fruit wall.

They are subdivided into two namely;

- Monocotyledons:** These have parallel leaf venation fibrous root system, scattered vascular bundles in the stem and only one seed coat e.g. maize, rice, grasses, bamboo etc.
- Dicotyledons:** These have net or reticulate venation ringed or organized vascular bundles in the stem, taproot system and two seed coats e.g. bean, mango, hibiscus etc.

Memory Tip: Samuel The Biologist Planted Sugarcane

CLASSIFICATION OF ANIMALS

Major phyla

Phylum : Protozoa

This phylum includes a great diversity of small, microscopic organisms. These are single celled eukaryotes. Their locomotion happens using pseudopodia, cilia or flagella.

The nutrition is either autotrophic or heterotrophic. They reproduce either asexually or by sexual methods. Eg: *Amoeba*, *Paramoecium*, *Plasmodium*.

Phylum : Porifera

These are multicellular, aquatic organisms. They have a cellular grade of construction without the occurrence of tissues. The sponges belonging to this phylum are characterised by the presence of a canal system in their body. The body wall contains spicules. They can reproduce both by asexual and sexual methods. Eg: Sponges.

Phylum : Coelenterata or Cnidaria

All coelenterates are aquatic animals. They are mostly marine. The body is radially symmetrical. The body wall is of two layers of cells. The outer layer is called the ectoderm. The inner layer, entoderm is separated from the ectoderm by a non-cellular mesogloea. The mesogloea

is a jelly-like substance. Due to the presence of two layers in the body wall, these are said to be **diploblastic animals**. Many coelenterates exhibit polymorphism. In this phylum, organisms exist in two different body forms namely, a polyp, and a medusa. The ectoderm contains stinging cells called **nematocytes (cnidoblasts)**. These cells when triggered can explosively penetrate prey and inject poison. The layers in the body wall contain several cells and tissues such as muscle cells epithelial tissues, gland-cells and sensory cells.

They reproduce both asexually and sexually. They are divided into three classes, namely **Hydrozoa**, **Scyphozoa** and **Anthozoa**.

In **Hydrozoa**, the animal has a dominant polyp body form and a reduced medusa stage. (e.g) *Hydra*, *Obelia*.

In **Scyphozoa** the medusa form is permanent. This group includes jelly fishes such as *Aurelia*. They swim in the surface waters. They have a bell shaped medusa stage.

The **Anthozoans** mostly remain as polyps. Their body cavity is divided by large radial partitions called **mesenteries**. (eg) sea-anemone and corals.

All animals of subsequent phyla show the following general characters.

1. All of them have three layers in the body wall. They are named as outer ectoderm, middle mesoderm, and inner endoderm. Thus they are called as **Triploblastic animals**.
2. The body is bilaterally symmetrical.

Phylum: Platyhelminthes

This phylum includes flatworms. These are acoelomates, without a body cavity called coelom. The alimentary canal is either absent or very simple. Excretion and osmoregulation occur through flame cells. These worms are mostly hermaphrodites, having both male and female reproductive organs in a single individual. Most of the members are parasites. It is divided into three classes, namely **Turbellaria**, **Trematoda** and **Cestoda**.

Class Turbellaria :- These are free living aquatic flatworms. The *Planaria* of this class shows characteristic regeneration.

Class Trematoda :- These are flukes living as parasites inside a host (en- doparasites). A protective cuticle covers the outer surface of the body. Flukes have suckers for attachment to the host tissues. The examples are *Fasciola* (liver fluke), *Schistosoma* (blood fluke).

Class Cestoda :- It includes all tape worms. These are internal parasites with a complex life history. The life cycle involves two hosts.

Their body characters are adaptations for parasitic life. Mouth and alimentary canal are absent. Food is absorbed through general body surface. The head is called the **scolex**. It has a ring of hooks and suckers for attachment to the host tissue. The body consists of several segments called **Proglottids**. (eg) sheep and cattle tape worms.

Phylum : Nematoda

These are the popular round worms. The body is narrow

and pointed at both the ends. There are no body segments. The body is covered by a thin cuticle. The body cavity is considered as a pseudocoelom. The alimentary canal is a straight tube. They reproduce sexually and the sexes are separate. There are several free living soil nematodes. Others are parasites. (eg) *Ascaris lumbricoides*.

In subsequent Phyla the animals show following general characters

1. There is a coelom within the mesoderm. Hence these are called as coelomates.
2. The body consists of a series of compartments. This phenomenon is called as **metameric segmentation**. They have a circulatory system providing internal transport.

Phylum: Annelida

These are worm like animals. The body segments are rings externally. Internally the segments are separated by **septa**. Externally the body is protected by a cuticle. Excretion and osmoregulation are achieved by ciliated tubules called **nephridia**. There is a central nervous system. The brain is formed of ganglia in the head region. The nerve cord is ventral in position. For the first time head formation or cephalization happens. These are bi- sexual and hermaphroditic. The larva is called the **trochophore**.

This phylum includes three Classes, namely **Polychaeta**, **Oligochaeta** and **Hirudinia**. The polychaetes are marine worms. They have a distinct head. There are pairs of lateral projections called **parapodia**. The examples are *Nereis* (ragworms), *Arenicola* (lugworm).

Earthworms are included in the Class **Oligochaeta**. The Class: **Hirudinia** includes leeches. These are blood suckers and ectoparasites. They have well developed suckers for attachment at anterior and posterior ends.

Phylum : Arthropoda

These are the most successful group of animals. They outnumber all other animals in population strength. The body is segmented. It is covered by a hard exoskeleton made of **chitin**. During growth the exoskeleton is shed (moulting of ecdysis). The legs or paired appendages are jointed. The head region has a pair of prominent compound eyes. Each compound eye is made up of several photoreceptor sub units called **Ommatidia**.

They have an open circulatory system without vessels. The body cavity is filled with a fluid called **haemolymph**. Such body cavity is known as **haemocoel**. These are unisexual, exhibiting sexual dimorphism. The young forms produced are invariably called the **larvae**. The larvae undergo metamorphosis and develop into adults.

This Phylum comprises five Classes:

Class Onychophora :- It includes small worm like *Peripatus*. *Peripatus* shows Annelidan and Arthropoda characters. Hence this may be considered as a connecting link between the two groups.

Class Crustacea :- The examples for this class are prawns, crabs and lobsters. The dorsal body surface is covered by a shield like carapace.

Class Myriapoda :- It includes centipedes and millipedes. These organisms have a distinct head and simple eyes. The centipedes have a pair of poison claws. The body consists of numerous segments, bearing pairs of legs.

Class Insecta :- It comprises the common insects. The body is divided into head, thorax and abdomen. In several insects, the adults have two pairs of wings on the thorax. Respiration happens through the tracheal system.

Class Arachnida :- It includes scorpions, spiders, ticks and mites. The body is divided into cephalothorax and abdomen. There are four pairs of legs attached to the cephalothorax.

Phylum Mollusca It is a very successful and diverse group of animals. Considered to be the second largest group of animals with regard to species number. These are soft bodied animals without segmentation. The body is divided into head, muscular foot and visceral mass. The body is covered by a mantle and a shell. Respiration happens through gills (ctenidia) in the mantle cavity. The most common larva is a trochophore larva. There are seven classes of which three are more prominent.

Class Pelecypoda or Bivalvia :- These are aquatic molluscs having bivalves. They burrow in mud and sand. The body is laterally compressed. (eg) mussels, clams, oysters.

Class Gastropoda :- These are either aquatic or terrestrial molluscs. They possess a spiral shell. The foot is large and flat. They have well developed head with tentacles and eyes. (eg) snails, slugs, and limpets.

Class Cephalopoda :- These are mostly marine. They are adapted for swimming. The foot is modified into eight to ten long tentacles in the head region. The shell is either internal or absent. (eg) Octopus, Loligo, Sepia.

Phylum Echinodermata :- These are marine organisms. While the adults are radially symmetrical the larvae remain bilaterally symmetrical. The mouth is on the lower surface. They have a water vascular system with tube feet. eg. star fishes, brittle stars, sea urchins and sea-cucumbers.

Phylum Chordata

This phylum derives its name from one of the common characteristics of this group namely the notochord (Gr. noton, back + L. chorda, cord). The animals belonging to all other phyla of the Animal Kingdom are often termed 'the non-chordates' or 'the invertebrates' since they have neither notochord nor backbone in their body. The backbone animals (vertebrates), together with a few closely related animals which do not possess a backbone, are included in this phylum. Most of the living chordates are familiar vertebrate animals. The chordates are of primary interest because human beings are members of this group.

Classification.

The Phylum Chordata is classified into four sub phyla:

Sub phylum 1. Hemichordata,

Sub phylum 2. Cephalochordata

Sub phylum 3. Urochordata

Sub phylum 4. Vertebrata.

First three sub phyla are collectively known as Protochordates. Since the members of these sub phyla do not have a cranium or skull they are also referred to as Acrania.

Protochordata (Acrania)

The protochordates are considered as the fore runners of vertebrata. The classification of the protochordates is based on the nature of the notochord.

Sub phylum : Hemichordata.

These are exclusively marine organisms. They are solitary or colonial forms. They mostly remain as tubicolous forms. The body is soft, vermiform, unsegmented, bilaterally symmetrical and triploblastic. The body is divisible into three distinct regions namely proboscis, collar and trunk. The body wall is composed of single layer of epidermal cells. The dermis is absent. They have no endoskeleton. A projection from pharynx, projecting inside the proboscis may be considered as notochord. They have a spacious coelom lined by coelomic epithelium. The alimentary canal is a straight tube running between mouth and anus. They are ciliary feeders. Sexes are separate. Examples: Balanoglossus, Saccoglossus.

Sub phylum : Cephalochordata.

Cephalochordates are small fish like marine chordates. The persistent notochord extends forward beyond the brain. Hence these are called cephalochordates. The epidermis is single layered. Paired fins are absent. Muscles, nephridia and gonads are segmentally arranged. The pharynx is large with numerous gills. It is a filter feeder. Example: Amphioxus.

Sub phylum : Vertebrata (Craniata)

This group is characterized by the presence of brain case or cranium and a vertebral column which forms the chief skeletal axis of the body.

The sub phylum vertebrata may be classified into two groups (i) Pisces and (ii). Tetrapoda.

Class : Pisces

Fishes are poikilothermic, aquatic vertebrates with jaws. The body is streamlined. It is differentiated into head, trunk and tail. Between head and trunk, the neck is absent. Locomotion is effected by paired and median fins. Examples: Shark, Catla.

Tetrapoda

The vertebrates with two pairs of limbs adapted for locomotion on land are known as tetrapods. The limbs are of pentadactyl type. The tetrapods are identified by a cornified outer layer of skin and nasal passages communicating with mouth cavity and lungs. The super

class Tetrapoda is divided into four classes namely. Amphibia, Reptilia, Aves and Mammalia.

Class :Amphibia

The living representatives of this class include frogs, toads, newts, salamanders and limbless caecilians.

Amniota

The tetrapods like reptiles, birds and mammals are referred to as amniotes. The amniotes have certain membranes associated with embryos inside the egg. It is an adaptation in terrestrial forms during development. These membranes are the amnion, chorion and allantois.

Class :Reptilia

Reptiles are represented by lizards, snakes, turtles, tortoises, alligators, crocodiles and the tuatara lizard, *Sphenodon punctatum*.

Class :Aves

Birds are one of the most interesting and widely known group of animals. Birds as a group exhibit a characteristic uniformity in structure.

Aves are warm blooded vertebrates with an exoskeleton of feathers forming a non-conducting covering to keep the body warm. The feet are covered with scales. The forelimbs are modified as wings and provided with feathers for flight. The hindlimbs are attached far forwards to balance the weight of the body. Examples : Pigeon, parrot, crow, sparrow, peacock, ostrich, penguin.

Class : Mammalia

The term "mammalia" was given by Linnaeus (1758) to that group of animals which are nourished by milk from the breasts of the mother. They are a successful group, for they adapt themselves readily to new situations and to new food habits. The class Mammalia is subdivided into three subclasses namely:

1. Sub class :Monotremata or Prototheria
These are primitive egg laying mammals Example : Spiny ant-eater, duck billed platypus.

2. Sub class :Marsupialia or Metatheria
These are popularly called as marsupials or pouched mammals. The young ones are born in an immature stage and migrate into the pouch on the mother's body. Further development is completed in the pouch or marsupium.
Example : Kangaroo

3. Sub class :Placentalia or Eutheria
In this group eggs develop within the uterus. The developing embryo receives nutrition through maternal blood circulation via the placenta.
Example : Elephant, tiger, lion, man, monkey, dog, cat, rat, bat.

Order Primates :

It is an order coming under the subclass Eutheria. This order is of interest because it includes man, besides lemurs, tarsiers, monkeys and apes. They inhabit chiefly the warmer parts of the world. This group stands first in the

animal kingdom in brain development. However, most of them are unspecialized and tree dwelling (arboreal).

Revision Questions

2.1. MONERA (Prokaryote) e.g. Bacteria and Blue-green algae

1. UME 98/2

The prokaryotic cell type is characterized by a

- A. Complex cytoplasm in which different regions are poorly defined
- B. Localization of different regions of the cell into tissues
- C. Collection of organelles and macromolecular complexes
- D. Simple cytoplasm with well defined regions

2. UME 2003/31

The pioneer Organisms in ecological succession are usually the

- A. Lichens
- B. Algae
- C. Ferns
- D. Mosses

3. UME 2003/4

The similarity among organisms belonging to the same group will be least within each

- A. order
- B. family
- C. species
- D. kingdom

4. UME 2010/8/TYPE: D

Which of the following groups of cells is devoid of true nuclei?

- A. Algae
- B. Monera
- C. Fungi
- D. Viruses

5. UME 2007/39

The chromosomes of members of the kingdom monera are within the

- A. nucleoplasm
- B. cytoplasm
- C. nucleus
- D. nucleolus

6. UME 2009/7

A blue-green alga is not a protophyte because

- A. it is aquatic
- B. its cells are prokaryotic
- C. it cannot move
- D. it is not a green plant

7. The bacteria are known for their:

- A. being photosynthetic
- B. being holophytic
- C. being unicellular
- D. being non-nucleated
- E. None of the above

8. Bacteria are quite common except:

- A. in the atmosphere
- B. in salt water
- C. in the soil
- D. in the alimentary tract of animals
- E. None of the above

9. The cell of bacterium differs from that of a typical plant by:

- A. the possession of semipermeable membrane
- B. the absence of the cytoplasm
- C. the absence of cellulose

- D. the presence of a nucleus
E. None of the above

10. Which of the following statements are true?
A. Bacteria add to the soil certain nitrogenous materials
B. Vinegar is produced by bacterial activities
C. The taste of butter is due to bacteria
D. Bacterial activities are involved in preparing and preservation of silage (food for cattle made from maize stems, leaves and cobs)
E. All of the above

11. Useful as bacteria may be, yet they cause a number of diseases. Which of these are not caused by bacteria?
A. Tuberculosis
B. Pneumonia
C. Lockjaw
D. Malaria
E. None of the above

12. The bacteria that produce food poisoning are
A. The floating bacteria
B. Putrefying bacteria
C. Denitrifying bacteria
D. Fruit bacteria
E. None of the above

13. The poisonous substance produced by bacteria is known as:
A. Auxins
B. Antibodies
C. Antitoxins
D. Toxins
E. Tetanus

14. After one has successfully resisted an attack of a particular disease such as chicken pox, a large quantity of antibodies remain in the body to prevent further attack by the same disease. This power to resist attacks is known as:
A. Innoculation
B. Vaccination
C. Resistance
D. Immunity

2.2. PROTISTA (Protozoan and protophyta) E.g. Amoeba and Euglena

15. UME 87/2
Which of the following features of Euglena is found only in animals
A. Paramylum granules
B. Flagellum
C. Pellicle
D. Pyrenoid

16. PCE 91/5
The structure in Amoeba which performs a similar function as the mammalian kidney is the
A. food vacuole
B. cytoplasm
C. plasmalemma
D. contractile vacuole

17. PCE 94/3
An organism that possesses both plant and animal characteristics is
A. Spirogyra
B. Amoeba
C. Paramecium
D. Euglena

18. UME 88/7
Spirogyra, Euglena and Chlamydomonas share many characteristics EXCEPT
A. Nutrition
B. Reproduction
C. Mobility
D. Irritability

2.3. FUNGI E.g. Mushrooms and Bread moulds

19. UME 2003/1
The umbrella-shaped fruiting body of a fully developed mushroom is the
A. mycelium
B. basidium
C. pileus
D. stipe

20. UME 87/3
An organism found on a bare rock surface has features of algae and fungi. The organism is
A. an epiphyte
B. a lichen
C. a bryophyte
D. a fern.

21. UME 95/3
One common characteristic of fungi, algae, mosses and ferns is that they
A. show alternation of generations
B. reproduce sexually by conjugation
C. produce spores that are dispersed
D. possess chlorophyll in their tissues.

22. UME 81/39
One common feature of the fungi, algae, mosses and ferns is that they
A. Are photosynthetic
B. Show alternation of generation
C. Reproduce by means of conjugation
D. Can survive dry conditions
E. Have no seeds.

23. UME 91/4
Which of the following are non-green plants?
A. Euglena
B. Fungi
C. Spirogyra
D. Angiosperms.

2.4. PLANTAE (PLANTS)

24. UME 89/6
A multinucleated body without internal cell boundaries is a characteristic of
A. bryophytes
B. fungi
C. algae
D. gymnosperms.

25. UME 94/5
Which is the correct order in an evolutionary sequence for the following plants groups?
A. Bacteria → Ferns → Algae → Mosses → Seed plants
B. Bacteria → Ferns → Mosses → Algae → Seed plants
C. Bacteria → Algae → Mosses → Ferns → Seed plants
D. Bacteria → Mosses → Algae → Ferns → Seed plants

26. UME/2000/5
Which of the following groups is the most advanced?
A. Pteridophytes
B. Bryophytes
C. Thallophytes
D. Gynospores.

27. PCE 2002/10 TYPE: 5
The first group of plants to live exclusively on land are the
A. pteridophytes
B. angiosperms
C. bryophytes
D. Gymnosperm

2.4.1. BRYOPHYTA (Mosses and Liverworts)

28. UME 84/3

Mosses, liverworts and ferns can be grouped together because they

- A. are all aquatic plants
- B. all grow in deserts
- C. are seedless plants
- D. have undifferentiated plant bodies
- E. all produce colourless flowers.

29. UME 85/8

Bryophytes are different from flowering plants because they

- A. live in moist habitats
- B. are small plants
- C. reproduce sexually and asexually
- D. have small leaves
- E. have no vascular tissues.

30. UME 2004/27 TYPE: 4

The absence of special food and water-conducting systems restricts the body size in

- A. Bryophytes and the Pteridophytes
- B. Thallophytes and the Pteridophytes
- C. Liverworts, Mosses and Ferns
- D. Algae, Liverworts and Mosses.

31. UME 2006/33

The dominant phase in the life cycle of a bryophyte is the

- A. gametangium
- B. sporophyte
- C. gametophyte
- D. prothallus

2.4.2. PTERIDOPHYTA (Ferns)

32. UME 90/5

The algae, bryophytes and pteridophytes are similar in that they

- A. are sea weeds
- B. have no vascular tissues
- C. require moisture for fertilization
- D. are macroscopic plants.

33. UME 91/6

Which of the following are differentiated into true roots, stems and leaves?

- A. Algae
- B. Schizophyta
- C. Pteridophyta
- D. Bryophyta

34. UME 2009/4

Alternation of asexual and sexual modes of reproduction is found in

- A. blue-green algae
- B. *Euglena*
- C. fern
- D. maize

35. UME 2006/3

The evidence that supports the advancement of ferns over mosses is derived from

- A. comparative anatomy
- B. molecular records
- C. biochemical similarities
- D. physiological records

36. UME/2010/4/TYPE: D

The dominant phase in the life cycle of a fern is the

- A. prothallus
- B. sporophyte
- C. antheridium
- D. gametophyte

2.4.3. GYMNOSPERMAE (Conifers)

37. UME 85/3

Which of the following is seed bearing?

- A. Mosses
- B. Whistling pine
- C. Algal filaments
- D. Liverwort
- E. Fern fronds.

38. UME 86/2

Which of the following pairs are fully adapted to terrestrial life?

- A. Ferns and algae
- B. Ferns and mosses
- C. Bryophytes and flowering plants
- D. Flowering plants and conifers

39. UME 86/4

Which of the following has cones?

- A. Angiosperm
- B. Gymnosperm
- C. Pteridophyte
- D. Bryophyte

40. UME 95/2

Production of naked seeds is a distinctive feature of the group of plant called

- A. grasses
- B. conifers
- C. legumes
- D. palms

41. PCE/03/23/TYPE: N

The production of naked seeds is a distinctive feature of

- A. palms
- B. grasses
- C. conifers
- D. legumes

2.4.4. ANGIOSPERMAE (Flowering Plants)

42. UME 87/7

Double fertilization is a unique feature of

- A. angiosperms
- B. bryophytes
- C. pteridophytes
- D. algae

43. UME 91/3

Angiosperms and gymnosperms belong to the plant group known as

- A. Schizophyta
- B. Bryophyta
- C. Pteridophyta
- D. Spermatophyta

2.5 ANIMALIA (Animals)

2.5.1 INVERTEBRATES

44. UME 88/6

Which of the following sets of organism represents the correct trend from simple to complex structural organization?

- 1. Mollusca
- 2. Platyhelminthes
- 3. Nematoda
- 4. Protozoan
- A. 1→2→3
- B. 4→3→2→1
- C. 4→2→1→3
- D. 1→3→2→4

45. UME 2008/8
The most abundant group of organisms in the animal kingdom is
A. mammalian B. aves
C. annelid D. insect

2.5.1.1. COELENTERATES (E.g. Hydra)

46. UME 86/3
Which of these animals is radially symmetrical?
A. Squid B. Hydra
C. Snail D. Cockroach

47. UME 89/6
A good example of a diploblastic organism is
A. Amoeba B. Hydra
C. Earthworm D. Roundworm.

48. UME 89/5
A characteristic of the phylum coelenterate is that
A. Most of them are marine
B. they possess a gut with a single opening
C. they possess numerous pores in their body
D. They are bilaterally symmetrical

49. UME 90/3
Hydra removes undigested food by
A. passing it through the anus
B. passing it through the mouth
C. means of contractile vacuole
D. egesting it through the body surface

50. UME 90/4
Which of the following groups of invertebrates reproduces by budding?
A. Arthropoda B. Annelida
C. Mollusca D. Coelenterata.

51. UME 91/5
Sting cells are normally found in
A. Flatworms B. Hydra
C. Snails D. Paramecium

52. UME 97/5
The cnidoblast cells found in Hydra are used for
A. Reproduction B. Offence and defence
C. Locomotion and nutrition D. Food collection

53. UME 2000/2 TYPE: M
Coelom is absent in the class of animals termed
A. Mollusca B. Reptilia
C. Arthropoda D. Coelenterata

54. UME/2006/38
One primitive feature of the coelenterates is the possession of
A. a dorsal mouth B. radial symmetry
C. bilateral symmetry D. a false root

55. Hydra is a coelenterate. It is because it has:
A. Tentacles B. Mesogloea
C. Cnidoblast D. Hypostome

56. That structure through which the mouth of the hydra opens is:
A. The hypostome B. the oral cone
C. the bulge D. the tentacles
E. the bud

57. The testes of the hydra are usually located
A. At the foot of the hydra B. At the basal disc or foot
C. One the tentacles D. Below the buds
E. Below the tentacles

58. The ovary is usually located:
A. A little distance above the basal disc or foot
B. on the hypostome
C. Above the buds
D. Just below the tentacles
E. Above the tentacles

59. The cnidoblasts, structures of offence and defence of the hydra are found in large quantities on:
A. the ectodermal cells
B. The endodermal cells
C. The hypostome
D. The tentacles
E. The foot

60. The interstitial cells of hydra are responsible for
A. Digestion B. Defence
C. Replacement of lost cells
D. Elongation of the hydra
E. None of these

61. The hydra receives impulses and stimuli through it:
A. Nerve net B. Sense cells
C. Nematocysts D. Flagellae
E. None of the above

62. Hydra lives in fresh water. Since the osmotic pressure of its body fluid is greater than that of the surrounding water, much water is taken into the body by osmosis. The excess water is removed by
A. The contractile vacuoles B. Cnidoblasts
C. Enteron D. Flagellae
E. None of the above

63. The small green plants that live inside hydra are called
A. Zoochlorellae B. Chlamydomonas
C. Spirogyra D. Euglena
E. None of the above

64. One important method by which the hydra moves from place to place is:
A. Running B. Crawling
C. Looping D. Rolling
E. Sliding

65. The gametes of the hydra are formed from:
A. the nematocyst cells
B. the gland cells
C. the interstitial cells
D. The roving cells of the mesogloea
E. None of the above

66. Hydra guards against self-fertilization by being
 A. Protogynous B. Hermaphroditic
 C. Bisexual D. Monoecious
 E. Protandrous

2.5.1.2. PLATYHELMINTHES

67. UME 82/40
 Flatworms and roundworms are said to be invertebrates because
 A. they are small animals
 B. they can live inside the vertebrates
 C. some of them are unicellular
 D. they have no backbones
 E. they are parasitic

68. UME 2003/5 TYPE: A
 Hermaphroditic reproduction can be found among the
 A. Annelids and molluscs
 B. Pisces and amphibians
 C. Coelenterates and platyhelminthes
 D. Arthropods and nematodes

2.5.1.3. NEMATODES (Roundworms)

69. Annelids differ from nematodes in that they
 A. exhibit bilateral symmetry
 B. are triploblastic
 C. are metamerically segmented
 D. possess complete digestive system

2.5.1.4. ANNELIDS (Earthworms)

70. UME 89/8
 Which of the following phyla have members with both internal and external segmentation?
 A. Platyhelminthes B. Nematoda
 C. Annelida D. Mollusca

71. UME 91/10
 In the earthworm, the cocoon is secreted by the
 A. chaeta B. prostomium
 C. peristomium D. clitellum.

72. UME 94/4
 The soil swallowed by the earthworm to form the worm cast in ground up in the
 A. Clitellum B. Prostomium
 C. Mouth D. Gizzard

73. UME 99/5 TYPE: D
 The organism that has a hydrostatic skeleton is
 A. Tilapia B. Hydra
 C. Mosquito larva D. Earthworm

2.5.1.5. ARTHROPODS.

74. UME 85/4
 Each of the following is an arthropod EXCEPT the
 A. Crab B. Spider
 C. Snail D. Millipede
 E. Cockroach

75. UME 93/7

- The most successful group of animals in terms of diverse of species is
 A. Mollusca B. Arthropoda
 C. Mammalia D. Platyhelminthes

2.5.1.6. CRUSTACEAN

76. UME 86/7
 Which of the following lacks chaetae, tentacles antennae?
 A. Snail B. Crab
 C. Millipede D. Earthworm

77. UME 87/8
 The crayfish is an arthropod because
 A. its body consists of a cephalothorax and an abdomen
 B. it has a pair each of antennae and antennules
 C. every segment of its body carries a pair of appendages
 D. its body is covered with an exoskeleton made of chitin.

78. UME 91/11
 The function of maxillipeds in Crayfish is to aid
 A. walking B. swimming
 C. feeding D. respiration

79. UME 78/23
 Which of these is NOT true of the insect? The possession of
 A. two pairs of antennae B. jointed appendages
 C. exoskeleton D. three pairs of legs
 E. segmented bodies

80. UME 82/26
 Which of these statements is NOT true of insects?
 A. They are arthropods
 B. Their body is divided into three distinct regions of the thorax and abdomen
 C. Their thorax comprises three segments. Only two which bear a pair of appendages each
 D. Respiration is by means of trachea.
 E. They undergo metamorphosis.

81. UME 98/7
 The ability of the cockroach to live in cracks and crevices enhanced by the possession of
 A. Wings and segmented bodies
 B. Compound eyes
 C. Claws on the legs
 D. Dorsal-ventrally flattened body.

2.5.1.7. ARACHNIDS

82. UME 88/13
 Lungbooks are used for respiration in
 A. Spiders B. Insects
 C. Millipedes D. Snails.

83. UME 89/11
 The palpal in spiders is used for
 A. grasping B. walking
 C. feeling D. web spinning.

84. UME 99/7/TYPE: D
The group of arthropods that has no antennae is the
A. Crustacea
B. Chilopoda
C. Arachnida
D. Diplopoda

85. UME 83/23
Centipede differs from a millipede by its
A. Colour
B. Numerous abdominal segments
C. Paired legs on each abdominal segment
D. Poison claws
E. Cylindrical body.

86. UME 88/14
Insects and millipedes have many features in common
EXCEPT
A. Exoskeleton
B. Jointed appendages
C. Compound eyes
D. Segmented body.

2.5.1.9. MOLLUSCS

87. UME 87/13
Which of the following is NOT a characteristic of snails?
A. Bilateral symmetry
B. Chitinous exoskeleton
C. Muscular foot
D. Soft unsegmented body in mantle

88. UME 88/5
Parasitic forms are NOT found among
A. Platyhelminthes B. Nematodes
C. Molluscs D. Annelids.

89. UME 89/12
The body of a snail is divided into head,
A. thorax and abdomen
B. visceral mass and abdomen
C. thorax and foot
D. visceral mass and foot

90. UME 91/12
The respiratory organ in the land snail is the
A. radula B. Mantle
C. tentacle D. foot

2.5.2. VERTEBRATES

91. PCE/95/3
The correct sequence that represent vertebrate
evolutionary trend is
A. fish→amphibians→birds→reptiles→mammals
B. fish→reptiles→Amphibians→birds→mammals
C. fish→birds→reptiles→amphibians→mammals
D. fish→amphibians→reptiles→birds→

92. UME 2001/2/TYPE: I
Amphibians are normally found
A. on dry land and in water

B. in water and on moist land
C. on moist land D. in water

93. UME 2001/3/TYPE: I
Viviparity occurs mainly in the
A. mammals B. reptiles C. aves D. amphibians

94. UME/2009/5
The first terrestrial vertebrates evolved from
A. pisces B. reptilian
C. amphibian D. mammalian

95. PCE/03/16/TYPE: N
Vertebrates having only one functional ovary are the
A. mammals B. toads C. reptiles D. birds

96. PCE/04/26/TYPE: 9
Which of the following vertebrates by the highest number
of eggs?
A. Birds B. Mammals
C. Reptiles D. Fishes

MISCELLANEOUS QUESTIONS (2)

97. Euglena may be classified as a plant because it
A. has chloroplasts B. has a gullet
C. lives in pond D. possess a flagellum
E. has a pellicle (UME/78/7)

98. Which of these animals is radially symmetrical?
A. squid B. Hydra C. Snail D. cockroach
(UME/86/3)

99. Which of the following organisms does not exist as a
single free living cell?
A. paramecium B. volvox C. amoeba
D. chlamydomonas (UME/2001/9/Type 7)

100. The hyphal wall of fungi is rigid owing to the
presence of
A. cell wall B. lignin
C. cellulose D. chitin
(UME/2005/3/Type D)

101. Angiosperms and gymnosperms belong to the class
A. schizophyta B. spermatophyta
C. pteridophyta D. bryophyta
(UME/2005/4/Type:D)

102. The leech and the earthworm belong to the
A. Molluscs B. crustaceans
C. arachnids D. annelids (UME/2005/6/type D)

103. The correct order of the evolutionary sequence is
A. bacteria→mosses→algae→seed plants
B. bacteria→algae→mosses→ferns→seed plants
C. bacteria→ferns→algae→mosses→seed plants
D. bacteria→ferns→mosses→algae→seed plants (PCE
03/18/Type: N)

104. In evolutionary trends, the pteridophytes are higher
than the bryophytes because of their possession

- A. Spore - bearing bodies
- B. Xylem and phloem vessels
- C. Photosynthetic pigments
- D. Parenchymatous cells (PCE/2004/2/Type:9)

105. Which of the following phyla has the basic characteristic of the body being divided into head, visceral hump and muscular foot?

- A. protozoa
- B. chordate
- C. mollusca
- D. platyhelminthes (PCE/93/6).

106. Which of the following group of animals is completely oviparous?

- A. Amphibia
- B. Aves
- C. Mammalia
- D. Reptilia (PCE/93/8)

107. A water medium is necessary for fertilization in

- A. fungi
- B. conifers
- C. ferns
- D. angiosperms

108. Which of the following exists in the form of a colony?

- A. spirogyra
- B. chlamydomonas
- C. volvox
- D. euglena (PCE/91/4)

109. Crab, grasshopper and millipede belong to the same phylum but different classes because they have

- A. exoskeleton with varied number of appendages
- B. endoskeleton with varied number of appendages
- C. exoskeleton with jointed appendages
- D. endoskeleton with jointed appendages (PCE/91/6)

110. Which of the following features is common to all insects?

- A. possession of stalked compound eye
- B. complete metamorphosis in the life cycle
- C. possession of jointed appendages
- D. possession of two pairs of wings (PCE 91/7)

111. Which of the following indicates the correct order of increasing structural complexity?

- A. Amoeba → Planaria → Hydra → Lumbricus
- B. Amoeba → Planaria → Lumbricus → Hydra
- C. Amoeba → Hydra → Planaria → Lumbricus
- D. Amoeba → Lumbricus → Planaria → Hydra (PCE/98/4)

112. Diploblastic body organization is found in

- A. Amoeba
- B. Obelia
- C. Lumbricus
- D. Periplaneta (PCE/98/5)

113. The earthworm is more advanced than the roundworm owing to its possession of

- A. pseudocoelom
- B. an alimentary canal
- C. true segmentation
- D. a hydrostatic skeleton (PCE/04/3/TYPE: 9)

114. A feature of annelids that distinguishes them from nematode is their possession of

- A. a three-layered body
- B. a segmented body
- C. an elongated body
- D. a cylindrical body (PCE/06/27/TYPE: A)

115. Which of the following flatworms is free living?

- A. Tapeworm
- B. Planaria
- C. Liver fluke
- D. Ascaris (PCE/07/4)

116. Seed-producing vascular plants belong to the

- A. phylum thallophyta
- B. phylum spermatophyta
- C. phylum pteridophyta
- D. phylum bryophyta (PCE/07/06)

117. In terms of the number of individuals, which of the following taxa is most inclusive

- A. order
- B. family
- C. class
- D. species (UME/2008/1)

118. unique feature of the gymnosperms is the production of

- A. naked seeds
- B. fruits without seeds
- C. multicellular spores
- D. seeds within the fruit (PCE/06/34/TYPE: A)

Understand the following after reading this chapter:

Spirogyra or any green filamentous algae
Rhizopus or Mucor
Brachymerium and Marchantia or any other Moss/Liverworts.
Dryopteris or Neprolepsis or any other fern.
Flowering Plants
Amoeba and Paramecium
Tapeworm (Taenia)

SPIROGYRA OR ANY OTHER GREEN

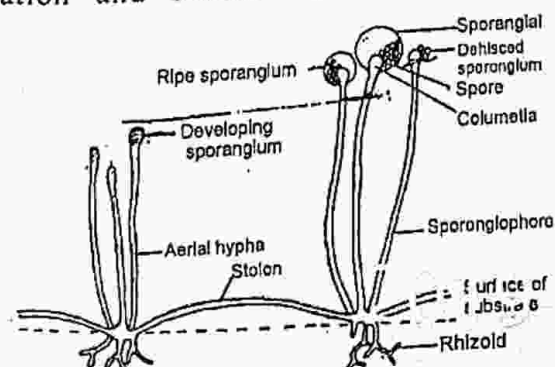
FILAMENTOUS ALGAE

Spirogyra is an algae. It floats in masses in ponds, lakes and slow-moving stream. It's a green plant which is composed of large similar cells joined end to end to form a filament. Each cell is cylindrical and contains one or more green ribbon-shaped chloroplasts formed into a characteristic helix. The chloroplast contains numerous colourless dense bodies called pyrenoids around which starch granules are found. Nutrition is autotrophic by photosynthesis. Reproduction is by fragmentation (asexual) and conjugation (sexual).

RHIZOPUS OR MUCOR

Rhizopus is a fungus. It is a saprophyte on moist substrates rich in carbohydrates e.g bread, soil and other decaying matter.

The body is a **mycelium** which is made up of white fine threads called **hyphae**. The hypha is coenocytic (i.e has no cross walls) and encloses a mass of cytoplasm with many nuclei and oil droplets. The mycelium is differentiated into branched **rhizoids** that penetrate the substrate, horizontal hyphae known as **stolons** that spread over the surface of the substrate, and erect **sporangiophores** (sporangial stalks) that develop in tufts above the rhizoid and bear the asexual reproductive structure, the **sporangia**. The mould spreads rapidly by producing stolons which penetrate the substrate at intervals to produce new rhizoids. Respiration is aerobic and is by diffusion of gases. Asexual reproduction is by spore formation and sexual reproduction is by conjugation.



MOSS/LIVERWORTS

They are found in damp shaded areas. The stem is erect and surrounded by thin leaves. At the base of the stem are hair-like out-growths, rhizoids. These anchor the plant and absorb water and mineral salts.

Reproduction (sexual): The sex organs are found at the apex of the stem. These are: **antheridia**, which produce large numbers of ciliated male gametes called **antherozoids**; **archegonia**, each of which contains a single female gamete, **ovum**.

When matured the antherozoids are released and they swim through the film of moisture on the leaves. They are attracted to the ovum by chemicals it secretes. The female gamete is eventually fertilized by a single male gamete. The resulting zygote develops within the female organ on the parent plant into a spore bearing structure called **sporophyte**. The sporophytes are produced in the wet season; each consists of a long stalk bearing an enlarged structure called the **capsule**. The capsule contains a spore-bearing tissue covered by a ring of tooth-like structures (**peristome**). The capsule dehisces at the tip when dry. Dried spores are dispersed by the peristome and are carried by air currents. The spores, which land on suitable moist habitat, develop into a branched filamentous structure called **protonema**. Protonema develops into new upright leafy moss plant.

Reproduction (asexual): could be by regeneration of leaf or other parts.

FERN

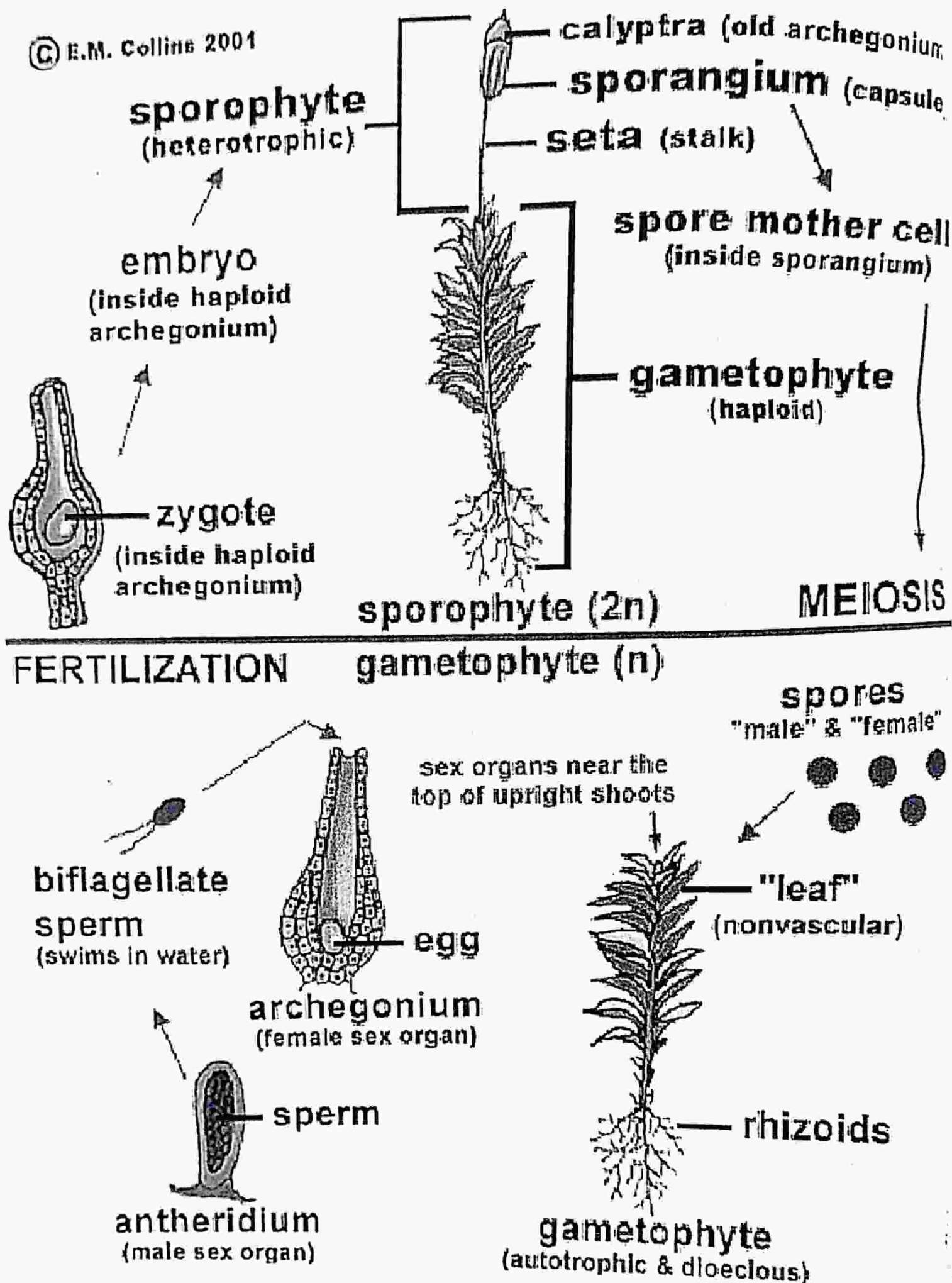
They are found in wet shady areas usually near streams or as epiphytes on palms or other trees.

The leaves called **fronds** are large and feather-like showing pinnate arrangement. They arise from buds on underground rhizome. The young leaves are tightly rolled showing circinate venation. The rhizome bears true root.

Reproduction: Groups of spore-bearing sporangia are borne on the underside of mature leaves. Each group is called **sorus** and is covered by an umbrella like **indusium**.

At maturity the sporangium dehisces and releases many spores which are dispersed by air currents. If a spore lands on moist soil, it germinates and develops into a small, flat, heart-shaped structure called **prothallus**.

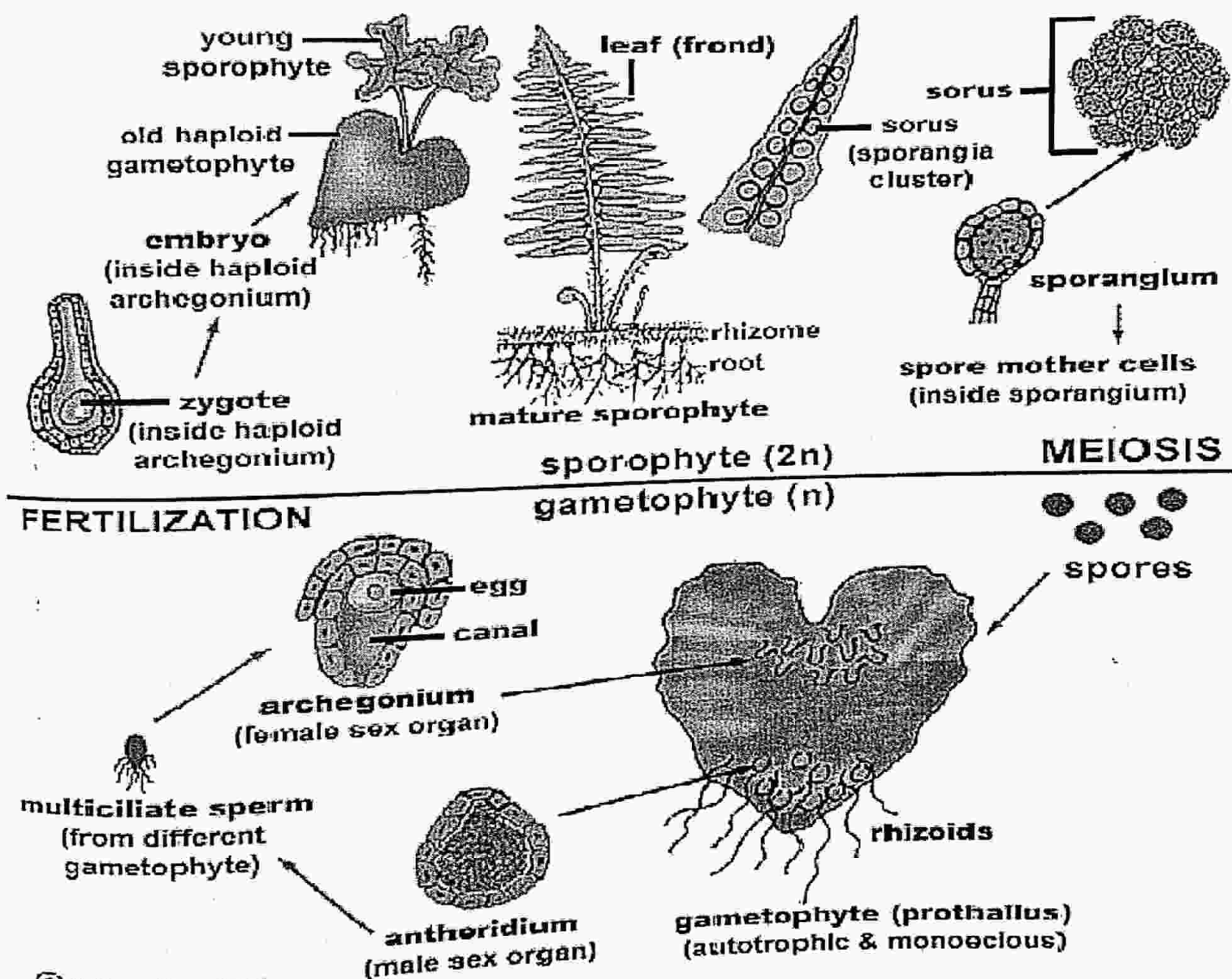
CHAPTER 3



The prothallus is green and bears many rhizoids. Archegonia develops towards the notch of the heart-shaped prothallus while the antheridia develops towards the conical part. The male gametes swim by the aid of flagella to the female ova. A fertilized ovum develops into a new fern plant.

Alternation of Generation

Both mosses and ferns show alternation of generation. This means that the asexual spore-bearing phase (sporophyte) alternates with the sexual gamete-bearing phase (gametophyte) in the life cycle. In the moss, the sporophyte grows on the gametophyte. In fern, the prothallus is the gametophyte while the fern plant is the sporophyte; both exist independently.



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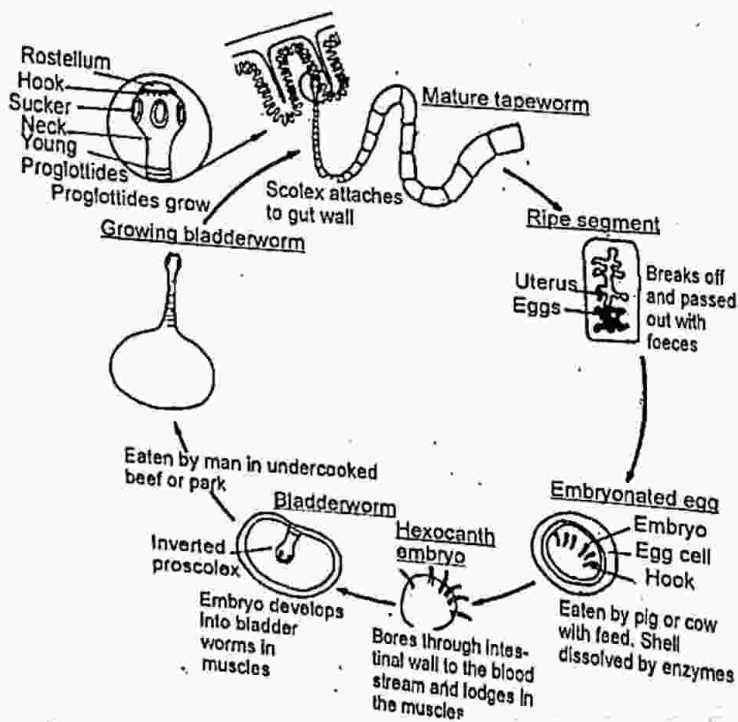
AMOEBA AND PARAMECIUM

AMOEBA

Nucleus:	Variable shape
Locomotion:	One
Food-getting:	By pseudopodia, giving slow speed
Ingestion:	Pseudopodia surrounding food
Egestion:	At any point
Digestion:	At any point
Excretion:	Any part of cytoplasm
Reproduction:	One contractile vacuole, variable location
Organ of defence:	By fission (asexual only)
	None

TAPEWORM (Taenia)

The tapeworm (*Taenia solium*) is a flat, ribbon or tape-like organism. It has a minute knob-like scolex or head. The scolex has four muscular suckers on the sides. The tip of the scolex is elevated to become rostellum. The rostellum is surrounded by hooks. A short "neck" connects the scolex with the rest of the body. The body consists of a series of proglottides about 1000 in number. Each proglottid is separated from the next by a transverse septum. The proglottides are identical in structure, but those at the posterior end are larger and most matured. A ripe proglottid with fertilized eggs breaks off from the rest of the tapeworm and passes out with faeces to soil. A man is the primary host. The eggs remain viable for weeks



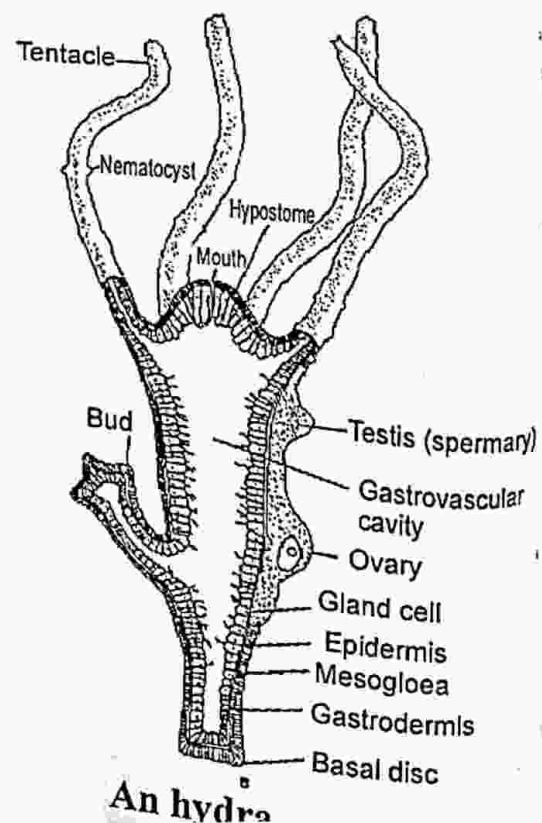
Structure and life cycle of tapeworm

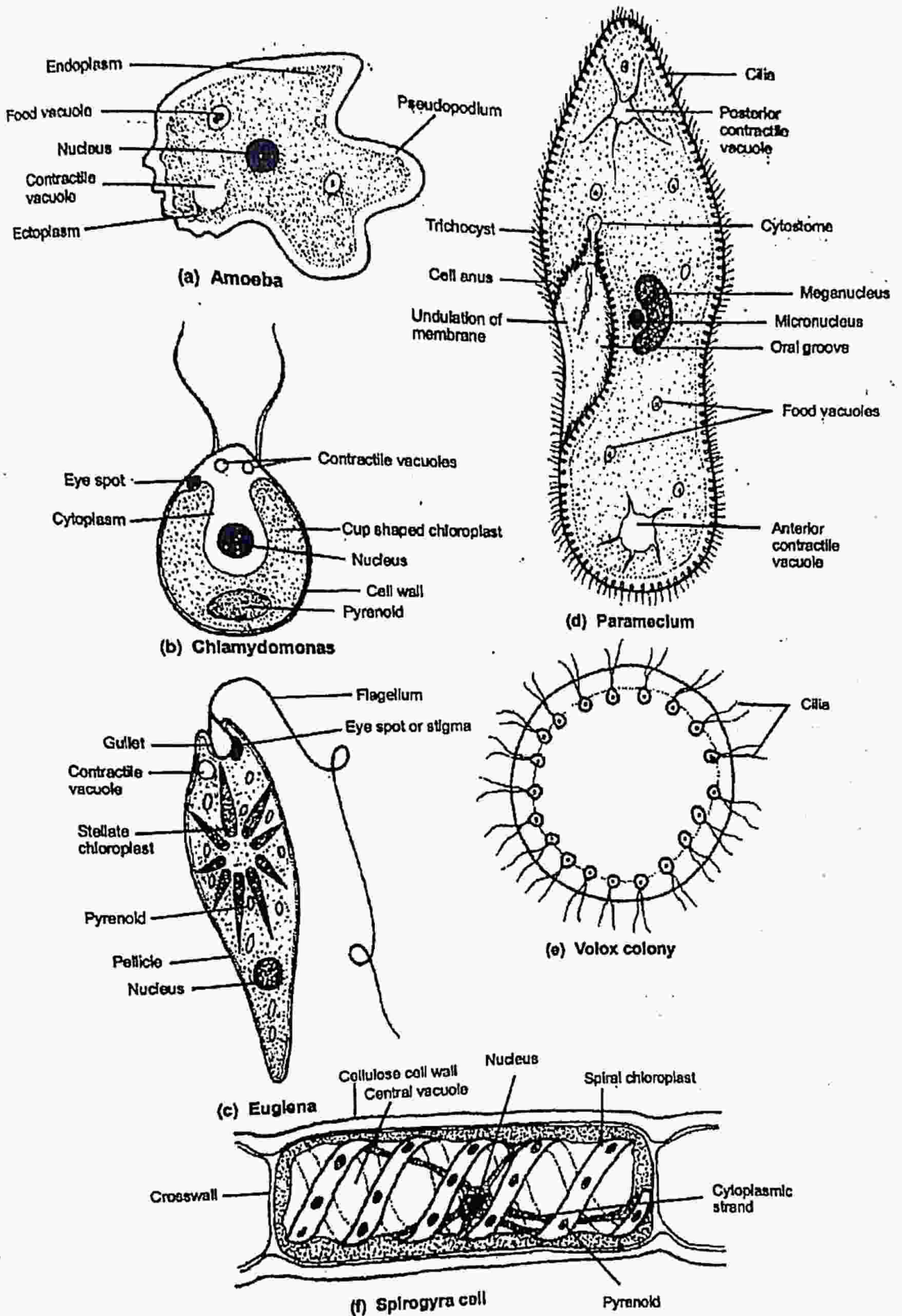
PARAMECIUM

Constant shape ensured by pellicle
Two, each with a separate function
By cilia, co-ordinated by sensory organelles giving rapid speed
Cilia in oral groove creating feeding current
Only through the mouth-pore
Only through anal-pore
Only along a definite path
Two contractile vacuole, each with six collecting canals, constant location
By fission and conjugation (asexual and sexual)
Trichocysts

in the soil. If a pig which is the secondary host eats the eggs, the eggs hatch into larvae in the small intestine. The larvae bore through the wall of the small intestine and enter the general blood and lymphatic circulation and carried to the muscles where they develop into cyst-like "bladderworms" or cysticerci.

A man becomes infected when he eats raw or inadequately cooked meat infected with bladderworm. The bladder worm passes into the small intestine and grows into an adult in 5 to 12 weeks and attaches itself into the wall of the small intestine by means of its hooks.





Revision Questions

3.1. SPIROGYRA OR ANY GREEN FILAMENTOUS ALGAE

1. UME/99/8/Type D

In which of the following organisms does each cell combine the functions of nutrition, reproduction and growth?

- A. Rhizopus B. Dryopteris
C. Brachymerium D. Spirogyra.

2. UME/86/6

Which of the following is NOT true of spirogyra?

- A. Reproduces by conjugation
B. Reproduces by fragmentation
C. Consists of branched filaments
D. Consists of unbranched filaments

3. UME/85/1

In spirogyra, the pyrenoid

- A. excretes wastes products
B. is suspended to cytoplasmic strands
C. is mainly used for respiration
D. makes the plant skinny to touch
E. usually contains starch

4. UME/80/2

One of the following statements is NOT true of spirogyra

- A. It is a simple multicellular plant.
B. During conjugation, it gametes, which are structurally and physiologically similar fuse to form a zygote
C. It possess spiral chloroplast which enables the plant to photosynthesize
D. It cells are protected by a layer of mucilage
E. There are pyrenoids along the chloroplast.

5. Some types of Spirogyra can be attached to rocks by means of:

- A. Roots B. Rhizoids
C. Tentacles D. Sticky hairs
E. Mucilage

6. The slimy and slippery nature of the Spirogyra is due to a covering of a substance known as

- A. Jelly B. Gelatin
C. Mucus D. Mucilage
E. Moss

7. The chloroplast of Spirogyra differ from other types in being:

- A. Spirally coiled with indented edges
B. Brown with dark spots
C. Ladder-like with many nucleic
D. Green and clustered around the nucleus
E. None of the above

8. Nutrition in Spirogyra is said to be

- A. Holozoic B. Saprophytic
C. Parasitic D. Symbiotic
E. Holophytic

9. In the asexual reproduction of spirogyra the growth of the cross wall to divide the mother cell into two daughter cells begins from

- A. the wall of cell and grows inward
B. from the centre of cell C. from the side wall
D. from within the cell E. None of the above

10. In the sexual reproduction of spirogyra, at the end of conjugation tube is formed the cell contents contracts to form:

- A. the zoospores B. the zygospores
C. the gametophytes D. the gametes
E. the sperms

11. Spirogyra is regarded as a multicellular plant because

- A. Its cells are linked together by cytoplasmic strands
B. Its cells are joined to form organs
C. The cylindrical cells are linked end to end
D. It contains large vacuole
E. None of the above

12. The nucleus of spirogyra is suspended by:

- A. the vacuoles B. the chloroplasts
C. the cytoplasm D. pyrenoids
E. none of these

3.2. RHIZOPUS OR MUCOR

13. The part of Mucor within the substance that it grows in is called

- A. The roots B. The mycelium
C. The rhizoids D. The holdfast
E. The nodules

14. The fine thread-like filaments that make up what is commonly termed the 'root system' of Mucor are:

- A. the hyphae B. the root hairs
C. the rhizoids D. the mycelia
E. the hyphen

15. The cell-wall of Mucor is made of

- A. Cellulose B. Hemi-cellulose
C. Fungal-cellulose D. Plant cellulose
E. Celluloid

16. In sexual reproduction of Mucor, fusion takes place between

- A. the two swollen heads of the hyphae
B. the two gametangia
C. the plus and negative strains
D. two nuclei, one from each gametangium
E. the gametes

17. The difference between the hypha and sporangiophore is that:

- A. The hypha has nuclei while the sporangiophore has none
B. Structurally, there is no difference
C. The hypha is more durable than the sporangiophore
D. The sporangiophore has funal cell wall, while the hypha has not
E. None of the above

18. JME/5/1

Which of the following is NOT true of Mucor? It

- A. contains chlorophyll
B. grows saprophytically
C. bears spores in sporangia
D. consist of hyphae
E. reproduces by binary fission

19. UME/87/9

The hyphae of Rhizopus

- A. does not contain cross walls
B. has no cross walls
C. is vascularised
D. stores oil globules

20. UME/93/9

The spores of Mucor

- A. water
B. insects
C. insects

21. PCE/2000/7

The hypha of Rhizopus

- A. septate and unbranched
B. non-septate and branched
C. non-septate and unbranched
D. septate and branched

22. UME/80/1

In mucor or Rhizopus

- A. starch
B. glycogen
C. oil

23. UME/83/43

In Rhizopus, the main

- A. glucose
B. glycogen
C. oil

3.3. BRACHYMYCETES OR ANY OTHER FUNGI

24. UME/85/10

In lower plants, the

- A. root hairs
B. hyphae
C. thalli

25. UME/90/7

In bryophytes, the

- A. gametophyte
B. protonema

26. UME/87/11

In the reproduction

- A. they live in moist places
B. they cannot reproduce asexually
C. the male gametes are flagellated
D. they produce zygotes

27. PCE/95/8

The asexual stage of

- A. as

2019-08

- A. contains chlorophyll
- B. grows saprophytically
- C. bears spores in sporangium
- D. consist of hypae
- E. reproduces by conjugation

19. UME 87/9
The hyphae of *Rhizopus* is said to be coenocytic because it

- A. does not contain chlorophyll
- B. has no cross walls
- C. is vascularised
- D. stores oil globules.

20. UME/93/9
The spores of *mucor* are dispersed by

- A. water
- B. wind
- C. insects
- D. explosive mechanism

21. PCE/2000/7/Type A
The hypha of *Rhizopus* is

- A. septate and uninucleate
- B. non-septate and uninucleate
- C. non-septate and multinucleate
- D. septate and multinucleate

22. UME/80/1
In *mucor* or *Rhizopus* carbohydrate is absorbed in the form of

- A. starch
- B. sucrose
- C. glucose
- D. glycogen
- E. arabinose

23. UME/83/43
In *Rhizopus*, carbohydrate is stored in the form of

- A. glucose
- B. paramylon
- C. glycogen
- D. starch
- E. oil

3.3. BRACHYMENIUM AND MARCHANTIA OR ANY OTHER MOSS/LIVERWORTS

24. UME 85/10
In lower plants like Mosses, the structure which performs the functions of roots of higher plants is called

- A. root hairs
- B. rootlets
- C. hyphae
- D. rhizoids
- E. thalli

25. UME 90/7
In bryophytes, sex organs are produced in the

- A. gametophyte
- B. rhizoid
- C. protonema
- D. sporophyte

26. UME 87/11
In the reproduction of mosses, water is essential because

- A. they live in moist habitats
- B. they cannot reproduce without water
- C. the male gametes must swim to fertilize the ovum
- D. they produce spores

27. PCE 95/8
The asexual stage in the life cycle of a moss plant is known as

- A. sporophyte generation
- B. gametophyte generation
- C. bryophyte generation
- D. pteridophyte generation

28. PCE 2000/5
The male and female reproductive organs of mosses and liverworts are respectively called

- A. antheridia and archegonia
- B. antheridia and oogonia
- C. paragymous and amphigymous
- D. spermatocyte and oocyte

3.4. DRYOPTERIS OR NEPROLEPSIS OR ANY OTHER FERN

29. UME 92/3
Which of the following correctly summarizes the life cycle of fern plant?

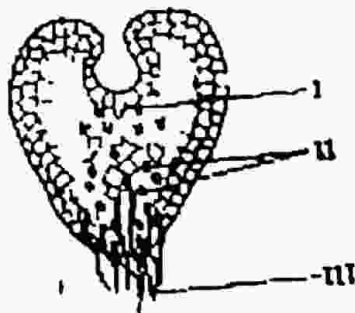
- A. Spores → prothallus → thallus → sporangium
- B. Male and female gametangia → zygospore
- C. Spores → thallus → spermatozo + ovum sporangium
- D. Prothallus → spermatozoid + egg cell → leafy plant → sporangium → spore

30. UME 97/7
Which of the following perform similar functions?

- A. Ascospores and ascocarp
- B. Antherozoids and rhizoids
- C. Sorus and indusium
- D. Strobili and inflorescence

31. UME 97/8
In ferns, the sporophyte

- A. develops from a haploid zygote
- B. reproduced sexually to produce spores
- C. is haploid and dependent on the gametophyte
- D. is diploid and independent of the gametophyte



32. UME/2001/10/Type T
The structures labeled II and III respectively are

- A. female organs and rhizoid
- B. male organs and rhizoid
- C. sporophyte and sori
- D. annulus and stalk of sporangium

33. UME/2001/11/Type T
In ferns, the structure above is equivalent to the

- A. zygote of moss
- B. saprophyte generation of a moss
- C. gametophyte generation of a moss
- D. sporangium of a moss

34. UME/84/5

Which of the following is INCORRECT

The prothallus of a fern.

- A. is a flanged heart-shaped structure
- B. is green because its cells contain chloroplasts
- C. is the dominant plant
- D. bears the sexual organs
- E. is attached to the ground by numerous rhizoids.

35. UME/87/4

In a plant exhibiting alternation of generations, the diploid multicellular stage is known as

- A. gametophyte
- B. spermatophyte
- C. holophyte
- D. sporophyte

36. PCE/93/4

Which of the following describes the function of an antherozoid?

- A. It facilitates the growth of the antheridium
- B. It regulates the formation of gametophyte
- C. It forms the hair-like structures called paraphyses
- D. It fertilizes the ovum in the archegonium to produce zygote

37. PCE/97/6

The spores that germinate from the sporangium of a fern grows into

- A. a thellus
- B. a prothallus
- C. an indusium
- D. an antheridium.

38. PCE/98/3

Which of the following has vascular tissue?

- A. *Pinaris*
- B. *Dryopteris*
- C. *Sporogira*
- D. *Rhizopus*

39. The sporangium of the fern is found in a chamber of the

- A. the epidermis
- B. the indusium
- C. the placenta
- D. theramenta
- E. none of the above

40. The annulus of the sporangium of the fern assists in the dispersal of spores by:

- A. Losing water from its cells
- B. Taking up water from the air
- C. Losing the entrapped air of its cells
- D. Collapsing in the face of strong winds
- E. Being attacked by insects

41. The sporangium is

- A. Biconcave in shape
- B. Plano-convex in shape
- C. Biconvex in shape
- D. Circular in shape
- E. None of the above

42. The stomium has

- A. Highly sclerotized cell walls
- B. thin-walled walls
- C. Overlapping cell walls
- D. tough rectangular cell walls
- E. Trap-like walls

43. The spore of the fern usually germinates into:

- A. the fern plant
- B. the prothallus
- C. the sporangium
- D. the sporophyte
- E. None of the above

44. The life history of the fern illustrates the phenomenon known as:

- A. Sporophyte generation
- B. Gametophyte generation
- C. Antheridial generation
- D. Archegonium – antheridial generation
- E. Alternation of generation

45. The gametophyte generation is

- A. the antheridia
- B. the spermatozoa
- C. the prothallus
- D. the ovum or egg
- E. the zygote

3.5. FLOWERING PLANTS

46. UME/94/3

Green plants are distinguished from other living organisms by their ability to

- A. make use of water
- B. make use of oxygen
- C. respond to sunlight
- D. manufacture their own food

47. UME/92/6

The flowering period of plants in a habitat is determined by

- A. duration of sunlight
- B. intensity and duration of rainfall
- C. relative humidity of the atmosphere
- D. temperature of the habitat.



48. UME/2005/8/Type D

The angle between the parts labeled II and III is the

- A. bud
- B. node
- C. internode
- D. axil

49. UME/2005/7/Type D

The part of the plant where photosynthesis is least likely to take place is

- A. I
- B. II
- C. III
- D. IV

50. UME/2000/3/Type M

Which of the following

2019-08-14 13:32

- A. Fibrous root system, branched network of veins and one seed leaf
- B. Fibrous root system, two seeds leaves and floral parts in threes
- C. One seed leaf, petals in threes or groups of threes and parallel venation of leaves.
- D. One seed leaf, net-veined leaves and petals in threes or multiples of three

51. The root hairs are found in:

- A. The region of cell division of roots
- B. Near the root caps
- C. The region of cell elongation
- D. The region of maturation
- E. The region of growth.

52. Some roots are modified for storage purposes. What is the name given to such roots?

- A. Rhizomes
- B. Tuberous roots
- C. Corms
- D. Swollen roots
- E. None of the above.

53. Which of these plants have storage roots?

- A. Yam
- B. Onion
- C. Cocoyam
- D. Cassava
- E. Canna Lily

54. Which of these plants produces breathing roots?

- A. The oil bean tree
- B. Silk cotton tree
- C. Para-rubber
- D. White mangrove
- E. None of the above.

55. Epiphytes produce aerial roots which can absorb water from the atmosphere. This feat is possible because the roots possess a spongy-like structure called:

- A. Xylem
- B. Spongen
- C. Velamen
- D. Velum
- E. None of the above.

56. Monocotyledonous plants do not increase in thickness or girth because they have:

- A. Diffused cambium
- B. Undifferentiated bundles
- C. No cambium
- D. Poor root system
- E. None of the above.

57. The area between two adjacent vascular bundles is known as:

- A. The pith
- B. The interfascicular cambium
- C. The medullary ray
- D. The vascular
- E. The cambium

58. What is the difference between the collenchyma and the parenchyma cells?

- A. Both the collenchyma and the parenchyma cortical cells
- B. The parenchyma cells contain chloroplasts while the collenchyma cells do not
- C. Collenchyma cells contain starch grains while the parenchyma cells do not
- D. Collenchyma cells have thickened cell walls those of the parenchyma cells are thin
- E. The collenchyma cells are nucleated while those of the parenchyma cells are not

59. The principal function of the pith of stems is:

- A. To protect the vascular bundles

- B. To give strength to the stem
- C. To manufacture food
- D. To accumulate air
- E. To store food for the plant

60. The structure that produces cork in the stem is:

- A. The phelloderm
- B. Phellogen
- C. Periderm
- D. Complimentary cells
- E. None of the above.

61. The meristematic structure of the stem of the plant is:

- A. The cambium
- B. Epidermis
- C. Phloem
- D. Pith
- E. None of the above.

62. That part of the leaf that secures it to the stem is:

- A. The peduncle
- B. The pedicel
- C. The petiole
- D. The patula
- E. The perianth

63. The leaves of the onion are modified into a storage organ known as:

- A. The rhizome
- B. The corm
- C. The tuber
- D. The stem
- E. None of the above.

64. The food stored by the onion is:

- A. Starch
- B. Sugar
- C. Protein
- D. Oils
- E. Fats

65. The plant Bryophyllum pinnatum can be propagated easily because:

- A. The leaf stores food
- B. It does not die
- C. The stem is stout
- D. It is immortal
- E. None of the above.

3.6. AMOEBA AND PARAMECIUM

66. UME 79/2

Amoeba moves by means of

- A. cilia
- B. flagella
- C. pseudopodia
- D. swimmerets
- E. setae

67. UME 88/9

What is the function of trichocyst in paramecium?

- A. Movement
- B. Defence
- C. Excretion
- D. Reproduction

68. UME 90/2

Which of the following characteristics is common to Amoeba and Paramecium

- A. Oral groove
- B. Trichocyst
- C. Contractile vacuole
- D. Cilia

69. PCE 91/5

The structure of Amoeba which performs a similar function as the mammalian kidney is the

- A. food vacuole
- B. cytoplasm
- C. plasmalemma
- D. contractile vacuole

70. The delicate skin covering the amoeba is called:

- A. Ectoderm
- B. Membrane
- C. Pellicle
- D. Epidermis
- E. Plasmalemma

71. The waste nitrogenous product of amoeba is:

- A. Ammonia
- B. Urea
- C. Urine

D. Uric acid E. Carbondioxide

72. Excretion in amoeba is effected through:
 A. its general body surface B. Its pseudopodia
 C. The contractile vacuole D. Its food vacuoles
 E. Its endoplasm

73. Amoeba feeds in the main on:
 A. Bacteria B. Other weak amoeba
 C. Microscopic plants D. Water fleas
 E. Sea weed

74. The purpose of the contractile vacuule is:
 A. To prevent desiccation of amoeba
 B. To regulate the intake of salts
 C. To expel excess water taken in by osmosis
 D. To encyst
 E. To increase the protoplasmic contents of amoeba

75. It has been discovered that those amoebae found in the sea have no contractile vacuule. This is because:
 A. Sea amoebae are different from fresh water types
 B. Sea amoebae do not eliminate waste materials
 C. Sea amoebae have cytoplasm whose osmotic pressure equals that of the surrounding water
 D. The contractile vacuule is not an essential structure
 E. None of the above

76. Amoeba encyst for purpose of:
 A. Reproduction
 B. Distribution and survival
 C. Maintaining their body fluid
 D. Defense against diseases E. None of the above

77. Amoebae encyst for purposes of:
 A. Reproduction
 B. Distribution and survival
 C. Maintaining their body fluid
 D. Defence against diseases
 E. None of these

78. Amoeba is economically important to man because:
 A. It is difficult to find
 B. It is found in ditches
 C. It pollutes man's source of water supply
 D. It can very easily be destroyed
 E. When in man it causes dysentery

79. Which of these characteristics identify Euglena as belonging to the plant kingdom
 A. The possession of single nucleus
 B. The possession of a pellicle
 C. The possession of chloroplasts
 D. The possession of granular cytoplasm
 E. None of the above

80. Which of these is not true of the Euglena as an animal?
 A. It has a contractile vacuole
 B. It has a flagellum
 C. It manufactures carbohydrates
 D. It has a gullet
 E. None of the above

81. The surface of the paramecium is covered with:
 A. Flagellae B. Cilia C. Trichocysts
 D. Bristles E. Spines

82. The tough skin surface of paramecium is known as:
 A. Plasmalemma B. Ectoderm
 C. Pellicle D. Neuroneme E. Epidermis

83. The paramecium takes its food through:
 A. the oral groove B. the undulating membrane
 C. the food vacuole D. the general body surface
 E. None of the above

84. The paramecium eliminates its waste products through:
 A. the contractile vacuoles
 B. the food vacuoles
 C. the oral groove
 D. the temporary anus
 E. None of the above

85. The contractile vacuules in the paramecium are used mainly for:
 A. the elimination of nitrogenous wastes
 B. the elimination of carbon dioxide
 C. the control of the body fluid
 D. the storage of salts
 E. the elimination of undigested food

3.7. TAPEWORM (Taenia)

86. UME 80/5

Which one of the following animals is NEVER secondary host of tapeworms?

A. cow B. fish C. pig D. man E. bat

87. UME 81/23

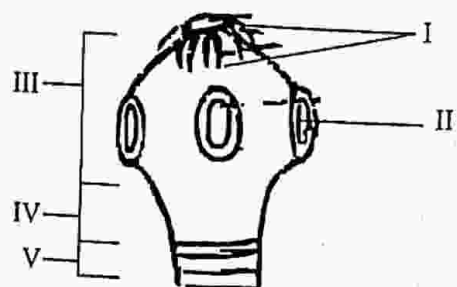
The two types of human tapeworm can be distinguished by the presence or absence of

A. scolex B. hook C. head D. sucker E. proglottids

88. UME 87/12

In tapeworm, the two structures that run throughout the length of the body are the

A. nerve cord and the excretory duct
 B. sperm duct and the nerve cord
 C. genital pore and the excretory duct
 D. sperm duct and the genital pore.



89. PCE 95/6

The parts labeled I and II are the

A. Head and the neck B. Testis and the scolex
 C. Ovary and the rostellum D. Hook and the scolex

90. PCE/95/7

The body of the organism lengthens when new proglottids are produced at the part labeled

- A. I B. II C. IV D. V

91. PCE/98/7/TYPE: C

Tapeworm feeds by absorbing

- A. blood from the host's intestine
B. tissues of the host's intestine
C. partly digested food in the host's intestine
D. undigested food in the host's intestine

92. The tape worm is usually segmented. Each of the segments is known as:

- A. Strobila
B. proglottid
C. Embryophore
D. Cysticercus E. Hydatid

93. The most conspicuous system absent in the tape worm is:

- A. The excretory system
B. The nervous system
C. The alimentary system
D. The system responsible for anchorage
E. None of the above

94. The region from where new segments arise in the tape worm is:

- A. the posterior end of the worm
B. the head of the worm
C. the anterior region above the suckers
D. some distance behind the suckers
E. the middle portion of the worm

95. The tape worm is anchored in the intestine of its host by means of:

- A. the suckers B. the rostellum
C. the body spines D. Its teeth
E. The curved hooks

96. The eggs in the mature proglottid of the tape worm are stored in:

- A. the receptaculum B. the vagina
C. the uterus D. the embryo sac
E. the oviduct

97. The secondary host of the tape worm *Taenia solium* is:

- A. the man B. the dog
C. the snail D. the pig
E. the hen

98. When the hexacanth embryo of the tape worm gets into its host it bores into

- A. the muscles B. the blood stream
C. the heart D. the walls of the alimentary canal
E. the liver

99. One of the effects of the attacks of tape worm is:

- A. The irritation of the gut B. Sleepiness
C. Frequent stooling D. The loss of appetite
E. None of the above

MISCELLANEOUS QUESTIONS(3)

100. In ferns, the adult plant is the

- A. capsule B. rhizome C. gametophyte
D. sporophyte (PCE/2005/45/Types: P)

101. The vegetative structure of most fungi is

- A. mycelium B. thallus
C. conidium D. sporangium (PCE/96/2)

102. The lichen is a composite organism that consists of a fungus living symbiotically with

- A. a bacterium B. a protozoan
C. a virus D. an alga (PCE/96/12)

103. In bryophytes, sex organs are produced in the

- A. gametophyte B. rhizoid
C. protonema D. sporophyte (UME/90/7)

104. The prothallus of a fern is equivalent to the gametophyte generation of a moss because it

- A. is inconspicuous B. has rhizoids
C. bears sexual organs D. is multicellular (UME/88/8)

105. The essential structural difference between *Hydra* and tapeworm is that while *Hydra*

- A. has tentacles, tapeworm is parasitic
B. is diploblastic, tapeworm is triploblastic
C. has mouth, tapeworm feeds by suckers
D. has mesoderm, tapeworm has mesogloea (UME/92/5)

106. Which of the following groups of plants shows the correct trend from simple to complex forms?

- A. schizophyta → bryophyte → thallophyta → pteridophyta
B. thallophyta → Schizophyta → bryophyte → pteridophyta
C. schizophyta → thallophyta → bryophyte → pteridophyta
D. pteridophyta → bryophyte → thallophyta → schizophyta
PCE/94/2

CHAPTER 3

1.D	2.C	3.E	4.B	5.B
6.D	7.A	8.E	9.A	10.D
11.C	12.C	13.B	14.C	15.C
16.B	17.B	18.A	19.B	20.B
21.C	22.C	23.C	24.D	25.A
26.C	27.A	28.A	29.D	30.D
31.D	32.B	33.B	34.C	35.D
36.D	37.B	38.B	39.B	40.B
41.C	42.B	43.B	44.E	45.C
46.D	47.A	48.D	49.D	50.C
51.D	52.B	53.D	54.D	55.C
56.C	57.C	58.D	59.B	60.B
61.A	62.C	63.E	64.B	65.E
66.C	67.B	68.C	69.D	70.E
71.A	72.A	73.A	74.C	75.C
76.B	77.B	78.E	79.C	80.C
81.B	82.C	83.E	84.E	85.C
86.D	87.B	88.A	89.D	90.C
91.C	92.B	93.C	94.D	95.E
96.C	97.D	98.B	99.A	100.D
101.A	102.D	103.A	104.C	105.B
106.C				

CHAPTER 4

COMMON INSECTS IN OUR ENVIRONMENT

Understand the following after reading this chapter:

Mosquito
Cockroach
Housefly
Termite

GENERAL CHARACTERISTICS OF INSECTS

1. The body is segmented
2. The body is divisible into three-head, thorax and abdomen
3. The body is covered by chitinous exoskeleton
4. Each thoracic segment bears a pair of jointed.
5. On the head are a pair of antennae and a pair of compound eyes
6. Respiration is by means of a network of open air tubes called trachea; these open outside as spiracles
7. One pair of wings is present on each of the last two thoracic segments. In some insects one or more of these may be absent, reduced or modified.
8. The young develops through a series of changes in form and in structure before maturing into an adult; this is known as **metamorphosis**

Complete metamorphosis has four stages: egg, larva, pupa, adult; the larva is completely different from the adult, e.g. butterfly.

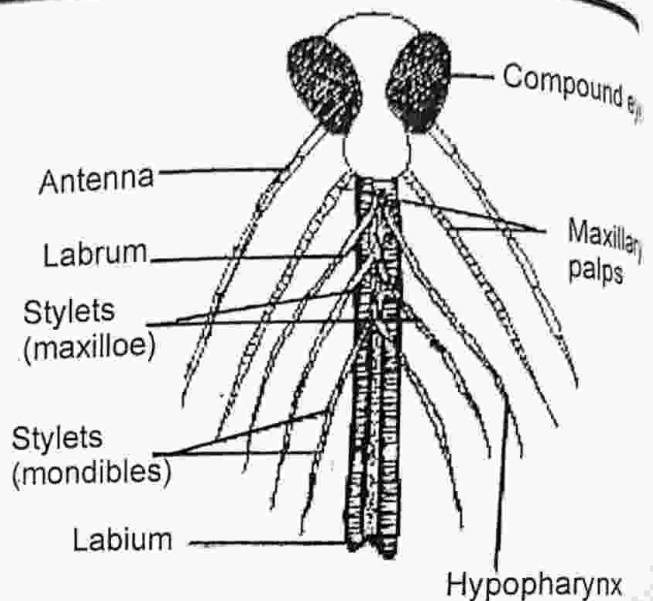
Incomplete metamorphosis has three stages: egg, nymph (larva), adult; the nymph resembles the adult except that it is smaller, wingless and sexually immature e.g. Termite.

MOSQUITO

It is slender in shape. It has only a pair of wings for flight and the other is reduced to *balancers*. The head bears one compound eyes, and a mouth modified into a long tube called *proboscis* for piercing and sucking. There are 8 visible abdominal segments.

Economic importance

They pollinate flowers in their search for nectar both male and female. In addition, the female requires a blood meal to stimulate egg production; in doing this they become vector of disease organisms. The female *Anopheles* mosquito transmits *Plasmodium*, a protozoan parasite causing malaria. The female *Culex* mosquito transmits *Wuchereria bancrofti*, a roundworm, causing filariasis (elephantiasis). The female *Aedes* mosquito transmits a virus causing yellow fever.



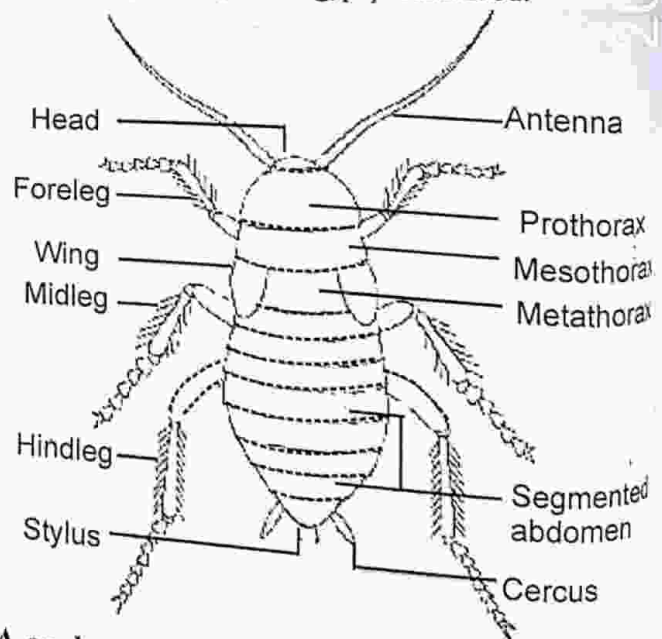
Mouthpart of mosquito

COCKCROACH

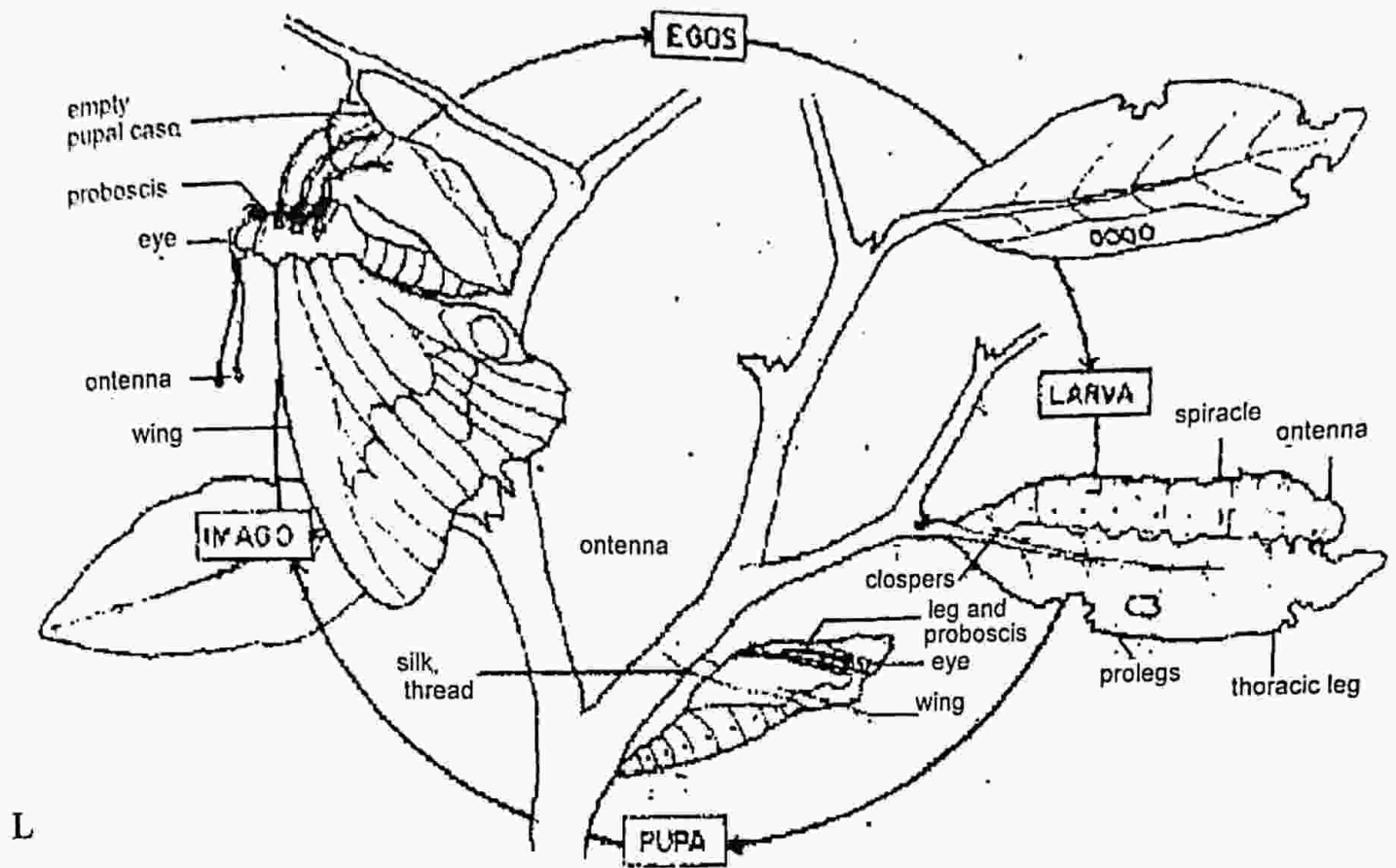
It has a dorso-ventrally flattened and oval body. It has long antenna for taste sensation. Body divided into *prothorax*, *thorax* and *abdomen*. It has 2 pairs of wings. Abdomen has 10 segments which 8 are visible. The 10th segment bears a pair of jointed *Cerci* covered with setae (hairs) to detect air movement. In addition, the male has two unjointed *stylets*.

Economic importance

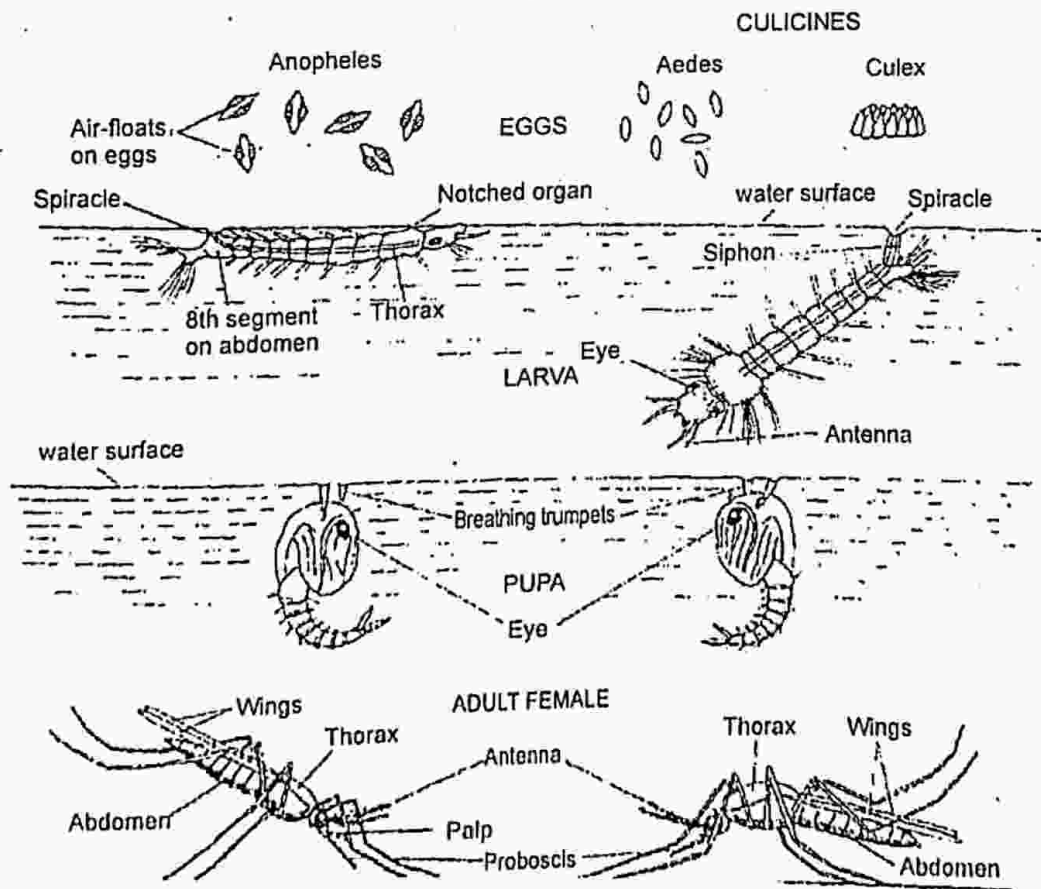
They destroy books, clothing, paper and food.



A cockroach



Life cycle of a Butterfly



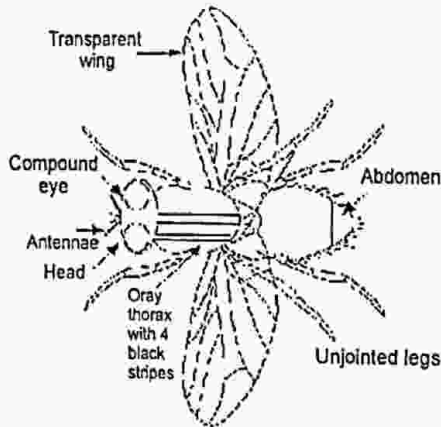
Life cycle of *Anopheles* and *Culex* mosquitoes

HOUSEFLY

It has hairy body. The head has a pair of compound eyes and three small simple eyes, a pair of short antennae. There is one pair of membranous wings. The second is reduced as balancers. There are four visible segments on the abdomen. The mouth is modified as a tubular *proboscis* for sucking up food.

Economic importance

They carry disease-organisms, causing diseases like cholera, dysentery and typhoid. They also transmit eggs of tapeworm and roundworms. They pollinate some foul-smelling flowers. The maggots (larvae stage) speed up decay process of organic matter.



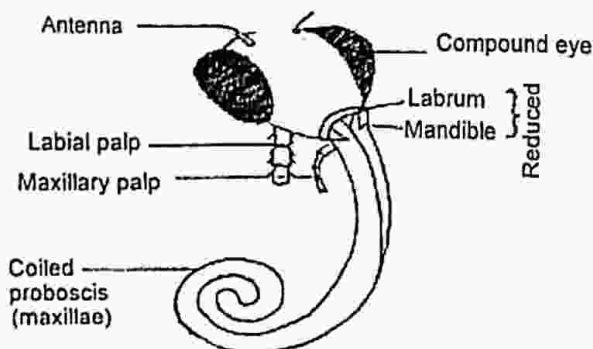
A common housefly

BUTTERFLY

It has a slender body and two pairs of large wings covered with scales. The head has a pair of large compound eyes and antennae which are club-shaped at the tips. The mouth is modified into a long coiled tube called *proboscis* which can be uncoiled during use.

Economic importance

Caterpillars (larval stage) eat leaves of crops and forest trees thereby causing food losses. Adult butterflies pollinate many flowers.



Mouthpart of butterfly

1. PCE 2003/41/Type N

The part labeled I is the

- A. maxillary palp
- B. proboscis
- C. mandible
- D. stylet

2. PCE/2003/42/Type N

The mouth part is adapted for

- A. cutting
- B. biting
- C. cutting and sucking
- D. piercing and sucking

3. UME/2004/30/Type 4

The larval stage of a mosquito is called

- A. caterpillar
- B. maggot
- C. wriggler
- D. grub

4. UME/2002/6/Type Y

Which oil is poured into the breeding site of mosquitoes, it

- A. deprives the larvae of water
- B. kills the adult
- C. suffocates the pupae
- D. Slows down egg development

5. UME 95/6

Which of the following is transmitted through mosquito bite?

- A. Filariasis
- B. Typhus
- C. Plague
- D. Schistosomiasis

6. PCE/96/7

The saliva injected into the wound during mosquito bite principally to

- A. transmit diseases to the host
- B. digests the host's muscle
- C. prevent the host's blood from clotting
- D. kills germs contained in the host's blood

7. PCE/06/30/TYPE: A

The mosquito that transmits elephantiasis in humans is the

- A. male culex
- B. female aedes
- C. male aedes
- D. female culex

8. The anopheles mosquito lays its eggs on water. How are these eggs able to float?

- A. the eggs are oily
- B. the eggs have 'floats'
- C. the eggs are weightless
- D. the eggs are laid in rafts
- E. the eggs have internal air chambers

9. What is the organ of respiration found in the larvae of mosquito?

- A. The trachea
- B. The spiracle
- C. The siphon
- D. Body bristles
- E. Mouth brushes

10. How is movement effected by the pupa of the mosquito when disturbed?
- The pupa sinks by sheer weight of body
 - It moves by means of its tail fins
 - It moves by taking in water through its trumpets
 - The pupa rounds up when disturbed
 - It moves away by the twisting of its abdominal region

11. The name of the malarial parasite transmitted by the mosquito is

- Plasmania
- Plasma
- Filaria
- Plasmodium
- Sporozoites

12. In what way is the life cycle of the mosquito different from that of the cockroach?

- The embryos of the mosquitoes are not protected
- All adult mosquitoes have wings while some adults of the cockroach have no wings
- The mosquito has no nymphal stage while the cockroach has no larva pupal stages
- The cockroach is a domesticated insect
- None of the above reasons

4.2. COCKROACH

13. UME/98/7

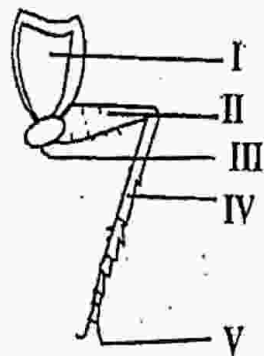
The ability of the cockroach to live in cracks and crevices is enhanced by the possession of

- wings and segmented body
- compound eyes
- claws on the legs
- dorso-ventrally flattened body.

14. PCE/91/8

Which of the following can be found in the thoracic region of the cockroach?

- cerci
- elytra
- styles
- labium



15. PCE 2004/8/Type 9

The structure labeled I is the

- tibia
- coxa
- torsal
- claw

16. PCE 2004/9/Type 9

The part used for gripping is labeled

- I
- III
- IV
- V

17. Considering the mouthparts of the cockroach, the organ of mastication is:

- the maxilla
- the labrum
- the mandible
- the stylets
- the labium

18. What is the external difference between a male and female cockroach?

- The female is usually smaller in size

- The female does not fly but rather prefers to run
- The female has no styles
- The male is lighter in colour
- The female usually carries its egg-case about.

19. Which of these is the most appropriate definition of metamorphosis?

- It is the development of the insect from the larval into the pupal stage
- The growth of the insect from the egg to the adult
- It is the transformation of the insect from the pupa to the adult
- It is all changes in the development of an insect-from the embryo to the adult
- None of the above

20. The eggs of the cockroach are usually are usually enclosed in a hard chitinous case called:

- the puparium
- the cocoon
- the otheca
- the ootheca
- None of the above

21. What is the name given to the young cockroaches that have just emerged from the egg-case?

- Grubs
- Larvae
- Nymphs
- Imagines
- None of the above

21. What is the difference between new emergent and adult cockroaches?

- The young cockroach has no legs
- The young cockroach does not feed
- The young cockroach is wingless
- The adult cockroach has many more abdominal segments than the emergent one
- There is no different between the two

22. The cockroach feeds on a number of items which may items which may include meat, paper, wood, etc. It is said to be

- Carnivorous
- Insectivorous
- Omnivorous
- Herbivorous
- None of the above

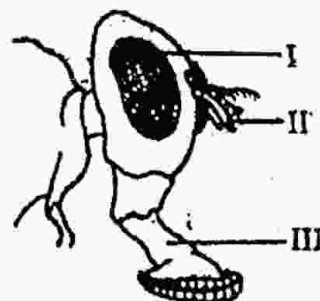
23. The cercus of the cockroach is:

- a tactile organ
- a direction-finding hair
- an organ of defence
- an auditory organ
- None of the above

24. The cockroach has its two antennae removed. Which of these functions do you think the insect will be unable to perform?

- it will not see
- it will not run well
- It will not detect the direction of wind
- It will not be able to locate its food
- None of the above

4.3. HOUSEFLY



25. UME 99/7 Type B
The structure labeled II is used for
A. tasting B. feeling C. biting D. sucking

26. UME 99/8 Type B
The structure labelled III represents the
A. mandible B. palp C. proboscis D. labium

27. UME/84/1
The mouthparts of the housefly are adapted for
A. lapping and sponging
B. sucking and chewing
C. piercing and sucking
D. chewing and lapping
E. biting and chewing.

28. PCE 97/8
One of the function performed by the hook-like spine at the anterior end of housefly larva is to
A. secrete digestive enzymes
B. attach itself to objects
C. sting its attacker
D. assist the larva in crawling

29. What is the name given to the hind wings of the housefly?
A. Drumstick B. Halteres
C. Halteers D. Elytra
E. None of the above

30. What is the function of the balancers in the housefly?
A. To keep it flying
B. To prevent loss of balance
C. To aid a good take-off
D. Serves no useful purpose
E. To prevent rolling from side to side when in flight

31. Which of these diseases cannot be transmitted by the housefly?
A. Diarrhoea B. Typhoid
C. Dysentery D. Malaria

4.4. TERMITE

32. PCE/93/7
The insect that has the ability to digest cellulose is the
A. Termite B. Butterfly
C. Mosquito D. Housefly

33. UME/86/10
When the original king and queen of termites die, they are replaced by
A. The king and queen of another colony
B. Some adult reproductive from the same colony
C. Some adult which are specially fed to breed
D. Developing nymphs nurtured as secondary reproductive.

34. UME/98/8
The caste of termites that lacks pigmentation is the
A. king B. worker C. soldier D. queen

35. PCE/99/3/Type: 3

The caste of the termite that causes the greatest damage to man is
A. queen B. soldier C. worker D. king

36. PCE/92/50

Which of the following should be killed in order to destroy a termite colony?
A. the king B. the queen
C. the workers D. the soldiers

37. PCE/93/7

The insect that has the ability to digest cellulose is the
A. termite B. butterfly
C. mosquito D. housefly

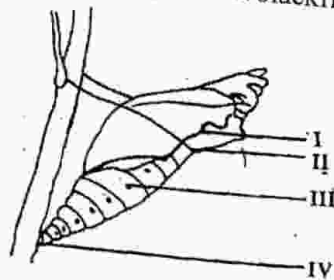
38. The home of termites is known as
A. the mole hill B. the habitation
D. the termitarium E. the sanatorium

39. The termite colony has a number of individuals such as the queen, king, soldiers and workers. The queen is known by her:
A. Prominent and extended abdomen
B. Sclerotized cuticle
C. Large head
D. Poorly developed brain
E. Reduced legs

40. The termites are important because they:
A. Transmit diseases
B. Destroy wood and other materials
C. Eat honey
D. Attack animals
E. None of the above

4.5. BUTTERFLY

41. UME/2003/11/Type A
An insect whose economic importance is both harmful and beneficial is the
A. butterfly B. mosquito C. blackfly D. testsefly



42. UME/2000/1/Type: M
The part that will develop into an organ for feeling is labeled
A. IV B. III C. II D. I

43. UME/2000/2/Type: M
The part labelled II is the
A. silk thread B. thorax C. forewing D. antenna



44. PCE/99/3/Type: 3
The structure of the termite colony is
A. Head B. Soldier C. Worker D. King

45. PCE/92/50
The diagram shows the structure of a termite colony
A. butterfly B. soldier C. worker D. king

46. Which of the following is not a butterfly?
A. The possum B. The possum C. The possum D. The possum E. The possum

47. Which of the following is not a butterfly?
A. The possum B. The possum C. The possum D. The possum E. The possum

MISCELLANEOUS

48. The joint which are seen in the housefly are
A. maxilla B. antenna C. antenna D. antenna

49. Which of the following is not a butterfly?
A. Mesquite B. Housefly C. Housefly D. Housefly

50. The structure absent in female housefly is
A. anal styles B. anal cerci C. anal cerci D. anal cerci

51. Which of the following is not a butterfly?
A. Mosquito B. Termite C. Termite D. Termite

52. The group of insects showing metamorphosis is
A. houseflies B. cockroaches, grasshoppers
C. house flies, bees D. aphids, grasshoppers

53. The larval stage of a butterfly is
A. Caterpillar B. Wiggler C. Wiggler D. Wiggler

54. Which of the following is not a butterfly?
A. Grasshopper B. Housefly C. Housefly D. Housefly

55. The only caste of termites that can feed themselves is
A. Reproductive B. Nymphs C. Nymphs D. Nymphs

44. PCE/2002/3/Type:5
The structure labelled II is the
A. Head
C. Compound eye

B. Ear
D. Maxillary palp.

45. PCE/2002/4/Type:5

The diagram is the mouth parts of a

A. butterfly B. cockroach C. housefly D. mosquito

46. Which of these characteristics marks out clearly the butterfly?

- A. The possession of club-like antennae
- B. The possession of two pairs of dissimilar wings
- C. The possession of the proboscis
- D. The possession of very tiny legs
- E. The possession of scales

MISCELLANEOUS QUESTIONS (4)

47. Which of the following spreads malaria in Nigeria?

- A. Anopheles Mosquito B. Culex mosquito
- C. Aedes mosquito D. Tsetse fly
- E. Housefly (UME/80/3)

48. The jointed structures in insects that bears organs which are sensitive to touch, smell and vibration is the

- A. maxilla B. labium
- C. antenna D. abdomen (UME/2001/4/Type: T)

49. Which of the following is a typical filter feeder?

- A. Mosquito larva B. Mosquito adult
- C. Housefly larva D. Housefly adult (PCE/92/10)

50. The structures found in male adult cockroaches but absent in females are the

- A. anal styles B. antennae
- C. anal cerci D. proboscis (PCE/2005/47/Type:P)

51. Which of the following insects lays its eggs in horny, purse-like cases?

- A. Mosquito B. Butterfly
- C. Termite D. Cockroach (UME/95/7)

52. The group of insects that undergoes complete metamorphosis is

- A. houseflies beetles and cockroaches
- B. cockroaches, grasshoppers and bees
- C. house flies, beetles and butterflies
- D. aphids, grasshoppers and butterflies (UME/97/9)

53. The larval stage of a mosquito is called

- A. Caterpillar B. Maggot
- C. Wiggler D. Grub

(UME/2004/30/Type:4)

54. Which of the following is destructive at the developmental and adult stages?

- A. Grasshopper B. Mosquito
- C. Housefly D. Termite (PCE/2000/9/Type:A)

55. The only caste in the termite colony whose members can feed themselves are

- A. Reproductive B. Workers
- C. Nymphs D. Soldiers

(UME/2005/49/Type: D)

56. The stage in the history of a moth responsible for the destruction of agricultural crops is the

- A. nymph B. imago
- C. pupa D. caterpillar (UME/2005/10/Type:D)

57. The process of shedding the exoskeleton of an arthropod is known as

- A. metamorphosis B. instars formation
- C. ecdysis D. tagmosis (UME/2007/42)

58. Which of the following is common to the mosquito, housefly and butterfly?

- A. they undergo complete metamorphosis
- B. they are parasites of man
- C. their immature stages are aquatic
- D. their adults have two pairs of wings (UME/2007/43)

59. The dorsal part of insect is covered by

- A. pleurum B. tegmen
- C. sternum D. tergum (PCE/96/6)

60. The correct sequence that describes the insect segmented leg is

- A. coxa → trochanter → femur → tibia → tarsus
- B. coxa → stemum → pleuron → femur → tarsus
- C. coxa → femur → trochanter → tibia → tarsus
- D. coxa → pleuron → trochanter → femur → tarsus (PCE/97/7)

61. A specie with a body divided into three regions, two large compound eyes, two short antennae and a pair of wings is a

- A. spider B. cockroach
- C. housefly D. butterfly (PCE/98/6)

62. An example of a nocturnal insects is the

- A. termite B. blackfly
- C. cockroach D. housefly (PCE 06/28 TYPE: A)

63. The skin covering the dorsal aspect of the insect is called:

- A. the pleuron B. the tergum
- C. the steron D. the ternum E. the sternum

64. The cuticle of the insect is made up of a substance known as:

- A. Wax B. Chitin C. Cutin
- D. Sclerotin E. None of the above

65. On which of these are the tracheal openings of insects found?

- A. the sternum B. the abdominal walls
- C. the thoracic walls D. the pleuron
- E. None of these

66. The thorax of insects is divided into segments namely: prothorax, meso-thorax and metathorax. Which of these bears the first pair of wings or the elytra?

- A. the prothorax B. the mesothorax
- C. the metathorax D. no segment in particular
- E. None of the above

67. The abdominal region of insects differs from the thoracic region in this respect:

- A. it is always bigger than the thorax
- B. it is segmented
- C. it has spiracles
- D. it has no locomotory structures
- E. It is laterally flattened

68. The metamorphosis of insects is either complete or incomplete. Which of these insects shows an incomplete form of metamorphosis?

- A. the bee
- B. the beetle
- C. the tsetse fly
- D. the mosquito
- E. None of the above

69. What are the trumpets?

- A. A sound producing apparatus found on the thorax of certain insects
- B. Respiratory structures of the housefly
- C. Respiratory structures found only in the pupa of an insect
- D. A sound-producing apparatus of the mosquito
- E. None of the above

70. What is the difference between the adult cockroach and mosquito?

- A. The mosquito is a harmful insect while the cockroach is not
- B. The cockroach is protected by the cuticle while the mosquito is not
- C. The mosquito has no hind wings
- D. The cockroach cannot eat blood
- E. None of these reasons

71. What is the main function of the adult insect?

- A. To fly about
- B. To distribute the insect
- C. To eat up predators
- D. To provide food for other animals
- E. To reproduce

72. The larvae of all beetles resemble the adults in one respect and this is:

- A. Both have biting mouthparts
- B. Both possess the anal cerci
- C. Both possess abdominal
- D. Both possess an equal number of abdominal segments
- E. None of the above reasons

73. Moulting in insects is

- A. the change in colour of insects
- B. the emergence of young larvae from eggs
- C. the casting away of old skin in insects
- D. the emergence of the adult from the pupa
- E. none of the above

74. Insects shed their skin:

- A. to remove waste materials in the body
- B. To prevent the attack of diseases
- C. To transform the insect
- D. To allow for growth
- E. None of the above reasons

75. It is discovered that a once-active insect cannot fly early in the morning. But as the sun rises in the sky it takes to wings. This implies that:

- A. Insects do not fly in the morning
- B. In the mornings the muscles that control the wings of insects are non-functional
- C. High temperature is a vital factor in the activity of insects
- D. The blood of insects freezes up at night
- E. None of the above reasons

CHAPTER 4

1. B	2. D	3. C	4. C	5. A
6. C	7. D	8. E	9. B/C	10. B
11. D	12. C	13. D	14. B	15. B
16. D	17. C	18. C	19. E	20. D
21. C	22. C	23. C	24. D	25. B
26. C	27. A	28. D	29. B	30. E
31. D	32. A	33. B	34. B	35. C
36. B	37. A	38. D	39. A	40. B
41. A	42. D	43. A	44. C	45. A
46. A	47. A	48. C	49. A	50. A
51. D	52. C	53. C	54. A	55. B
56. D	57. C	58. A	59. D	60. A
61. C	62. C	63. B	64. B	65. B
66. B	67. D	68. A	69. E	70. E
71. E	72. A	73. C	74. D	75. C

Understand the following after reading this chapter:

Bony fish
Toad
Lizard
Bird
Small mammal

Vertebrates are animals with vertebral column or backbones, internal skeleton, bilateral symmetry, brain in cranium, paired fore and hind limbs, well developed sense organs and 'closed' circulatory system. They include: Pisces (Fishes), Amphibians, Reptiles, Aves (Birds) and Mammals.

CHARACTERISTICS AND ADAPTATION

1. **Pisces (Fishes):** They are all aquatic, variable body temperature (poikilothermic), overlapping scales pointing

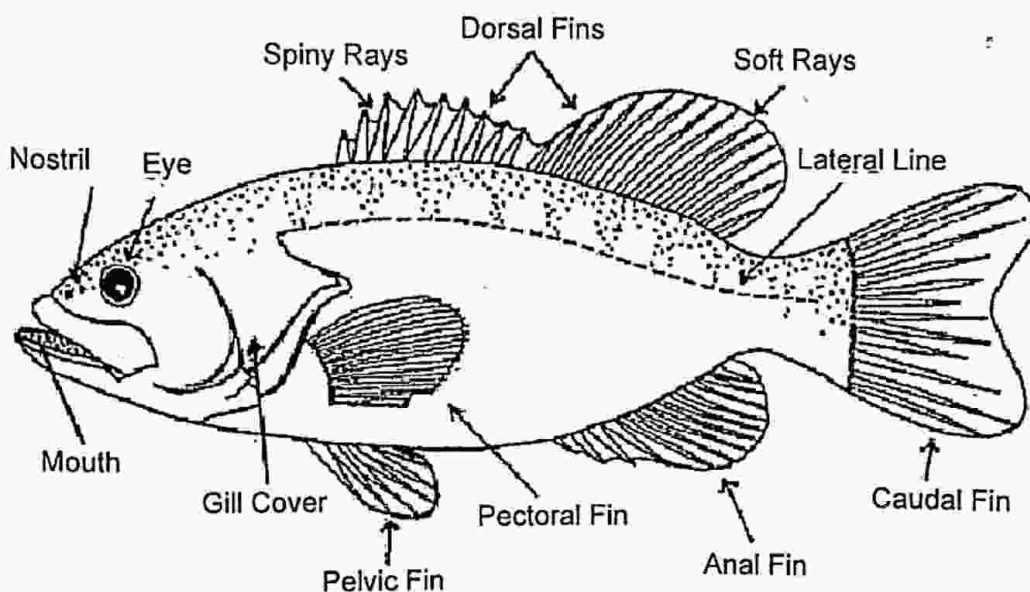
backwards, endoskeleton, gills used for respiration. They have paired and unpaired fins for swimming. They reproduce by laying eggs (oviparous) and external fertilization. e.g. Tilapia, Salmon, Shark etc.
Adaptation of Tilapia (bony fish) to:

a. **Movement:** Streamlined shape. Paired and unpaired fins for fast movement and stability.

Muscles in tail region used for movement. Absence of neck. Slimy body for locomotion and escape.

b. **Protection:** Greyish in colour blending with surrounding to give protective camouflage. Body covered with scales for protection.

Lateral line to detect vibrations or changes in water pressure, lidless eye to give a wide range of vision. Nostrils to detect chemical substances (smell). Swim bladder for buoyancy. Vascularized gills for respiration in water.



External structure of a bony fish

2. **Amphibians:** They are both terrestrial and aquatic. They spend most of their life time on land but usually return to water during breeding season. They are poikilothermic. Immovable eyelids, oviparous, gills used for respiration during larval stage and lungs, skin and mouth for respiration at adult stage. Their bodies are divided into head and trunk, e.g. toad, frog, salamander etc.

Adaptation of Toad (*Bufo regularis*)

A. **Movement:** Stream lined shape, absence of neck and tail suitable for hopping, fore limbs for landing, long and powerful hind limbs for propulsion.

b. **Feeding:** Carnivorous- absence of teeth, anteriorly fixed and sticky tongue with mucous. Ability to swallow prey alive.

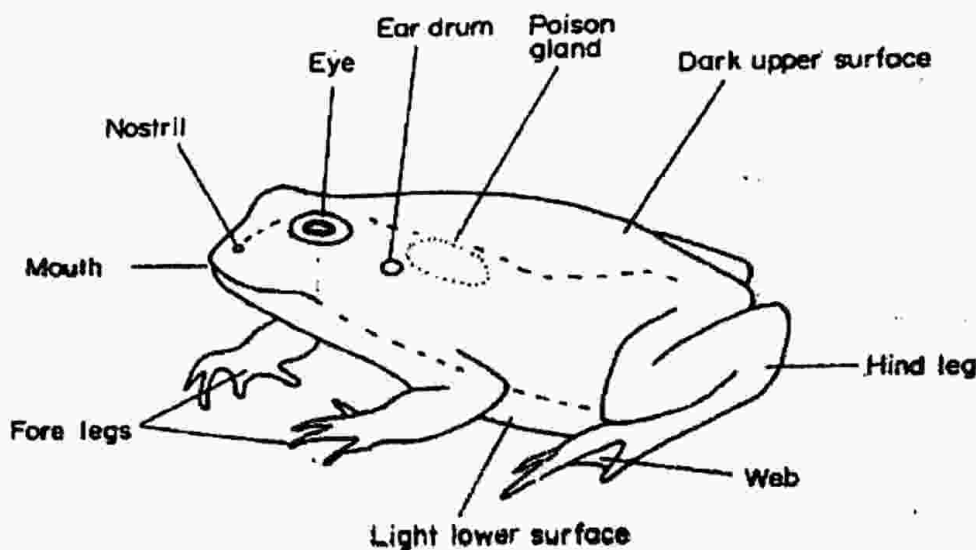
c. **Protection (tadpole stage):** Jelly coat provides buoyancy to eggs for absorption of light, protection from predators, prevention from injury, attack of bacteria and fungi and also for oxygen circulation. Tail fin with muscles or movement in water.

(Adult stage): Brown, rough and warty skin for protective camouflage and to reduce water loss on land. Ability to fill lungs with air to scare away

enemies. Poisonous secretion from gland under skin making it unpalatable for predators like snakes. Well developed and positioned eyes and tympanum to give a wide range of vision and detect vibrations of preys and predators. Webbed hind limbs for swimming in water during

breathing. Ability to breathe through skin, lungs and mouth cavity. Can carry out anaerobic respiration.

3. Reptiles: They are mostly terrestrial. Poikilothermic. Dry sharp pointed overlapping scales, dry skin, long tail, fore and hind limbs with five digits ending in claws. Well developed eyes with movable eyelids, have



External structure of a toad

two pairs of walking limbs. Their bodies are divided into head, neck and trunk. Homodont dentition. Internal fertilization and oviparous. e.g. tortoise, turtle, snakes, lizards etc.

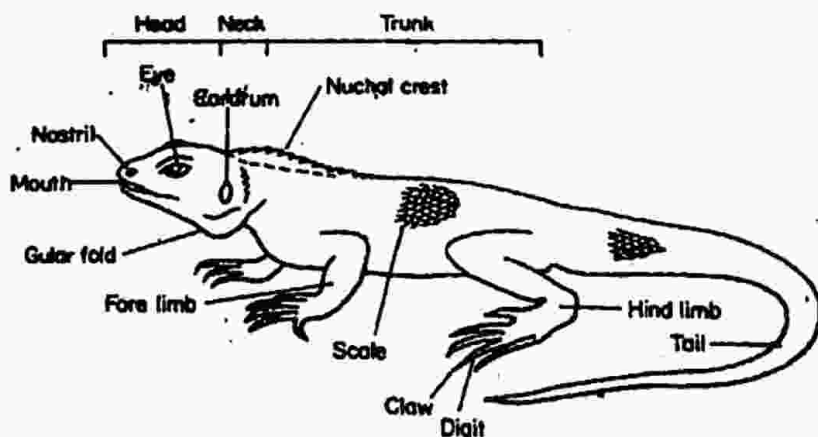
Adaptation of Lizard (*Agama agama*)

Stream lined shape. Powerful jaws to hold prey. Homodont dentition to crush preys. Well developed eyes and tympanum to afford a wide range of vision and to detect vibration respectively.

Nostrils for smell and respiration, lungs for respiration on land. Flexible neck to turn to all directions. Dry scaly skin to

prevent dessication on land. Clawed digits to hold on to surfaces during movement. Ability to complete life cycle on land and also exhibit internal fertilization. Ability to blend with environment, power of regeneration of tail, leathery shells can absorb oxygen and water and also protect eggs.

4. Aves (Birds): They are mostly terrestrial and their bodies are covered with feathers. They are homoeothermic i.e. maintain constant body temperature. Periodic moulting. Skull extended to form beak. Stream lined shape.



External structure of a lizard

Hind limbs covered with scales, fore limbs modified to form wings, feet adapted for feeding and to habitat; movable eyelids with a nictitating membrane. Body divided into head, neck, trunk, and a short tail. Internal fertilization. e.g. ostrich, kiwi, fowl, duck etc.

Adaptation of Birds for flight

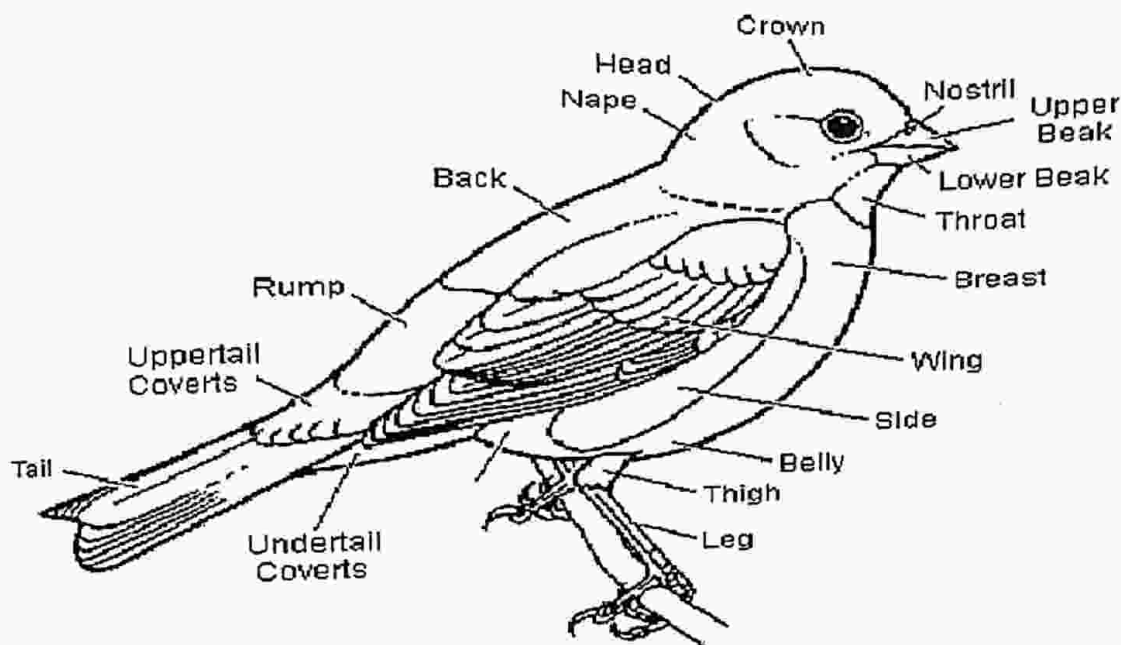
Dry skin without sweat glands, body covered with feathers, wing and tail feathers for flight. Feathers act as insulators, fusion of skeleton to provide strength. Light hollow bone to reduce weight, sternum modified into keel

for attachment of muscles. Streamlined shape, large air sacs in lungs, fusion of pelvic girdle with backbone. Flexible neck for easy movement.

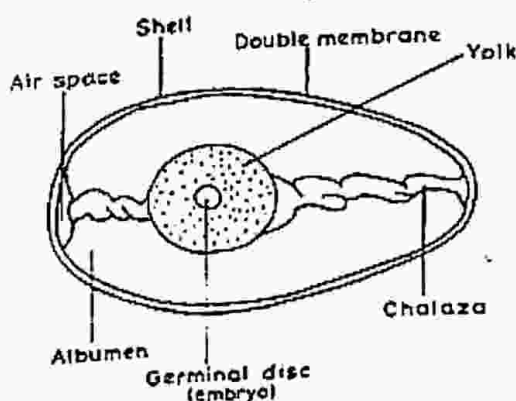
Types of feathers: Contour (covering body and giving shape). Contour made up of quill and covert feathers. Covert to cover body and quill found on wing and tail for flight.

Down feather gives insulation. Short shafts and loose barbs. No barbules.

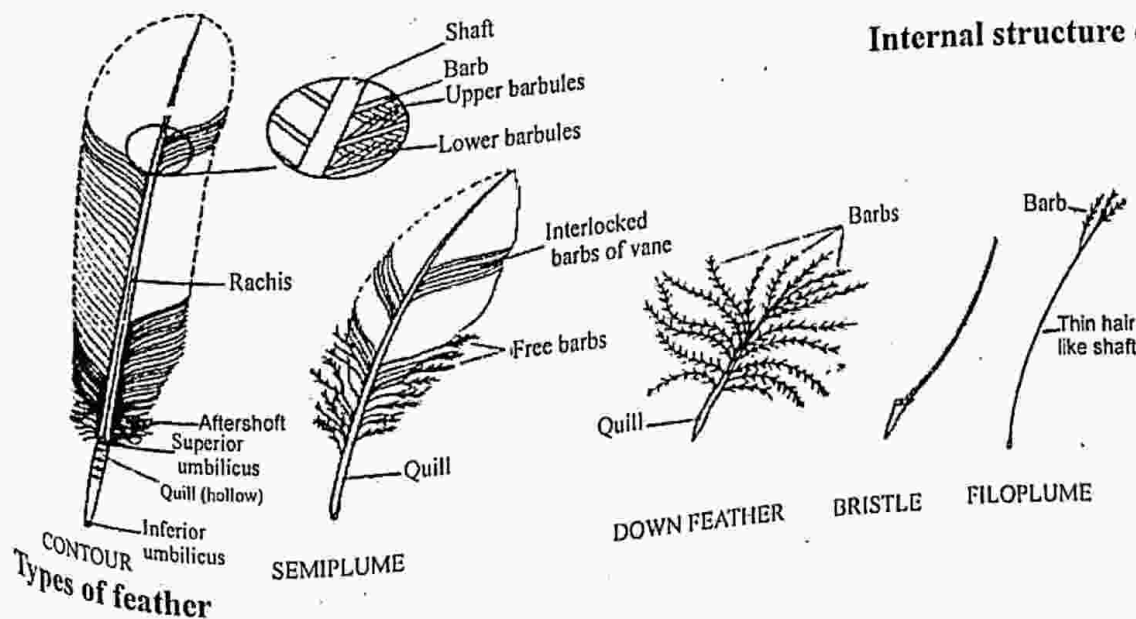
Filoplume feather has slender shaft with tuft of loose barbs at end. They are specialized sensory structures that aid in the operation or movement of other feathers for flight, insulation or display. Others include: Semiplumes and Bristles.



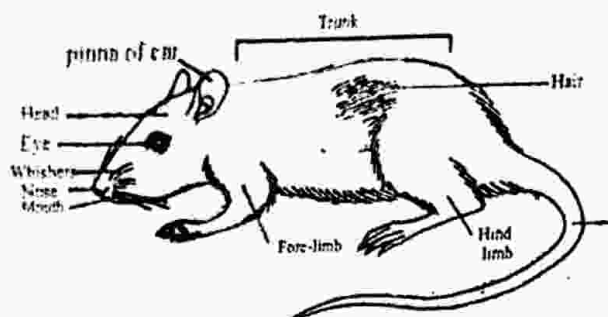
External structure of a bird



Internal structure of egg



5. Mammals: Their bodies are covered with hairs. They have mammary glands and give birth to their young ones alive (viviparous). They are homeothermic. They have sweat gland, heterodont dentition, external ears (pinnae), large and complicated brain. Coelom is divided by diaphragm into thorax and abdomen. e.g. man, lion, bat, rat etc.



External structure of a rat

Revision Questions

5.1. BONYFISH

1. UME 86/9

Fishes are cold-blooded because their body temperature is

- A. constantly low
- B. constantly high
- C. dependent on that on their surroundings
- D. regulated at will

2. UME 94/9

Which of the following combinations differentiates a bony fish from a cartilaginous fish?

- I. Presence of gills
 - II. Absence of gills slits
 - III. Possession of bony skeleton
 - IV. Possession of laterally compressed body
 - V. Possession of dorso-ventrally compressed body
- A. I, II and III B. I, II and IV
C. II, III and IV D. II, III and V

3. UME/94/10

Which is the most important adaptation of a bony fish in water?

- A. the possession of a streamlined shape
- B. The presence of overlapping scale
- C. The covering of the body by thin film of slime
- D. The possession of a caudal fin

4. UME/82/17

The main function of swim bladder is for

- A. swimming B. detecting sound
- C. buoyancy D. breathing
- E. reproduction

5. UME/84/14

The dorsal and anal fins of fish are used for

- A. upward movements
- B. controlling rolling movements
- C. downward movements
- D. steering
- E. buoyancy

6. UME/85/5

In fish, the sense organs which detects movement in water are located within the

- A. gills
- B. operculum
- C. nostrils
- D. median fins
- E. lateral lines

7. UME/86/12

The fins making up the limbs of the bony fish are

- A. caudal and ventral
- B. ventral and pelvic
- C. pelvic and pectoral
- D. pectoral and dorsal

8. UME/91/15

The gills rakers of fishes take part in

- A. feeding
- B. respiration
- C. swimming
- D. diffusion

9. UME/2008/10

Which of the following is used mainly for balancing in fish?

- A. the caudal fin
- B. the pectoral fin
- C. the anal fin
- D. The dorsal fin

10. PCE/96/9

Which pair of adaptive features enables the lung fish to live in different environment?

- A. Gills and fins
- B. Gills and swim bladders
- C. Barbells and fins
- D. Barbells and nostrils

11. PCE/97/9

In tilapia, the fins lying just behind the gill covers are

- A. caudal
- B. pectoral
- C. pelvic
- D. anal

12. The osmotic pressure of sea water is higher than that of the body fluid of the fish living in it. The cartilaginous fish despite this are at home in the sea. This is because they retain a large quantity of urea in their body fluids. The purpose of the urea is:

- A. To prevent the cartilaginous fish from desiccation
- B. To assist the cartilaginous fish in excretion
- C. To raise the osmotic pressure of the body fluid of the cartilaginous fish to that of the surrounding water
- D. To neutralize the effect of salt drunk by the cartilaginous fish
- E. None of the above

13. The cartilaginous fish differ from the bony fish

- A. in having no anal fin
- B. in having internal fertilization
- C. in having no claspers
- D. in having no lateral system
- E. in having no vertebrae

14. The operculum is possessed by:

- A. the deep-sea fish
- B. the bony fish
- C. all types of fish
- D. Fishes with the swim bladder
- E. All fishes other than cartilaginous ones

15. The fish will roll from side to side if:

- A. the pectoral fins are cut off
- B. the caudal fin is cut off
- C. the ventral fin is cut off
- D. the dorsal, ventral and anal fins are cut off

16. The propulsive forward movement of the fish is due to:
 A. the flapping of the pelvic fin
 B. the movement of the operculum
 C. The side to side movement of the tail
 D. The up and down movement of the pectoral fin
 E. None of the above

17. Many fish have swim bladders which are used for varied purposes. For which of these do the fish not employ the swim bladder?
 A. Respiratory purposes
 B. Maintenance of body position in water
 C. Production of gases
 D. Sound detection in water
 E. Storage of excretory material

18. Why are the olfactory lobes of the brain of the fish much more developed than any other part?
 A. They control the life of the fish
 B. They correlate the activities of the fish
 C. The fish depends much on its sense of smell for existence
 D. The olfactory lobes are anteriorly placed and hence come into new environment before any other part of the body
 E. None of the above reasons

19. With reference to the fish, the cerebellum controls:
 A. Their movement
 B. Their muscular activities
 C. Their escape reaction
 D. Reason and thought
 E. None of the above

20. Which of these is not true of the fish living in fast moving water?
 A. they are robust
 B. they have adhesive structures
 C. they are strong swimmers
 D. they are laterally flattened
 E. None of the above

5.2. TOAD

21. UME/83/28

In what order do the following structures develop during metamorphosis of the toad?

1. External gills 2. Internal gills 3. Forelimbs
 4. Hindlimbs 5. Mouth
 A. 1, 2, 3, 4, 5 B. 1, 5, 2, 4, 3
 C. 2, 1, 3, 4, 5 D. 5, 3, 4, 1, 2 E. 5, 4, 3, 2, 1

22. UME/78/29

Which of these is not associated with the tadpole stages of the toad or frog?

- A. V-shaped gland B. operculum
 C. external gills D. jelly
 E. shell

23. UME/88/12

The long and coiled intestine of a young tadpole is an adaptation to its.

- A. herbivorous diet
 B. carnivorous diet
 C. aquatic diet
 D. insectivorous habit

24. UME/90/11

In which of the following does external fertilization takes place?

- A. Toad B. Lizard
 C. Bird D. Cockroach.

25. PCE/91/9

The function of the layer of jelly which surrounds the eggs of frogs when laid is to

- A. provide mineral salts for the developing eggs
 B. provide food for the developing eggs
 C. protect the eggs from being eaten by aquatic animals
 D. disallow gaseous exchange

26. PCE/99/1/Type: E

The poison glands found in the toad produce a substance which

- A. kills its predators
 B. Protects it against predators
 C. keeps the body surface shiny and moist
 D. blends it with the surrounding

27. PCE/92/9

What function does the nictitating membrane perform?

- A. it gives the body a streamline shape and enables the animal to move easily through water
 B. it is pulled upwards to cover the eyes of the animal when there is danger
 C. it secretes materials that moisten the body of the animal
 D. it help reduce water loss from the body of the animal

28. Which of these characteristics may NOT be regarded as amphibian?

- A. Smooth and moist skin
 B. Presence of scales
 C. Permeable skin
 D. A four-digit fore limb
 E. None of the above

29. A frog such as *Xenopus* is placed in a basin of salt water. It dies a moment later and analysis shows that it died of desiccation. This means:

- A. *Xenopus* cannot live long in water
 B. *Xenopus* has a permeable skin
 C. *Xenopus* prefers land to water
 D. *Xenopus* has no mucous glands in its skin
 E. None of the above

30. The moveable thin film of skin that moves over the eyes of the toad is:

- A. the conjunctiva
 B. the eyelash
 C. the cornea
 D. the membrane oculi
 E. the nictitating membrane

31. That gland in the skin of the toad that produces a milky white poisonous fluid is

- A. the paraxoid gland B. the mucus gland
 C. the parotid gland D. the parotoid gland
 E. the milk gland

32. The amphibians have different respiratory surfaces. Which of these is not a respiratory surface?
A. the skin B. the lining of the cloaca
C. the lungs D. the lining of the buccal cavity
E. the webs of the feet.

33. Other than its slight build the male toad is distinguished from the female by:

- A. The possession of the five digits in the fore limb
B. The possession of a black breeding pad
C. The possession of a penis
D. The possession of a dark skin beneath the chin
E. None of the above

34. With regard to the toad, which of these may not be considered functions of the jelly to the eggs in them?

- A. It protects the eggs from injuries
B. It prevents the attack of bacteria
C. It prevents overcrowding
D. It protects eggs from attack by fleas
E. It provides the eggs and hence the developing tadpoles with mineral salts

35. Looking at the early life of the toad in water, which of these structures is not essential for a successful life?

- A. Ventral sucker
B. External and internal gills
C. Ciliated skin
D. Elongated tail
E. None of the above

36. With reference to the toad, which of these is true of metamorphosis?

- A. The change from the tad-pole to the adult toad
B. The development of the lungs
C. The process involving the internal rearrangements of the blood vessels in the toad
D. The total development of the toad from the eggs to the adult
E. None of the above

5.3. LIZARD

37. UME/80/6

Some of the features of an animal are scales, teeth, nares and backbone. The animal is likely to a

- A. Toad B. Bird
C. Lizard D. Rat
E. Bat

38. UME/83/6

Which of the following structures is NOT found in the female Agama Lizard?

- A. Pre-anal pads B. Eardrum
C. Gular fold D. Nasal Scale
E. Nuchal crest.

39. UME/2004/46/Type 4

One adaptation of reptiles to water loss is the presence of

- A. long nails B. long sticky tongues
C. keratinous scales D. claws on limbs

40. PCE/96/8

Lizards are able to breed on land because they produce

- eggs covered with a
A. thin leathery shell
C. thick slimy mucus
B. thick leathery shell
D. thin slimy mucous

41. UME/2007/15

In lizards, the lowering of the gular fold is used to

- A. catch insects B. attract mates
C. defend their territory D. frighten enemies

42. PCE/91/10

A major characteristic of reptiles is that they

- A. are four-legged
B. have homodont dentition
C. have long narrow bodies
D. have hair under their scales

43. Which of these is not true of the reptiles?

- A. Their skins have hard and horny epidermal scales
B. They have no external ears
C. They undergo no metamorphosis
D. They lay large yolky eggs which are protected by
E. Fertilization is external

44. Which of these can be used to separate the reptiles from the amphibians?

- A. the presence of claws on digits
B. the presence of a neck
C. Viviparity
D. The presence of a tail
E. none of the above

5.4. BIRD

45. UME/95/9

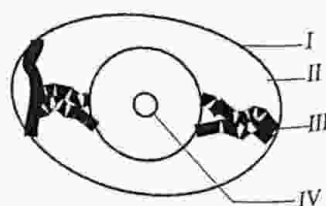
Birds maintain their body temperature with the help of their

- A. blood which maintains constant temperature
B. feathers which cover the body
C. skin which conserves moisture
D. veins which transport fluid of constant temperature
E. all body tissue

46. UME/84/32

Which of the following adaptations is NOT connected with the flight of birds?

- A. Stream lined shape
B. Presence of powerful muscles
C. Reduced body weight
D. Broad sternum
E. Webbed feet.



47. UME/2001/14/Type T

The structure that holds the yolk in position is labeled

- A. II
B. III
C. IV
D. I

8. UME/91/10

The part labeled

- A. albumen
B. yolk

9. PCE/91/10

Which of the following is not a function of the egg white?

- A. Down formation
B. Quill formation

50. UME/71/10

Which of the following is not a function of the egg white?

- A. Wing formation
C. Inferior formation

51. UME/2001/14/Type T

The peak of the egg is labeled

- A. scooping
B. catching
C. picking
D. boring

52. UME/2001/14/Type T

Parental care is shown by

- A. snails
C. birds

53. Which of the following is not a function of the egg white?

- A. They have a large surface area
B. Their bodies are small
C. They have a large number of eggs
D. The number of eggs is large
E. None of the above

54. Unlike the reptiles, birds have

- A. Non-glandular skin
C. without stratum corneum
E. None of the above

55. The feathers of a bird are called

- A. the pteridium
C. the apteria
E. the tract

56. Which of the following is not a function of the bird's feathers?

- A. Filoplumes
B. Contour feathers
C. Model feathers
D. Down feathers
E. Plumules.

57. The feathers of a bird are called

- A. Providing a protective covering
B. Providing a means of locomotion
C. Insulation and temperature regulation
D. Colouration and display
E. All of the above

48. UME/2001/15/type: T
The part labeled IV is the

- A. albumen
- B. germinal disc
- C. yolk
- D. air space.

49. PCE/96/10

Which of the following feathers of a bird make a warm light covering?

- A. Down feathers
- B. Covert feathers
- C. Quill feathers
- D. Filoplumes

50. UME/78/31

Which of these is NOT a part of the feather of a bird?

- A. Wing
- B. Aftershaft
- C. Inferior umbilicus
- D. Vane E. Barbule

51. UME/2008/11

The beak of a duck is structurally adapted for

- A. scooping and sieving food
- B. catching and grasping food
- C. picking and cracking food
- D. boring and sucking food

52. UME/2010/5/TYPE: D

Parental care is exhibited by

- A. snails
- B. earthworms
- C. birds
- D. toads

53. Which of these is not true of the birds?

- A. They have air sacs
- B. Their bodies may be covered with epidermal scales
- C. They have no teeth
- D. The number of digits of the fore limbs are reduced
- E. None of the above

54. Unlike the toad, the skin of the bird is:

- A. Non-glandular
- B. without blood vessels
- C. without stratified epithelia loose
- E. None of the above

55. The feathers on the skin of birds are arranged on rows. Those tracts without feathers on the body of birds are called:

- A. the pterium
- B. the apteridia
- C. the apteria
- D. the pterylea
- E. the tract

56. Which of these is not connected with the feather system of the bird?

- A. Filoplumes
- B. Contour feathers
- C. Model feathers
- D. Down feathers
- E. Plumules.

57. The feathers serve the birds in:

- A. Providing a protective and a water-repellent covering
- B. Providing a good device for flight
- C. Insulation and hence a good device for body temperature regulation
- D. Colouration and sex display
- E. All of the above

58. The only true skin gland of the bird is:

- A. the sebaceous gland
- B. the sweat gland
- C. the avicarian gland
- D. the lipid gland
- E. the uropygial gland

59. The adaptations of the fish-eating birds are:

- A. The possession of beaks with serrated edges
- B. The possession of hook like structures on the palate and tongue of birds to hold prey
- C. The possession of long necks
- D. The possession of long beaks
- E. None of the above

60. The adaptation of the birds associated with water are:

- A. Possession of long legs
- B. Possession of webbed feet
- C. Possession of oily feathers
- D. Possession of flipper like wings
- E. Possession of sharp claws

5.5. SMALL MAMMALS

61. UME/78/8

A traveler on a fishing trip landed a type of animal with hairs on its body. This animal could have been

- A. an amphibian
- B. a reptile
- C. shark
- D. a mammal
- E. seagull

62. UME/2004/28/Type: 4

A peculiar characteristics of mammals is that they

- A. have sebaceous glands
- B. have teeth
- C. are warm blooded
- D. have lungs

63. UME/2004/38/Type: 4

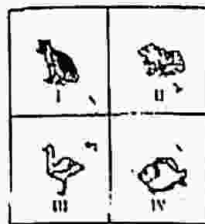
Rodents gnaw on food with their

- A. strong jaws
- B. flat-ridged teeth
- C. chisel-like front teeth
- D. molar teeth

64. UME/84/10

The mammalian organ through which nourishment and oxygen diffuse into a developing embryo is called

- A. amnion
- B. chorion
- C. umbilical cord
- D. oviduct
- E. placenta



65. UME 2002/7/Type: Y

The correct evolutionary sequence of the organisms represented is

- A. I → III → II → IV
- B. II → III → IV → I
- C. III → II → I → IV
- D. IV → II → III → I

CHAPTER 5.

1. C	2. Nil	3. A	4. C	5. B
6. E	7. C	8. B	9. B	10. B
11. B	12. C	13. B	14. B	15. D
16. C	17. E	18. C	19. B	20. B
21. B	22. E	23. A	24. A	25. C
26. B	27. B	28. B	29. B	30. E
31. C	32. B	33. E	34. E	35. C
36. D	37. C	38. A	39. C	40. A
41. B	42. B	43. E	44. B	45. B
46. E	47. D	48. B	49. A	50. A
51. A	52. C	53. E	54. A	55. C
56. C	57. E	58. E	59. A	60. B
61. D	62. A	63. C	64. E	65. D
66. D	67. A	68. A	69. A	70. B
71. A	72. C	73. D	74. B	75. C
76. A	77. B	78. B	79. D	80. B

INTERNAL STRUCTURE OF ORGANISMS

Understand the following after reading this chapter:

Flowering plants (roots, stems and leaves)
Mammals

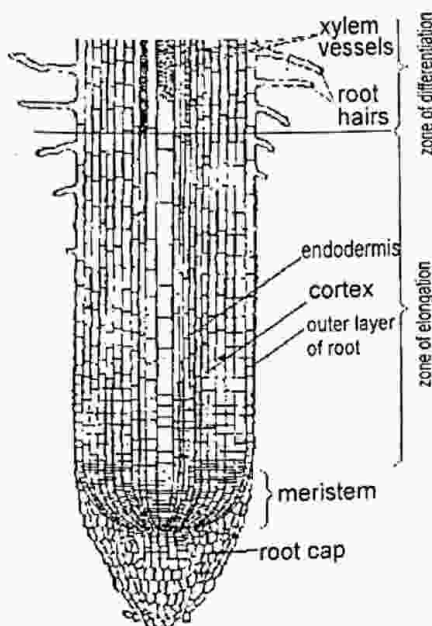
INTERNAL STRUCTURE OF ROOTS

The piliferous layer is a single row of cells, which, in young roots, may project to form root hairs used for absorption of water and salts. The cortex is wide and consists of loose parenchyma cells. The innermost cortex is a layer of tightly arranged rectangular cells, the endodermis.

The lateral walls of the endodermal cells are sealed with suberin thereby forcing water and solutes to pass through the cell's protoplasm where the movement of dissolved material is regulated.

Inside the endodermis is the pericycle consisting of fibres and a central stele of vascular tissues. The xylem is in the centre and has many radiatory arms (protoxylem points). The phloem occurs between these arms. During secondary growth, cambium is developed where the xylem and phloem meet. There is no pith.

In the monocot root, there is a wide pith around which is a layer of xylem and phloem tissues alternately arranged.



Longitudinal section of a dicot root

outer side of the phloem is a pericycle made up of thickened fibres. The innermost region is the pith, it is parenchymatous.

In monocot stems the cortex is narrow. The vascular bundles are scattered in the parenchymatous ground tissue. The bundles are smaller and more numerous towards the outside, while those towards the centre are fewer and bigger. Each vascular bundle is surrounded by a sheath of strengthening sclerenchyma cells. There is no cambium.

During secondary growth in the dicot stem, inter-fascicular cambium is formed in the pith ray, linking all the cambium cells into a continuous ring. The cambium then divides and produces new xylem cells in the inside and new phloem cells on the outside. Simultaneously a cork cambium is developed just below the epidermis. This divides to produce layers of water proof cork cells which inter form the bark. Perforations called lenticles are developed on the bark to allow movement of air in the stem.

INTERNAL STRUCTURE OF LEAVES

The leaf surface is covered by the upper and lower epidermis. The epidermis is perforated by stomatal pores; they are more numerous on the lower epidermis. Stomata allow exchange and gases for respiration and photosynthesis, as well as movement of water vapour during transpiration. In general, the stomata open in the presence of light and close in the dark.

The mesophyll is the photosynthetic tissue between the upper and lower epidermis. It consists of an upper palisade parenchyma and spongy parenchyma. The palisade cells are cylindrical and are arranged in one or more rows perpendicular to the epidermis. The spongy mesophyll consists of loosely arranged cells with large air spaces. All mesophyll cells contain chloroplasts.

Small vascular bundles called veins, arise from the vascular bundle in the mid-rib and form a network in the leaf tissue. Veins conduct water, mineral salts, and food, and also mechanically support the mesophyll tissue.

In the leaves of monocots, the mesophyll not differentiated into distinct palisade and spongy parenchyma layers.

MAMMALS

1. The body cavity is divided into the chest and abdomen by muscular sheet – the diaphragm
2. Organs in the chest include: Lungs, heart (covered by pericardium). Organs in the abdomen include: Stomach, liver, kidney intestines.
3. Embryonic development occurs in the uterus of mother and the young are born alive. No eggs are laid.
4. The parents care for their young; the mothers feed the young on milk from mammary gland

INTERNAL STRUCTURE OF STEMS

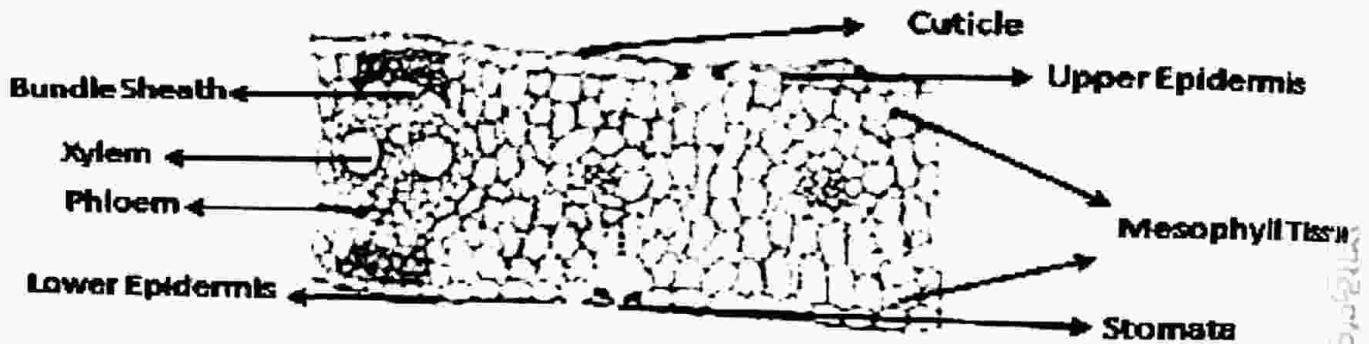
The outermost layer is the epidermis. Within this is a wide cortex predominately parenchymatous. Some of the cells contain chloroplasts and are photosynthetic. In some plants, an outer hypodermis layer made up of collenchyma is demarcated; it provides rigidity to the stem to withstand swaying in the wind.

The innermost layer of the cortex is the endodermis; it is a single layer of rectangular-shaped cells containing starch granules. The vascular tissues are organized into separate bundles arranged in a radial pattern. Each consists of an inner xylem, an outer phloem and a cambium in the middle. On the

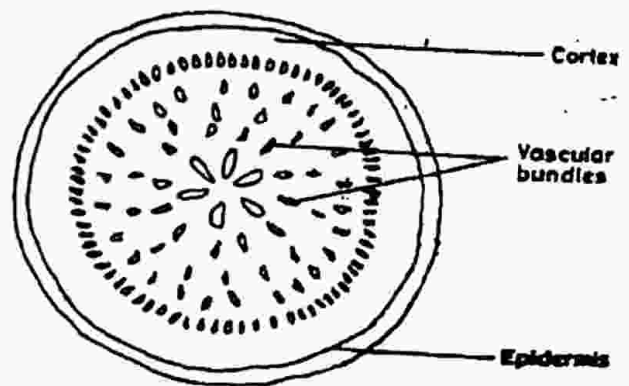
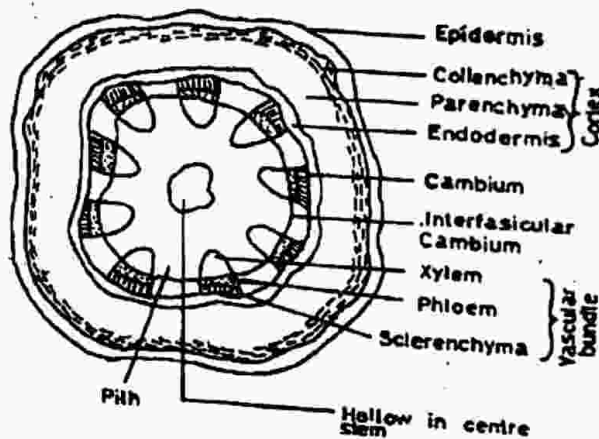
Internal Structure of Dicot leaf



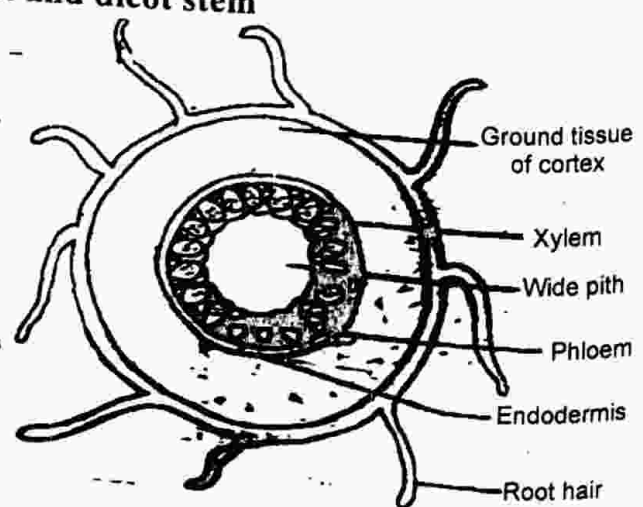
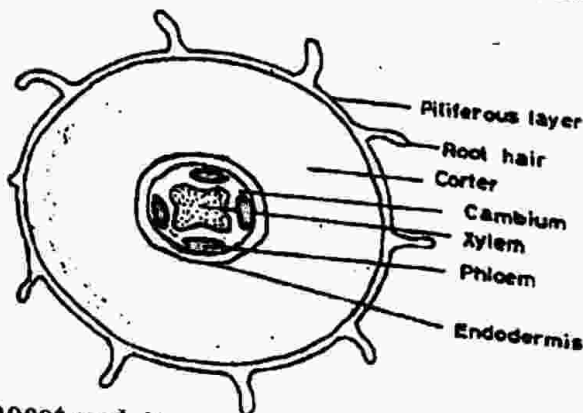
Internal Structure of Monocot leaf



Internal structure of dicot and monocot leaf



Monocot and dicot stem



Monocot and dicot root

Revision Questions

6.1.1. ROOT, STEM AND LEAVE

1. UME/84/27

Which of the following statements is NOT true of the piliferous layer of a root? It

- A. has a very thin cuticle
- B. is the outermost layer of the cortex
- C. may bear root hairs
- D. breaks down as root ages
- E. is replaced by cork in old roots.

2. UME/85/13

Which of the following is common to a dicotyledonous stem and a monocotyledonous root?

- A. medullary ray
- B. central pith
- C. wide cortex
- D. narrow cortex
- E. pericycle fibers

3. UME/86/13

The stem differs from the root in having the xylem and phloem stands

- A. on the same radii
- B. scattered
- C. on alternate radii
- D. towards the pith

4. UME/87/15

The correct sequence of tissues in the anatomy of a young dicotyledonous stem from the inside to the outside is

- A. pith, phloem, cambium, xylem, parenchyma, collenchyma and epidermis
- B. xylem, phloem, cambium, cortex, endodermis, collenchyma, and epidermis
- C. pith, xylem, cambium, phloem, collenchyma, parenchyma, and epidermis
- D. phloem, xylem, cambium, cortex, endodermis, collenchyma and epidermis

5. UME/87/16

Secondary thickening is initiated in dicotyledonous stem by the

- A. Xylem parenchyma
- B. Secondary phloem
- C. Endodermis
- D. Cambium

6. UME/88/17

In the transverse section of the leaf of maize vascular bundles are arranged in

- A. a row
- B. one circle
- C. alternate positions
- D. two circles

7. UME/89/15

In a dicotyledonous stem, each companion cell is found beside the

- A. Endodermal cell
- B. Xylem vessel
- C. Sieve tube
- D. Pericyclic fibre.

8. UME/90/16

Which of the following cells is thin-walled and living at maturity?

- A. collenchyma
- B. sieve tubes
- C. xylem vessel
- D. sclerenchyma

9. UME/92/11

The flow of air and water in or out of the mesophyll layer of a leaf is controlled by the

- A. stomata
- B. lenticels
- C. air spaces
- D. guard cells

10. UME/92/13

The major site of photosynthesis in the leaf is the

- A. palisade parenchyma
- B. mesophyll parenchyma
- C. upper epidermis
- D. lower epidermis

11. UME/98/11

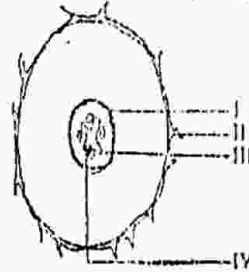
Which of the following structures is capable of producing more tissues in the stem of a herbaceous flowering plant?

- A. Epidermis
- B. Pericycle
- C. Xylem
- D. Cambium

12. UME/2001/21/Type T

The veins of the leaf are formed by the

- A. vascular bundles
- B. cambium cells
- C. palisade tissue
- D. spongy mesophyll



13. UME 2002/41 Type: Y

The function of absorption is performed by the structure labeled

- A. I
- B. II
- C. III
- D. IV

14. UME/2002/42/Type: Y

The structure labeled I represents the

- A. phloem
- B. Xylem
- C. cortex
- D. pericycle.

15. UME/2004/6/Type: A

In the internal structure of plants, a wide pith in the centre of common to

- A. dicot roots and monocot stems
- B. dicot stems and monocot stems
- C. dicot stems and monocot roots
- D. dicot roots and monocot roots

16. UME/2007/23

Monocot stems differ from dicot stems in that monocots have

- A. no cambium
- B. no pith
- C. fewer vascular bundles
- D. phloems with parenchyma

17. UME/2006/17

In the transverse section of a dicot stem, the region lying between the endodermis and the vascular bundle is the

- A. parenchyma
- B. pericycle
- C. phloem
- D. hypodermis

18. UME/2008/13

In a dicotyledonous stem, the zone between the epidermis and the pericycle is the

- A. cortex B. stele C. xylem D. phloem

19. UME/2009/6

In plants, the structures that play roles similar to the arteries and veins of animals are the

- A. xylem and phloem B. root hairs and xylem
C. lenticels and phloem D. roots and stems

20. UME/2010/11/TYPE: D

Which of the following is true of the transverse section of a dicot stem?

- A. the xylem is more interiorly located than the phloem
B. the cambium lies between the cortex and the vascular bundles
C. the vascular bundles are randomly scattered within the cortex
D. the epidermis is completely encircled by the cortex

6.2. MAMMAL

21. UME/84/10

The mammalian organ through which nourishment and oxygen diffuses into a developing embryo is called

- A. amnion B. chorion C. umbilical cord
D. oviduct E. placenta

22. UME/2002/35/Type Y

Which of the following pairs of organs is located in the anterior half of the mammalian body cavity?

- A. Kidneys and lungs B. Heart and ovary
C. Lungs and heart D. Kidneys and heart

23. UME/2006/7

The organ situated in the pericardial cavity of a mammal is the

- A. heart B. Liver
C. Stomach D. spleen

24. UME/2006/9

In rabbits, the chamber of the heart that receives oxygenated blood from the lungs is the

- A. left ventricle B. left auricle
C. right ventricle D. right auricle

25. UME/2006/16

The mammalian vein which starts with and ends in a capillary network is the

- A. pulmonary vein B. mesenteric vein
C. renal vein D. hepatic portal vein

26. UME/2010/20/TYPE: D

The sheet of muscle that separates the thoracic and the abdominal cavities is the

- A. intercostals muscle
B. pleural membrane
C. pericardium
D. diaphragm

27. The mammals are distinguished from the amphibians by

- A. the possession of scales
B. the possession of a tail
C. the possession of the mammary glands
D. the possession of five digits in the limbs
E. the possession of glands in the skin

28. The teeth of the mammal are

- A. Monodont B. Bonodont
C. Heterodont D. Dipodont
E. None of the above

29. With reference to the mammal, what is the perineum?

- A. it is a mesentery in the body
B. it is a fold in the abdominal cavity
C. it is the membrane covering the heart
D. it is a partition separating the anus from the urinogenital aperture
E. It is an organ

30. The ear of the mammal is almost like the ear of the bird except for:

- A. the presence of auditory meatus
B. the absence of the tympanic membrane
C. the absence of the utriculus
D. the absence of the cochlea
E. the presence of a fold of skin called the pinna

31. Hair regulates the body temperature of the mammal by

- A. Causing the goose flesh
B. Preventing heat radiation from the body
C. Acting as an insulator
D. Preventing the loss of sweat from the body
E. None of the above

32. A rabbit went through an operation. On recovery, it was discovered that it could not digest fat. It is also seen that one of the ducts in its body had been tied up. This duct is

- A. the dorsal aorta B. the bile duct
C. the oesophageal duct D. the celiac duct
E. none of the above

33. The pancreatic duct opens into this region of the intestines

- A. the ileum B. the colon
C. the duodenum D. the pyloric region
E. the caecum

34. This structure has a blind ending:

- A. the wind pipe B. the caecum
C. the colon D. the spleen
E. the gall bladder

35. This structure produces enzymes from digestion. It is also a ductless gland

- A. the walls of the stomach B. the liver
C. the gall bladder D. the pancreas
E. None of the above

36. The kidney is distinguished from the amphibian kidney by

- A. the kidney is larger
B. the kidney is more complex
C. the kidney is more numerous
D. the kidney is more numerous
E. the kidney is more numerous

37. This is the structure which is more blood-rich than the other structures

- A. it is the heart
B. it is the liver
C. it is the spleen
D. it is the kidney
E. None of the above

38. When the food is taken into the mouth, it is first

- A. the pylorus
B. the spleen
C. the small intestine
D. the cardiac stomach
E. the rectum

39. The thyroid gland is

- A. the oesophagus
B. the auricle
C. the trachea
D. the aorta
E. the gall bladder

40. Which of the following is not a part of the digestive system?

- A. the mesentery
B. the liver
C. the diaphragm
D. the lungs
E. None of the above

41. The blood in the liver is

- A. the liver
B. the spleen
C. the spleen
D. the spleen
E. None of the above

42. Considering the regions of the digestive system, which is solely

- A. the duodenum
B. the colon
C. the colon
D. the oesophagus
E. the oesophagus

43. The bladder is

- A. the urethra
B. the urethrum
C. the ureter
D. the spermatic duct
E. the vas deferens

36. The is the biggest organ in the body
 A. the kidney
 B. the stomach
 C. the liver
 D. the heart
 E. the spleen

37. This structure is responsible for the formation of urea;
 A. the kidney
 B. the spleen
 C. the gall bladder
 D. the liver
 E. the bladder

38. When there is a great muscular activity in the animal, which is these structures will be called upon to supply more blood so the muscles
 A. it is the heart
 B. it is the spleen
 C. it is the lungs
 D. it is the liver
 E. None of the above.

39. Hydrochloric acid is produced in:
 A. the pyloric region of the stomach
 B. the spleen
 C. the small intestine
 D. the cardiac region of the stomach
 E. the rectum

40. The thyroid gland is located on this structure
 A. the oesophagus
 B. the auricle of the heart
 C. the trachea
 D. the aortic arch
 E. the gall bladder

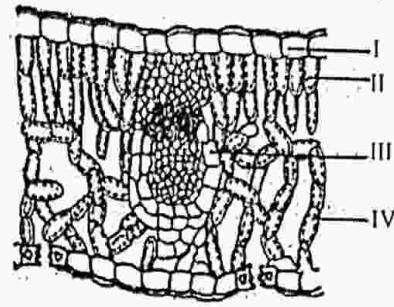
41. Which muscle structure in the body is tranversed only by the oesophagus and the blood vessels?
 A. the mesentery
 B. the liver
 C. the diaphragm
 D. the lungs
 E. None of the above

42. The blood cells have a certain amount of iron. When they are destroyed, the iron is stored in
 A. the liver
 B. the gall bladder
 C. the spleen
 D. the pancreas
 E. None of the above

43. Considering the alimentary canal which of these regions is solely for the absorption of water?
 A. the duodenum
 B. the ileum
 C. the colon
 D. the stomach
 E. the oesophagus

44. The bladder stores urine which is carried into it from the kidney through
 A. the urethra
 B. the urethrum
 C. the ureter
 D. the spermatic duct
 E. the vas deferens

MISCELLANEOUS QUESTIONS (6)



45. The major site for photosynthesis is labeled?
 A. I
 B. II
 C. III
 D. IV (PCE/2000/11/Type:A)

46. The function of the structure labeled II is to
 A. allow gaseous exchange between the tissues of the leaf
 B. allow light penetrate the leaf
 C. prevent bacterial and fungal infections of the leaf
 D. transport water and mineral salts to the leaf
 (PCE/2000/12/Type:A)



47. The structure labeled I represent the
 A. parenchyma
 B. endodermis
 C. cortex
 D. collenchyma (PCE/2003/35/Type:N)

48. Water and mineral salts are transported in the structure labeled
 A. II
 B. I
 C. IV
 D. III (PCE/2003/36/Type:N)

49. In the root vascular system, the stele is directly surrounded by the
 A. Pericycle
 B. cortex
 C. Endodermis
 D. parenchyma (UME/2005/14/Type: D)

50. In which part of a leaf are stomata found
 A. in the palisade layer
 B. in the spongy mesophyll layer
 C. between two guard cells on the epidermis
 D. guard cells and air spaces (PCE/91/12)

51. The components of the photosynthetic tissues of a leaf are the
 A. palisade and spongy mesophyll
 B. upper and lower epidermis
 C. sclerenchyma and collenchyma tissues
 D. guard cells and air spaces (PCE/2001/10/Type:4)

52. The part of the stem between the epidermis and the vascular bundles is called the

- A. cortex
- B. collenchyma
- C. parenchyma
- D. endodermis (PCE/98/11/type C)

53. In plants, the cell-type that die off at maturity is

- A. sclerenchyma
- B. parenchyma
- C. collenchyma
- D. cambium (PCE/94/11)

54. In the leaf, each stoma is usually enclosed by kidney-shaped cells called

- A. endodermis cells
- B. border parenchyma cells
- C. guard cells
- D. phloem cell (PCE/95/12)

55. The part of the internal structure of the leaf where photosynthesis occurs most is the

- A. spongy mesophyll
- B. palisade mesophyll
- C. lower epidermis
- D. upper epidermis (PCE/96/11)

CHAPTER 6

1. A	2. B	3. A	4. C	5. D
6. B	7. C	8. B	9. D	10. B
11. D	12. A	13. B	14. D	15. C
16. A/B	17. B	18. A	19. A	20. A
21. E	22. C	23. A	24. B	25. D
26. D	27. C	28. C	29. D	30. E
31. C	32. B	33. C	34. B	35. D
36. C	37. D	38. A	39. D	40. C
41. C	42. A	43. C	44. C	45. B
46. B	47. B	48. D	49. A	50. C
51. A	52. A	53. A	54. C	55. B

CHAPTER 7 NUTRITION

Understand the following after reading this chapter:

Modes of nutrition
Plant nutrition
Animal nutrition

NUTRITION

Processes by which organisms obtain and use the nutrients required for maintaining life.

Nutrients - Substances that are needed for

Sources of chemical potential energy (ATP)

Sources of molecules and atoms which are used for building up cells

Inorganic Nutrients - An inorganic substance in which the molecules of ions are relatively small and which does not contain carbon. E.g. H_2O , O_2 , CO_2 and nitrate ions.

Organic Nutrients - In an organic substance the molecules are usually relatively large and contain carbon. E.g. carbohydrates, fats and protein

TYPES OF NUTRITION

Different types of organism will use different types of nutrition. There are two types of nutrition used.

Autotrophic

The use of inorganic nutrients

Carried out by all plants, algae and some bacteria

Organisms which use this type of nutrition are known as AUTOTROPHS

They need an energy source which is usually the sun but can be chemical energy

Heterotrophic

The use of a mixture of organic and inorganic nutrients

Organisms which use this form of nutrition are known as heterotrophs.

Holozoic Nutrition

This is a form of heterotrophic nutrition when the nutrients that are taken into the body are passed down some form of alimentary canal. Holozoic organisms can be divided into herbivores, carnivores, omnivores and scavengers.

Saprophytic or Saprobiontic Nutrition

Organisms that use this type of nutrition are known as saprophytes or saprobionts

They don't invade living tissue

They feed on dead or rotting matter

They secrete digestive enzymes directly onto the food this called extracellular digestion and occurs outside the cell. They then reabsorb the digested products back into their cells.

These organisms are involved in decomposing and decay. They are really important in the recycling of nutrients.

Mutualistic Nutrition

These organisms are involved in symbiotic relationships

An example is the bacteria *Rhizobium* with legumes

The bacteria are nitrogen fixing, they convert nitrogen and hydrogen to ammonia.

They are found in the soil and in the root nodules of legumes.

The plants use the ammonia produced by the bacteria is used by the plants to make amino acids.

The bacteria benefits from this relationship because they gain carbohydrate and ATP from the plant.

The plant benefits because the ammonia produced by the bacteria allows them to flourish in nitrogen deficient soil.

Parasitic Nutrition

This is a relationship between parasite and host. The parasite is dependent on the host for nutrition and protection. The parasite will generally do the host harm.

Parasites are highly adapted for this type of nutrition. There are two main types of parasite ectoparasites which lived outside the organism and endoparasites which live inside the organism.

Carnivorous Plants

Carnivorous plants are plants that derive some or most of their nutrients (but not energy) from trapping and consuming animals or protozoans, typically insects and other arthropods. Carnivorous plants have adapted to grow in places where the soil is thin or poor in nutrients, especially nitrogen, such as acidic bogs and rock outcroppings. Examples of carnivorous plants include: Venus Flytraps, Sundews, Pitcher Plants a.k.a. Monkey Cups, North American Pitcher Plants, and Bladderworts.

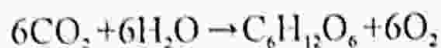
PHOTOSYNTHESIS

Plants use light energy from the sun as their external energy source, this type of autotrophic nutrition is photosynthesis.

During photosynthesis carbon dioxide and water are used to manufacture carbohydrates. Light energy is trapped by chlorophyll and used to generate ATP (cyclic photophosphorylation) and also to split water (the light reaction); the hydroxyl (-OH group) is recombined to make oxygen (which is excreted); the hydrogen is later used to fix carbon dioxide and reduce it to form carbohydrates. Plants manufacture all the other organic molecules (DNA, vitamins, lipids etc. etc.) that they need from these carbohydrates and very small quantities of

mineral ions obtained from the soil.

Carbon dioxide + water $\rightarrow$ (light + chlorophyll) $\rightarrow$ glucose + oxygen



The biochemistry of photosynthesis.

There are two basic stages to photosynthesis.

1. The Light Dependent Reaction

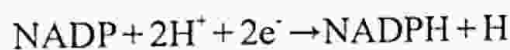
This takes place in the thylakoids of a chloroplast.

During this stage light energy is trapped by chlorophyll molecules in the light harvesting stage. This raises the energy level of electrons in the chlorophyll molecules. In the process of non-cyclic photophosphorylation this energy is used to convert NADP to NADPH + H⁺ and ADP to ATP.

Non-cyclic photophosphorylation involves photosystems II and I.

During this process water is split into oxygen - a waste gas (most of which will be released from the plant) - and into hydrogen ions and electrons.

These are used to generate NADPH + H⁺ and ATP. In cyclic photophosphorylation only photosystem I is involved and only ATP is produced.



2. The Light Independent Reaction

This takes place in the stroma of a chloroplast. The process is known as The Calvin Cycle.

Carbon dioxide fixation takes place.

Initially the carbon dioxide combines with ribulosebiphosphate (RuBP), a compound containing five carbon atoms.

The enzyme RuBisCO catalyses this reaction. Thus life on Earth largely depends on this one chemical!

This reaction can also be called a carboxylation, since carbon dioxide is being added: it gives a six carbon intermediate compound, which instantly splits into 2 molecules of glycerate-3-phosphate (GP) a 3-carbon compound.

The GP is then phosphorylated (phosphate is added), using the ATP generated in the Light-dependent reaction and then reduced to glyceraldehyde-3-phosphate (GALP)

These reactions use up the NADPH + H⁺ and ATP which were produced in the Light Dependent reaction. NADP and ADP are regenerated.

Most of the GALP is used to regenerate RuBP, but some is used to produce 6 carbon sugars (monosaccharides), which can then be used to produce a range of organic molecules.

These include: -

1. Starch, a polysaccharide, stored temporarily in the leaf, or in organs such as potato tubers for more long-term

storage.

2. Sucrose, a disaccharide transported to other parts of the plant in the phloem.
3. Lipids, for energy storage, in seeds, for example.
4. Amino acids and proteins - the manufacture of these requires nitrogen, the source of this is normally nitrate ions from the soil.

The factors that affect the rate of photosynthesis and their interactions:

- a. **Temperature** - higher temperatures raise the speed of the light-independent reactions, up to about 35-40°C (= gross photosynthetic rate); it has little effect on the speed of the light reactions. However, they also raise the speed of respiration in the plant and so net photosynthesis slows down at higher temperatures. Peak crop yields are at about 25°C
- b. **Carbon Dioxide** - normally 360 ppm, growers raise this to about 1000 ppm in greenhouses to raise yields. Peak photosynthesis is at even higher concentrations, but it is not economical to produce this. N.B. Higher world CO₂ levels = better plant growth.
- c. **Light** - the more the better (brightness); 16-20 hours a day (duration); blue and red must be in balance (i.e. filament lamps are no good!)
- d. **Water** - often a limiting factor - plants shut their stomata long before they wilt. Shut stomata = no gas exchange and no photosynthesis!

How leaves are adapted to carry out photosynthesis effectively:

stomata for gas exchange (largely on the underside to maximise palisade cells)

thin to allow gas to reach mesophyll cells quickly

moist to allow gasses to dissolve

waterproof cuticle to reduce transpiration and so water stress

good transport system

xylem - to bring water and minerals

phloem - to carry away the products of photosynthesis

large surface area - to trap light

palisade mesophyll - near top to trap maximum light

spongy mesophyll to aid in light absorption

leaves arranged in space so as not to shade each other.

Transparent cuticle/upper epidermis to allow maximum light penetration

MINERAL NUTRIENTS

Mineral nutrients are the inorganic matters or chemicals in the soil released during weathering of rocks. They are grouped into two, based on the quantities required of them by plants;

1. **Macro-nutrients:** These are minerals required in large quantities e.g. sodium, potassium, phosphorus, nitrogen, calcium, sulphur, magnesium.

2. **Micro-nutrients or trace elements:** These are minerals required in small quantities e.g. iron, manganese

molybdenum, zinc, copper, boron, cobalt. Animals obtain minerals indirectly by feeding on plants or their products.

Name	Function in plant	Deficiency symptoms
Nitrogen	Proteins, amino acids	Light green to yellow appearance of leaves, especially older leaves; stunted growth; poor fruit development.
Phosphorus	Nucleic acids, ATP	Leaves may develop purple coloration; stunted plant growth and delay in plant development.
Potassium	Catalyst, ion transport	Older leaves turn yellow initially around margins and die; irregular fruit development.
Calcium	Cell wall component	Reduced growth or death of growing tips; blossom-end rot of tomato; poor fruit development and appearance.
Magnesium	Part of chlorophyll	Initial yellowing of older leaves between leaf veins spreading to younger leaves; poor fruit development and production.
Sulphur	Amino acids	Initial yellowing of young leaves spreading to whole plant; similar symptoms to nitrogen deficiency but occurs on new growth.
Iron	Chlorophyll synthesis	Initial distinct yellow or white areas between veins of young leaves leading to spots of dead leaf tissue.
Copper	Component of enzymes	Poor growth.
Manganese	Activates enzymes	Intervinal yellowing or mottling of young leaves.
Zinc	Activates enzymes	Intervinal yellowing on young leaves; reduced leaf size.
Boron	Cell wall component	Leaf tips become yellow followed by necrosis. Leaves get a scorched appearance and later fall off.
Molybdenum	Involved in N fixation	Poor growth.
Chlorine	Photosynthesis reactions	May affect growth.

ANIMAL NUTRITION

ENZYMES

Enzymes are biological catalysts, made of protein, that speed up the rate of reactions. They can be breakdown (degradation) or build up (synthesis) reactions.

Substrate (what you start with) $\Rightarrow$ Enzyme (biological catalysts) $\Rightarrow$ End Product (what you end up with)

Enzyme rate is affected by temperature and pH. All enzymes work best at 37°C.
 Pepsin works best at pH 2.5 and catalase works best at pH 9.
 Enzymes are therefore said to be "specific" and work best at their "optimum".
 Enzymes work like a "lock and key". The substrate is the key and the enzyme is the lock.
 The enzyme remains unchanged at the end of the experiment and can be reused.

Substrate	Enzyme	End Product	Synthesis (S) / Degradation (D)
Starch	Amylase	Maltose	D
Hydrogen Peroxide	Catalase	Water and oxygen	D
Protein	Pepsin	Peptides	D
Fat	Lipase	Fatty acids	D
Glucose - 1 - phosphate	Phosphorylase	Starch	S

Amylases, proteases and lipases are group of enzymes that digest carbohydrates, proteins and lipids respectively.

FOOD SUBSTANCES

The main classes of food are as follows:

Carbohydrates:

This nutrient is an organic compound composed of carbon, hydrogen and oxygen.

Function:

It is used as an energy resource, essential in respiration to release energy.
 It is used in creating the cellulose, the substance forming

cell walls of plant cells.
 Carbohydrates are 3 types:

Monosaccharides:

The smallest and simplest form.
 Water soluble.
 Chemical formula $C_6H_{12}O_6$.
 Examples: Glucose-Fructose-Galactose.
 Sources: Fruits, Honey.

Disaccharides:

- Each molecule consists of two Monosaccharide joined together
- Water soluble
- Examples: Lactose, Sucrose, Maltose
- Sources: Table sugar, Milk.

Polysaccharides:

- Each molecule has many joined monosaccharide forming a long chain.
- Insoluble in water
- Examples: Starch-Glycogen-Cellulose
- Sources: Bread, Potatoes, Pasta, Cellulose in plant walls and Glycogen in livers.

Monosaccharide and Disaccharides are sugars, they are reducing for Benedict's reagent, except for the disaccharide sucrose, it is non-reducing. Polysaccharides are not considered as sugars and don't have a sweet taste. Excess polysaccharides are stored in the liver and muscles.

Lipids (Fats):

These are composed of carbon, hydrogen and oxygen. But their ratios are different than that of carbohydrates. One fat molecule is made of a glycerol unit and three molecules of fatty acids.

Fats are essential in a diet because they are needed to:

- Release high amounts of energy
- Make cell membranes
- Store them under the skin to insulate heat.
- Forming a layer of fats around organs to protect them from damage
- Storing energy (better than glycogen)

When fats are respired, they produce about twice as much energy as carbohydrates.

Proteins:

These are also organic compounds; they contain the elements Carbon, Hydrogen, Oxygen, Nitrogen and sometimes Phosphorus or Sulfur.

A molecule of protein is a long chain of simpler units called amino acids.

These amino acids are linked together by what's called "peptide bond".

Types of protein:

- Animal Protein:** It contains the most biological value because it contains all essential amino acids (Meat, Milk, Fish, Eggs etc.)
- Plant Protein:** It contains a lower biological value to humans because it contains fewer essential amino acids (Cereals, Peas, Beans etc.)

Needs of proteins:

- Making and new body cells
- Growth and repair
- Making enzymes (they are proteins in nature)
- Build up hormones
- Making antibodies

Although proteins are needed in high amounts, the body will only absorb as much as needed, so excess protein is deaminated in the liver and excreted as urea.

Vitamins: These are organic, soluble substances that should be present in small amount in our diets, they are very important though. Most of the amount of vitamins in the bodies was taken in as nutrients, the body its self can not make few Vitamins, so we have to have to get them from organisms that make them, such as plants. Each type of Vitamin helps in chemical reactions that take place in the cells.

Types of Vitamins:

Vitamin C: This is present in most fruits and vegetables, specially citrus fruits like lemon and oranges, however, it is damaged by heating so if these foods have no value of Vitamin C if they are eaten cooked. Vitamin C is essential for the formation of Collagen, a protein that functions as cementing layer between cells, Vitamin C also increases immunity.

Vitamin D: This is present in fish oils, egg yolk, milk and liver. Unlike Vitamin C, Vitamin D is made by animals as well as plants, this occurs when the skin is exposed to the Ultra Violet Rays of the sun. Vitamin D plays a big role in absorbing Calcium from the small intestine and depositing it in bones. So it is responsible for having healthy bones.

Minerals (Inorganic Ions):

These are a lot of types, each needed in small quantities. Iron and Calcium are the most important minerals, and they are needed in higher amounts.

Types of Minerals:

Calcium: This mineral is needed for the formation of bones and teeth as they are made of calcium salts, it also helps in blood clotting and transmission of nerve impulses. Good sources of the mineral Calcium are milk, dairy products and hard water.

Iron: This mineral is needed for the formation of the pigment haemoglobin which is essential for the transport of oxygen around the body in red blood cells. Good sources of Iron include red meat specially liver and green vegetables.

Roughages (Fibre):

Although roughages are not even absorbed by the body, they are a very important nutrient in our diet. Roughages are mostly cellulose, which is the substance that makes up the cell walls of plants we eat. We humans, have no enzyme that could digest cellulose, that means that roughages enter the body from the mouth, go through the digestive system, and out through the anus unchanged. But as it goes through the digestive system, roughages take space in the gut to give the gut muscles something to push against, this process of pushing the food through the gut is called peristalsis, without roughages peristalsis is very slow and weak. Quick and strong peristalsis means that food stays in the alimentary canal for a shorter period, this prevents harmful chemicals of certain foods from changing the DNA of cells of the alimentary canal causing cancer, so roughages also helps stay away from cancer. Roughages are found in leafy vegetables.

Water:

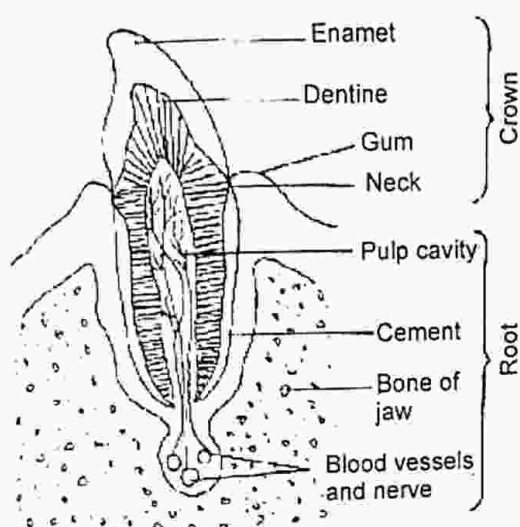
About 70% of your weight is water. Water is perhaps a very essential nutrient we should take in. The functions of water include:

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is a solvent which reactants of metabolic reactions are dissolved in.
 It makes up most of the blood plasma which red blood cells, nutrients, hormones and other materials are carried in.
 It helps in lowering the body temperature in hot conditions by secreting it as sweat on the skin, the sweat evaporates using heat energy from the body, thus lowering the temperature.

THE MAMMALIAN TOOTH

Structure of tooth: -**enamel**: the strongest tissue in the body made from calcium salts -**cement**: helps to anchor tooth -**pulp cavity**: contains tooth-producing cells, blood vessels, and nerve endings which detect pain. -**dentine**: calcium salts deposited on a framework of collagen fibres -**neck**: in between crown and root, it is the gums

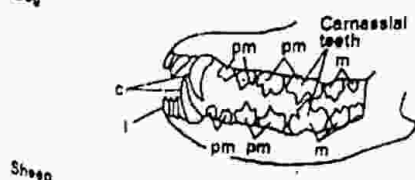


Section through a tooth

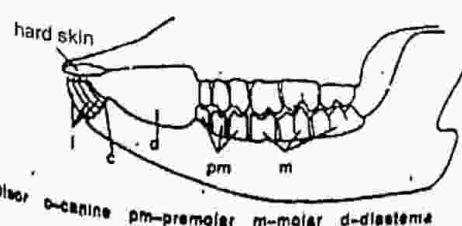


Man

Dog



Sheep



Teeth and dentition in mammals

DENTITION AND TYPES OF TEETH

Dentition is the number and arrangement of teeth in the mouth. The teeth are used for holding, cutting, biting and grinding food. A tooth is made of three parts namely: crown, neck and root. A longitudinal section of a tooth shows the enamel, dentine pulp cavity and cement. The types of teeth in mammals are:

1. **Incisors**: these have been sharp flat edges suitable for cutting and biting food. They are found at the centre in front of the jaw.

2. **Canines**: These are pointed and are strong teeth for tearing food.

3. **Pre-Molars**: These have broad crown sharp edges and flat tops for cutting and grinding.

4. **Molar**: These have broad surface with projections for crushing bones and flesh.

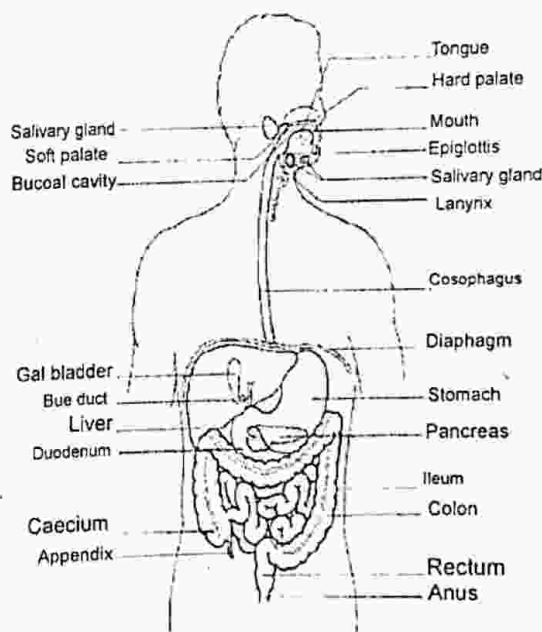
Carnassial teeth: found in advanced carnivores, it is the last upper premolar and the first lower molar. Dental formula is the figurative and numerical expression of each half of the upper and lower jaws. Examples:

Man - $I2/2' C1/1' Pm2/2' M3/3 = 32$

Dog - $I3/3' C1/1' Pm4/4' M2/3 = 42$

Rabbit - $I2/2' C0/0' Pm3/2' M3/3 = 28$

DIGESTIVE SYSTEM



Alimentary canal of man

The digestive system consists of a tube which leads from mouth to anus. The digestive system is made up of the oesophagus, the stomach, small intestine and large intestine. Each of these parts has a different function and a different structure. Food and fluids which are ingested are pushed through the digestive system by a process of peristalsis.

Parts of the Digestive System

Stage 1 - The Mouth

Food first enters the digestive system in the mouth. Here is it initially broken down both mechanically (through chewing) and chemically through the enzymes in saliva, salivary amylase (ptyalin) which breaks down starch.

Starch (ptylin)→Maltose

Stage 2 - The Oesophagus

This is a tube which connects the mouth to the stomach, food travels down this tube by peristalsis. There are a few mucus secreting glands but only small villi.

Stage 3 - The Stomach

The stomach is effectively a bag of muscle which can expand there are three layers of muscle which churns the contents of the stomach. Food is stored in the stomach for a few hours during which time it becomes a liquid known as chyme. The mucosa of the stomach wall has no villi but does have structures which are known as gastric pits which are associated with gastric glands. The gastric glands secrete hydrochloric acid to kill bacteria. The enzymes pepsin and rennin are secreted to digest proteins. The chyme is gradually released into the small intestine gradually through a large muscular sphincter known as the pyloric sphincter.

Protein (pepsin)→Peptide

Milk (renin) →Curdled milk

Stage 4 - The Small Intestine

The small intestine is made up of three parts the duodenum the jejunum and the ileum. In the duodenum digestion continues and two separate secretions assist this. The pancreas secretes pancreatic juices which contain carbohydrases (amylases), proteases and lipases. Bile is secreted from the liver and contains bile salts which aid lipid digestion and sodium hydrogen carbonate which neutralize the stomach acid. The jejunum and the ileum are very similar they are where final digestion occurs, glands in the mucosa and sub mucosa secrete enzymes, mucus and sodium hydrogen carbonate.

Duodenum:

Starch (amylase)→Maltose

Protein (peptidase)→Polypeptide

Lipid (lipase)→Fatty acid and glycerol

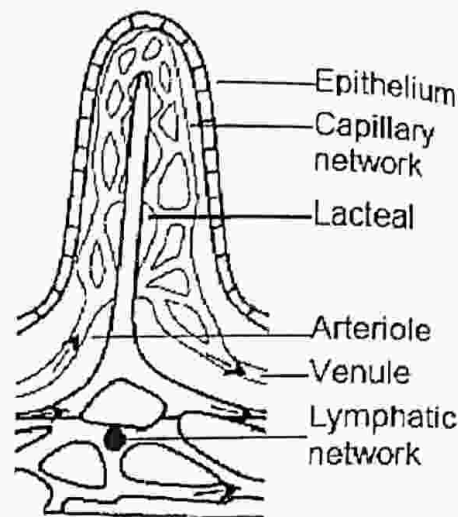
Stage 5 - The Large Intestine

The large intestine is made up of the caecum, appendix, colon and rectum. There are numerous villi and mucus secreting glands. Water is reabsorbed from the large intestine and semi solid faeces are formed.

ABSORPTION

Absorption is the movement of digested food molecules through the wall of the intestine into the blood or lymph. The small intestine is the region for the absorption of digested food.

The small intestine is folded into many villi which increase the surface area for absorption. One villus will have tiny folds on the cells on its outside called **microvilli**. More surface area means more absorption can happen.



Structure of a villus

THE LIVER

The liver is a reddish-brown organ lying against the posterior surface of the diaphragm and partly covering the stomach.

Functions of the Liver

1. Regulation of blood sugar
2. Deamination of proteins
3. Detoxication of harmful drugs and chemicals
4. Manufacture of blood plasma proteins
5. Storage of irons
6. Storage of vitamins A and D
7. Production of bile
8. Reservoir for blood production

EXPERIMENTS ON NUTRITION

Experiment: 1 Test for starch (cooked starch)

Materials required. Cooked starch, iodine solution, white tile, a dropper.

Method. A few drops of the test material (cooked starch) are placed on different spots of a white tile with the aid of a dropper. Iodine solution is added to each drop of the starch solution.

Observation. A bluish black colour appears.

Conclusion. A bluish black colour with the iodine solution shows that starch is present.

Experiment :2 Tests for sugars

(a) reducing sugars (e.g. glucose)

Materials required. Fehling's solution, Benedict's solution, glucose (a reducing sugar), test tube, Bunsen burner.

Method. Some glucose is placed in a test tube and a few drops of Fehling's solution are added to the glucose. The mixture is then heated to boiling point; OR Into glucose in a test tube, some drops of Benedict's solution are added and the mixture is warmed for some time.

Observation. There is orange (brick) red precipitate with Fehling's solution or Benedict's solution.

Conclusion. The orange red precipitate indicates the presence of a reducing sugar.

Non-reducing sugars (e.g. sucrose)

Materials required. Sucrose, water, HCl, test tube, NaOH, Fehling's solution.

Method. Some sucrose is put in a test tube and a few drops of water and HCl are added to the sucrose. Next, some drops of NaOH are added to neutralise the acid. Finally, Fehling's solution is added.

Observation. There is an orange red precipitate.

Conclusion. The test material is sucrose.

Experiment : 3(a) Test for fats and oils.

Materials required. Test material (e.g. groundnut seeds), filter papers.

Method. The test material is placed in between folds of filter papers. The test material is then pressed. Next, the paper is held before light.

Observation. The spots with fats become translucent.

Conclusion. The test material contains fats or oils. If the translucent spots are washed with benzol or ether, the spots clear off.

Experiment : 3(b) test for fats and oils using Sudan (III) solution.

Materials required. Test material, Sudan (III) solution, test tube.

Method. The test material is put in a test tube and a few drops of Sudan (III) solution is added. Heat the mixture.

Observation. A red colouration appears before boiling. A black precipitate is formed on boiling.

Conclusion. The test material contains fats or oils.

Experiment : 4 Test for protein (e.g. milk, egg albumen)

Materials required. Egg albumen (or milk), test tube, NaOH solution, CuSO₄ solution.

Method.

1. **Millon's test.** Some egg albumen or milk is put in a test tube and Millon's reagent is added to it. Heat the mixture. OR
2. **Biuret test.** Some milk or egg is put in a test tube. Drops of NaOH and CuSO₄ are added to the test material. Both solutions are made dilute.
3. **Xanthoproteic test.** Some milk or egg albumen is put in a test tube and concentrated HNO₃ is added and warmed. The mixture is allowed to cool under a running tap and concentrated NH₃ is added.

Observation.

Millon's test. There is a deep red colouration of the test

material.

Biuret's test. There is a violet colouration.

Xanthoproteic test. There is a yellow colouration which on cooling and adding NH₃ becomes orange.

Conclusion. The test material is protein.

Experiment: 5 To demonstrate that ptyalin is necessary for the digestion of starch.

Materials required. A solution of enzyme, a solution of starch, sodium bicarbonate, small size beaker, 3 test tubes, water bath, iodine solution, a white tile, a dropper and labels.

Method. The enzyme solution is made by taking some water into the mouth and turning it for some time. This is done after the mouth has been thoroughly washed with warm water to remove any food substances. The water in the mouth is poured into a small beaker. The starch is made by adding a little cold water into powder starch and stirring it. Some boiled water is then added. This gives the starch solution. The solution is left to cool. A little starch solution is poured into 3 test tubes, labelled 1, 2, and 3 and they are treated with the enzymes (ptyalin) as follows:-

Test tube 1. Into this tube, some saliva is poured and sodium bicarbonate is added to provide an alkaline medium.

Test tube 2. Into this tube, no enzyme is added.

Test tube 3. Into this tube, boiled saliva is added, i.e. the tube contains starch and killed enzyme.

The three tubes are transferred to a boiling water bath. The temperature of the solution in the test tube is maintained around 37°C which is the body temperature of mammals. The mixture in the test tube is stirred continually. After some time, the solution in each test tube is tested for starch with iodine on white tile. The test tube is repeated every 30 minutes for 1 hour, 30 minutes. Next, the content of the first tube is boiled with Benedict's solution.

Observation. The second test tube which contained only starch, continued to give a blue-black colour with iodine, i.e. No change. The third test tube which contained killed enzyme gives the same result as in the second tube i.e. no change. The first test tube which contained the enzyme and the alkaline, showed gradual changes from blue-black to purple and then to light brown. A brick red colour appeared when the first test tube content was boiled with Benedict's solution.

Conclusion. This first test tube shows that saliva contains an enzyme (ptyalin) which acts on starch. It converts starch into complex sugar (maltose) which on boiling with Benedict's solution gave a brick red colour. In other tube (2 and 3) where there were no changes, means that either the enzyme was absent or destroyed.

Experiment: 6 To test for leaf for starch.

Materials required. A leaf, a beaker, 70% alcohol, test tube, water bath, iodine solution, a white tile.

Method. A leaf (e.g. cassava leaf) is poured at the end of a

sunny day. The cells of the leaf are killed by boiling for five minutes in a beaker. Some alcohol (70%) is poured into a test tube and is warmed in a water bath the killed leaf is then transferred into the test tube. Heat is continually applied to the test tube until the alcohol is close to boiling and the green colour of the leaf will gradually disappear and the whole leaf will have become decolourised. The leaf is removed from the test tube and washed with hot water for about five minutes to soften it. The leaf is then tested for starch with iodine solution. The test is done on a white tile.

Observation. There is a blue-black colour.

Conclusion. The blue-black colour shows the presence of starch.

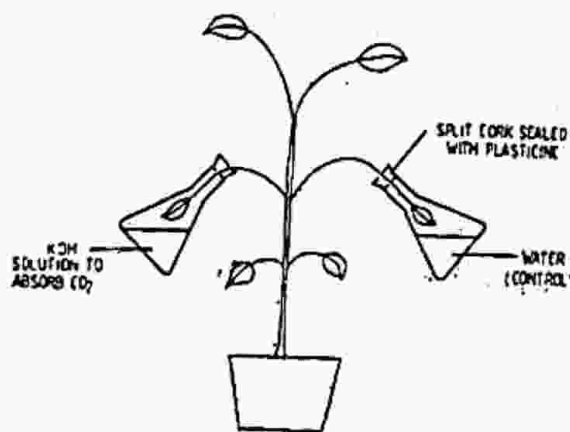
Experiment: 7 To demonstrate that carbon (IV) oxide is necessary for photosynthesis.

Material required. Two young potted plant, two bell jars, a beaker, sodium hydroxide solution, and candle sticks.

Method. Two young potted plants are kept in the dark such as inside a cupboard. This is to remove the starch from the leaves. After this period the plant are brought from the cupboard and the bell jar is placed over each plant as shown in the above figure. A small beaker which contains sodium hydroxide (NaOH) is placed beside one of the plants while a burning candle in the other. The NaOH is to remove carbon (IV) oxide under the first bell jar and the burning candle is to remove oxygen under the second bell jar but to leave carbon (IV) oxide. Both experiments are then exposed to light for some time. After this period a leaf from each plant is plucked and tested for starch as in experiment 7.

Observation. The leaf from the second experiment gave a bluish black colour with iodine. There was no change in the leaf from the first experiment.

Conclusion. The bluish black colour indicates the present of starch grains. This shows that carbon (IV) oxide is necessary for photosynthesis.



Experiment: 8 To demonstrate that light is essential for photosynthesis.

Material required. A young potted plant, paper clips, strips of brown paper, iodine solution, and white tile.

Method. A young potted plant is left in the dark for about 48 hours to remove all starch in the leaves. To ensure that starch is not present, some leaves are tested two strips of brown paper are taken and a hole of letter S is made in each. Next, a leaf is selected from the plant (without plucking it). Carefully, the brown papers are placed over the leaf, one on

each side of the lamina. The papers are then clipped together by means of paper clip as shown in fig. 40.6. The potted plant is then exposed to adequate sunlight for a period of 4 hours. After this period, the leaf is plucked and tested for starch as in experiment 7.

Observation. The area of the leaf which received sunlight gave a bluish black colour. Letter S is printed on the leaf. Other areas of the leaf were dark.

Conclusion. The areas of the leaf which gave a bluish black colour contained starch grains and hence, a letter S with starch grains was printed on the leaf. This shows that light is necessary for photosynthesis.

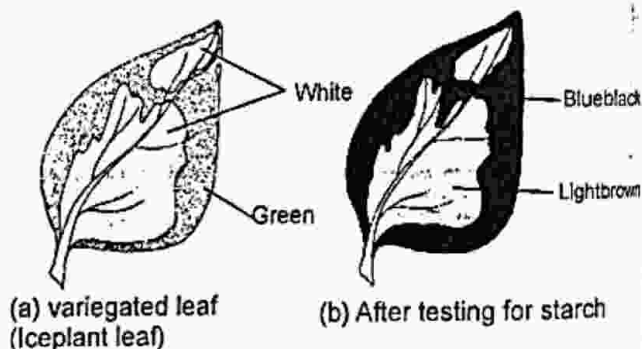
Experiment: 9 To demonstrate that only green areas of a leaf can photosynthesize.

Materials required. A variegated leaf, iodine solution, a white tile, water bath, test tube, 70% alcohol.

Method. A variegated leaf is selected from a plant around the laboratory at the end of a bright day. All the area of the leaf which bear green colour are carefully mapped out. The leaf is treated and tested for starch with iodine as in experiment 7.

Observation. Only the area that is green gave blue-black with iodine.

Conclusion. The blue-black colour shows the present of starch grain in the green areas. This shows that only the green area of the leaf can photosynthesize.



Experiment: 10 To demonstrate that oxygen is given out during photosynthesis.

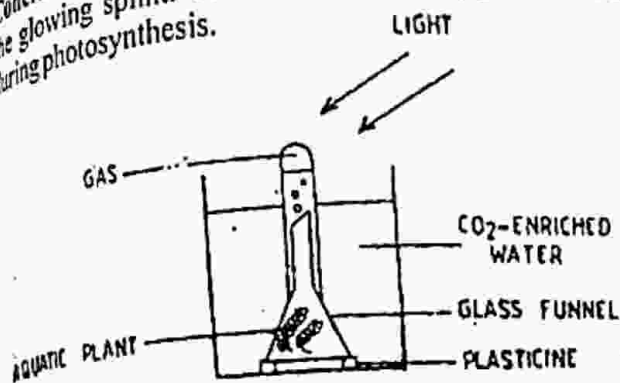
Material required. A beaker, funnel, test tube, water weed (e.g. Elodea).

Method. A beaker is filled with water, almost mid-way. Carbon (IV) oxide is bubbled into this water until it is clearly saturated. Some water weeds (e.g. Elodea) are placed in water and a funnel is placed over the weeds. The rim of the funnel is raised from the bottom of the beaker to allow for free circulation of air. Next, medium sized test tube which nearly filled with water is used to cover the stem of the funnel as shown in fig. 40.8. After this, the setup is kept in a place where it can receive sufficient sunlight.

Observations: Bubbles of a gas are noticed to rise up the funnel. The gas displaces the water in the test tube and collects at the upper end. When the test tube was carefully removed (after sufficient gas has collected in the tube) and

tested with a glowing splint, the splint is rekindled.

Conclusion: The gas in the tube is oxygen which rekindled the glowing splint. This shows that oxygen is liberated during photosynthesis.



Revision Questions

7.1. MODES OF NUTRITION

1. UME/83/30

Which of the following statements is NOT true of insectivorous plants?

- A. They obtaining part of their food by trapping and feeding on insects
- B. They attract insect simply because of pollination
- C. They grow in soil poor in nitrogenous salts
- D. They can supplement their nitrogen supply by feeding on insects
- E. Examples include butterworts, sundews and pitcher plants.

2. UME/90/17

The mode of nutrition in which digestion is extracellular is

- A. holophytic
- B. parasitic
- C. holozoic
- D. saprophytic

3. UME/92/12

Fungi are heterotrophic because they

- A. have no leaves
- B. lack roots
- C. are filamentous
- D. lack chlorophyll

4. UME/99/14/Type: D

The modes of nutrition in *Nitrobacter*, *Ascaris* and *Homo sapiens* respectively are

- A. Photosynthetic, parasitic and holozoic
- B. Chemosynthetic, parasitic and holozoic
- C. Photosynthetic, parasitic and heterophytic
- D. Chemosynthetic, holophytic and holozoic

5. UME/2002/28/Type: Y

Examples of organisms in which extracellular digestion occurs are

- A. fungus, lorchanthus and housefly
- B. rhizopus, sponges and earthworm
- C. roundworm, tapeworm and hydra
- D. rhizopus, housefly and hydra

6. UME/2002/31/Type: Y

The capture and digestion of insects by a pitcher plant is a special form of nutrition termed

- A. autotrophic
- B. heterotrophic
- C. chemosynthetic
- D. saprophytic

7. UME./2002/36/Type: Y

The mode of nutrition exhibited by a tapeworm is

- A. symbiotic
- B. saprophytic
- C. parasitic
- D. holozoic

8. PCE/2004/12/Type: 9

The possession of haustoria by some plants is an adaptation for

- A. parasitic life
- B. saprophytic life
- C. carnivorous plant
- D. symbiotic life

9. UME/2008/15

Insectivorous plants trap and kill their prey to derive

- A. phosphorus
- B. calcium
- C. nitrogen
- D. zinc

10. UME/2009/15

Which of the following is an example of a carnivorous plant?

- A. Hydra
- B. Bladderwort
- C. Yeast
- D. Spirogyra

11. UME/2010/15/TYPE: D

The mode of nutrition of sundew and bladderwort can be described as

- A. saprophytic
- B. holozoic
- C. chemosynthetic
- D. autotrophic

12. UME/2010/25/TYPE: D

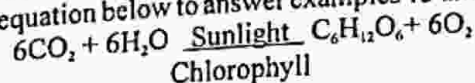
Chemosynthetic organisms are capable of manufacturing their food from single inorganic substances through the process of

- A. denitrification
- B. reduction
- C. phosphorylation
- D. oxidation

7.2. PLANT NUTRITION

7.2.1. PHOTOSYNTHESIS

Use the equation below to answer examples 13 and 14



13. UME/78/2

The process represented by the above equation is

- A. proteinsynthesis
- B. respiration
- C. photosynthesis
- D. transpiration
- E. translocation

14. UME 78/3

The oxygen given off during the process is derived from:

- A. sunlight
- B. chlorophyll
- C. carbon dioxide
- D. atmosphere
- E. water

15. UME/90/18

The first step in the process of photosynthesis is the

- A. activation of the chlorophyll
- B. photolysis of water
- C. reduction of carbondioxide
- D. formation of sugar

16. In photosynthesis, the following processes are part of the light reaction except
 A. Transfer of radiant energy which reduces a co-factor
 B. absorption of radiant energy by chlorophyll
 C. utilization of energy in the electron transfer chain from ATP
 D. formation of glucose using energy from NADPH
 E. splitting of water into its component parts.

17. UME/87/20

- In standard experiment to show that oxygen is given off during photosynthesis, sodium bicarbonate is used to
 A. neutralize the acid in water
 B. supply mineral salt to water plant
 C. supply carbon dioxide for photosynthesis
 D. kill micro-organisms in water.

18. UME/78/34

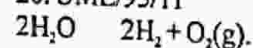
Which of these is a direct photosynthetic product?

- A. Glucose B. Starch
 C. Protein D. Fats
 E. Lactose

19. UME/89/18

- If a healthy potted plant is continuously kept in dim light?
 A. the rate of respiration may equal that of photosynthesis
 B. more carbon dioxide and water are taken in
 C. respiration may be halted
 D. the volume of oxygen released increases

20. UME/95/11



The equation above represents a part of the light stage of photosynthesis. Which of the following must be present for this reaction to occur?

- A. Enzyme and light energy
 B. carbondioxide and light energy
 C. light energy and chlorophyll
 D. chlorophyll and enzyme

21. UME/98/12

The manufacture of carbohydrates by plants takes place only

- A. the leaves B. the green stems
 C. chlorophyllous parts D. flowering plants

22. UME/99/15/Type D

The dark reaction of photosynthesis involves

- A. Fixation of carbon (iv) oxide to give a six-carbon sugar
 B. Fixation of carbon (iv) oxide with the help of oxygen
 C. Use of carbon (iv) oxide to produce glucose using ATP
 D. Fixation of carbon (iv) oxide on chlorophyll using Hydrogen

23. PCE/97/14

The dark stage of photosynthesis takes place in the

- A. stoma B. stroma C. guard cell D. lamellae.

24. UME/2007/20

Photosynthetic pigments include
 A. chlorophyll and carotenoids
 B. chloroplasts and cytochromes

- C. melanin and haemoglobin
 D. carotenoids and haemoglobin

25. Chlorophyll absorbs from sunlight more of:

- A. Infra-red rays B. Violet rays C. Red rays
 D. Yellow rays E. Green rays

26. The metallic ion present in chlorophyll is:

- A. Iron B. Copper
 C. Magnesium D. Carotone
 E. Sodium

27. One of the yellow pigments of chlorophyll is:

- A. Carotene B. Lycopene C. Xanthophyll
 D. Hydrocarbon E. Chlorophyll

28. Carbon dioxide gets to the chloroplast of cells in:

- A. Liquid form
 B. In gaseous form
 C. In bicarbonate form
 D. As carbonic acid
 E. None of the above

7.2.2. MINERAL NUTRIENTS

9. UME 79/47

The element Nitrogen is utilized in

- A. formation of ATP
 B. formation of glucose
 C. formation of amino acids
 D. photosynthesis E. None of the above

30. UME/79/48

Magnesium is utilized in the formation of

- A. ATP B. glucose C. amino acids
 D. chlorophyll E. fats

31. UME/85/36

A young plant showing yellowing leaves is likely to be deficient in

- A. calcium
 B. magnesium
 C. potassium
 D. boron
 E. molybdenum

32. UME/82/4

Some of the major elements required by plant are

- A. Potassium, Nitrogen, Phosphorus, Sodium, Calcium
 B. Nitrogen, Phosphorus, Molybdenum, Sodium, Calcium
 C. Potassium, Phosphorus, Molybdenum, Sodium, Calcium
 D. Potassium, Nitrogen, Iron, Sodium, Calcium
 E. Potassium, Nitrogen, Phosphorus, Silicon, Calcium

33. UME/99/16/Type D

Yellowing of the leaves in a symptom associated with

- A. iron, calcium and magnesium
 B. nitrogen, sulphur, potassium
 C. sulphur, phosphorus and iron
 D. magnesium, nitrogen and iron.

34. UME/89/21
The seedling in a rice field were found to have thin lanky growth with reddish leaves and poor root development. This is because; the soil lacks
A. sulphur B. phosphorus C. potassium D. iron

35. UME/97/13
Trace elements are required by plants mainly for the
A. formation of pigments and enzymes
B. production of energy and hormones
C. manufacture of carbohydrates
D. manufacture of proteins.

36. UME/98/13
In a water culture experiment, a plant showed poor growth and yellowing of the leaves. These may be due to deficiency of
A. Copper B. Iron C. Magnesium D. Calcium

37. UME/2007/25
Stunted growth and poor root development are a result of a deficiency in
A. sulphur B. phosphorus
C. calcium D. iron

38. In the synthesis of protein, the essential substance added to simple sugars is:
A. Magnesium B. Iron C. Nitrogen
D. Hydrogen E. Boron

7.3. ANIMAL NUTRITION

7.3.1. FOOD SUBSTANCES

39. Food is defined as :
A. Substance that contain starch and protein
B. Those substances which when eaten supply fats and oils
C. The material requirements of organisms
D. Substance containing potential energy
E. None of the above

40. UME/79/13
Which of these is not an enzyme?
A. Pepsin B. Gastrin
C. Amylase D. Chemotrypsin
E. Trypsin

41. UME/80/8
Human beings require vitamins in their diet because
A. contain carbohydrates and fats
B. prevent kwashiorkor
C. stimulate the alimentary canal
D. digest proteins in the body
E. influence many important chemical processes in the body

42. UME/81/41
Which of these substances is likely to be deficient in the diet of a person having goiter?
A. potassium B. calcium C. iodine
D. sodium E. phosphorus

43. UME/82/6
Kwashiorkor is caused by severe deficiency of
A. water B. oil C. drugs D. proteins E. sugars.

44. UME/83/9
If an organic compound has its Hydrogen: Oxygen ratio as 2:1, it is likely to be
A. a protein B. a carbohydrate
C. a fat D. a fatty acid and glycerol
E. an amino acid.

45. UME/85/16
If calcium is deficient in food, this may cause
A. anaemia B. retarded growth
C. sterility D. goiter E. beri-beri

46. UME/86/16
The main organic substances found in the human body are
A. carbohydrates, proteins and salt
B. salts, fats and proteins
C. fats, carbohydrates and proteins
D. salts, fats and carbohydrate

47. UME/87/18
The vitamin which is important in the formation of the retina pigment is
A. vitamin A B. vitamin B.
C. vitamin C. D. vitamin D.

48. UME/95/13
Which vitamin plays an important role in blood clotting?
A. Vitamin A B. Vitamin K
C. Vitamin B₆ D. Vitamin C

49. UME/2000/22/Type: M
The greatest amount of energy will be obtained by the oxidation of 100kg of
A. meat B. butter C. sugar D. biscuit

50. UME/2010/14/TYPE: D
Which of the following is lacking in the diet of a person with kwashiorkor?
A. proteins B. carbohydrates
C. minerals D. vitamins

51. Animals depend for their existence on carbohydrates, proteins, fats and oils, mineral salts, water and vitamins. Vitamins are:
A. Deficiency diseases caused by insufficient food
B. Some kind of food
C. Substances essential for life
D. Substance present in food and which are responsible for good health and growth
E. Substance necessary for body metabolism.

52. Vitamin A, a fat soluble substance, found in milk, liver oils and green vegetables when absent in food not only cause night blindness, but also:
A. Blindness B. Loss of blood
C. Weak bones D. Dry and scaly skin
E. Loss of appetite

53. The anti-sterility vitamin is:
 A. Vitamin K B. Vitamin E C. Vitamin C
 D. Vitamin D E. None of the above

54. Mineral salts are essential for the maintenance of:
 A. Growth of the tissues
 B. Healthy growth of the muscles
 C. Osmotic pressure of the body fluids
 D. Efficient excretory systems
 E. Body temperature

55. Milk sugar is:
 A. Simple sugar B. Maltose C. Sucrose
 D. Cane sugar E. Lactose

56. The glucose of the blood is known as:
 A. Simple sugar B. Liver sugar
 C. Glycogen D. Blood sugar
 E. Blood carbohydrates

7.3.2. FOOD TESTS

57. UME/79/22

A sugar solution was boiled with Fehling's solutions A and B and the colour remained blue. The sugar tested was

- A. glucose B. maltose
 C. fructose D. sucrose
 E. lactose.

58. UME/80/18

In testing for glucose the necessary reagent and the condition under which the reagent reacts best are

- A. Fehlin's reagent, in the cold
 B. Millon's reagent, boiled
 C. Fehling's reagent in acid medium, heated
 D. Iodine solution, boiled
 E. Fehling's reagent in neutral or alkaline medium, boiled.

59. UME/85/15

Food substance which produces red colouration with sudan III contains

- A. protein B. sugar
 C. starch D. cellulose
 E. fat

60. UME/86/20

The extract from a food substance reacting with sodium hydroxide and copper sulphate solutions will produce violet to purple coloration if

- A. fats are present
 B. carbohydrate is present
 C. protein is present
 D. reducing sugar is present

61. UME/2000/21/Type:M

The production of violet colouration, when dilute NaOH solution is added to a solution of food substance, followed by drops of 1% CuSO₄ solution while making indicates the presence of

- A. protein
 B. carbohydrates
 C. fats
 D. reducing sugar

62. UME/2001/24/Type:T
 When specimen X is mixed with few drops of iodine solution, the appearance of a blue black colour confirms that X is

- A. galactose B. starch
 C. sucrose D. glucose

63. UME/2010//16/TYPE: D

When a mixture of a food substance and Benedict solution was warmed, the solution changed from blue brick-red. This indicates the presence of

- A. fatty acid B. sucrose
 C. amino acid D. reducing sugar

7.3.3. THE MAMMALIAN TOOTH

64. Which of the following is the dental formula of a man

- A. i. 2/1, c0/0 pm 3/2, m3/3
 B. i. 0/3, c0/1, pm3/3, m3/3
 C. i2/2, c1/1, pm2/2, m3/3
 D. i3/3, c1/1, pm4/4, m2/3
 E. i2/2, c1/1, pm2/3, m3/3

65. UME/81/8

The kind of teeth used for tearing food materials is the fang

- A. canine B. canine C. premolar
 D. carnassial E. incisor

66. UME/83/46

Which of the following statements about the dentition of man is INCORRECT? Man has

- A. more molars than incisors
 B. no diastema
 C. the same number of teeth on upper and lower jaws
 D. a total of thirty-two teeth
 E. a total of sixteen molars.

67. UME/84/36

The presence of a diastema in the jaw bone indicates that the mammal lacks the teeth suitable for

- A. tearing
 B. chewing
 C. cutting
 D. chewing and tearing
 E. cutting and grinding

68. UME/86/22

Evidence that a tooth is a living part of the mammalian body can be found within the

- A. gum
 B. pulp cavity
 C. cement
 D. enamel



69. UME/89/16
Which of the structures is the molar
A. 1 B. 2 C. 3 D. 4

70. UME/89/17
What is the function of 3?
A. cutting of large pieces of food
B. seizure and tearing of prey
C. grinding of food
D. tearing of flesh only

71. UME/91/22
The crown of the mammalian tooth is covered with
A. cement
B. dentine
C. caries
D. enamel

72. UME/2002/23/TYPE:Y
The carnassial teeth of a carnivorous animal consists of the
A. last upper premolar and the first lower molar
B. last upper molar and the last lower molar
C. first upper premolar and the first lower molar
D. first upper molar and the first lower molar

73.4. MAMMALIAN ALIMENTARY CANAL

73. UME/78/40
The mammalian stomach can carry out a number of processes except
A. absorption of glucose
B. secretion of hydrochloric acid
C. secretion of gastric juice
D. churning of food
E. production of trypsin

74. The first part of the stomach to receive food from the oesophagus is:
A. The duodenum
B. The jejunum
C. The pyloric zone
D. The ileum
E. The cardiac zone

75. UME/79/4
Which of these is the terminal portion of the alimentary canal of a mammal?
A. Oesophagus
B. Stomach
C. Rectum
D. Colon
E. Appendix

76. UME/81/15
Which of the following is NOT a function of the liver? A.
Regulation of blood sugar
B. Storage of iron
C. Formation of bile
D. Breakdown of excess amino acids
E. Excretion of urea from the blood

77. UME/81/49
The correct route for a piece of bread eaten by a mammal is
A. Oral cavity → oesophagus → stomach → duodenum
→ ileum → rectum
B. Oral cavity → stomach → oesophagus → duodenum
→ ileum → rectum
C. Oral cavity → oesophagus → duodenum → stomach → ileum
→ rectum
D. Oral cavity → oesophagus → ileum → stomach → duodenum
→ rectum
E. Oral cavity → oesophagus → stomach → ileum → duodenum
→ rectum

78. That portion of the alimentary system into which the pancreatic juice is poured is:
A. The duodenum B. The pancreatic zone
C. The small intestine D. The pyloric zone
E. The colon

79. UME/82/39
If the gall bladder of a man is removed by surgery, which of the following process will be most seriously affected?
A. digestion of fats and oils
B. formation of urea
C. digestion of starch
D. conversion of proteins
E. storage or release of urine.

80. UME/86/21
The three important organs that are situated close to the stomach are
A. kidney, liver and gall bladder
B. pancreas, liver and kidney
C. liver, kidney and spleen
D. gall bladder, pancreas and spleen

81. UME/87/19
Which of the following lists of organs is directly involved in nutrition?
A. oesophagus, bronchus, stomach, pancreas and anus
B. spleen, pharynx, duodenum, jejunum, and rectum
C. teeth, oesophagus, ileum, lungs and large intestine
D. salivary gland, liver, stomach, villi and colon

82. UME/91/20
One of the accessory organs of the digestive system is the
A. kidney B. spleen
C. liver D. lung

83. UME/97/12
The part of the stomach nearer the gullet is called the
A. epiglottis B. cardiac sphincter
C. duodenum D. pyloric sphincter

84. UME/80/15
Which of the following food substances is digested in the stomach?
A. carbohydrates
B. fats and oils
C. fats and proteins
D. proteins
E. carbohydrates and fats

85. UME/2006/14
Which of the following produces both hormones and enzymes?

- A. ileum
B. pancreas
C. gall bladder
D. kidney

86. UME/2008/14
The order of passage of food in the digestive system is A.
ileum - caecum - large intestine - rectum
B. ileum - colon - caecum - rectum
C. large intestine - ileum - caecum - rectum
D. colon - caecum - ileum - rectum

87. UME/2010/26/TYPE: D
The part of the human gut that has an acidic content is the
A. duodenum
B. ileum
C. colon
D. stomach

88. That flap of tissue that prevents food from straying into the larynx is:
A. The glottis
B. The arytenoids
C. The cricoid
D. The epiglottis
E. The palate

7.3.5. NUTRITIONAL PROCESS

89. UME/79/34
The digestive enzyme that coagulates proteins in milk is
A. ptyalin
B. pepsin
C. rennin
D. trypsin
E. amylase

90. UME/79/50
The arrangements below are steps in protein digestion. Which is the correct sequence?
a-polypeptides, b-protein, c-amino acids, d-peptones
A. a → b → c → d
B. c → d → a → b
C. b → c → a → d
D. b → d → a → c
E. b → a → c → d

91. UME/82/22
In the enzymic reactions: starch → sugar. Starch is referred to as the
A. substrate
B. product
C. enzyme
D. enzyme-substrate complex
E. reaction mixture.

92. UME/84/13
Which of the following statement is NOT true of enzymes?
They
A. are proteins
B. need co-factors to activate them
C. are sensitive to hydrogen ion concentration
D. are specific in their action
E. can withstand high temperature.

93. UME/85/17
Partially digested food ready to leave the stomach is referred to as
A. chyme
B. curd
C. glycogen
D. paste
E. roughage

94. UME/88/20
Which of the following will be digested first, if ingested, the same time?
A. cooked beans
B. cooked rice
C. cod liver oil
D. roasted beef

95. UME/89/19
The pancreatic juice contains the enzymes amylopsin,
A. pepsin and trypsinogen
B. rennin and steapsin
C. steapsin and trypsinogen
D. steapsin and ptyalin.

96. UME/90/20
The organ which secretes digestive enzymes as well as hormone is the
A. liver
B. salivary glands
C. pancreas
D. spleen.

97. UME/97/21
The process of deamination is essential for the
A. digestion of protein
B. secretion of bile
C. formation of urea
D. formation of antibody

98. UME/2007/26
The pancreas secretes enzymes for the digestion of
A. fats, proteins and carbohydrate
B. fats, vitamins and cellulose
C. fats, carbohydrates and vitamins
D. proteins, cellulose and minerals

MISCELLANEOUS QUESTIONS (7)

99. The mode of feeding in Amoeba and Hydra is
A. heterotrophic
B. holophytic
C. autotrophic
D. symbiotic
(UME/2001/8/Type 7)

100. The gall bladder of a mammal has a duct connected to the
A. duodenum
B. liver
C. pancreas
D. small intestine
(UME/2004/1/type: 4)

101. The part of the mammalian digestive system where absorption of materials takes place is the
A. duodenum
B. colon
C. ileum
D. oesophagus (UME/2003/19/type)

102. Metabolic production of urea is carried out in the
A. urinary bladder and kidney
B. pancreas
C. kidney and malpighian tubule
D. liver (UME/2003/22/Type:A)

103. The element common to protein, carbohydrate and lipid is
A. Hydrogen
B. Sulphur
C. Nitrogen
D. Phosphorus (UME/91/21)

104. The crown of the mammalian tooth is covered with
A. cement
B. dentine
C. caries
D. enamel (UME/91/22)

105. The blood vessel which carries blood from the alimentary canal to the liver is the

A. hepatic artery
C. hepatic portal vein
(UME/92/15)

B. hepatic vein
D. mesenteric artery

106. Which enzymes are contained in the pancreatic juice?

- A. ptyalin, lipase and pepsin
- B. maltase, erepsin and trypsin
- C. renin, sucrase and lipase
- D. amylase, lipase and trypsin (UME/92/22)

107. Interveinal chlorosis is normally associated with the deficiency of

- A. magnesium
- B. potassium
- C. iron
- D. calcium (UME/93/14)

108. Osmic acid boiled with a solution of food substances gave a black precipitate. This indicates the presence of

- A. fats and oils
- B. proteins
- C. aminoacids
- D. starch (UME/93/15)

109. Which of the following is an autotrophic mode of nutrition?

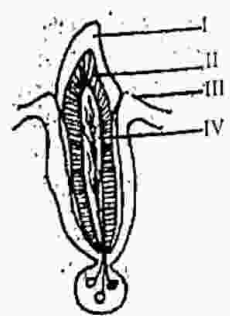
- A. saprophytism
- B. chemosynthesis
- C. parasitism
- D. commensalisms (PCE/2005/7/type:P)

110. In millon test, when the reagent is added to a protein food item, a white precipitate is produced which turns

- A. blue on heating
- B. yellow on heating
- C. green on heating
- D. red on heating (UME/98/14)

111. The modes of nutrition in *Nitrobacter*, *Ascaris* and *Homo sapiens*, respectively are

- A. photosynthetic, parasitic and holozoic
- B. chemosynthetic, parasitic and holozoic
- C. photosynthetic, parasitic and heterotrophic
- D. chemosynthetic, holophytic and holozoic (UME/99/14/Type: D)



112. The section that is extremely sensitive to heat and cold is labeled

- A. IV
- B. III
- C. I
- D. II (PCE/2005/16/Type:P)

113. The part labeled II is the

- A. dentine
- B. cement
- C. enamel
- D. gum (PCE/2005/17/Type:P)

114. Autotrophic nutrition involves

- A. only unicellular animals
- B. the use of complex organic substances as starting materials
- C. the production of energy rich organic compounds
- D. deactivation of green pigments (PCE/91/13)

115. A cleaning agent containing lipase will be most effective in removing stains caused by

- A. proteins
- B. fats
- C. mud
- D. pollen grains (PCE/92/17)

116. The sulphur bacteria which use the energy from acidation of substances to fix carbon are called

- A. holozoic heterotrophs
- B. chemosynthetic autotrophs
- C. parasite heterotrophs
- D. photosynthetic autotrophs (PCE/92/15)

117. The dark reaction of photosynthesis involve the

- A. reduction of carbon dioxide
- B. absorption of water and carbon dioxide
- C. photolysis of water
- D. conversion of starch into sugar (PCE/92/13)

118. Sterility in man may be caused by a deficiency of

- A. vitamin A
- B. vitamin C
- C. vitamin D
- D. vitamin E (PCE/93/16)

119. Osteomalacia is a symptom of the deficiency of

- A. vitamin A
- B. vitamin C
- C. vitamin D
- D. vitamin K (PCE/96/14)

120. The disease that results from eating mainly polished rice is

- A. cancer
- B. cholera
- C. beri-beri
- D. dysentery (PCE/98/12/TYPE: C)

121. Deamination in the liver involves the

- A. storage of iron from the haemoglobin
- B. conversion of poisonous substances to harmless ones
- C. conversion of excess amino acids to glucose and urea
- D. storage of excess fat and sugar (PCE/06/TYPE: A)

CHAPTER 8

TRANSPORT

Understand the following after reading this chapter:

Need for transportation
Mammalian circulation system
Plant vascular system
Media, process and mechanism of transport

NEED FOR TRANSPORTATION

Surface Area: Volume Ratio

Ratio of SA: Vol. ratio in a small organism is greater than in a large organism

Smaller organisms have the advantage of a relatively large absorbing surface

Larger organisms have a low SA: Vol. ratio, their skin not very permeable to gases.

Thus large organisms need special systems (e.g. circulatory systems) to help them transport the substances and special absorbing substances such as lungs or gills to increase the surface area for absorption. The ratio decreases with increasing size of the organism

The Mammalian Circulatory System

The circulatory system is made up of blood, blood vessels, lymph and the heart.

Composition of Blood

Plasma: 1. Approximately 55% of blood is made up of plasma, the straw-coloured liquid portion of blood; it is 90% water and 10% dissolved molecules (mainly plasma proteins).

2. These can be divided into three types:

- Albumins** - these help to regulate water potential, by maintaining normal blood volume and pressure. They are the most common plasma protein.
- Immunoglobins (antibodies)** - These are very large proteins that target infection and so cause infected or foreign cells to be attacked by white blood cells (WBC's). Together with the WBC's they form the immune system.
- Fibrinogen** - these are tightly coiled proteins that unwind to form a blood clot.

Other molecules include: glucose, amino acids, vitamins, minerals (mainly NaCl), urea, CO_2 , hormones.

BLOOD CELLS

Red Blood Cells

Tiny (8μ) biconcave disc-shaped cells (thus large SA).

Do not have nucleus, mitochondria, ribosomes.

Cell full of haemoglobin - binds O_2 (and CO).

Made in the bone marrow - live about 120 days.

Destroyed and recycled by the liver.

White Blood Cells (leucocytes)

These are colourless cells and possess a nucleus. They function in defending the body against

pathogens.

Phagocytes - 'granulocytes' 'feed' on pathogens by phagocytosis.

Monocytes are one form of phagocytes.

Lymphocytes - 'agranulocytes' - produce antibodies, the specific defence proteins.

Made in bone marrow and lymphatic tissue.

Platelets

Responsible for clotting of the blood

Responsible for repair of damaged tissue - release the hormone **platelet growth factor**.

Short life - under 7 days.

Made in bone marrow

Functions of Blood System

Transport: to and from tissue cells

Nutrients to cells: amino acids, glucose, vitamins, minerals, lipids (as lipoproteins).

Oxygen: by red blood corpuscles (as *oxyhaemoglobin*).

Wastes from cells: urea, CO_2

Temperature Regulation: by altering the blood flow through the skin.

Immunity: protection against pathogens - clotting; phagocytes, lymphocytes and antibodies distributed in blood.

Communication: hormones distributed to all parts of the body in the blood.

Defence: clotting following a wound

BLOOD VESSELS

These are channels or tubes which convey blood to all parts of the body. They include the arteries, veins and capillaries.

Arteries - carry oxygen rich blood (except pulmonary artery) away from the heart at a high pressure. They have thick walls and no valves.

Veins - carry blood without oxygen (except pulmonary vein) towards the heart at a low pressure. They have thinner walls and valves.

Capillaries

Capillaries are the link between arteries and veins - where exchange with tissues occurs.

The capillary wall is one cell thick and somewhat porous - ideal to allow materials to pass in and out. All tissue cells very close to a capillary so exchange is very efficient.

Exchange at the capillaries is by diffusion, mass flow and active transport.

Blood flow in capillaries is slow giving enough time for effective exchange.

One type of cell, thus a **tissue**.

Differences Between Artery and Vein

Artery	Vein
i. It has thick/muscular wall	It has thin/less muscular wall.
ii. It has elastic wall	Its wall is non-elastic
iii. It carries blood away from the heart	It carries blood to the heart
iv. It carries oxygenated blood, except the pulmonary artery	It carries deoxygenated blood, blood except the pulmonary vein
v. Blood in it is pink or bright red in colour	Blood in it is dark-re in colour
vi. It is situated deep in the muscles	It is superficially located or situated
vii. It has small lumen	It has large lumen
viii. pressure is high	pressure is low
ix. pulse is readily detectable	pulse is not readily detectable
x. it has no valve except semilunar valves	it has valves

Closed System of Blood Vessels

The blood does not make direct contact with the tissue cells.

The blood is retained in the blood vessels.

A closed system is very responsive to the change needs of the organs and is highly efficient.

Double Circulation

The double circuit prevents mixing of oxygenated and deoxygenated blood. Therefore oxygen supply is highly efficient.

Pulmonary Circulation: deoxygenated blood flows from the heart to the lungs

oxygen is taken on and carbon dioxide is excreted, oxygenated blood flows from the lungs back to the heart.

Systemic Circulation: oxygenated blood flows from the heart to the organ systems of the body.

oxygen is delivered and carbon dioxide is taken on, deoxygenated blood flow from the organs systems back to the heart.

Portal System

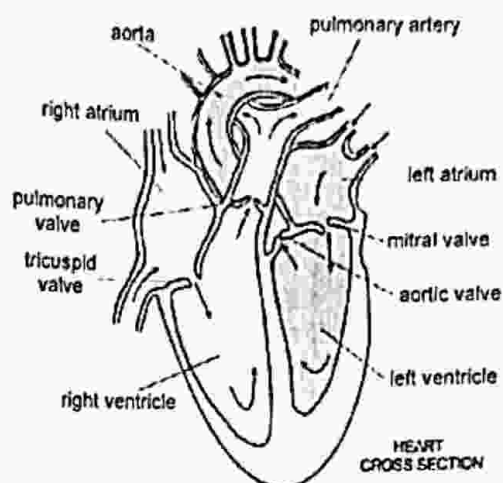
A portal blood vessel has a set of capillaries at each end.

The hepatic portal vein carries blood rich in absorbed nutrients from the capillaries in the gut to capillaries in the liver.

Hepatic artery supplies liver with oxygen

Hepatic vein takes all the blood away from the liver, back to heart.

The Heart



Internal structure of the heart

The heart is a muscular pumping organ situated between lungs in the thoracic cavity. It is made up of specialized muscles called cardiac muscles and protected by the pericardium. The heart consists of four chambers. The two upper chambers are called auricles while the two lower chambers are known as ventricles and are separated by septum into left and right. The left ventricle has a thicker wall in order to generate enough pressure on contraction needed to pump blood through the aorta. Each pumping action of the heart is known as a heart-beat.

Summary of circulation in the heart

Superior vena cava → right auricle → through tricuspid valve → right ventricle → pulmonary artery → lungs → pulmonary vein → left auricle → through bicuspid valve → left ventricle → aorta → all parts of the body.

Action of heart Two phases

Diastole: auricles contract and ventricles relax. The heart fills with blood under low pressure from the veins.

Systole: auricles relax and ventricles contract. The chambers of the heart are emptying of blood.

THE LYMPHATIC SYSTEM

A collection of special drainage vessels receiving excess tissue fluid.

Once the tissue fluid enters the lymphatic capillaries it is called lymph.

Lymph nodes (e.g. tonsils) filter the lymph and produce lymphocytes.

The lymph vessels have many valves, but low pressure.

The lymph is moved along by the squeezing action of: the skeletal muscles, pressure changes in the thorax during breathing and by the rhythmic contraction of the lymph vessel walls.

Lymph re-enters the blood just before the right atrium.

Functions of the Lymphatic System:

Circulatory role

Return the excess tissue fluid to the blood: this maintains blood volume, pressure and concentration.

Collect and deliver the absorbed lipids from the small intestine to the blood

Defence role

The lymph nodes filter out pathogens in the lymph. Production and 'export' of lymphocytes to the blood system for general distribution. Detection of antigens and production of specific antibodies.

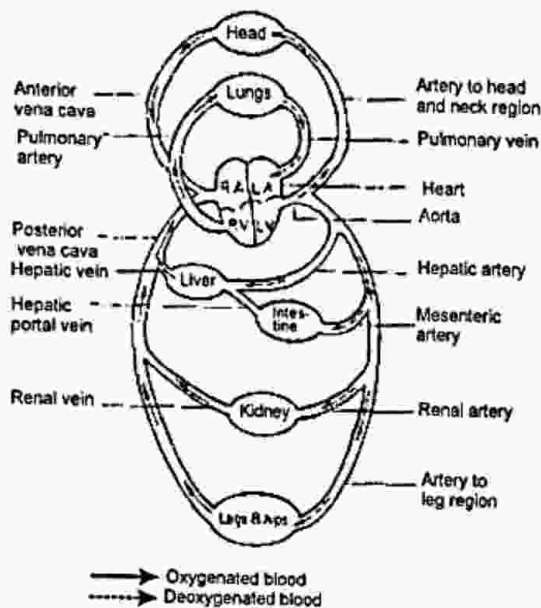


Diagram of blood circulation in mammals

Transport Systems in Plants

The plants have two transport systems:

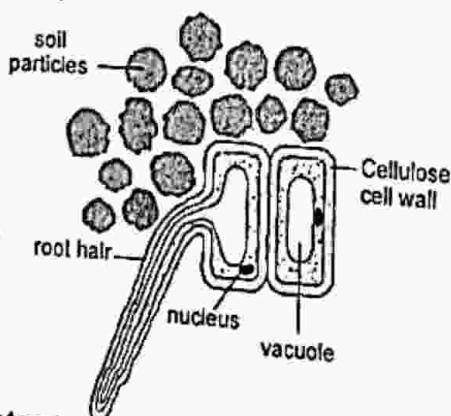
1. Xylem which carries water and minerals, and
2. Phloem which carries the food materials which the plant makes (Phloem also carries the hormones made by the plants in their root and shoot tips).

The transport of materials in a plant can be divided into two parts:

- (i) Transport of water and minerals in the plant, and
- (ii) Transport of food and other substances (like hormones) in the plant.

Transport of Water and Minerals:

The root hair cell:



Path of water from root hair to xylem vessels 66

Function: to absorb water and minerals from the soil. They have an elongated shape for more surface area. The path of water in plants is as follows:

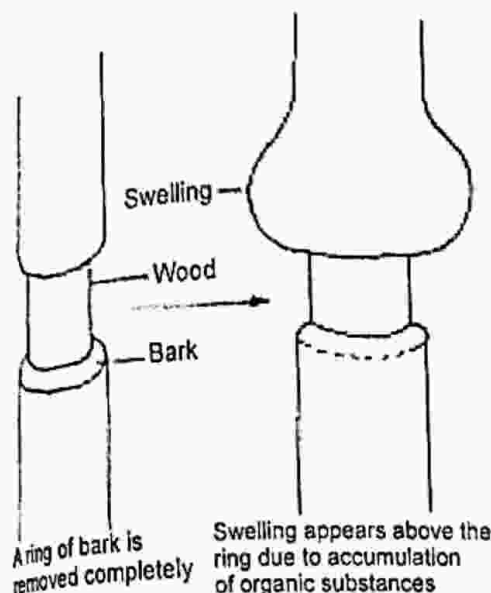
1. Water enters the root hair cell for the moist soil because the water potential is higher in the soil, than in the cytoplasm.
2. Water passes through the cortex cells by osmosis but mostly by "suction".
3. Water (and dissolved substances) is forced to cross the endodermis.
4. Water enters the xylem then leaves when it gets to the mesophyll cells. The uptake of water is caused by water loss in leaves through the stoma lowering the water potential in leaves, then water moves from xylem to enter leaf tissues down a water potential gradient, then water moves up the stem in the xylem due to the tension (because of cohesion / sticking of water molecules to each other) caused by water loss from the leaves, and ends with the gain through roots. The upward flow of water is called the transpiration stream. The minerals present in soil dissolve in water to form an aqueous solution. So, when water is transported by the root of the plant to its leaves, then the minerals dissolved in it also get transported along with it.

Transport of Solutes:

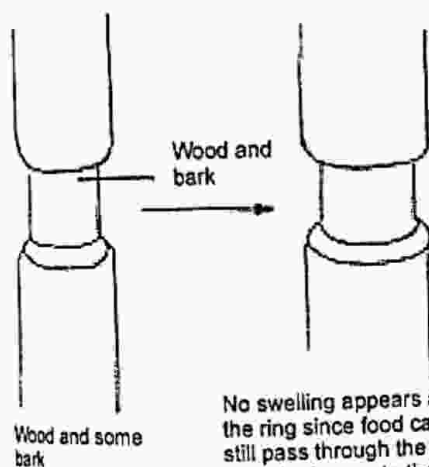
The phloem contains a very concentrated solution of dissolved solutes, mainly sucrose, but also other sugars, amino acids, and other metabolites. This solution is called the sap, and the transport of solutes in the phloem is called translocation.

Unlike the water in the xylem, the contents of the phloem can move both up or down a plant stem, often simultaneously. It helps to identify where the sugar is being transported from (the source), and where to (the sink).

- During the summer sugar is mostly transported from the leaves, where it is made by photosynthesis (the source) to the roots, where it is stored (the sink).
- During the spring, sugar is often transported from the underground root store (the source) to the growing leaf buds (the sink).
- Flowers and young buds are not photosynthetic, so sugars can also be transported from leaves or roots (the source) to flowers or buds (sinks).



Swelling appears above the ring due to accumulation of organic substances which moved down the phloem from the leaves

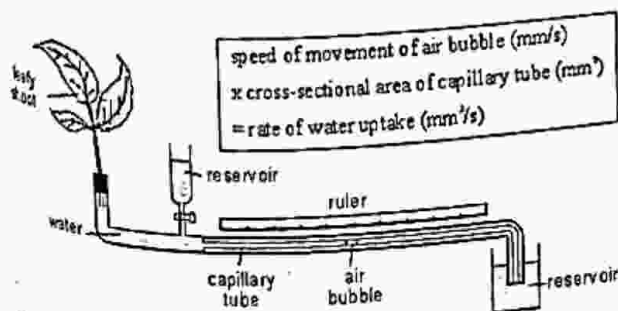


No swelling appears above the ring since food can still pass through the phloem from the leaves to the roots

To show that the phloem tissue transport food substances downwards from the leaves

Factors affecting Transpiration

The rate of transpiration can be measured in the lab using a **potometer** ("drinking meter"):



A potometer actually measures the rate of water uptake by the cut stem, not the rate of transpiration; and these two are not always the same. During the day plants often transpire more water than they take up (i.e. they lose water and may wilt), and during the night plants may take up more water than they transpire (i.e. they store water and become turgid). The difference can be important for a large tree, but for a small shoot in a potometer the difference is usually trivial and can be ignored.

The potometer can be used to investigate how various environmental factors affect the rate of transpiration.

- **Light.** Light stimulates the stomata to open allowing gas exchange for photosynthesis, and as a side effect this also increases transpiration. This is a problem for some plants as they may lose water during the day and wilt.
- **Temperature.** High temperature increases the rate of evaporation of water from the spongy cells, and reduces air humidity, so transpiration increases.
- **Humidity.** High humidity means a higher water potential in the air, so a lower water potential gradient between the leaf and the air, so less evaporation.
- **Air movements.** Wind blows away saturated air from around stomata, replacing it with drier air, so increasing the water potential gradient and increasing transpiration.

Many plants are able to control their stomata, and if they are losing too much water and their cells are wilting, they can close their stomata, reducing transpiration and water loss. So long periods of light, heat, or dry air could result in a decrease in transpiration when the stomata close.

Movement across Cell Membranes

Cell membranes are a barrier to most substances, and this property allows materials to be concentrated inside cells, excluded from cells, or simply separated from the outside environment. Obviously materials need to be able to enter and leave cells, and there are five main methods by which substances can move across a cell membrane: Diffusion, Osmosis, Passive Transport, Active Transport; and Vesicles.

Diffusion (or Simple Diffusion)

Diffusion is the movement of molecules of a substance from a region of higher concentration to a region of lower concentration till equilibrium is reached.

For example, the movement of oxygen from lungs to bloodstream and absorption of nutrients from small intestine into bloodstream etc. When there is no net movement, equilibrium is achieved.

Factors affecting the rate of diffusion:

1. Amount of substances on either side (i.e. the concentration)
2. Permeability of membrane
3. Temperature (movement of particles, increased temperature usually increases the speed of the particles, according to the kinetic theory of collision)
4. Distance to be travelled
5. Size of the particles

Biological significance of diffusion

1. It is responsible for the exchange of gases between living organisms and their environment.
2. It is responsible for the exchange of materials between the mother and foetus through the placenta.
3. It is responsible for movement of materials from cell to cell along the phloem during translocation.

- It is responsible for even distribution of substances admitted into the body.

Osmosis

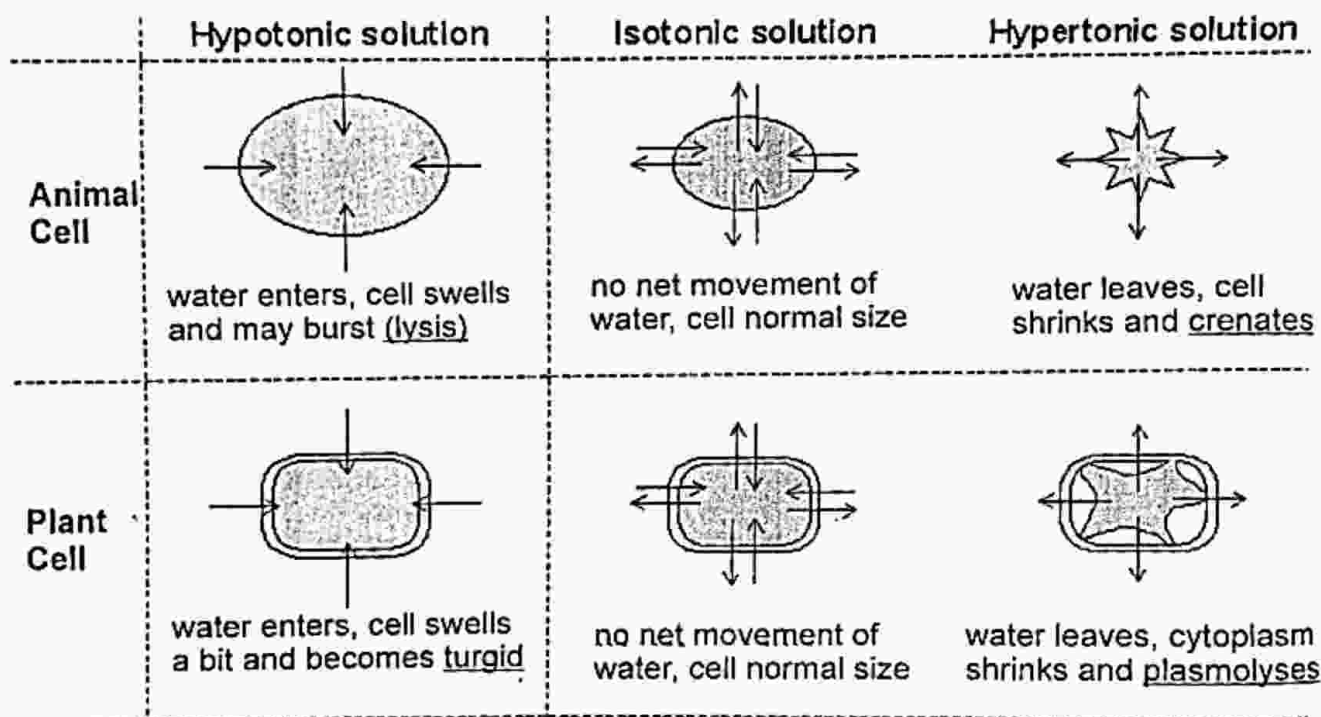
Osmosis is a process which involves the movement of water molecules or any solvent across a semi-permeable membrane from a weaker to a stronger solution. A semi-permeable membrane is a membrane which allows certain solvents to pass through while disallowing others. E.g. cellophane paper, fish bladder egg membrane and ectoplasm of plant cells. The contents of cells are essentially solutions of numerous different solutes, and the more concentrated the solution, the more solute molecules there are in a given volume, so the fewer water molecules there are. Water molecules can diffuse freely across a membrane, but always down their concentration gradient, so water therefore diffuses from a dilute to a concentrated solution.

Osmotic Pressure (OP). This is an older term used to describe osmosis. The more concentrated a solution, the higher the osmotic pressure. It therefore means the opposite to water potential, and so water move from a low to a high OP. Always use $\square$ rather than OP.

Cells and Osmosis. The concentration (or OP) of the solution that surrounds a cell will affect the state of the cell, due to osmosis. There are three possible concentrations of solution to consider:

- Isotonic solution** a solution of equal OP (or concentration) to a cell
- Hypertonic solution** a solution of higher OP (or concentration) than a cell
- Hypotonic solution** a solution of lower OP (or concentration) than a cell

The effects of these solutions on cells are shown in the diagram:



These are problems that living cells face all the time. For example:

- Simple animal cells (protozoans) in fresh water habitats are surrounded by a hypotonic solution and constantly need to expel water using contractile vacuoles to prevent swelling and lysis.
- Cells in marine environments are surrounded by a hypertonic solution, and must actively pump ions into their cells to reduce their water potential and so reduce water loss by osmosis.
- Young non-woody plants rely on cell turgor for their support, and without enough water they wilt. Plants take up water through their root hair cells by osmosis, and must actively pump ions into their cells to keep them hypertonic compared to the soil. This is particularly difficult for plants rooted in salt water.

Biological Significance of Osmosis

- It is responsible for the absorption of water from the soil by the root hairs
- It is responsible for the movement of water from one

cell to another

- It is responsible for the turgidity of cells
- It is responsible for the re-absorption of water in the kidney tubules.

Passive Transport (or Facilitated Diffusion).

It is the movement of particles across a membrane with the help of a molecule in the membrane (globular protein in bilayer of cell membrane).

The energy comes from the kinetic energy of the molecules involved.

For example, the movement of ADP into mitochondria

Active Transport (or Pumping).

It is the process by which molecules of a substance move from areas of lower concentration of that substance to areas of higher concentration Against a concentration gradient

For example: Absorption of mineral salts by the root hair cells, absorption of glucose and amino acids by cells in the small intestine of humans

Bulk Movement via vesicles

Endocytosis:

Substances are being imported into cells via vesicles

Phagocytosis = cell eating

Pinocytosis = bulk transportation of liquid (when taking in of nutrients)

Exocytosis:

Substances being exported out of cells

Summary of Membrane Transport

Method	Uses energy	Uses proteins	Specific	Controllable
Diffusion	N	N		
Osmosis	N	N	N	N
Passive Transport	N	Y	Y	N
Active Transport	Y	Y	Y	Y
Vesicles	Y	N	Y	Y

EXPERIMENTS ON TRANSPORT

Experiment :1 To demonstrate that leaves give out water vapour (by using cobalt chloride paper).

Materials required. A young potted plant, cobalt chloride solution, glass slides, filter papers.

Method. two filter papers are dipped in a solution of cobalt chloride. As these papers become soaked, they gradually turn blue. They are then dried and put aside for the experiment. A leaf, preferably the one at the lower part of the stem of a potted plant is selected. The two cobalt chloride papers are placed on the leaf, one on the papers and rubber binds are used to tie the glass slides together. By placing a glass on the leaf, vapour from outside is cut off. After this, the potted plant is placed near a window to receive sunlight. Both surfaces of the leaf are then checked from time to time.

Observations. The cobalt chloride paper on each surface of the leaf turns pink.

Conclusion. Anhydrous cobalt chloride paper is blue when dry but turns pink when wet. Since the paper turns pink, it means that the vapour from the leaf must have made them so. Therefore, leaves give off water vapour.

NOTE: the degree of pinkness on both leaf surfaces is either same or different. This is due to the distribution of stomata on the different surfaces.

Experiment :2 To demonstrate that leaves give out water vapour (with a potted plant).

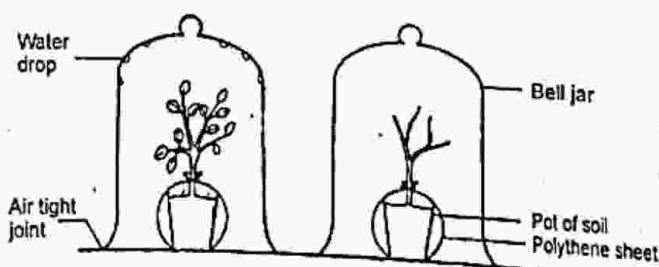
Materials required. A young potted plant, polythene bag, a bell jar, glass plate, Vaseline, anhydrous copper (II) tetraoxosulphate (VI).

Method. A young potted plant is collected and a polythene is used to cover the pot and the base of the stem of the plant as shown in Fig. 40.9. A bell jar is placed over the pot. The bell jar is stood on a glass plate whose surface has been rubbed with Vaseline. A second experiment with no plant is

set up to serve as a control. Both experiments are then exposed to sunlight, for some hours. If there are drops of liquid on the inner side of the bell jar, the liquid is tested with an anhydrous copper (II) tetraoxosulphate (VI).

Observations. Drops of colourless liquid are noticed on the inner side of the bell jar with a plant. No liquid in the control. When the liquid is tested with an anhydrous copper (II) tetraoxosulphate (VI), it is noticed that the white anhydrous copper (II) tetraoxosulphate (VI) turned blue.

Conclusion. The white anhydrous copper (II) tetraoxosulphate (VI) turning blue shows that the colourless liquid is water.



Experiment :3 To demonstrate the loss in weight of a potted plant.

Materials required. A young potted plant, polythene bag, weighing instrument, note book.

Method. A young potted plant is obtained and its soil is properly moistened. A polythene bag is used to cover the pot and part of the stem as shown in Fig. 40.9. After this, the set-up is weighed and the initial weight is recorded. Next, the potted plant is placed close to a window. More weighing is made and recorded every 30 minutes for a day. The weights are then carefully studied.

Observations. There is a loss in weight of the potted plant.
Conclusion. The loss in weight is due to water vapour given off from the plant. The water vapour must have come from the parts of the plant which were not covered. This shows that there is loss in weight of a potted plant.

Experiment :4 To demonstrate the rate of water loss during transpiration.

Potometer is the name of the apparatus which is used for this experiment. The apparatus is of various designs. One type is the one shown in Fig. 40.10. the apparatus is used mainly to measure the rate at which water is absorbed by a plant. However, this rate approximately equals the rate at which water is lost during transpiration. Because of this, one can conveniently, by means of a photometer determine the rate of transpiration in a plant shoot.

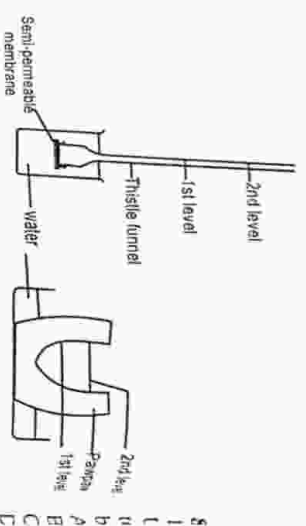
A healthy plant shoot is cut under water. It is fitted into a rubber stopper and the rubber stopper is inserted in C. The apparatus is filled with water by turning the tap of the funnel. After this, the setup is left for some time for the plant to absorb water. Air will enter the apparatus at A, once absorption of water begins. The air is noticed as a bubble in the graduated tube. The bubble moves along and the quantity of water absorbed is then measured. When the bubble gets to somewhere close to B, the tap of the funnel is then turned so that water can again fill the graduated tube. In this way the experiment can be repeated many times. With this apparatus, the rates of transpiration in different external conditions, e.g. in a dark room, in laboratory, out of door in the field, etc. are compared.

Experiment :5(a) To demonstrate osmosis with a non-living membrane.

Materials required. A cellophane or parchment paper or a pig's bladder, thistle funnel, sugar solution, a beaker.

Method. A semi-permeable membrane such as cellophane or parchment paper or a pig's bladder is tied around the mouth of a thistle funnel. Some sugar solution is poured into the bulb of the funnel. The level of the sugar solution is then noted. A beaker is collected and water is poured into it until it is about half way filled. Next, the thistle funnel is immersed in the water inside the beaker as shown in Fig. 40.13A. A control experiment is also prepared but the bulb of the funnel contains distilled water instead of sugar solution. Both experiments are then left for some hours.

Observations. There is rise in liquid up the stem of the thistle funnel of the first experiment. No rise in liquid in the control experiment.
Conclusion. The rise in the liquid up the stem must have been due to water molecules which moved from the beaker into the thistle funnel through the non-living membrane. This shows that osmosis has taken place.



Experiment :5(b) To demonstrate osmosis with a living membrane.

Materials required. A yam or pawpaw, sugar solution, trough of pure water.

Method. A yam tuber or pawpaw is taken. A lump is cut off the tuber. The lump is cut in such a way that it is a base for it. Part of the base is peeled, say about an inch down. A cavity or hollow is made in the yam. The cavity is made deep down to the base. Next, the tissues are washed with tap water to remove any debris. Some sugar solution is added into the cavity which has been made out of the yam or pawpaw. The yam or pawpaw is then stood in a dish of pure water. A control experiment is also prepared but the cavity contains distilled water instead of sugar solution. Both experiments are left for some hours.

Observations. There is a rise in liquid up the cavity in the first experiment. No rise in the control experiment.

Conclusion. The same conclusion as with a non-living membrane.

Experiment :6 To demonstrate plasmolysis.

Materials required. Filaments of *Spirogyra* plant, microscope, plain slides, 30% of sodium chloride solution.

Method. A few cells of *Spirogyra* plant are placed on a plain slide. They are examined under a compound microscope. The structure of the cells are then noted. A few drops of 30% sodium chloride solution are added to the cells and they are further examined.

Observations. The protoplasm shrinks and separate from the cell wall. A tiny space appears between the protoplasm and the cell wall. Further shrinkage occurs and the protoplasm finally rounds up.
Conclusion. The changes as observed above are due to loss of concentration than that of the cell. Because of this, water is lost to the surrounding medium from the cell. This shows that plasmolysis has taken place.

Revision Questions

NEED FOR TRANSPORT

1. UME/98/16
 a. All organisms transport essential nutrients directly to all parts of their bodies by the process of diffusion. **False**
 b. In small organisms, they have a large surface area to volume ratio.
 c. In large organisms, they have a small surface area to volume ratio.
 d. In large organisms, they have a large surface area to volume ratio.
 e. In large organisms, they have a small surface area to volume ratio.

2. UME/2003/16/Type A.

- Organism I, II, III and IV have surface area/volume ratios of 1:2, 1:3, 1:4 and 1:5 respectively. The organism that is likely to have the most complex transport system is
 A. III B. IV C. II D. I

MAMMALIAN CIRCULATORY SYSTEM

(heart, arteries, veins, and capillaries)

1. UME/83/36

- Which of the following statements is the mammalian circulatory system is NOT true?
 A. Blood in the pulmonary artery is richer in oxygen content than blood in the pulmonary vein.
 B. The blood in the hepatic portal vein is the richest in food substances.
 C. Blood flow is controlled by valves in veins.
 D. Arteries are generally thicker and larger than veins.
 E. Fibrin helps in the formation of blood clot.

2. UME/84/37

- Normally the flow of blood is never from
 A. artery to arterioles
 B. arterioles to capillaries
 C. capillaries to venules
 D. arterioles to the arteries

3. UME/86/23

- Blood circulation in a mammal is said to be double because
 A. it passes twice through the heart in a complete cycle.
 B. it moves in both arteries and veins.
 C. it circulates in both the heart and other organs.
 D. the heart contains auricles and ventricles.

4. UME/99/22

- The lymphatic system of mammals rejoins the blood circulatory system at the
 A. hepatic vein B. subclavian vein
 C. renal vein D. common iliac vein

5. UME/83/50

- The path taken by glucose from the ileum to the heart is
 A. ileum → hepatic portal vein → hepatic artery → vena cava → heart
 B. ileum → hepatic portal vein → hepatic artery → vena cava → heart
 C. ileum → hepatic portal vein → hepatic artery → vena cava → heart
 D. ileum → hepatic portal vein → vena cava → heart
 E. ileum → hepatic portal vein → vena cava → heart

6. UME/78/43

- A carbohydrate molecule in the right ventricle of the heart is pumped into the cell of the toe of man. Which of these structures is it unlikely to pass through en-route?
 A. pulmonary artery
 B. lungs
 C. heart
 D. liver
 E. aorta

7. UME/81/46

- The hepatic portal vein is unique because it
 A. carries deoxygenated blood
 B. begins and ends with capillaries
 C. is the longest vein in mammals
 D. carries digested food
 E. is the shortest vein in mammals.

8. UME/91/28

- Deoxygenated blood flows into the right and left lungs through the
 A. pulmonary vein B. vena cava
 C. pulmonary artery D. subclavian vein

9. UME/85/19

- The vein which returns blood from the head and arms to the heart is called
 A. aorta B. inferior vena cava
 C. superior vena cava D. pulmonary vein
 E. pulmonary artery.

10. UME/98/17/Type C

- The mammalian heart chamber that pumps blood through the longest distance is the
 A. right auricle B. right ventricle
 C. left ventricle D. left auricle

019-08-14

- B. the lymphatic system
- C. pulmonary circulation
- D. the pulmonary artery

17. UME/2010/31/TYPE: D

The blood component that has the greatest affinity for oxygen is the

- A. leucocytes
- B. erythrocytes
- C. thrombocytes
- D. lymphocytes

8.3. PLANT VASCULAR SYSTEM

18. PCE/91/18

The path of translocation of mineral ions in the plant is through the

- A. cortex
- B. phloem
- C. pith
- D. xylem

19. UME/99/20/Type D

Substances manufactured by the leaves are transported to other parts of the plant through the

- A. xylem
- B. companion cells
- C. sieve tubes
- D. cambium.

20. UME/2001/27/Type: T

Salts and water are absorbed in the roots and transported to the leaves by

- A. diffusion through the xylem tissues
- B. osmosis through the phloem tissues
- C. diffusion through the phloem tissues
- D. osmosis through the xylem tissues.

21. UME/2003/14/Type: A

If water that has been coloured red is poured at the base of wilting plant, it will appear as a red stain in the cells of the

- A. phloem
- B. parenchyma
- C. xylem
- D. epidermis

8.4. MEDIA, PROCESS AND THE MECHANISM OF TRANSPORT

22. UME/83/11

Which of the following statements is NOT true of mammalian erythrocytes?

- A. They have haemoglobin
- B. They appear yellow when looked at singly
- C. They are disc-shaped
- D. The cells are more numerous than leucocytes
- E. They have nuclei at maturity.

23. UME/84/40

Which of the following sequences represents the process of blood clotting?

- 1. Fibrin forms a network of threads
- 2. Red blood cells are caught and a clot is formed
- 3. Fibrinogen in plasma changes into soluble fibrin
- 4. Blood is exposed to air

A. 4, 3, 2, 1

B. 4, 3, 1, 2

C. 2, 1, 4, 2

D. 1, 2, 3, 4

E. 3, 1, 2, 4

24. UME/85/18

The function of lymph nodes is to

- A. supply oxygen
- B. filter out bacteria
- C. form red blood cells
- D. supply amino acids
- E. supply simple sugars

25. UME/87/21

One cubic centimeter of lymph is richer than an equal volume of blood in

- A. erythrocytes
- B. leucocytes
- C. amino acid
- D. glucose.

26. UME/93/21

The main function of blood in mammals is to transport

- A. excretory materials from tissues
- B. carbondioxide from lungs to tissues
- C. digested food from cell the body tissues
- D. oxygen to the lungs

27. UME/97/11

In mammals, the exchange of nutrients and metabolic products occur in

- A. lymph
- B. lungs
- C. heart
- D. liver

28. PCE/95/19

Which of the following materials is NOT carried by the blood plasma?

- A. dissolved food
- B. oxygen
- C. waste products
- D. protein

29. PCE/2001/17/Type: 4

The medium of transportation in plants is the

- A. ectoplasm
- B. cell sap
- C. lenticel
- D. lymph

30. UME/80/25

A potometer is used to determine the rate at which a shoot

- A. respire
- B. ~~sinks~~ eventually die
- C. loses weight
- D. absorbs water
- E. transpires.

31. UME/80/30

If the bark and phloem tissues of a woody shoot are peeled off by ringing, the whole plant will eventually die because:

- A. water does not reach the leaves
- B. water and salts remain below the rough portion
- C. there is a withdrawal of water from the root by soil
- D. manufactured food does not reach the roots.
- E. the roots store too much water.

32. UME/81/21

The rate of transpiration of a leafy shoot would be highest under

- A. damp, cold, still air

3. damp, warm, moving air.
 4. dry, warm, moving air
 5. dry, cold, moving air
 6. dry, warm, still air.

32. UME/83/35

Which of the following will NOT allow osmosis to take place?

- A. pig's bladder
 C. parchment paper
 E. cow's bladder

- B. cellophane
 D. transparent polythene

33. UME/86/24

Which is the correct order of water loss from the leaf? 1. Mesophyll 2. Veins 3. Substomatal cavity

4. Stomata
 A. 3-2-1-4
 B. 2-3-1-4
 C. 2-1-3-4
 D. 1-2-3-4

35. UME/89/9

Mineral salts can be absorbed into the roots by

- A. osmosis only
 B. osmosis and diffusion
 C. diffusion and active transport
 D. imbibition only

36. UME/90/22

If a ring of bark and phloem is removed from a stem, the

- A. plant dies immediately
 B. plant dies after two days
 C. movement of food is not affected
 D. movement of mineral salts is hardly affected.

37. UME/97/17

Oozing out of water from the leaves of plant in a humid environment is known as

- A. Transpiration
 C. Pinocytosis
 B. Osmosis
 D. Guttation.

38. PCE/95/20

What is the correct equation of water movement in plants?

- A. Root hairs - cortex - phloem vessels - leaves
 B. Root hairs - xylem vessels - cortex - leaves
 C. Root hairs - phloem vessels - cortex - leaves
 D. Root hairs - cortex - xylem vessels - leaves

39. PCE/2002/41/Type:5

In higher plants, the mass flow of food and water within the vascular tissues to all parts of the plant is known as

- A. translocation
 C. absorption
 B. diffusion
 D. conduction

40. UME/2008/19

What will happen when two equal sized pieces of unripe pawpaw labelled x and y are dropped into equal volumes of concentrated salt solution and distilled water respectively?

- A. pawpaw x will become turgid
 B. both will increase in size
 C. pawpaw y will become turgid
 D. both will decrease in size

MISCELLANEOUS QUESTIONS (8)

41. The main function of blood in mammals is to transport
 A. excretory materials from tissues
 B. carbon dioxide from lungs to tissues
 C. digested food from all the body tissues
 D. oxygen to the lungs (UME/93/21)

42. Salts and waters are absorbed in the roots and transported to the leaves by
 A. diffusion through the xylem tissues
 B. osmosis through the phloem tissues
 C. diffusion through the phloem tissues
 D. osmosis through the xylem tissues
 (UME/2001/27/Type: T)

43. Serum differs from blood plasma because it
 A. contains blood cells and fibrinogen
 B. contains soluble food and mineral salts
 C. lacks blood protein, fibrinogen
 D. lacks blood cells and albumin
 (UME/2000/24/Type: M)

44. The function of the fluid-filled pericardium is to
 A. reduce the friction caused by the pumping movements of the heart
 B. supply the heart with oxygen and nutrients
 C. prevent disease organisms from attacking the heart
 D. reduce the intensity of the pumping action of the heart
 (UME/2005/18/Type: D)

45. Excess water in plants is excreted as water vapour and as droplets respectively through
 A. respiration and guttation
 B. transpiration and guttation
 C. photosynthesis and guttation
 D. guttation and condensation
 (UME/2005/22/Type: D)

46. The dark-red colour of blood is determined by it's
 A. plasma
 C. pigment
 B. protein
 D. cells (PCE/2002/29/Type: 5)

47. An organism with double circulation is the
 A. duck
 C. toad
 B. rat
 D. lizard (PCE/2003/46/Type: N)

48. Substances manufactured by the cells are transported to other parts of the plant through the
 A. xylem
 B. companion cells
 C. sieve tubes
 D. cambium (UME/99/20/Type: D)

49. The opening and closing of the stoma are regulated by
 A. transpiration
 C. diffusion
 B. respiration
 D. osmosis (UME/2007/29)

50. Poison from a snake bite on a man's leg is carried to his heart through the
 A. anterior vena cava
 C. hepatic portal vein
 B. posterior vena cava
 D. hepatic artery (PCE/92/19)

51. The blood cells responsible for body defense against disease are the
 A. erythrocytes B. platelets
 C. fibrinogens D. leucocytes (PCE/92/20)
52. The total force by which the cells of the root absorb water from its surrounding is known as
 A. osmotic pressure B. suction pressure
 C. turgor pressure D. water pressure
 (PCE/93/19)
53. The tube that carries blood from the heart to the liver is called
 A. pulmonary vein B. hepatic vein
 C. hepatic artery D. pulmonary artery
 (PCE/93/21)
54. The return of blood to the heart from the main arteries is prevented by the
 A. semi-lunar valve B. bicuspid valves
 C. tricuspid valve D. mitral valves (PCE/96/15)
55. The primary protein that keeps the blood plasma in osmotic equilibrium with the cells of the body is the
 A. albumin B. Myoglobin
 C. myosin D. actin (PCE/97/16)
56. The mammalian heart chamber that pumps blood through the longest distance is the
 A. right auricle B. right ventricle
 C. left ventricle D. left auricle (PCE/98/17)
57. An organism with an open transport system is
 A. grasshopper B. earthworm
 C. rat D. lizard (PCE/07/14)
58. That is turgor pressure?
 A. It is the force that forces water into a cell
 B. It is the force of equilibrium in a cell
 C. It is the force set up by the cell to prevent entry of water into it
 D. It is the force exerted on the wall of cells
 E. None of the above
59. What is transpiration?
 A. it is the loss of water by plants
 B. it is a process in which water escapes through some parts of plants
 C. it is the evaporation of water from the leaves of plants
 D. it is the loss of water from plants in vapour form
 E. None of the above
60. What is guttation?
 A. This is the loss of water by leaves
 B. it is the water found on plants early in the morning
 C. It is the water lost by young coleoptiles of maize
 D. it is the exudation of water in liquid form from plants in water-saturated atmosphere
 E. The exudation of water from leaf tips
61. A young plant is uprooted from the garden by mistake. It is immediately replanted. The plant wilts but its healthy state is restored three days later. What was wrong with the plant?
 A. Transpiration had stopped in the plant
 B. The leaves were scorched by the sun
 C. The xylem vessels responsible for water uplift in plants were defective
 D. It lacked an efficient root system
 E. The rate of water absorption from the soil was slow due to the destruction of the root hairs
62. A cassava plant is ringed by the removal of a complete circle of its bark about one inch wide. It is seen that at the end of the season no tuberous roots have formed. This shows that:
 A. The phloem cells do not transport food materials
 B. The xylem vessels can transport manufactured food
 C. Xylem vessels are not as important as phloem cells
 D. Manufactured food of plants are transported to areas of storage in the roots through the bark cells
 E. None of the above
63. The experiment in question 62 shows that transpiration water gets from the roots to the leaves through the
 A. Xylem vessels B. Wood fibres C. The cambium
 D. Pith cells E. Some bark cells
64. With regard to defence, which of these is untrue of the blood?
 A. blood, by forming a mesh-work of fibrin, can prevent loss of blood
 B. Blood induces the red blood cells to fight bacteria
 C. Phagocytes of the blood eliminate unwanted material from the body
 D. Blood can produce antibiotic effect on the body
 E. None of the above
65. How does the blood regulate the body temperature of an animal?
 A. It stores the heat for the body
 B. As the origin of heat, it releases it into the body at regular intervals
 C. It dissipates the heat round the body
 D. It lowers the amount of heat in the body
 E. None of the above
66. The liquid part of blood is called
 A. The matrix B. The serum
 C. The lymph D. The plasma
 E. None of the above
67. Which of these cells in the blood is responsible for the storage of haemoglobin?
 A. the white blood corpuscles
 B. the red blood corpuscles
 C. the blood plasma D. the blood platelets
 E. The blood corpuscles
68. What protein structure in the blood is responsible for blood clotting?
 A. Lecithin
 B. Fibrinogen

C. Blood platelets
D. Heparin
E. Thrombin

69. What is the membranous structure that encloses the heart?
A. the pleural cavity
B. the diaphragm
C. the pericardium
D. the mesentery

70. Why are the ventricles of the heart more muscular than the auricles?
A. The ventricles store blood while auricle do not
B. The auricles originated from the ventricles
C. The auricles give rise to the valves, e.g. the tricuspid and bicuspid valves
D. The ventricles are responsible for blood circulation round the blood
E. None of the above

71. What is the common name for all blood vessels which carry blood into the heart?
A. the arteries
B. the pulmonary veins
C. the veins
D. the vena cava
E. None of the above

72. Why is the blood vessels from the right ventricle carrying impure blood to the lungs called the pulmonary artery?
A. It is exceptional
B. It is a main blood vessels
C. It carries blood away from the heart
D. The blood is partly pure
E. None of the above reasons

73. What is the chief characteristics of the capillaries?
A. capillaries are found in the tissues of the body
B. they possess very thin walls
C. they contain the lymph fluid
D. they are easily destroyed
E. They cannot be seen

74. What is the name of the blood vessel carrying blood to the head of the mammal?
A. The subclavian artery
B. The anterior mesenteric artery
C. The pulmonary artery
D. The caput artery
E. The carotid artery

75. The coeliac artery supplies blood to:
A. the stomach
B. the intestines
C. the rectum
D. the spleen
E. the liver and the stomach

76. What is a portal vein?
A. This is vein that sends its blood through another organ before getting to the heart
B. This is a bottle-necked system
C. This is a vein that carries food substances
D. This is a blood system found only in the rat
E. None of the above

77. Trace the movement of a blood cell from the right kidney to the left arm
A. Right kidney → Posterior vena cava → Heart → Lungs → Aorta → Left arm
B. Right kidney → Posterior vena cava → Heart → Lungs → Heart → Aorta → Left arm
C. Right kidney → Liver → Posterior vena cava → Heart → Lungs → Heart → Aorta → Subclavian artery → Left arm
D. Right kidney → Posterior vena cava → Heart → Lungs → Heart → Aorta → Subclavian artery → left arm
E. None of the above

78. What is the importance of the lymph nodes?
A. the nodes store iron
B. the nodes store fats and oils
C. the nodes manufacture white blood corpuscles
D. the nodes produce vitamin D and A for the body
E. None of the above

79. The pigments of the lymph responsible for the transport of oxygen is
A. Haemoglobin
B. Lumphane
C. Lymphatics
D. Lymph corpuscles
E. None of the above

80. Fatty acids and glycerol enter the blood stream through
A. vena cava
B. aorta
C. carotid valve
D. left jugular vein
E. none of the above

81. Oxygen and digested food are transported to the legs and gonads through
A. iliac artery
B. mesenteric vein
C. renal vein
D. hepatic portal vein
E. vena cava

CHAPTER 8

1. B	2. B	3. A	4. D	5. A
6. B	7. E	8. D	9. B	10. C
11. C	12. C	13. D	14. A	15. C
16. A	17. B	18. B	19. C	20. D
21. C	22. E	23. B	24. B	25. B
26. A	27. A	28. B	29. B	30. E
31. D	32. C	33. D	34. C	35. C
36. D	37. D	38. D	39. A	40. C
41. A	42. D	43. C	44. A	45. B
46. C	47. B	48. C	49. D	50. B
51. D	52. A	53. C	54. A	55. A
56. C	57. A	58. D	59. D	60. D
61. E	62. A	63. A	64. B	65. C
66. D	67. B	68. B	69. C	70. D
71. C	72. C	73. B	74. E	75. E
76. A	77. D	78. C	79. E	80. E
81. A				

CHAPTER 9

RESPIRATION

Understand the following after reading this chapter:

Outline of the process and its significance
Respiratory organs and surfaces
Mechanism of gaseous exchange
Aerobic respiration
Anaerobic respiration

RESPIRATION

Respiration is the process of releasing energy from the breakdown of glucose. Respiration takes place in every living cell, all of the time and all cells need to respire in order to produce the energy that they require.

What is the energy is used for?

The energy produced during respiration is used in many different ways, some examples of what it is used for are:

- Working of muscles
- Growth and repair of cells
- Building larger molecules from smaller ones i.e. proteins from amino acids
- Allowing chemical reactions to take place
- Absorbing molecules in active transport
- Keeping body temperature constant
- Sending messages along nerves

Types of Respiration

Respiration is divided into two phases: External (gaseous) respiration and internal (cellular) respiration.

RESPIRATORY SURFACES OR ORGANS

Respiration surfaces or organs are the media of gaseous exchange between organism and their surrounding

Organism	Respiratory Surface
1. Protozoa	Body Surface
2. Earthworm	Skin
3. Fishes, tadpoles	Gills
4. Insects	Trachea
5. Spiders	Lung-book
6. Land vertebrate	Lungs (Alveoli)

CHARACTERISTICS OF RESPIRATORY SURFACES OR ORGANS

Property of surface	Reason
Thin (ideally one cell thick)	short distance to diffuse
Large surface area	many molecules can diffuse at the same time
Moist	cells die if not kept moist
Well ventilated	concentration gradients for oxygen and carbon dioxide are kept up by regular fresh supplies of air
Close to blood supply	gases can be carried to/from the cells that need/produce them

MECHANISM OF GAS EXCHANGE

In Insects

Gas exchange is accomplished mostly by simple diffusion through the cell walls. Air enters the spiracles, and moves through the tracheal system. Each tracheal tube ends in a moist tracheole, a specialized cell for exchanging gas with another cell in the body.

When air reaches the tracheole, oxygen dissolves into the tracheole liquid. Through simple diffusion, oxygen then moves to the living cell and carbon dioxide enters the tracheal tube. Carbon dioxide, a metabolic waste, exits the body through the spiracles.

In Toads/Frogs

All frogs start life as aquatic tadpoles, breathing underwater through internal gills and their skin. Then they most develop into land animals with lungs for breathing air. But in all stages breathing is controlled by pulsing the throat. Most frogs lose their gills when they undergo metamorphose.

Frogs breathe with their mouths closed. Their throat movement pulls air through the nostrils to the lungs. They breathe out with body contractions.

Lungs can also help in water. Filling the lungs with air gives frog better buoyancy, making it float more easily.

Frogs can also breathe through their skin, with tiny blood vessels, capillaries, under the outer skin layers. The African 'Hairy' frog, *Trichobatrachus robustus*, has small projections on their back legs. This is because of the extra oxygen needs at this time.

In Fishes

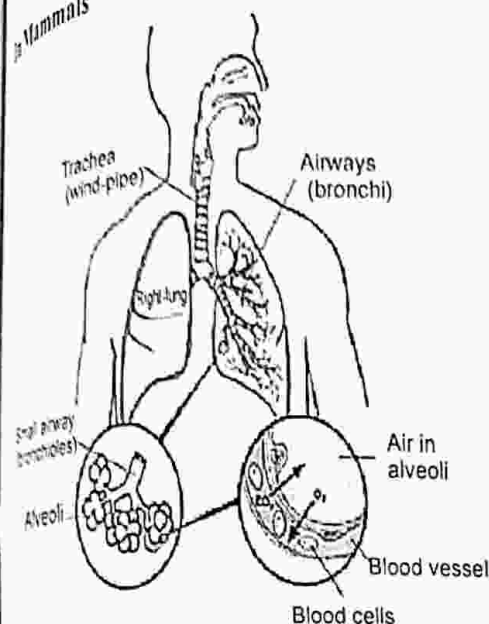
Fish obtain oxygen that is dissolved in the water. This occurs via the gills. Gills are feathery structures inundated with blood vessels which allow for a large surface area in contact with the oxygen present in the water.

Contraction of the mouth muscles lowers the floor of the mouth lowering its atmospheric pressure. Water (containing dissolved oxygen) moves into the mouth and at the same time the operculum remain closed.

The mouth then open and the operculum muscles relax causing them to bulge open; this increases the volume of the mouth cavity and lowers the pressure in the gill region.

Water from the cavity moves into the gill region due to the reduced pressure; and bathes the gill filaments. Oxygen diffuses into the blood capillaries due to its higher concentration in the gill region than the blood capillaries. It combines with hemoglobin and is transported as oxyhaemoglobin to the respiring tissues.

Carbon dioxide which is of higher concentration in the blood than the gill filaments diffuses into the gills and is exhaled through the water that moves out when the operculum opens.



Breathing in:

1. external intercostal muscles contract – pulls rib cage upwards and outwards
2. diaphragm muscles contract – diaphragm moves upwards
3. Lung volume increases – and pressure falls (Boyle's law: pressure and volume are inversely proportional)
4. Air rushes in – to equalize pressures.

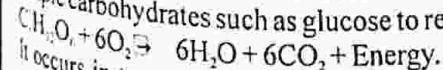
Breathing out:

1. external intercostal muscles relax – rib cage falls downwards and inwards
2. diaphragm muscles relax – returns to dome shape
3. Lung volume decreases – and pressure increases
4. Air is forced out

Inhaled (inspired) air: 21% oxygen, 0.04% carbon dioxide, 78% nitrogen and water vapour vary to climate.
Exhaled (expired) air: 18% oxygen, 3% carbon dioxide, 78% nitrogen, and saturated water vapour.

INTERNAL RESPIRATION

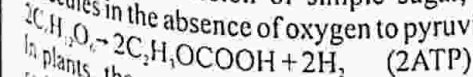
Cellular respiration is the oxidation and decomposition of simple carbohydrates such as glucose to release energy.



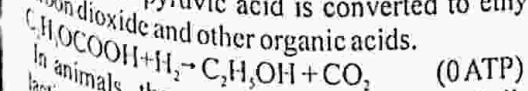
It occurs in two main stages viz; Glycolysis and Krebs's cycles.

GLYCOLYSIS (ANAEROBIC RESPIRATION)

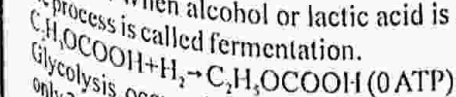
involves the conversion of simple sugar, e.g. glucose molecules in the absence of oxygen to pyruvic acid.



In plants, the pyruvic acid is converted to ethyl alcohol, carbon dioxide and other organic acids.



In animals, the pyruvic acid is converted directly into lactic acid. When alcohol or lactic acid is the end product, the process is called fermentation.

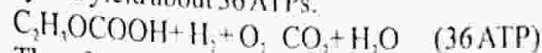


Glycolysis occurs in the cytoplasm of the cell. It yields only 2ATPs.

Anaerobic respirations occur in yeast, during vigorous exercise, some intestinal bacteria, germinating seeds. Fermentation in Yeast (a fungus) yields alcohol and carbon dioxide. The alcohol from the process is used to brew beer

and wine and the carbon dioxide to bake bread.

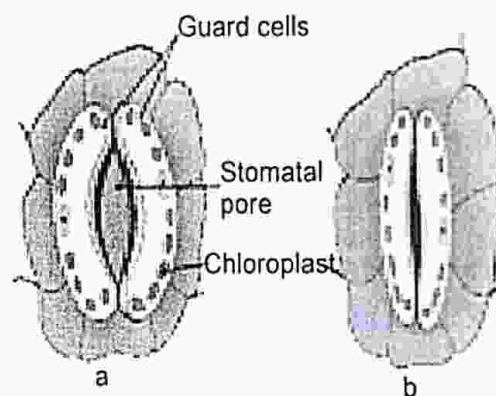
KREB'S CYCLE (AEROBIC RESPIRATION) involves the use of oxygen to completely breakdown pyruvic acid in the mitochondria to release energy. Krebs's cycles yield about 36 ATPs.



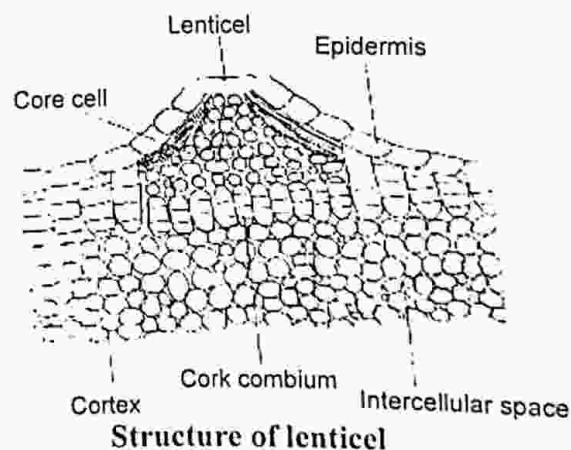
Therefore, a complete oxidation of one molecule of glucose yields a maximum of 38 ATPs. The reactions in glycolysis and Krebs's cycle are each catalyzed by specific enzymes.

RESPIRATION IN PLANTS

Plants respire by means of openings on the leaves called stomata and lenticles which lead to the intercellular spaces and mesophyll cells.



An open and a closed stomata



Structure of lenticel

EXPERIMENTS ON RESPIRATION

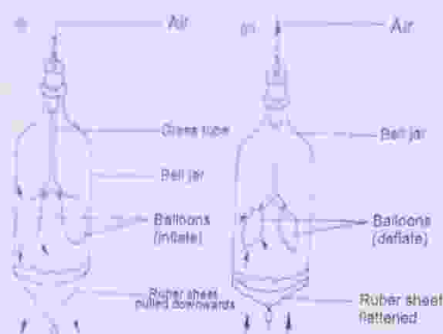
Experiment : 1 To demonstrate that carbon (IV) oxide is given out by man during expiration.

Materials required: Two conical flasks, rubber stopper, rubber tubes, glass tubes, lime water.

Method: The apparatus is set up as shown fig. 40.1. Both flasks contain the same amount of lime water. Air is breathed in and out through the mouth piece. The air goes into the flasks 1 and expired air passed through flasks 2.

Observations: In flasks 1, the lime water turns slightly milky, and in flasks 2, the lime water turns milky, but the milky is more pronounced.

Conclusion: The slight milky colour in flask 1 is due to the traces of carbon (IV) oxide present in the ordinary air. The heavier milky colour in the flask 2 is due to the fact that expired air contains carbon (IV) oxide and is of a higher percentage than ordinary air. This to show that carbon (IV) oxide is given out by man during expiration.



Experiment to demonstrate inhalation and exhalation

Experiment: 2 To demonstrate that carbon (IV) oxide gives out by a mammal during expiration.

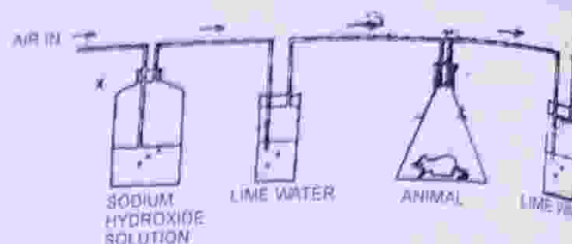
Materials required: 1 conical flask, 2 jars, bent glass tubes, capillary tube, rubber stopper, caustic soda solution, lime water and a living small mammal such as a rat.

Method: The apparatus is set up as shown in fig. 40.2. The first jar (1) contains caustic soda solution. The second and third jars contain lime water. The conical flask contains a well fed rat. The third jar is connected to a filter pump or aspirator. A second experiment is set up to serve as control. The control does not have the small mammal. Air is sucked slowly into the flask by means of a filter pump or an aspirator. The caustic soda in the first jar is to free the air of any carbon (IV) oxide and it is confirmed by the lime water in conical flask makes use of the air (O_2) from the second jar and the air it exhales passes into the 3rd jar.

Observations: As the air passes through the jar and flask it is observed.

- That the caustic soda in the first jar turned milky.
- That the lime water in the second jar did not show any change in colour.
- That the lime water in the third jar also turned milky.
- That the lime water in the third jar (the control experiment) remained unchanged in colour.

Conclusion: Since the air inhaled by the rat was free of carbon (IV) oxide and the lime water in the third jar turned milky, it follows that the rat must have exhaled carbon (IV) oxide which passed into the third jar. This shows that carbon (IV) oxide is given out by a mammal during expiration.



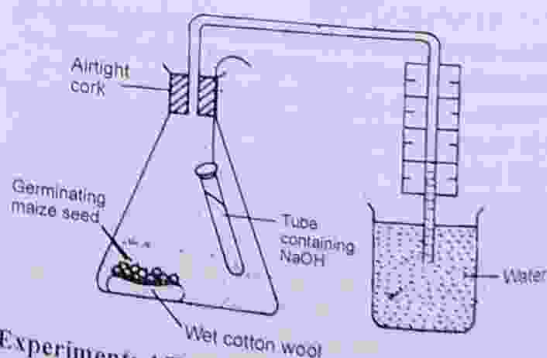
Experiment: 3 To show that germinating seeds or plants utilise oxygen.

Materials required: A beaker, 1 conical flask, 1 test tube, a glass tube, germinating maize seed, caustic soda solution.

Method: The experiment is set up as shown in fig. 40.3. Some maize seed which has been soaked for a few days and kept in a conical flask. Caustic soda solution is poured in a test tube and it is placed in the flask. The caustic soda solution removes carbon (IV) oxide which is formed by the germination seed, from the flask. The mouth of the flask carries a rubber stopper with a glass tube. The end of the glass tube is directed into a beaker which contains water. This set up is noted as experiment 1. Second experiment to serve as control, is set up but the flask contains dead grains (boiled grains). Both experiments are left for a while. Next a glowing splint is dipped into both flasks to test for oxygen.

Observation: Experiment 1. As the germinating seed develops, a rise in the level of water in the glass tube takes place. The glowing splint goes out. Control experiment. There is no rise in the tube. The glowing splint is rekindled.

Conclusion: In the first experiment where the glowing splint goes out, indicates that oxygen is absent there and must have been utilised by the germinating seed in respiration.



Experiment: 4 To demonstrate that germinating seeds produce heat energy.

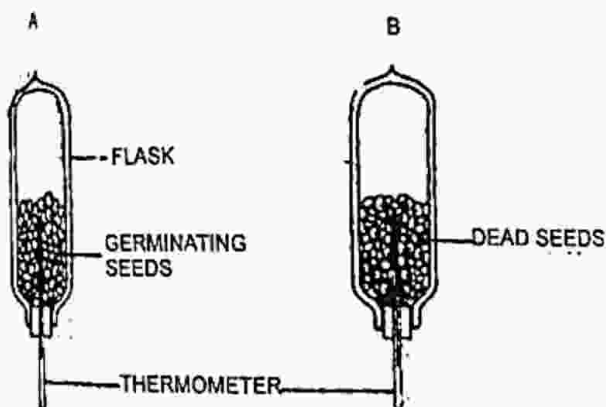
Materials required: Two vacuum flasks, some living seeds, thermometers, cotton wool.

Method: Some living seeds which have been soaked for some time are put in a vacuum flask. A thermometer is placed among the seeds and the thermometer is supported by placing in cotton wool round it in the mouth of the flask.

stand experiment, like the one above, to serve as control, setup. This vacuum flask however, contains dead seeds. The seeds are disinfected to prevent bacteria and fungi growing on them so that heat is not produced. Readings on the thermometer are taken each day for about seven days.

Observation: The thermometer in the first vacuum flask shows a rise in temperature while that of the second vacuum flask remains more or less unchanged.

Conclusion: The rise in temperature of the thermometer in the first vacuum flask shows that germinating seeds produce heat energy.



Experiment: 5(a) To demonstrate respiration in yeast cell.

Materials required: Brewer's yeast, glucose, lime water, 2 flat bottom flasks, rubber bungs, cotton wool bent glass tubes, test tubes.

Method: Some amount of brewer's yeast is mixed with 500ml of 10% glucose in a large flat bottom flask. The mouth of the glass is then fitted with a rubber bung that carries a glass tube. The glass tube dips into a test tube which contains some lime water. The mouth of the tube is covered by some cotton wool. A similar experiment (control) is set up but the glucose in the flask does not contain yeasts. Both experiments are left in a warm place.

Observation: Within a short time, the mixture of the glucose and yeasts froths. There is odour of alcohol. The lime water in the test tube turns milky.

Conclusion: The milkiness in the test tube is due to carbon (IV) oxide produced by the yeast cells during respiration. This is confirmed by the control experiment where the lime water remains unchanged. This shows that carbon (IV) oxide is liberated by the yeast cells.

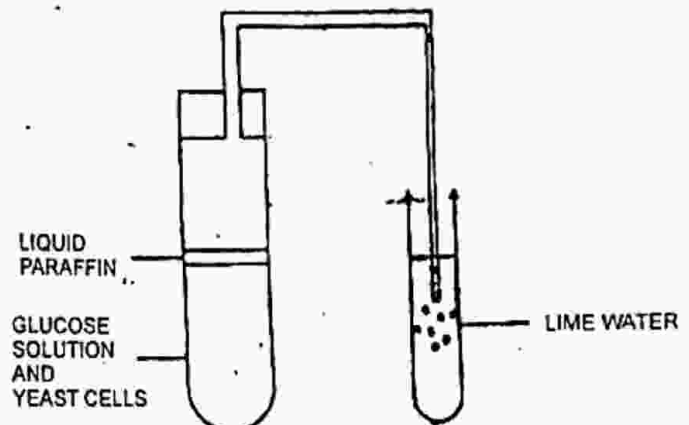
Experiment: 5(b) To demonstrate respiration in yeast cell.

Materials required: Some quantity of fresh palm wine (which contains yeast cells), a round bottom flask, rubber stopper, glass tube, test tube and lime water.

Method: Some fresh palm wine is poured into a round bottom flask. A rubber stopper which carries a glass tube is inserted into the mouth of the flask. The free end of the tube is directed into a test tube which contains lime water. The experiment is then left for some time.

Observation: The lime water in the test tube goes milky.

Conclusion: The milkiness is due to carbon (IV) oxide produced by the activities of the yeast cells. This shows that carbon (IV) oxide is liberated by the yeast cells.

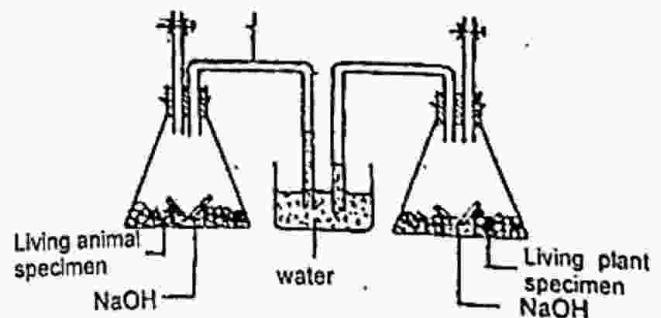


Revision Questions

9.1. OUTLINE OF RESPIRATORY PROCESS AND SIGNIFICANCE

1. UME/92/17

In the absence of oxygen, the pyruvic acid produced during glycolysis is converted to CO₂ and A. water B. glycerol
C. ethanol D. citric acid



2. UME/94/19

The most appropriate title for the set up is

- A. quantitative measurement of respiration in plant and animals
- B. measurement of respiration rates in living organisms
- C. comparison between photosynthesis and respiration
- D. comparison of respiratory rates in plants and animals.

3. UME/94/20

The part labeled J is called

- A. porosimeter
- B. photometer
- C. manometer
- D. auxanometer.

4. UME/95/16

The end product of glycolysis in plants and animals is

- A. pyruvic acid B. citric acid
C. aspartic acid D. malic acid.

5. PCE/2004/19/Type:9

The significance of respiration in organisms is the

- A. oxidation of food and release of useable energy
B. breakdown of waste materials for elimination
C. breathing in of oxygen and breathing out of carbon (IV) oxide
D. build-up of useful materials in the body

9.2. RESPIRATORY ORGANS AND SURFACES

6. UME/81/22

What is the name of the respiratory organ of the crab?

- A. Tubules
B. Trachea
C. Spiracles
D. Lungs
E. Gills

7. UME/82/14

A grasshopper respire by means of

- A. lung – books
B. gills
C. lungs
D. antennae
E. tracheal tubes.

8. UME/82/12

The glottis is the opening which leads to the

- A. oesophagus B. larynx C. nostrils
D. pharynx E. mouth

9. UME/85/23

Identify which of the following are characteristics of the vertebrate respiratory surface

1. Moist 2. Vascularized 3. Semi permeable
4. Freely-permeable 5. Dry
A. 1, 2, 3 B. 1, 2, 5
C. 2, 3, 5 D. 2, 4, 5
E. 1, 3, 5

10. UME/88/22

In the mammalian respiratory system, exchange of gases occurs in the

- A. lungs B. bronchi
C. bronchioles D. alveoli

11. UME/88/13

Lung books are used for respiration in

- A. spiders B. insects
C. millipedes D. snails

12. UME/92/16

Gaseous exchange in Aves occurs in the

- A. tracheoles B. bronchi
C. air sacs D. trachea.

13. PCE/2003/50/Type:N

The main respiratory organ of whales is the

A. skin

C. spiracles

B. lungs

D. gills

9.3. THE MECHANISM OF GASEOUS EXCHANGE

9.3.1. INSECTS

14. PCE/2004/20/Type:9

The correct pathway for the movement of air in respiration in insects is

- A. spiracle → trachea → tracheoles → body tissues
B. trachea → spiracle → tracheoles → body tissues
C. air sacs → spiracle → trachea → body tissues
D. spiracle → air sacs → trachea → body tissues.

9.3.2. FISH

15. PCE/2004/10 Type:9

Gaseous exchange in the bony fish is carried out through the

- A. nostrils B. lateral lines
C. gills D. skin

9.3.3. TOADS

16. PCE/2001/18 Type:T

The method of respiration found in an adult toad are

- A. cutaneous, buccal and olfactory
B. cutaneous and pulmonary
C. buccal and pulmonary
D. cutaneous, buccal and pulmonary

17. PCE 07/15

Gaseous exchange in tadpoles takes place in the

- A. lungs
B. gills
C. mouth
D. operculum

9.3.4. MAMMALS

18. UME/82/38

Which of the following statements is TRUE of inhalation of air by man?

- A. the ribs are not raised
B. diaphragm is raised
C. intercostal muscles relaxes
D. pressure of the thoracic cavity increases
E. size of thoracic cavity increase.

19. UME/86/27

Which of the following statements is NOT correct with respect to inhalation in mammals? The

- A. intercostal muscles contract
B. diaphragm is raised
C. ribs are raised
D. pressure of thoracic cavity decreases.

20. UME/2006/5

An increase in air pressure in the lungs is due to the

- A. increase in the volume of the thoracic cavity
B. upward movement of the ribs
C. relaxation of the diaphragm
D. contraction of intercostals muscles

...lower in temperature
...has more oxygen
...has more water vapour

20 TYPE:

ME of muscle that separates the thoracic and the abdominal cavities is the

- 1. apical muscle
- 2. pericardium
- 3. diaphragm
- 4. intercostal muscle

21 PCE/99/14

When the intercostal muscles contract during inspiration, they move

- 1. upward and forward
- 2. downward and inward
- 3. upward and inward
- 4. downward and forward

22 PCE/06/14

Diffusion of oxygen in the mammalian lung is enhanced by

- 1. presence of haemoglobin on the red blood cells
- 2. low diffusion gradient between gases in the alveoli
- 3. low concentration of oxygen in the blood
- 4. absence of haemoglobin in the red blood cells

23 PLANTS

24 UME/95/17

During respiration, air circulates round plant tissues via the

- 1. lenticels
- 2. stomata
- 3. guard cells
- 4. intercellular spaces

26 UME/78/11

The function of lenticels is

- 1. to remove excess water in the plant
- 2. to absorb water from the atmosphere
- 3. for gaseous exchange
- 4. to absorb light

27 UME/99/23

In woody stems, gaseous exchange takes place through the

- 1. metacarpals
- 2. lenticels
- 3. stomata
- 4. vessels

28 PCE/95/21

Investigation

- 1. decrease in solute concentration
- 2. increase in solute concentration in the guard cell
- 3. presence of a pore.

30. UME/2006/12

The formation of water in the tissue respiration results from the

- 1. breakdown of water molecules
- 2. reduction of oxygen by hydrogen
- 3. reduction of carbon (iv) oxide
- 4. combination of water molecules

9.4. AEROBIC RESPIRATION

31. UME/87/22

The oxidative part of the respiratory process takes place in the

- 1. mitochondria
- 2. ribosomes
- 3. endoplasmic reticulum
- 4. golgi bodies.

32. UME/2001/16/Type: T

The gas produced during tissue respiration can be identified by using

- 1. calcium hydroxide
- 2. calcium carbonate
- 3. copper sulphate
- 4. sodium hydroxide

33. UME/2002/40/Type: Y

Tissue respiration is important for the

- 1. absorption of oxygen into the alveoli
- 2. release of carbon (iv) oxide into the lungs
- 3. release of energy for body use
- 4. exhalation of carbon (iv) oxide from the lungs

34. PCE/2001/19/Type: 4

The life-saving respiratory technique applied to persons who are nearly drowned, electrocuted or shocked is

- 1. anaerobic respiration
- 2. artificial respiration
- 3. aerobic respiration
- 4. tissue respiration.

35. UME/2004/14/Type: 4

The anaerobic fermentation of a glucose molecule yields

- 1. pyruvic acid and alcohol
- 2. 38 ATP molecules
- 3. water and carbon (iv) oxide
- 4. 2 ATP molecules and alcohol

9.4. ANAEROBIC RESPIRATION

36. UME/81/25

In a bakery, yeast is added to flour to make bread rise. This is possible because yeast produces

- 1. alcohol
- 2. oxygen
- 3. carbon dioxide
- 4. energy

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- A. the addition of saccharin
- B. the addition of sugar
- C. the addition of whisky or local gin
- D. some complex hydrolysis reactions
- E. anaerobic respiration of an organism

38. UME/84/18

Anaerobic respiration in yeast produces

- A. carbon dioxide and ethanol
- B. carbon dioxide and water
- C. carbon dioxide and oxygen
- D. carbon dioxide and glucose
- E. ethanol and water.

39. UME/85/22

Which of the following events does NOT occur during anaerobic respiration of glucose?

- A. Muscle cells produces lactic acid
- B. Carbon dioxide is produced
- C. Milk bacteria produce lactic acid
- D. Energy is not produced
- E. Germinating seeds produced alcohol.

40. UME/86/28

The equation that can be used to summarize the process of anaerobic breakdown of sugar is

- A. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
- B. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
- C. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$
- D. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{Energy}$

41. UME/87/23

Fatigue of leg muscles may occur after riding many kilometers on a bicycle because of

- A. insufficient glucose
- B. excess carbon dioxide
- C. excess protein
- D. insufficient oxygen

42. UME/97/19

Anaerobic respiration differs from aerobic respiration by the production of

- A. less amount of energy and water
- B. greater amount of energy and alcohol
- C. less amount of energy and alcohol
- D. greater amount of energy and water.

43. UME 2009/20

The enzymes of the glycolytic pathway are located in the

- A. mitochondria
- B. gastric juice
- C. plasma
- D. cytoplasm

44. UME 2010/19/TYPE: D

Yeast respire anaerobically to convert simple sugar to carbon (iv) oxide and

- A. acid
- B. oxygen
- C. water
- D. alcohol

MISCELLANEOUS QUESTIONS (9)

45. The surface of an alveolus in a mammal is well supplied with tiny blood vessels known as

- A. capillaries

- B. arteries
 - C. arterioles
 - D. venules
- (UME/2004/20.Type:4)

46. The anaerobic fermentation of a glucose molecule yields

- A. pyruvic acid and alcohol
- B. 38ATP molecules
- C. Water and Carbon dioxide
- D. 2TP molecules and alcohol (UME/2004/14/Type:4)

47. A test tube containing yeast in glucose solution was suspended in a covered conical flask containing alkaline pyrogallol. The bubbles of carbon (iv) oxide produced indicate that the yeast cells are

- A. respiring in the absence of oxygen
 - B. liberating oxygen on their own
 - C. living and consuming oxygen
 - D. being killed by the alcohol produced
- (UME/2005/20/Type: D)

48. When yeast respire anaerobically, it converts simple sugars to carbon (iv) and

- A. oxygen
- B. acid
- C. alcohol
- D. water (UME/2001/19/type: T)

49. The respiratory surface in amphibians are modified for gaseous exchange by being

- A. rough and tough
- B. moist and rough
- C. moist and tough
- D. moist and vascularised (PCE/2002/27/Type:5)

50. The tiny holes in the cuticle of insects used for gaseous exchange are known as

- A. tracheoles
- B. book lungs
- C. spiracles
- D. tracheae (PCE/96/16)

51. Which of the following uses diffusion as the principle method of gaseous exchange?

- A. lizard
- B. grasshopper
- C. rat
- D. earthworm (UME 2007/32)

52. The main respiratory substrates of all cells are

- A. oligosaccharides
 - B. disaccharides
 - C. polysaccharides
 - D. monosaccharides
- (PCE/92/22)

53. During respiration, the incomplete oxidation of glucose to produce pyruvic acid is known as

- A. Krebs cycle
- B. aerobic respiration
- C. glycolysis
- D. autolysis (PCE/93/22)

54. The life-saving respiratory technique applied to person who is nearly drowned, electrocuted or shocked is

- A. anaerobic respiration
- B. artificial respiration

55. The word respiration when simply used means the taking in and expulsion of air from the lungs. But in the widest sense means

- A. Removal of energy
- B. Loss of energy by food
- C. Provision of energy
- D. Oxidation of food to provide energy
- E. None of the above

56. Apart from the stomata, through which other way can oxygen enter the shoot of the plant?

- A. Oxygen can enter the shoot of the plant by simple diffusion
- B. It can enter through any hole the stem
- C. It can enter through cork splits of stems
- D. It can enter through the lenticels
- E. It can enter through the scars of leaves

57. Respiration is a process that makes use of oxygen releasing carbon dioxide. A plant respiring in darkness releases quite a large quantity of carbon dioxide. This means:

- A. That respiration is a dark process
- B. That respiration takes place only in the night
- C. That respiration can be effective in the dark
- D. That when plants are respiring, carbon dioxide can only be released in the night
- E. None of the above

58. Roots can only take in oxygen when it is:

- A. In gases state
- B. Liquid state
- C. In chemical combination with other elements
- D. In solution with water
- E. None of the above

59. What is the name of the structure through which air gets into the body of the insects?

- A. Trachea
- B. Tracheoles
- C. Spiracle
- D. Ostia
- E. None of the above

60. The only section of the respiratory system of the insect that is not cuticularized is:

- A. The trachea
- B. Tracheole
- C. Spiracle
- D. Stigmata
- E. None of the above

61. The most essential element in the respiratory structure of the fish is:

- A. The gill
- B. The gill rakers
- C. The gill arch
- D. The gill filament
- E. The gill ray

62. In the bony fish, that large plate that covers the respiratory structure is called:

- A. The gill cover
- B. The spiracle
- C. The gill slit
- D. The operculum
- E. None of the above

63. The passage of water over the gills is due to:

- A. The closing of the mouth of the fish
- B. The pressure of water on the operculum
- C. The horizontal disposition of the gill filaments
- D. The raising of the floor of the mouth
- E. The shutting and raising of the floor of the mouth of the fish

64. The respiratory apparatus of the bird is made complex by the possession of:

- A. Air bladder
- B. Diaphragm
- C. Air sacs
- D. Hollow bones
- E. None of the above

65. Of what importance are the air sacs to bird?

- A. The air sacs are useful in gaseous exchange
- B. They are useful in the storage of air
- C. They are blood purifiers
- D. They are speed accelerators
- E. They sustain the sound box of birds

66. In man, the nasal cavity is separated from the buccal cavity by a bone called:

- A. The turbinal bone
- B. The palate complex
- C. Soft palate
- D. The tongue bone
- E. Hyoid apparatus

67. The trachea of the respiratory system is fortified by:

- A. Tiny bones
- B. Cartilage bones
- C. Elastic skin
- D. Ligaments
- E. Glottis

68. The tiny air spaces found in the lungs of mammals are called:

- A. Air sacs
- B. Alveoli
- C. Tracheoles
- D. Bronchioles
- E. Pleura

69. The auditory structures (vocal cords) of mammals are located in

- A. The trachea
- B. The glottis
- C. The wind-pipe
- D. The larynx
- E. The lung

70. What structures are responsible for the intake and output of air from the lungs?

- A. The air sacs and trachea
- B. The larynx and bronchi
- C. The ribs and intercostals muscles
- D. The ribs and the diaphragm
- E. None of the above

71. To take in air into the lungs the diaphragm must be:

- A. Dome-shaped
- B. Flattened
- C. Oblique
- D. Normal
- E. None of the above

72. What happens to air in the lungs?

- A. It diffuses into lungs
- B. It goes into solution at the respiratory surface
- C. It goes into solution with haemoglobin
- D. It is compressed into liquid
- E. None of the above

73. Oxygen is taken round the body as

- A. Oxyhaemoglobin
- B. Haemoglobin
- C. Haem
- D. Blood
- E. Lymph

CHAPTER 9

1. C	2. D	3. C	4. A	5. A
6. E	7. E	8. B	9. A	10. D
11. A	12. C	13. B	14. A	15. C
16. D	17. B	18. E	19. B	20. C
21. D	22. D	23. A	24. C	25. D
26. C	27. C	28. D	29. C	30. B
31. A	32. A	33. C	34. B	35. D
36. C	37. E	38. A	39. D	40. D
41. D	42. C	43. D	44. D	45. A
46. D	47. A	48. C	49. D	50. C
51. D	52. D	53. C	54. B	55. D
56. D	57. E	58. D	59. C	60. B
61. D	62. D	63. E	64. C	65. B
66. B	67. B	68. B	69. D	70. E
71. B	72. E	73. A		

CHAPTER 10

EXCRETION

Understand the following after reading this chapter:

- Types of excretory structures
- Excretory mechanism
- Excretory products of plants and animals

EXCRETION

Excretion is the elimination of waste products of metabolism from the cells.

Metabolism, on the other hand, is the sum total of chemical activities in the cell. It involves **anabolism** (simple molecules are built up into larger complex one e.g. photosynthesis) and **catabolism** (complex molecules are broken down into smaller, simpler ones e.g. respiration).

EXCRETORY ORGANELLES/ ORGANS IN ANIMALS

Animal	Organelle/Organ
1. Protozoa	Contractile vacuole
2. Flat worms	Flame cell
3. Roundworms	Nephridia
4. Insects	Malpighian tubules
5. Fishes	Gills
6. Land vertebrates	Kidney, skin, lung, liver and *large intestine.

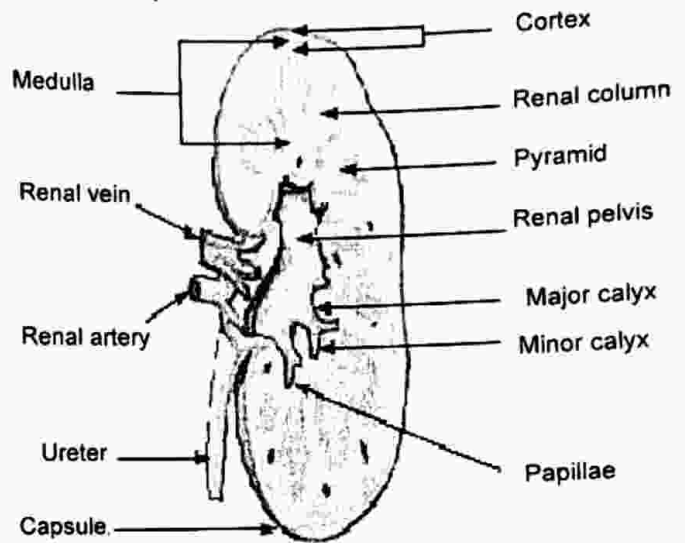
*excrete iron and calcium.

THE HUMAN EXCRETORY SYSTEM

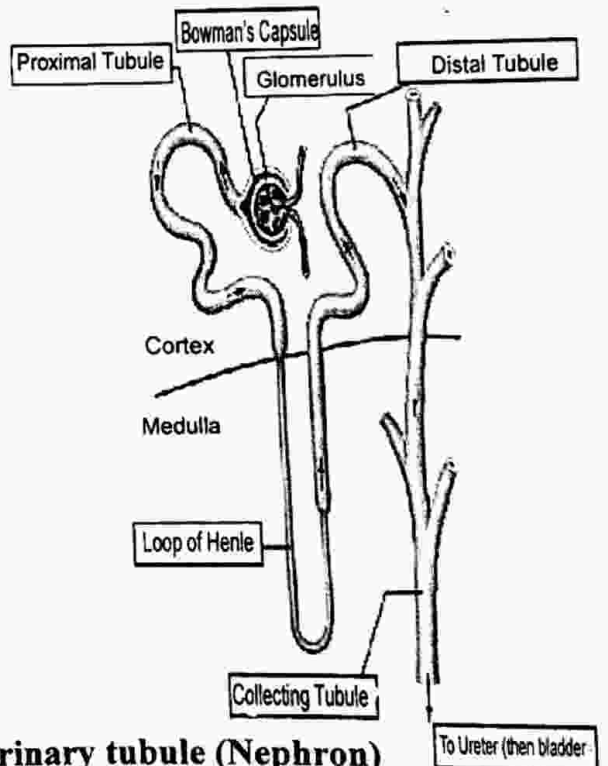
The urinary system is made-up of the kidneys, ureters, bladder, and urethra. The nephron, an evolutionary modification of the nephridium, is the kidney's functional unit. Waste is filtered from the blood and collected as urine in each kidney. Urine leaves the kidneys by ureters, and collects in the bladder. The bladder can distend to store urine that eventually leaves through the urethra.

The Nephron

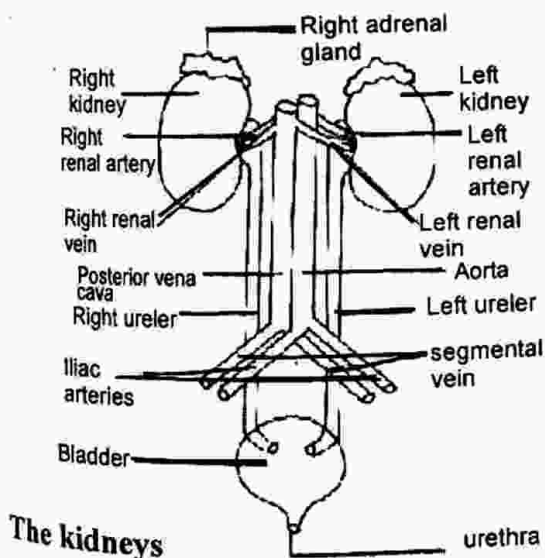
The **nephron** consists of a cup-shaped capsule containing capillaries and the **glomerulus**, and a long **renal tube**. Blood flows into the kidney through the renal artery, which branches into capillaries associated with the glomerulus. Arterial pressure causes water and solutes from the blood to filter into the capsule. Fluid flows through the **proximal tubule**, which include the **loop of Henle**, and then into the **distal tubule**. The distal tubule empties into a collecting duct. Fluids and solutes are returned to the capillaries that surround the nephron tubule.



Longitudinal section of the kidney



The urinary tubule (Nephron)



The kidneys

The nephron has three functions:

1. Glomerular filtration of water and solutes from the blood.
2. Tubular reabsorption of water and conserved molecules back into the blood.
3. Tubular secretion of ions and other waste products from surrounding capillaries into the distal tubule.

Nephrons filter 125 ml of body fluid per minute; filtering the entire body fluid component 16 times each day. In a 24 hour period nephrons produce 180 liters of filtrate, of which 178.5 liters are reabsorbed. The remaining 1.5 liters forms urine.

Urine Production

1. Filtration in the glomerulus and nephron capsule.
2. Reabsorption in the proximal tubule.
3. Tubular secretion in the Loop of Henle.

Components of the Nephron

- Glomerulus: mechanically filters blood
- Bowman's Capsule: mechanically filters blood
- Proximal Convoluted Tubule: Reabsorbs 75% of the water, salts, glucose, and amino acids
- Loop of Henle: Countercurrent exchange, which maintains the concentration gradient
- Distal Convoluted Tubule: Tubular secretion of H ions, potassium, and certain drugs.

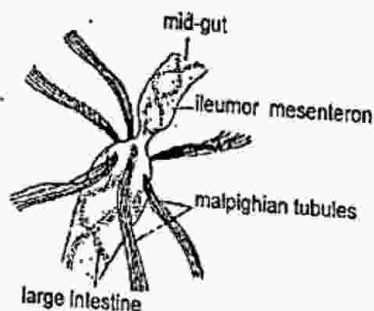
Kidney Function

Kidneys perform a number of homeostatic functions:

1. Maintain volume of extracellular fluid
2. Maintain ionic balance in extracellular fluid
3. Maintain pH and osmotic concentration of the extracellular fluid.
4. Excrete toxic metabolic by-products such as urea, ammonia, and uric acid.

EXCRETORY SYSTEM IN INSECTS

Body fluids are drawn into the Malpighian tubules by osmosis due to large concentrations of potassium inside the tubule. Body fluids pass back into the body, nitrogenous wastes empty into the insect's gut. Water is reabsorbed and waste is expelled from the insect.

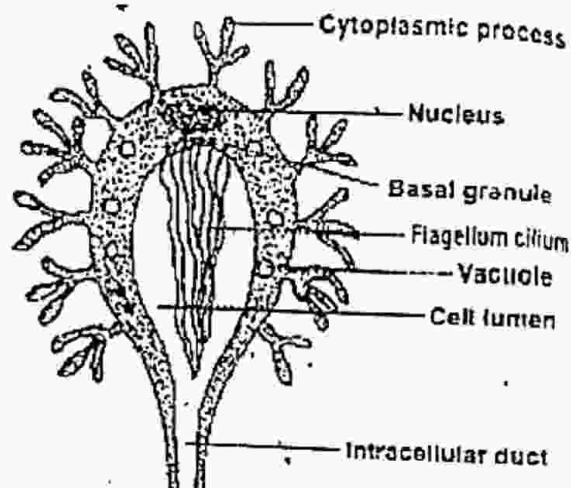


Malpighian tubule

EXCRETORY SYSTEM IN FLAT WORMS

The simplest tubular excretory system is the flame-cell system of flatworms (Phylum Platyhelminthes). These animals have neither circulatory systems nor coeloms and so the flame-cell system must regulate the contents of the interstitial fluid directly.

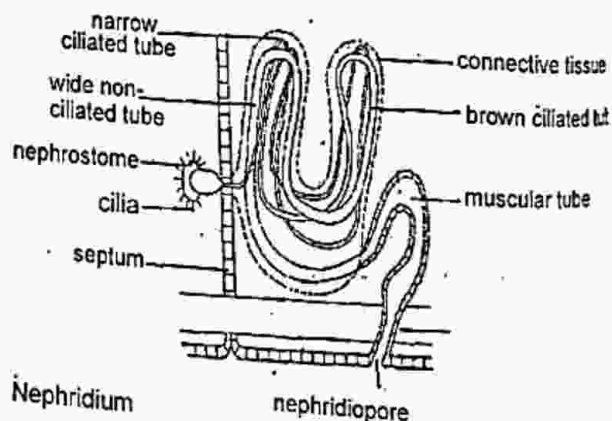
The apparatus consists of a branched system of tubules ramifying throughout the body. Each of the smaller tubules at the tips of this excretory tree is capped by bulbous cell called a flame cell. Interstitial fluid bathing the tissues of the animal passes through the flame cell and enters the tubular system.



A flame cell

EXCRETORY SYSTEM IN EARTHWORMS

In earthworm excretion occurs through nephridium. The excretory system in earth worm is closely associated with its closed circulatory system. Each segment of earthworm consists of a pair of nephridia or in some species cluster nephridium. The coelomic fluid or body fluid enters the nephridium through a membrane which is ciliated called nephrostome. The body fluid after reabsorption from the nephridia opens into a large bladder which opens exterior through nephridiopore. Ammonia and water and waste products excreted by nephridia while CO₂ excreted through general body surface.



EXCRETION IN PLANT

Plants excrete water, oxygen and carbon dioxide by diffusion through the stomata and lenticels.
Waste products such as tannins, mucilage, gum, alkaloids etc. are excreted at the bark of the stem or the leaves which are removed through peeling and leaf fall respectively.

Revision Questions

1. TYPES OF EXCRETORY STRUCTURES

1. UME/79/16

Which of the following is NOT an excretory organ?

- A. Lungs
- B. Kidney
- C. Leaf
- D. Large Intestine
- E. Skin

2. UME/92/18

The excretory organ in insects is

- A. Kidney
- B. Malpighian tubule
- C. Flame cell
- D. Nephridium

3. UME/93/8

Flame cells are the

- A. Excretory system of worms
- B. Excretory and respiratory systems of flatworms
- C. Secretory system of flatworms
- D. Excretory system of flatworms

4. UME/98/20

In which of the following groups of animal is the malpighian tubule found?

- A. Lizards, snakes and frogs
- B. Crickets, houseflies and grasshoppers
- C. Millipedes, centipedes and scorpions
- D. Earthworms, roundworms and flatworms

5. PCE/00/18/TYPE: A

Nephridium is used as an excretory structure in

- A. crustaceans
- B. annelids
- C. platyhelminthes
- D. coelenterates

10.2. EXCRETORY MECHANISM

10.2.1. KIDNEYS

6. UME/80/22

Which part of the following parts of the mammalian body is most closely associated with the production of urine?

- A. Malpighian capsule
- B. Urinary bladder
- C. Ureter
- D. vas deferens
- E. Urethra

7. UME/80/29

During excretion of urea there is also a corresponding reabsorption of water into the blood stream. This reabsorption takes place in

- A. Uriniferous tubules
- B. Bowman's capsule
- C. Glomerulus
- D. Malpighian capsule
- E. Renal artery

8. UME/87/24

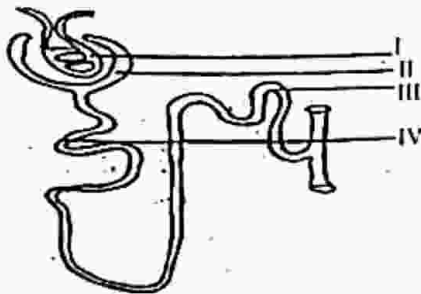
The function of the loop of Henle is to

- A. increase the flow of urine
- B. concentrate amino acids in the kidney tissue
- C. concentrate sodium chloride in the medulla of the kidney
- D. increase the volume of urine

9. UME/88/24

In the kidney, the malpighian corpuscle is located in the

- A. medulla
- B. hilum
- C. cortex
- D. pelvis



10. UME/2000/10/Type M

The parts labeled I and II make up the

- A. glomerulus
- B. convoluted tubules
- C. malpighian body
- D. Bowman's capsule

11. UME/2000/11/Type M

In mammals, reabsorption of salt takes place in

- A. IV
- B. III
- C. II
- D. I

12. UME/2006/15

Mammals are capable of producing hypertonic urine mainly because of reabsorption in the

- A. Bowman's capsule
- B. urethra
- C. Ureter
- D. loop of henle

13. UME/2007/28

The correct sequence of the movement of urea during urine formation is

- A. glomerulus → Bowman's capsule → convoluted tubule → henle's loop → collecting tubule
- B. glomerulus → Bowman's capsule → convoluted tubule → henle's loop → convoluted tubule → collecting tubule
- C. convoluted tubule → glomerulus → henle's loop → Bowman's capsule → collecting tubule
- D. convoluted tubule → Bowman's capsule → henle's loop → glomerulus → collecting tubule

14. UME/2008/21

In insects, the structure that performs the same function as the kidney in man is the

- A. nephridium
- B. flame cell
- C. malpighian tubule
- D. trachea

15. UME/2010/22 TYPE: D

The outer layer of the kidney where the bowman's capsules are found is the

- A. pelvis
- B. medulla
- C. pyramid
- D. cortex

16. PCE/01/21/TYPE A

The structure that connects the kidney with the urinary bladder is the

- A. urethra B. ureter C. pelvis D. pyramid

10.2.2. SKIN

17. UME/88/25

The skin, through the sweat glands, functions as

- A. an excretory organ
B. a respiratory organ
C. a sensory organ
D. a protective organ



18. UME/2004/15/Type 4

The function of the part labeled III is to

- A. produce oil for the skin
B. carry blood and nitrogenous waste
C. contract to pull the hair erect
D. conduct nervous impulses

19. UME/2004/16/Type 4

The sweat gland is the structure labeled

- A. IV B. III C. II D. I

20. UME/86/31

In the mammalian skin, melanin and keratin are contained in the

- A. sebaceous gland
B. sweat gland
C. subcutaneous gland
D. malpighian layer

21. UME 2010/22/TYPE: D

The oily substance that lubricates the mammalian hair to keep it flexible and water repellent is secreted by the

- A. sebaceous glands
B. fatty cells
C. granular layer
D. sweat glands

10.3. EXCRETORY PRODUCTS OF PLANTS AND ANIMALS

10.3.1. PLANTS

22. UME/84/30

Which of the following is NOT a waste product of plants?

- A. Tannins
B. Oxygen
C. Carbondioxide
D. Sap
E. Alkaloids.

23. UME/88/23

The main waste products formed in plants are

- A. Alkaloids, tannins and resins

B. Water, alkaloids and carbondioxide

C. Oxygen, carbondioxide and alkaloids

D. Water, carbondioxide and oxygen

24. UME/90/24

Excretory products responsible for the red, purple and blue colours of flowers are called

- A. alkaloids B. tannins C. anthocyanins D. resins

25. UME/94/22

Which of the follow waste products in plants is excreted through the stomata and lenticels?

- A. Carbondioxide B. Alkaloids
C. Tannins D. Anthocyanins

26. PCE/2000/19/Type A

The excretory products of higher plants which is also used for another metabolic process is

- A. nitrogen B. oxygen
C. carbon (iv) oxide D. water vapour

27. UME 2007/31

The waste product of plants used in the conversion of hid to leather is

- A. gum B. alkaloid
C. tannin D. resin

28. UME/82/43

Which of the following is NOT associated with excretion in mammals

- A. Glomerulus B. Urea
C. Faeces D. Carbondioxide
E. Bowman's Capsule

29. PCE/2003/40/Type N

The major excretory product of insect is

- A. sodium chloride
B. ammonia
C. Urea
D. Uric acid

MISCELLANEOUS QUESTIONS (10)

30. Reabsorption of chemical solutes in the kidney takes place in

- A. Bowman's capsule B. Tubules
C. Glomerulus D. Nephron (PCE/95/22).

31. The urinary tubules of the kidney function through

- A. Osmosis and diffusion
B. active transport and osmosis
C. ultra filtration and selective reabsorption
D. active transport and cytoplasmic streaming (UME/2005/21/ype D)

32. The kidney is connected to the bladder by the

- A. Cortex B. Urethra
C. Urinary bladder D. Ureter
(PCE/2005/19/Type P)

33. The excretory products in plants include

- A. gum, uric acid and mucilage
B. tannin, anthocyanin and resin

carotyanin, sweat and tannin
mucin, mucilage and urea (PCE/2005/12/type P)

32. In the mammalian kidney, the Bowman's capsules are
A. Cortex
B. Medulla
C. Pelvis
D. Cortex (PCE/94/21)

33. In a young human embryo, the excretory function is
performed by the
A. chorion
B. allantois
C. amnion
D. embryonic membrane
(PCE/92/21)

34. Excretion is defined as:
A. The voiding out from the body of obnoxious substances
such as faeces.
B. The elimination of only urea from the body
C. The elimination of water and salts from the skin as
sweat.
D. The elimination from the body of poisonous by products
of metabolism
E. None of the above

37. The excretory structures in flatworms are the
A. contractile vacuoles
B. nephridia
C. flame cells
D. kidneys (PCE/93/23)

38. One a cold day, more urine is produced because
A. little sweat is produced
B. metabolic rate is reduced
C. water intake is more
D. water absorption is less (PCE/96/17)

39. Substances from the glomerulus pass to the Bowman's
capsule by
A. selective reabsorption
B. diffusion
C. absorption
D. ultrafiltration
(PCE/03/32/TYPE: N)

40. The correct sequence in the formation of urine in the
mammalian kidney is
A. secretion reabsorption ultrafiltration
B. reabsorption ultrafiltration secretion
C. ultrafiltration reabsorption secretion
D. ultrafiltration secretion reabsorption
(PCE/04/22/TYPE: 9)

41. Which of these is not an excretory products?
A. Urea
B. Ammonia
C. Carbon dioxide
D. Water
E. None of the above

42. How is urea formed in the body?
A. The ammonia obtained from the deamination of amino-
acids reacts with carbon dioxide.
B. Urea is a product of uric acid decomposition
C. Urine reacts with water to produce urea
D. Proteins are broken down into urea
E. None of the above

43. Which of these is not an excretory organ?
A. The kidney
B. The skin
C. The liver
D. The lungs
E. None of the above

44. What is the name given to that part of the kidney in
which the pyramids are found?
A. The medulla
B. The Cortex
C. The pyramidal zone
D. The pelvis
E. None of the above

45. The outer zone of the kidney is:
A. The medulla
B. The periphery
C. The cortex
D. The capsular zone
E. The cortical membrane

46. Which of these structures is in no way connected with
the kidney?
A. The malpighian capsule
B. The glomerulus
C. The pyramid
D. The renal vein
E. None of the above

47. What is the function of the pyramids of the kidney?
A. The pyramids store fat
B. The pyramids bear the collecting tubes of the kidney
C. The pyramids strengthen the openings of the collecting
canals
D. None of the above reasons

48. In what way is the lung as excretory organ?
A. It is the responsible for the elimination of carbon
dioxide
B. It is responsible for the loss of the nitrogen of the air
from the tissues
C. It eliminate water from the tissues of body
D. It regulates salt metabolism
E. None of the above

49. On a cool and wet day it is observed that a large
quantity of urine is passes out by a large of students. This is
because
A. The kidney is over-active on wet days
B. Not much water is lost from the body by way of sweat
C. The uriniferous tubules can no longer absorb water
D. The convoluted tubules and the loop of Henle are
usually inactive on wet days
E. None of the above

50. In a very young embryo, the excretory materials are
stored in:
A. The bladder
B. The yolk
C. The allantois
D. The placenta
E. The embryonic membranes

51. In what way is the liver involved in excretion?
 A. It eliminates iron from the body
 B. It stores urea
 C. It converts urea into urine
 D. It manufactures urea
 E. It converts uric acid into urea

52. In the insect the organ of excretion is:
 A. The hindgut
 B. The hepatic caeca
 C. The crop
 D. The salivary gland
 E. The malpighian tubules

53. The function of the malpighian layer of the skin is:
 A. To protect the skin
 B. To separate the epidermis from the dermis
 C. To support the skin
 D. To produce new epidermal cells
 E. None of the above reasons

54. The excretory product of insects is:
 A. Urine
 B. Urea
 C. Ammonia
 D. Urates
 E. Uric acid

55. In insects, a large proportion of nitrogenous waste is stored in:
 A. The reproductive system
 B. The fat bodies
 C. The blood
 D. The gut
 E. The crop

56. In which of these processes occurring in insects is nitrogenous waste NOT lost?

- A. Ecdysis
- B. Pupation
- C. Moulting
- D. Metamorphosis
- E. Shedding of pupal case

57. How is an excretory material lost in the amoeba?

- A. Through the contractile vacuole
- B. Through the general body surface
- C. Through the food vacuoles
- D. Through the temporary anus
- E. None of the above

58. In the fish, apart from the kidney another excretory structure is:

- A. The skin
- B. The gills
- C. The lining of the buccal cavity
- D. The lateral lines
- E. None of the above

59. The sebaceous gland of the skin produces

- A. Urea
- B. Water
- C. A gelatinous substance
- D. A salt
- E. An oily substance

CHAPTER 10

1. C	2. B	3. D	4. B	5. B
6. A	7. A	8. C	9. C	10. C
11. A	12. D	13. B	14. C	15. D
16. B	17. A	18. B	19. A	20. D
21. A	22. D	23. D	24. C	25. A
26. C	27. C	28. C	29. D	30. B
31. C	32. D	33. B	34. D	35. B
36. E	37. C	38. D	39. D	40. C

CHAPTER 11

SUPPORT AND MOVEMENT

Understand the following after reading this chapter:

Supporting tissues in plants
Movement in plants
Supporting tissues in animals
Types and functions of skeleton

SUPPORTING TISSUES IN PLANTS

Specialized cells and tissues are used to provide support in plants. They include parenchyma, collenchyma, sclerenchyma and xylem tissues.

Parenchyma tissue: consists of loosely-packed, thin-walled rounded cells with air spaces between them. It is the principal tissue of the cortex and pith.

1. It makes up the ground tissue of many organs and therefore provides support.
2. It stores starch and other substances.
3. Support the plant by turgor.
4. Conducts gases and water across the plant.

Sclerenchyma tissue: the walls of the sclerenchyma cells are evenly thickened with lignin.

1. It strengthens and supports the plant.

Collenchyma tissue: the walls of the collenchyma cells are thickened at the corners.

1. It provides support for young stems and leaves to the lignin it contains.
2. It provides mechanical support to the plant.

Xylem tissue or wood: the conducting cells in the xylem are the dead elongated lignified tubes called vessels and tracheids.

1. It strengthens the tissues
2. It conducts water and other solutes



Parenchyma



Collenchyma



Sclerenchyma

Supporting tissues in plants

MOVEMENT IN PLANTS

Movement in plants nearly always involves growth.

Tropism: This is the response of parts of plants towards or away from a unilateral stimulus. Tropic movements are directional and positive if organism moves towards the stimulus and negative if it moves away from it. The response to light is phototropism, to water, hydrotropism,

to gravity, geotropism and to touch, thigmotropism. Tropism is easily demonstrated in actively growing structures. Shoots of plants are positively phototropic, roots of plants are positively hydrotropic and positively geotropic.

Nastism: This is the non-directional response of parts of plants of plants to a diffuse (general) stimulus. It may not involve growth. Examples: (1) Petals of sun-flower which open in light and close in the dark; (2) Leaves of plants (e.g. Oxalis) which spread out in light but fold up in the dark (3) Collapse of *Mimosa pudica* plant to touch this is also referred to as sleep movement.

Taxism: This is the response of a whole organism toward or away from stimulus. Tactic movements are directional and positive if organism moves towards the stimulus and negative if it moves away from it. Examples are phototaxis, chemotaxis and thermotaxis.

SKELETON AND SUPPORTING TISSUES IN ANIMALS

Skeleton is the bony framework which provides support for organisms.

Types of Skeleton

1. **Exoskeleton:** This is the outer supporting tissues of the body. It is found mostly in arthropods.
2. **Endoskeleton:** This is the inner supporting tissue of the body. It is found mostly in vertebrates.
3. **Hydrostatic skeleton:** It is made up of fluid that keeps the body turgid. Example is found in earth worm.

Functions of Skeleton

1. It provides support for the body
2. It provides rigidity and shape for the body
3. It provides a base for the attachment of muscles
4. Bones store salts, notably, calcium and phosphorus salts
5. Red blood cells are produced in the marrow of the long bones
6. It protects important organs of the body
7. It enhances movement of the body

In mammals, the endoskeleton is divided into two viz: the axial skeleton and the appendicular skeleton.

Axial Skeleton

The axial skeleton comprises the skull, vertebral column, ribs and sternum. The skull is made up of the cranium which houses the brain, the facial skeleton which anchors the nose, eyes and the muscles of the cheek and the jaws.

Each bone of the vertebral column is called a vertebra. The bones of the vertebra column is classified into five, based on their characteristic features.

Thus:

- | | | |
|-----------------------|---|--------------|
| 1. Cervical vertebrae | — | Neck region |
| 2. Thoracic vertebrae | — | Chest region |

- | | | |
|---------------------|----|--------------------|
| 3. Lumbar vertebrae | -- | Upper trunk region |
| 4. Sacral vertebrae | -- | Lower trunk region |
| 5. Caudal vertebrae | - | Tail region |

(Memory Tip: Come To London Study Commerce)

Cervical vertebrae

The atlas and axis are the first and second cervical vertebrae respectively. They possess special modification

Atlas

1. It has a highly reduced centrum
2. It has a highly reduced spine
3. It has a large neural canal
4. It has two large depressions on its anterior surface which fit into two projections at the base of the skull, so, enabling the head to nod
5. It has reduced transverse process

Axis

1. It has a prominent spine
2. It has reduced transverse process
3. It has a broad and flat centrum
4. It has a large neural canal
5. It has an elongated odontoid process which fits into the neural canal of the atlas, thus, allowing rotational movement of the head

Typical/Ordinary Cervical Vertebra

1. It has a centrum
2. It has a short neural spine
3. It has branched transverse processes
4. It has a large neural canal
5. It has vertebral foramina

Thoracic Vertebra

1. It has a long neural spine projected backwards
2. It has reduced transverse processes that articulate with the ribs.
3. It has articular facets on the centrum and transverse processes which articulate with the head and tubercle of ribs respectively

Lumbar vertebra

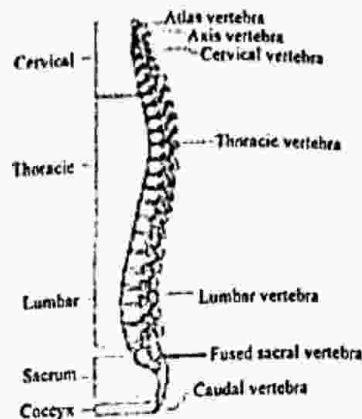
1. It has a short and flat neural spine
2. It has well developed transverse processes
3. It has a thick centrum
4. It has a well developed pre-and-post-zygapophyses

Sacral vertebrae

1. The sacral vertebrae consist of fused bones
2. The transverse process of the first and second vertebrae articulate with the pelvic girdle
3. The neural canal diminishes gradually from the first vertebra until it disappears in the last.

Caudal vertebra

1. Usually reduced to form small centrum bones
2. It lacks neural canal



Vertebral column of man

Ribs and Sternum

The ribs are long and curved bones consisting of two main parts; a bony part known as the vertebral part and a sternal part, made up of cartilage part which is attached to the sternum.

The ribs, sternum and the thoracic vertebrae form the rib cage which protect the heart and lungs.

Appendicular Skeleton

The appendicular skeleton consists of the pectoral girdle, the pelvic girdle, the forelimbs and the hind limbs

The Pectoral Girdle is made up of two halves with each consisting of a triangular bone called scapula or shoulder blade and the clavicles or collar bone

The Pelvic girdle consists of two halves known as innominate bone. Each consists of three fused bones namely, ilium, ischium and pubis.

The Forelimb is made up of a single bone nearest the body called humerus. The humerus is followed by the bones of the fore-arm called radius and ulna. The wrist comprises about 7-8 rod-like bones called carpals. The bones of the palm are called metacarpals, while the jointed one forming the fingers and thumb are called phalanges.

The Hind Limb consists of a single upper bone forming the thigh called femur and two bones of the lower leg called tibia and fibula. The small bones of the ankle are called tarsals. The rod-like bones of the sole of the foot are called metatarsals while the jointed bones of the toes are called phalanges.

Joint

A joint is a point where two or more bones meet. The bones are held together by a tissue called ligament. A joint is lubricated by the synovial fluid.

There are two types of joints, namely:

1. **Fixed or immoveable Joint:** This occurs between the flattened bones of the skull e.g. suture
2. **Moveable Joint:** This permits certain degree of movement. There are different types of moveable joints e.g.
 - i. **Hinge joint** e.g. elbow and knee joints
 - ii. **Ball and socket joint** e.g. hip and shoulder joints
 - iii. **Sliding or gliding joint** e.g. wrist and ankle joint
 - iv. **Pivot joint** e.g. neck where atlas rotate on odontoid process of axis.

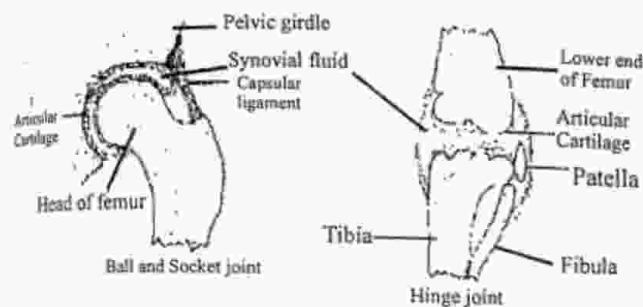
EXPERIMENT ON TROPISM

Experiment: 1 To demonstrate hydrotropism.

Materials required. A trough, top soil, a porous pot, seedlings.
Method. A trough is taken and it is two third filled with moist top soil. Some amount of soil is removed from the centre and an empty porous pot is placed where the soil has been removed. A few seedlings, say 2 or 3 are planted on the sides of the pot. The seedlings are planted about 6cm away from the pot. Without watering, the seedlings are then left for some days. Water is poured into the porous pot when the plants start to wither. A similar experiment to serve as control is also set up but the porous pot does not contain water. Both experiments are then left for many days. After this period, the young plants are dug out, exercising care so as not to damage the roots. The soil around the roots is carefully washed out to expose the roots.

Observations. The main root curves towards the direction of the porous pot. There is no curving of the root in the case of the control experiment. Here, the root is straight.

Conclusion. The curved root is in response to water which found its way out of the porous pot. This indicates that the root shows positive hydrotropism.



Two types of joints

Experiment: 2 To demonstrate geotropism.

(a) Negative geotropism (shown by most plant shoots)

Materials required. Bean seeds, container, soil, cupboard.

Method. A container is collected and top soil is packed inside it. Two or three bean seeds are sowed in the soil and the seeds are left to germinate. When the plants are at the seedling stage, the container is kept in a dark cupboard where the seedlings cannot receive light. The container is placed on its side and it is left like that overnight. The seedlings are then checked the following day.

Observations. The young shoot turns upwards.

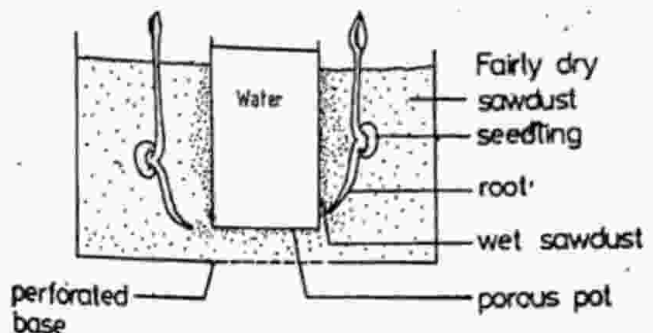
Conclusion. The upward turn shown by the shoots is as a result of negative response to the force of gravity.

(b) Positive geotropism (shown by most roots)

The above experiment is repeated. However, the container is left on its sides for about 3 days. After this period the soil around the main root is removed. It is noticed that the roots bend downwards in response to the force of gravity. The response by the root is described as **positive geotropism**.

Experiment: 3 To demonstrate phototropism.

Materials required and method. The material required for this demonstration is a pot of seedlings. The pot is placed beside a window, where light can get to the seedlings. They are left for a few days. After this period, the shoots are observed to bend towards the source of light. The side that is directed towards the light is the made to point towards another side, i.e. be directed away from the light source. The shoots again turn and face the light.
Conclusion. The turning towards the light by the shoots shows positive hydrotropism.



Experiment: 4 To demonstrate geotropic response of the shoot and root using a klinostat.

Materials required. A klinostat, a bean seedling.

Method. The cork of a klinostat is adjusted to a vertical plane, so that the bean seedling is horizontal. The apparatus is then rotated for a period of time and then stopped and left for sometime.

Observations. There was neither a bending of the shoot (plumule) upward, nor the root (radicle), downward, i.e. the shoot and root remained straight. But when the rotation stopped, the shoot and the root curved upwards and downwards respectively, after a time.

Conclusion. Both shoot and root did not respond to geotropic effect because the effects of the stimulus were diffused by the rotation of the cork disc. This shows that the force of gravity affects the growth of shoot and root. Secondly, the experiment proves that growth is due to unequal distribution of auxin.

NOTE: A klinostat is an instrument which supports the fact that phototropism and geotropism are due to unequal distribution of auxin in the shoot and root respectively. The instrument removes or neutralises the effects of directional stimuli of light and force of gravity on the plant organ. The parts of the klinostat include a clockwork mechanism. This bears a rod which carries a metal disc and to the metal disc is fitted a cork disc. During an experiment, seedlings

are pinned to the cork disc. Both discs can be adjusted to vertical or horizontal position. When the motor is on, the disc rotates slowly, thus turning the seedlings. Usually, there is one turn in every 15 minutes.

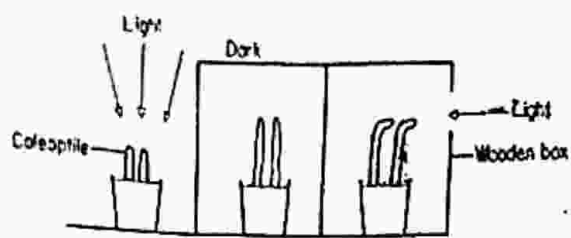
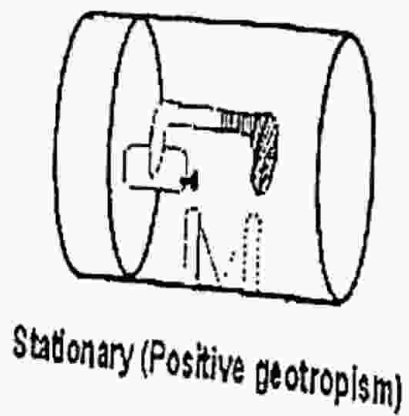
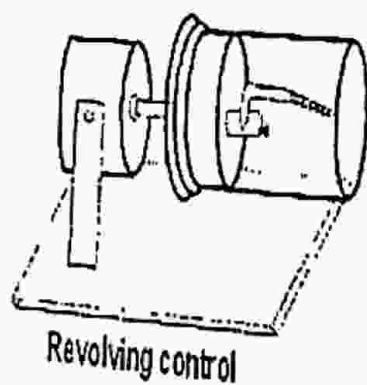
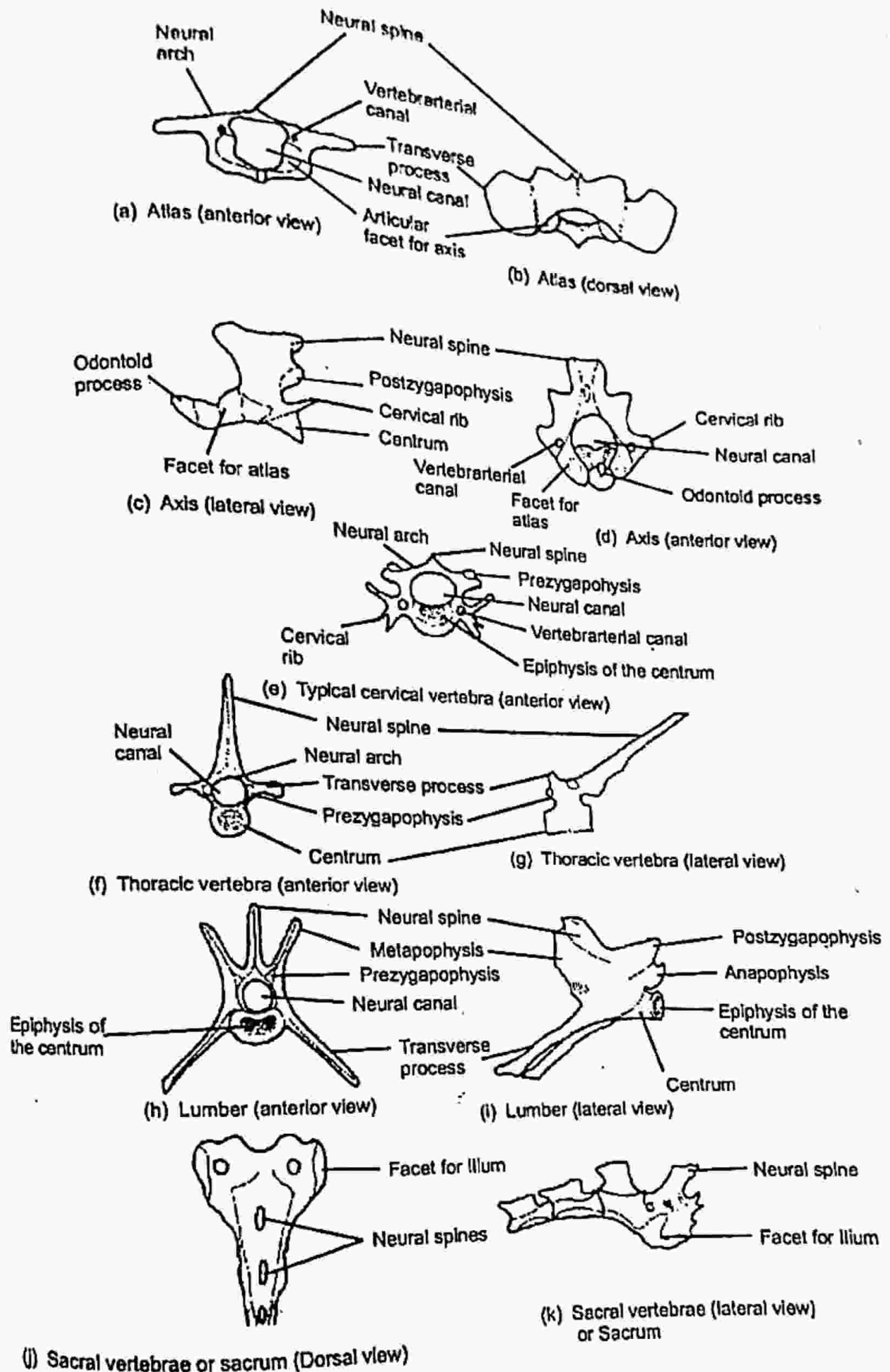
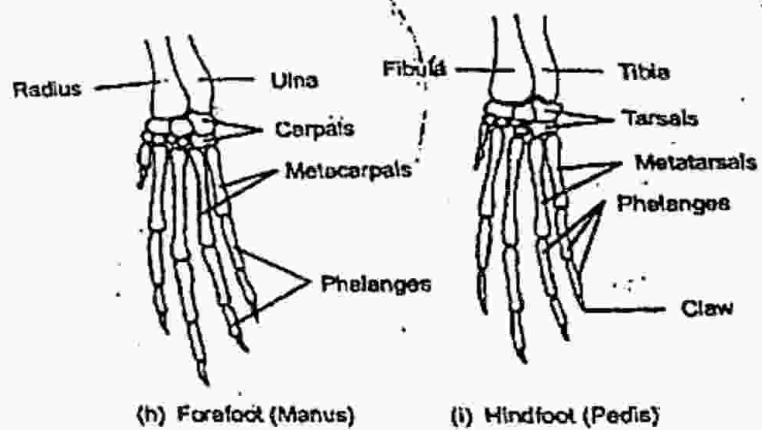
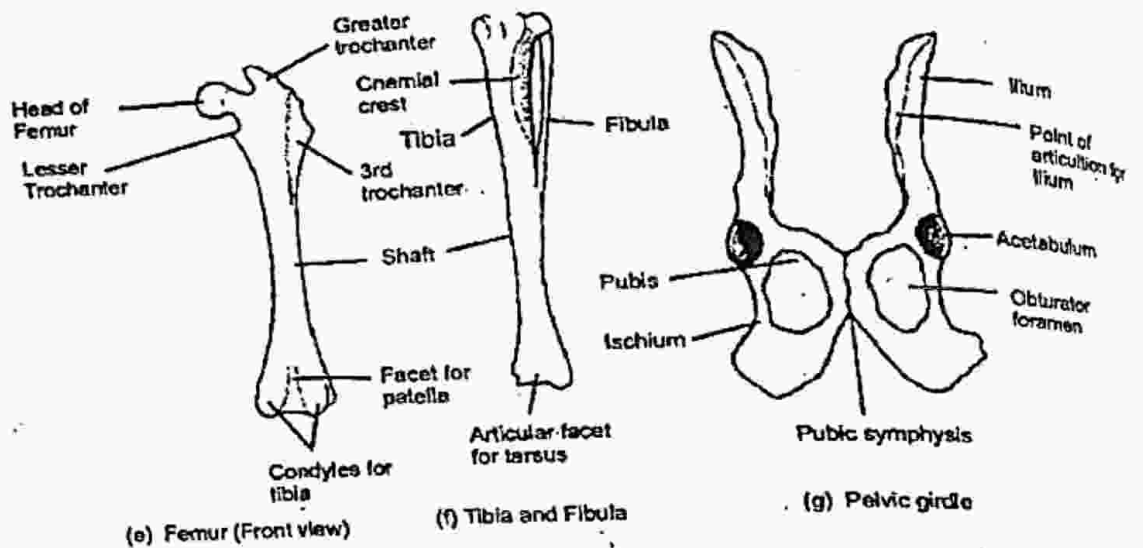
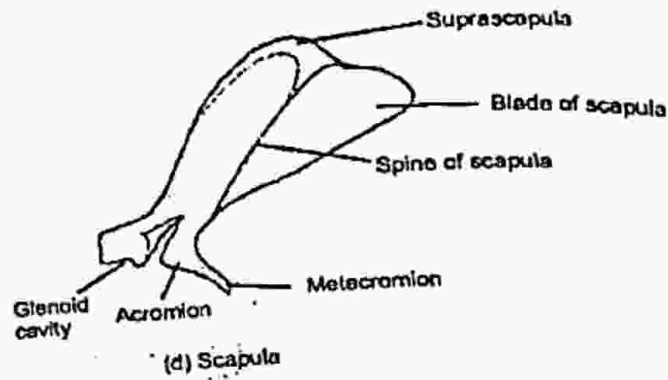
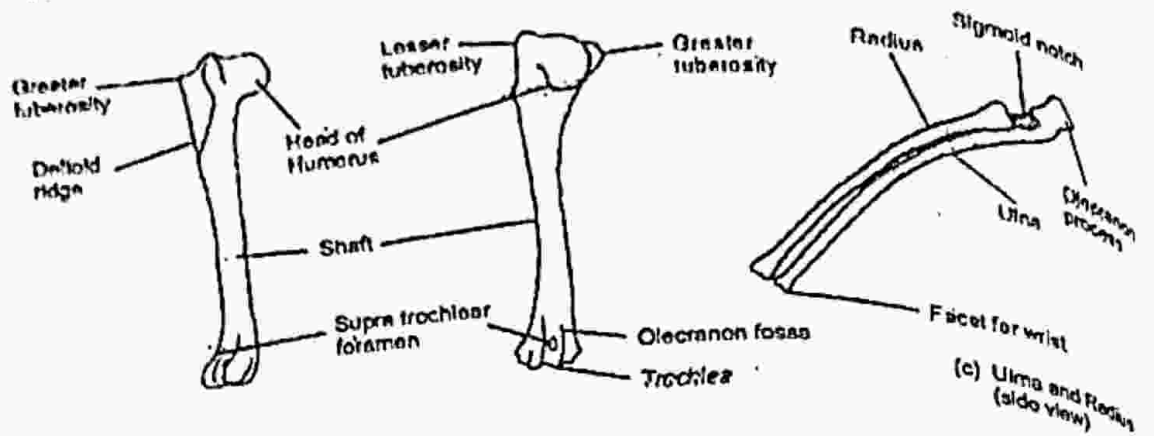


Fig. 65. Experiment to show phototropism





Parts of the axial skeleton (a-k)



Parts of Appendicular Skeleton (a-i)

Revision Questions

11.1. SUPPORTING TISSUES IN PLANTS

1. UME/79/3
Which of these tissues serves the function of support and water conduction?

- A. Parenchyma
- B. Sclerenchyma
- C. Phloem

- B. Collenchyma
- D. Xylem

2. UME/87/27
Sclerenchyma cells are lignified to
A. strengthen and support the plant
B. transport synthesized food
C. conduct water and salt
D. protect the plant from injury

3. UME/88/26
Parenchyma cells serve as supporting tissues when they
A. contain chloroplast
B. have crystals
C. become flaccid
D. become turgid

4. UME/94/17
The phloem parenchyma is sometimes used for
A. food storage
B. supporting the stem
C. production of the sieve tube
D. transporting water

5. PCE/2000/20/Type A
The plant tissue that is functionally similar to animal bones is
A. endodermal tissue
B. sclerenchymatous tissue
C. epidermal tissue
D. parenchymatous tissue

6. PCE/96/19
Which of the following contributes to the rigidity and strength of the stem?
A. Parenchyma
B. Cambium
C. Sclerenchyma
D. Xylem

7. UME 2008/22
A plant parenchyma cell also acts as a supporting tissue when it
A. becomes flaccid
B. contains crystals
C. becomes turgid
D. is pigmented

11.2. MOVEMENT IN PLANTS

8. UME 81/42
Growing yam tendrils climb for support. The growth response is
A. haptotropism
B. geotropism
C. phototropism
D. hydrotropism
E. chemotropism

9. UME/82/16
Roots of plants are normally
A. positively phototropic
B. negatively geotropic
C. negatively hydrotropic
D. positively hydrotropic

E. negatively chemotropic

10. UME/83/37
In a positive phototropic response of a coleoptile, the region of greatest curvature is brought about by the
A. Movement of auxins away from the region of curvature
B. Even distribution of auxins in all parts of the coleoptile
C. Inhibition of growth by auxins in the region of smaller curvature
D. Concentration of auxins in the region of curvature
E. Absence of auxins in the coleoptile

11. UME/88/27
Taxism differs from tropism because
A. the whole organism is affected
B. it is a directional movement
C. it is a response to multi-directional stimuli
D. part of the organism is affected

12. UME/91/30
The movement of the whole organism to an external stimulus is termed
A. tropism
B. a taxis
C. a nastic movement
D. a phototropic movement

13. UME/93/23
The response shown by the tips of the root and shoot of a plant to the stimulus of gravity is
A. haptotropism
B. phototropism
C. hydrotropism
D. geotropism

14. UME/95/20
When a healthy shoot of a flowering plant is illuminated from one side, auxins accumulate on the
A. non-illuminated side of the shoot
B. illuminated side of the shoot
C. upper side of the shoot
D. lower side of the shoot

15. UME/2003/25/Type A
The response of plants to external stimuli in a non-directional manner is known as
A. nastic movement
B. phototropism
C. tactic movement
D. geotropism

16. UME 2010/23/TYPER: D
Which of the following stimuli is likely to elicit a nastic response in an organism?
A. light intensity
B. chemical substances
C. gravity
D. touch

17. Tropism is defined as:
A. Movement to a source of light
B. Growth away from or to source of gravity
C. Response to water
D. Growth curvature to diffused light
E. Growth towards or from a unilateral stimulus

18. Nastic movement e.g. is seen in the opening of flowers and buds, is a response to the external stimulus of light. This stimulus, however, does not determine:
A. The opening of the flowers or buds
B. The direction of the movement of flowers and buds
C. The period of opening of the flowers and buds

- D. The duration of opening of the flowers and buds
E. None of the above

19. This is a good example of negative geotropism:

- A. The downward growth of roots of the palm tree
B. The prop roots of the screw pine
C. The roots of epiphytic plants
D. The breathing roots of mangrove trees
E. The clasping roots of pepper

20. Rhizomes and some stolons growing horizontally are

- A. Positively geotropic
B. Negatively geotropic
C. Nearly negatively geotropic
D. Neutral
E. None of the above

21. The klinostat is used in geotropic experiments to illustrate:

- A. Positive geotropism
B. How geotropic responses could be prevented
C. Negative geotropism
D. Horizontal geotropic curvature
E. None of the above

22. If young seedlings are illuminated from the top, the seedlings will:

- A. Bend towards the top
B. will bend sideways
C. Remain straight
D. Grow taller
E. None of the above

23. When the sensitive plant *Mimosa pudica* is touched its leaves drop. This is due to:

- A. The inherent nature of the plant
B. Contact stimulus
C. The activity of the pulvinus of leaf bases
D. Differential changes in turgor in the lower and upper sides of the pulvinus
E. None of the above

24. After a storm, the maize plants in a field are found blown down. A day after this, all the plants straightened up at some nodes of the stems. This is an advantage of:

- A. Negative geotropism
B. Negative phototropism
C. Wind activity
D. Hydrotropism
E. Positive geotropism

25. If a young seedling is stimulated by unilateral light, the seedling bends towards the source of the light. The accumulated hormones on the shaded side of the seedling caused

- A. Contraction of the cells on the shaded side of seedling
B. The elongation of the cells on the illuminated side
C. A greater elongation of the cells on the shaded side
D. The seedling to bend away from light
E. None of the above

26. The opening and closing of the stomata of leaves are controlled by a number of factors such as light, humidity of the surrounding air and water absorbed from the soil. Light

is responsible for:

- A. The evaporation of water from the cells
B. The formation of carbohydrates in the stomata
C. The formation of carbohydrates in the guard cells
D. The formation of the chloroplasts in the guard cells
E. None of the above

11.3. SUPPORTING TISSUES IN ANIMALS

27. Which of these is not true of the skeleton?

- A. The skeleton supports the body and its associated structures
B. The skeleton, especially the long bones, gives rise to the red blood cells
C. The skeleton serves as structure for the insertion of muscles
D. The skeleton, especially the skull and the ribs, protect the delicate structure within them
E. None of the above

28. Which of these is not an external skeleton?

- A. the scales
B. the carapace
C. the carpals
D. the hoofs
E. None of the above

29. Which of these make up the internal skeleton of an animal?

- A. Bone and cartilage
B. Ligament
C. Water
D. Elastin
E. None of the above

30. What type of skeleton has the earthworm?

- A. it is a bony skeleton
B. it is an internal skeleton
C. it is a hydrostatic skeleton
D. it is an external skeleton
E. None of the above

31. PCE/98/21/TYPE C

The difference between tendons and ligaments is that

- A. Tendons join bones to muscle while ligaments join bone to bone
B. Ligaments join bone to muscle while tendons join bone to bone
C. Both are strong muscle fibres but tendons are stronger
D. Ligaments are found in the hip bone while tendons are found in the shoulder bone

32. UME/84/15

Exoskeleton is NOT found in the

- A. maggot
B. mosquito larva
C. earthworm
D. caterpillar
E. termite

33. UME/84/39

The axial skeleton of a mammal does not include the bones of the

- A. skull
B. tail
C. limbs
D. back
E. neck

34. UME/85/29

The appendicular skeleton is made up of the

- A. limbs
B. skull and limbs

C. phalanges
D. ulna and radius
E. knuckles and limbs

35. UME/93/24

Which of the following is the correct order of the vertebrae along the spinal column?

- A. Axis → Atlas → Thoracic → Lumbar → Cervical → Sacral
- B. Atlas → Cervical → Axis → Thoracic → Lumbar → Sacral
- C. Atlas → Axis → Cervical → Thoracic → Lumbar → Sacral
- D. Axis → Atlas → Cervical → Thoracic → Sacral → Lumbar

36. UME/82/19

Which vertebra has a projection called odontoid process?

- A. Atlas
- B. Axis
- C. Thoracic
- D. Lumbar
- E. Caudal

37. UME/2000/16/TYP E M

The scapula and ischium are part of the

- A. pectoral girdle
- B. pelvic girdle
- C. appendicular skeleton
- D. hind limb

38. PCE/97/20

The function of zygapophysis is for the

- A. free movement of the bones
- B. support of the internal organs
- C. protection of the spinal cord
- D. articulation of successive vertebrae

39. PCE/97/19

The part of the typical vertebra through which spinal cord passes is the

- A. centrum
- B. neural arch
- C. neural canal
- D. vertebrarterial canal

40. PCE/99/24/Type E

The ball and socket joint allows movement

- A. only in one direction
- B. up and down with slight rotation
- C. in two directions
- D. in several directions

41. PCE/93/24

The type of joints found between the bones of the ankle is

- A. sliding or gliding
- B. ball and socket
- C. hinge
- D. rotatory

42. Which of the follow is NOT a function of the mammalian skeleton?

- A. Protection
- B. Respiration
- C. Transportation
- D. Support.

43. UME 2009/22

The axial skeleton is found in the

- A. skull, ribs, vertebral column and breastbone
- B. skull, humerus, vertebral column and ribs
- C. breastbone, clavicle, ribs and vertebral column
- D. femur, sternum, ulna and skull

44. PCE/96/18

The major difference between the atlas and the axis of the

cervical vertebrae is that only the axis possesses

- A. facets for attachment to the skull
- B. an odontoid process
- C. a long neural spine
- D. a large transverse process

45. Which of these structures is not common to both atlas and axis vertebrae?

- A. the vertebrarterial canals
- B. the neural arch
- C. the neural spine
- D. the postzygapophyses
- E. the centrum

46. What is the importance of the neural canal?

- A. it carries the neural fluid
- B. it carries the spinal cord
- C. it carries the neuron
- D. it carries the odontoid process of the axis bone
- E. none of the above

47. The atlas and axis bones differ from other cervical bones in this respect

- A. they have no neural spines
- B. they have no vertebrarterial canals
- C. they have no centra
- D. they have no transverse processes
- E. None of the above.

48. The thoracic bones are distinguished from other bones by the possession of

- A. long neural spines
- B. capitulum and tuberculum
- C. vertebrarterial canals
- D. demi-facets
- E. articular surfaces

49. What is the intervertebral disc?

- A. it is a pad in the centrum of a bone
- B. it is a pad between two centra
- C. it is cartilage bone in the body
- D. it is a soft pad of bone between the thoracic bones
- E. None of the above

50. The lumbar bones are large and stout. This is because

- A. they are the most important bones of the skeleton
- B. they carry the weight of the animal
- C. they are at the back
- D. they support the bones of the limbs
- E. They support the large muscles of the back

51. What is the characteristic of the sacrum?

- A. The bones are flattened out
- B. The transverse processes are small and slender
- C. The bones are fused together
- D. The neural canal is often very large
- E. None of the above

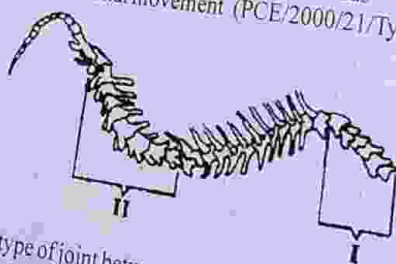
52. What is the name of the longest bone of the lower arm?

- A. It is the radius
- B. It is the ulna
- C. It is the tibia
- D. It is the intermedium
- E. It is the ulnare

53. The long bone of the thigh is:
 A. the thigh bone
 B. the humerus
 C. the femur
 D. the tibia

MISCELLANEOUS QUESTIONS (11)

54. Turgidity of cell in herbaceous plants is important as it
 A. provides mechanical support
 B. enables absorption of more water
 C. enables the intake of mineral salts
 D. prevents plasmolysis of the cells
 (PCE/2001/22/Type:4)
55. The liquid that serves as a lubricant to the mammalian joints is the
 A. amniotic fluid
 B. spinal fluid
 C. seminal fluid
 D. synovial fluid
 (PCE/2005/10/Type:P)
56. When bacterial swim from cold to warm region, this is known as
 A. negative chemotaxis
 B. positive thermotaxis
 C. positive phototaxis
 D. negative phototaxis
 (UME/2005/23/Type: D)
57. Hydrostatic skeleton is the type of supporting system found in
 A. mammals
 B. reptiles
 C. oligochaetes
 D. arthropods (UME/2005/23/Type: D)
58. Tropic response due to the stimulus of touch is known as
 A. geotropism
 B. phototropism
 C. thigmotropism
 D. hydrotropism (PCE/2001/23/Type: 4)
59. Taxism differs from tropism because
 A. only a part of the organism is affected
 B. the whole organism is affected
 C. it is a response to multi-directional stimulus
 D. it is a directional movement (PCE/2000/21/Type:A)
60. The type of joint between adjacent bones in the part labeled II is the
 A. hinge joint
 B. suture joint
 C. sliding joint
 D. ball-and-socket joint (UME/2004/8/Type:4)
61. The bones labeled II are called
 A. thoracic vertebrae
 B. lumbar vertebrae
 C. cervical vertebrae
 D. sacral vertebrae (UME/2004/9/Type:4)
62. The chitin in the exoskeleton of many arthropods is strengthened by
 A. calcium compounds
 B. organic salts
 C. lipids
 D. proteins (UME/99/26/Type:D)
63. The presence of endoskeleton is characteristic of
 A. invertebrata
 B. vertebrata
 C. insecta
 D. coelenterata (UME/2000/30/Type:Y)
64. When the forearm is raised, which muscle contracts in the process?
 A. cardiac muscle
 B. biceps
 C. triceps
 D. involuntary muscle (PCE/92/24)
65. The type of joints found between the bones of the
 A. sliding or gliding
 B. ball and socket
 C. hinge
 D. rotator (PCE/93/24)
66. The closing up of the lid of a pitcher plant on the of an insect is referred to as
 A. tactic movement
 B. tropic movement
 C. nastic movement
 D. heliotropic movement (PCE/96/22)
67. Which of the following is made up of dead cells
 A. cambium
 B. cork
 C. palisade
 D. meristem (PCE/98/20/TYPE: C)
68. Muscles are usually attached to bones by the
 A. cartilage
 B. ligaments
 C. membranes
 D. tendons (PCE/03/44/TYPE: N)
69. Stomata closes in the night because
 A. The air is dry
 B. No water is absorbed by the roots
 C. Transpiration is at a standstill
 D. There is no more formation of carbohydrates
 E. The chloroplasts of guard cells cease to function



CHAPTER 11

1. D	2. A	3. D	4. A	5. B
6. C	7. C	8. A	9. D	10. D
11. A	12. B	13. D	14. A	15. A
16. D	17. E	18. B	19. D	20. D
21. A	22. C	23. D	24. A	25. C
26. C	27. E	28. C	29. A	30. C
31. A	32. C	33. C	34. E	35. C
36. B	37. C	38. D	39. C	40. D
41. A	42. C	43. A	44. B	45. D
46. B	47. E	48. A	49. B	50. B
51. C	52. B	53. C	54. A	55. D
56. B	57. C	58. C	59. B	60. C
61. B	62. A	63. B	64. B	65. A
66. C	67. B	68. D	69. D	

CHAPTER 12

REPRODUCTION

Remember the following after reading this chapter:

Asexual reproduction
Sexual reproduction
Sexual reproduction in flowering plants
Reproduction in mammals
Courtship behaviour

REPRODUCTION

Reproduction is the process through which living organisms give rise to young ones of their kind. There are two types of reproduction viz, asexual and sexual reproduction.

ASEXUAL REPRODUCTION

Asexual reproduction is the formation of new individuals from the cell(s) of a single parent. It is very common in plants; less so in animals.

Budding

Some cells split via budding (for example baker's yeast), resulting in a 'mother' and 'daughter' cell.

Spore formation

Many multicellular organisms form spores during their biological life cycle in a process called sporogenesis. Plants and many algae on the other hand undergo sporic meiosis where meiosis leads to the formation of haploid spores rather than gametes. These spores grow into multicellular individuals (called gametophytes in the case of plants) without a fertilization event. These haploid individuals give rise to gametes through mitosis.

Fragmentation

Fragmentation is a form of asexual reproduction where a new organism grows from a fragment of the parent. Each fragment develops into a mature, fully grown individual. Fragmentation is seen in many organisms such as animals (some annelid worms, turbellarians and sea stars), fungi, and plants.

Natural Vegetative Propagation

All plant organs have been used for asexual reproduction, but stems are the most common.

Stems

In some species, stems arch over and take root at their tips, forming new plants. The horizontal above-ground stems (called stolons) of the

strawberry, for example, produce new daughter plants at alternate nodes.

Underground stems

- rhizomes
- bulbs
- corms and
- tubers

They are used for asexual reproduction as well as for food storage.

Irises and day lilies, for example, spread rapidly by the growth of their rhizomes.

Leaves

The typical example is the leaves of the common ornamental plant Bryophyllum (also called Kalanchoe). Mitosis at meristems along the leaf margins produce tiny plantlets that fall off and can take up an independent existence.

Roots

Some plants use their roots for asexual reproduction. The dandelion is a common example. Trees, such as the poplar or aspen, send up new stems from their roots. In time, an entire grove of trees may form — all part of a clone of the original tree.

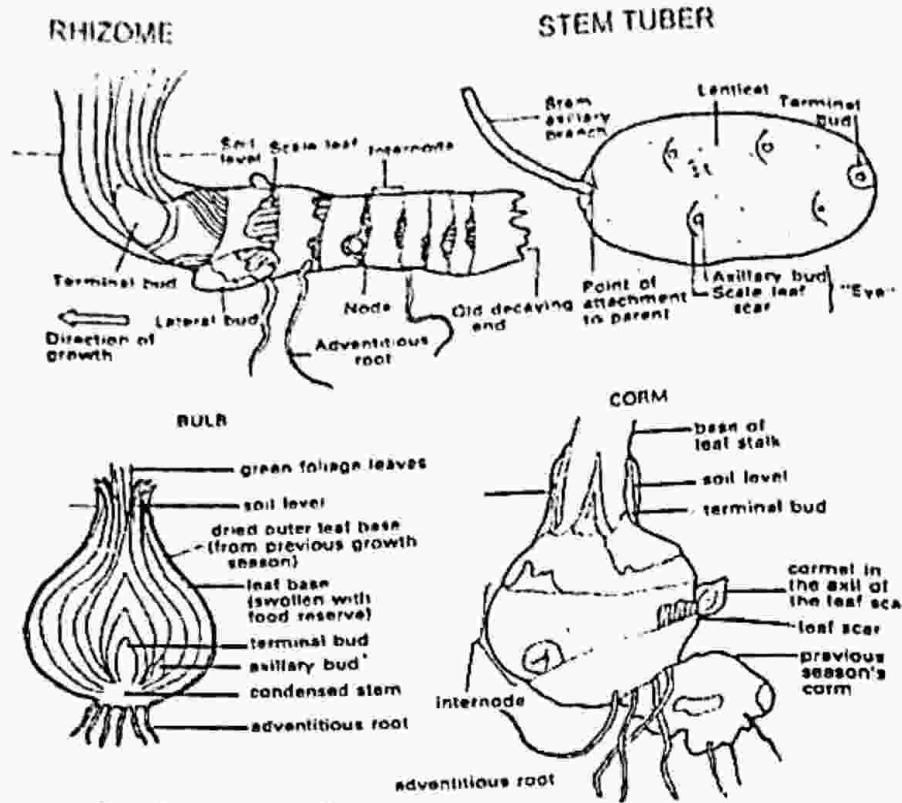
Artificial Vegetative Propagation

Commercially-important plants are often deliberately propagated by asexual means in order to keep particularly desirable traits (e.g., flower color, flavor, resistance to disease).

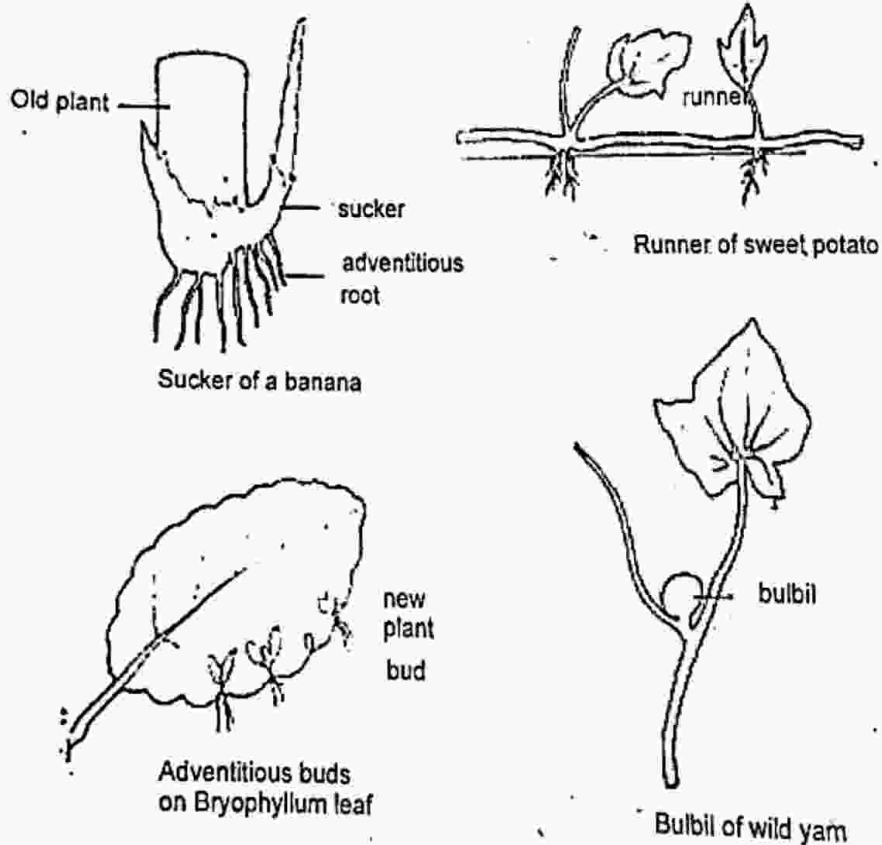
Grafting: In grafting 2 plants are used to develop a new plant with combined traits from the 2 parent plants. In grafting the scion is the above ground part of one plant. The scion is attached to the stock which is the rooted part of the second plant.

Cuttings: Cuttings are part of the plant that is cut off of the parent plant. Shoots with leaves attached are usually used. New roots and leaves will grow from the cutting. The shoot is cut at an angle. A growth promoter may be used to help with the growth of the roots

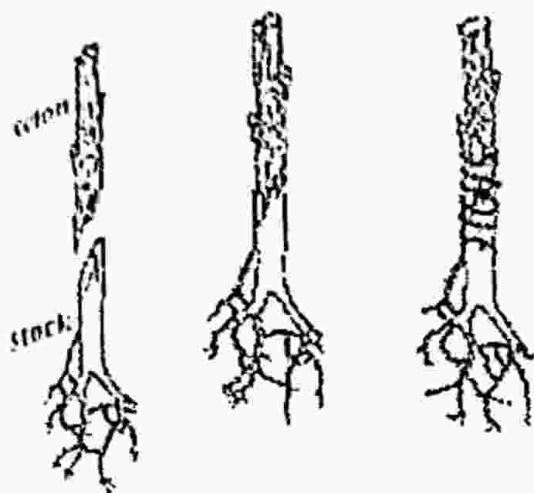
Layering: In layering a shoot of a parent plant is bent until it can be covered by soil. The tip of the shoot remains above ground. New roots and eventually a new plant will grow. These plants can then be separated.



Vegetative propagation by means of rhizome, stem tuber, bulb and corm



Vegetative propagation by means of sucker, runner, brophyllum and bulbil



ASEXUAL REPRODUCTION IN ANIMALS

Fission

An important form of fission is binary fission but there is some other type of fission that called multifission. In binary fission, the parent organism is replaced by two daughter organisms, because it literally divides in two.

In multiple fission, the nucleus of the parent cell divides by mitosis several times, producing several nuclei. The cytoplasm then separates, creating multiple daughter cells.

Budding

Here, offspring develop as a growth on the body of the parent. In some species, e.g., hydra, jellyfishes and many echinoderms, the buds break away and take up an

independent existence. In others, e.g., corals, the buds remain attached to the parent and the process results in colonies of animals. Budding is also common among parasitic animals, e.g., tapeworms.

Fragmentation

As certain tiny worms grow to full size, they spontaneously break up into 8 or 9 pieces. Each of these fragments develops into a mature worm, and the process is repeated.

Parthenogenesis

In parthenogenesis ("virgin birth"), the females produce eggs, but these develop into young without ever being fertilized. Parthenogenesis occurs in some fishes, several kinds of insects, and a few species of frogs and lizards. It does not normally occur in mammals because of their imprinted genes. However, using special manipulations to circumvent imprinting, laboratory mice have been produced by parthenogenesis.

Cell Division by Mitosis

Mitosis is a type of cell division that produces genetically identical cells. During mitosis DNA replicates in the parent cell, which divides into two new cells, each containing an exact copy of the DNA in the parent cell. The only source of genetic variation in the cells is via mutations.

Interphase



- This is when the cell is not dividing, but is carrying out its normal cellular functions.
- chromatin not visible
- DNA, histones and centrioles all replicated
- Replication of cell organelles e.g. mitochondria, occurs in cytoplasm.

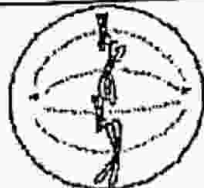
Prophase



chromosome

- chromosomes condense and become visible – this prevents tangling with other chromosomes.
- Due to DNA replication during interphase, each consists of two identical sister chromatids connected at the centromere
- centrioles move to opposite poles of cell
- nucleolus disappears
- phase ends with the breakdown of the nuclear membrane

Metaphase



- spindle fibres (microtubules) connect centrioles to chromosomes
- chromosomes align along equator of cell and attaches to a spindle fibre by its centromere.

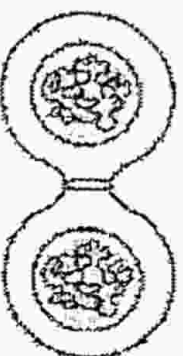
Anaphase



Telophase



Cytokinesis



First Division	Second Division
Interphase I chromatin not visible DNA & proteins replicated	Interphase II • Short • no DNA replication • chromosomes remain visible.
Prophase I • chromosomes visible • homologous chromosomes join together to form a <u>bivalent</u> • chromatids <u>cross over</u> (chiasmata—see below)	Prophase II • centrioles replicate and move to new poles.
Metaphase I • bivalents line up on equator	Metaphase II • chromosomes line up on equator
Anaphase I • chromosomes separate (not chromatids—centromere doesn't split)	Anaphase II • centromeres split • chromatids separate.
Telophase I • nuclei form • cell divides • cells have 2 chromosomes, not 4 chromatids.	Telophase II • 4 haploid cells, each with 2 chromatids • cells often stay together to form a <u>tetrad</u> .

Meiosis and Sexual Reproduction

Sexual reproduction is the production of offspring from two parent using gametes. The cells of the offspring have two sets of chromosomes (one from each parent), so are diploid. Sexual reproduction involves two stages:

- **Meiosis**—the special cell division that makes haploid gametes
- **Fertilisation**—the fusion of two gametes to form a diploid zygote

There are differences between meiosis II and mitosis:

- DNA will not replicate before meiosis II
- The sister chromatids in meiosis II will likely not be identical due to crossing over
- The number of chromosomes at the center of the cell in meiosis II is half the number of chromosomes in mitosis.

Meiosis results in 4 genetically different haploid nuclei

SEXUAL REPRODUCTION

Sexual reproduction: the process involving the fusion of haploid nuclei (23 chromosomes) to form a diploid zygote (46 chromosomes) and the production of genetically dissimilar offspring.

Advantages: produces genetically different offspring so they don't all die from change in the environment.
Disadvantage: it takes lots of time and energy, good characteristics can be lost.

Sexual reproduction in humans

Testes: have many coiled tubes which produce **sperm**, and the cells between tubes produce **testosterone**.

Scrotum: holds testicles

Sperm duct: carries sperm from testicles to urethra.

Prostate gland: makes **seminal fluid** (semen = 99.5% seminal fluid and 0.5% sperm) (*actually it makes prostate

Pollination
Many plants favour cross-pollination, so pollen must be transferred to the stigma of another plant if sexual reproduction is to take place. Some flowers rely on the wind to carry pollen grains others rely on insects.

Self-pollination is where the pollen is transferred to the stigma of the same flower or the stigma of another flower of the same plant. Self-pollination is obviously more reliable, particularly if the nearest plant is not very close. A potential drawback is that both gametes come from the same parent. If the plant is well adapted to a stable environment, the production of uniform offspring may be advantageous. However, inbreeding will result and if there are disadvantageous recessive characteristics in the parent, they are much more likely to be exposed than if the plant is cross-pollinated. Cross-pollination is less reliable and more wasteful because self-pollination, but it is genetically favourable because genes are transferred and variation increases.

Strategies to favour cross-pollination:

- **Diocious plants:** Some plants have male and female - they have separate flowers. Only female - they have only female flowers. **Monococious plants:** Some flowers have both male and female. So, self-pollination is possible. **Protandry:** Anthers on some plants mature before the stigma, so self-pollination is therefore not possible. **Proteogyny:** The stigma matures before the anthers, so self-pollination is possible. **Self-incompatibility:** Pollination on the pollen tube does not grow well, if self-fertilisation takes place.

For those plants that cross-pollinate, some are wind pollinated, others are insect pollinated. Here are some of the differences:

Wind pollinated flowers:	
Feature:	Insect pollinated flowers:
Petals: Small inconspicuous, sometimes absent. If present, not brightly coloured.	Large, brightly coloured, conspicuous and attractive to insects.
Scent: None.	Often scented.
Nectary: Absent.	Present.
Pollen: Produced in large quantities, light, smooth pollen grains.	Less produced, pollen grains larger, sculptured and attachment to insects and to stigma.
Anthers: Move freely, so pollen is easily dispersed.	Fixed to filaments and positioned to come into contact with visiting insects.
Stigma: Large often branched and feathery, hanging outside the flower to trap pollen.	Small enclosed within the flower, positioned away from contact with visiting insects.

Fertilisation

If the pollen grain lands on a compatible stigma, a pollen tube will grow so that eventually the egg cell, hidden away in the embryo sac, can be fertilised. A tube emerges from the grain, its growth being controlled by the tube nucleus at the tip of the tube. It may grow downwards in response to chemicals made by the ovary (a response known as chemotropism).

During the growth and extension of the tube, the generative nucleus, behind the tube nucleus, divides by mitosis to produce **2 male haploid gametes**. The pollen tube enters the ovule through the micropyle and penetrates the embryo sac wall. The tip of the tube bursts open, the tube nucleus dies and what follows is called **double fertilisation**.

1 male gamete fuses with the egg cell to produce a **diploid zygote**.

1 male gamete fuses with both the polar nuclei to produce

Stigma

Style

Ovule

Ovary wall

Placenta

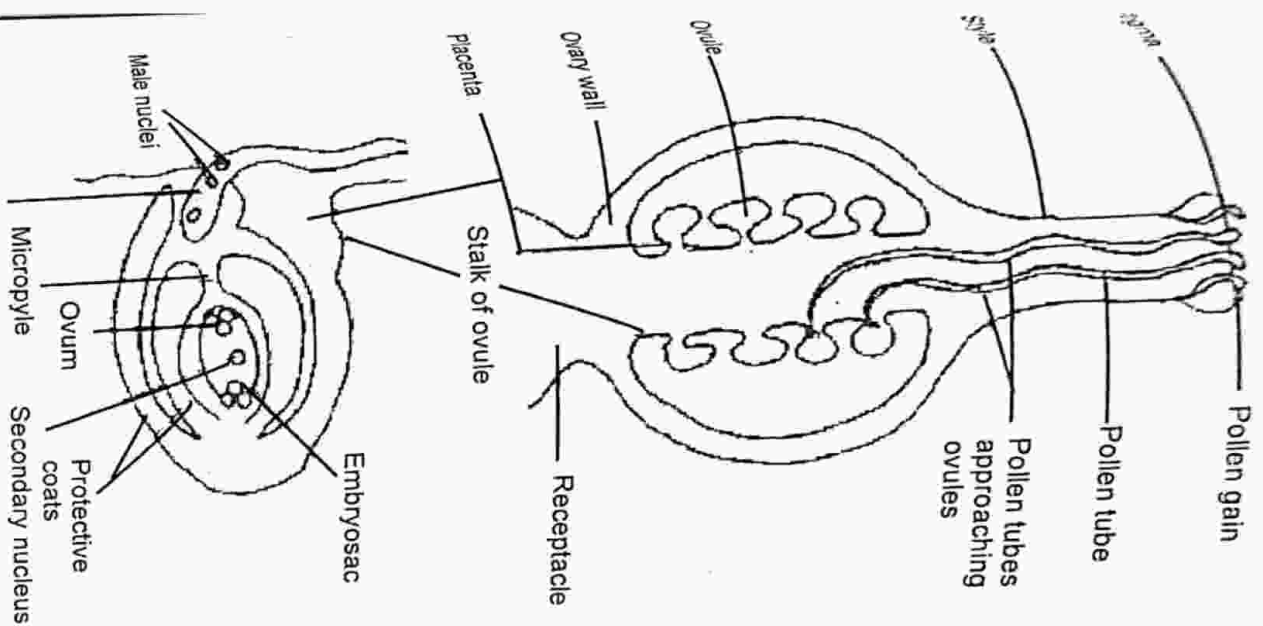
Male nuch

Fertilizati

The following

the triploid primary endosperm nucleus. Immediately after fertilisation, the ovule is known as seed.

1. The
2. Th
3. Tc
4. Tt
5. Td
6. Tv



Fertilization in flower and structure of ovule

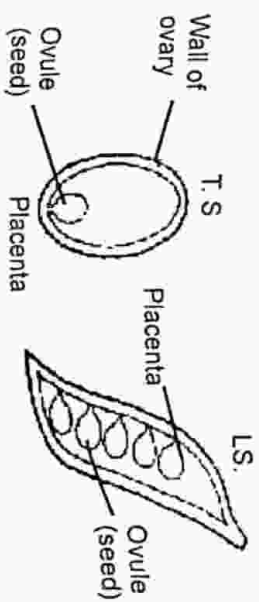
The following happens:

1. The zygote divides many times by mitosis to form the embryo.

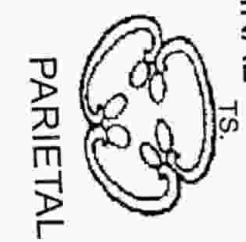
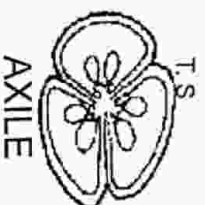
Why they can be brightly coloured and sweet; animals will eat them and scatter the seeds either at the time of eating or when they are passed out of the gut in defecation, unharmed.

The arrangement of ovules within the ovary is called **Placentation**. The main types of placentation are:

1. Marginal e.g. Crotalaria
2. Parietal e.g. Pawpaw
3. Axile e.g. Tomato
4. Free-central e.g. Waterleaf



MARGINAL



Types of placentation

TYPES OF FRUITS

There are two types of fruits viz:

1. **True fruits:** These are fruit developed from the ovary alone e.g. mango, orange etc
2. **False fruit:** These are developed from the ovary and other floral parts such as the calyx, sepal or receptacle e.g. apple, pear etc

CLASSIFICATION OF FRUITS

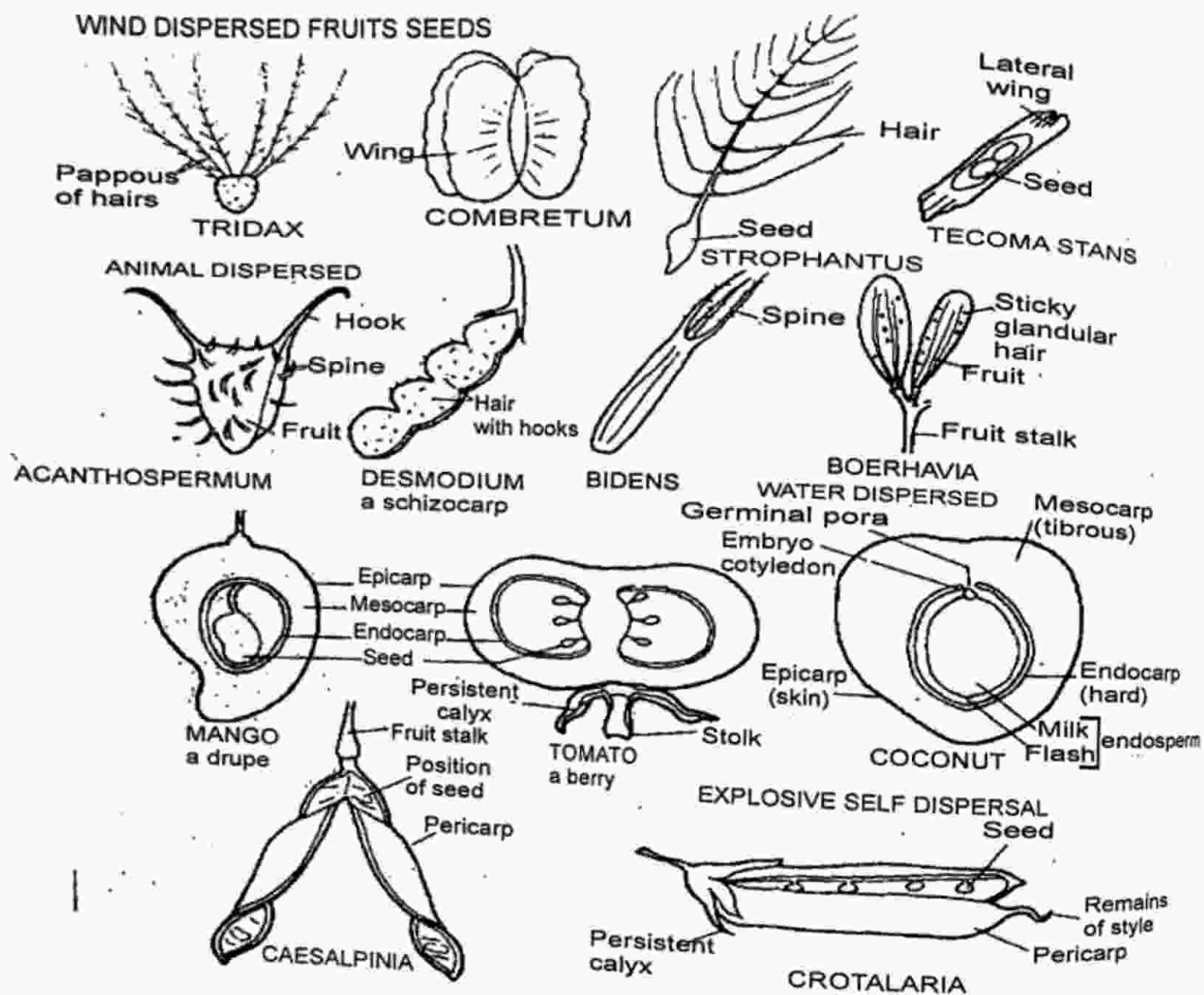
Fruits are broadly classified into two as: **Dry fruits** and **Fleshy fruits**.

is the bristle wing like structure of the seed-fruit e.g. *Tridax*.

Water dispersal (Hydrochory) - this is for plants that can float on water e.g. coconut. Most of the magrove species are dispersed by this means.

Animal dispersal (Zoochory) - this can be epizoochory or endozoochory. In epizoochory the seed is carried by animals accidentally on their body because the seed/fruit of the plants has structures that made them adhesive on animal skin/fur e.g. *Triumfetta*, *Desmodium*. In endozoochory the plant has very attractive fruit that can be eaten by the animal. The seed will be also consumed by the animal but not digestable e.g. guava, tomato.

Mechanical dispersal - dispersal caused by dehiscence of fruit. The fruit is usually pressurized and when ripe will explode. A good example is the fruit of okro - *Hibiscus esculentus*.



Dispersal of fruits

Revision Questions

SEXUAL REPRODUCTION AND BUDDING

1. UME 81/16

Which of the following reproduces vegetatively by

- A. fragmentation
- B. budding
- C. binary fission
- D. vegetative propagation

2. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

3. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

4. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

5. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
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- D. vegetative propagation

6. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

7. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

8. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

9. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

10. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

11. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

12. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

13. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

14. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

15. UME 81/16

Which of the following is a vegetative feature of

- A. binary fission
- B. sexual reproduction
- C. fragmentation
- D. vegetative propagation

10. UME 85/33
Which of the following is a

- A. bulb
- B. tap root
- C. rhizome
- D. root tuber

11. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

12. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

13. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

14. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

15. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

16. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

17. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

18. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

19. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

20. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

21. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

22. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

23. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

24. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

25. UME 85/30

Which of the following is a

- A. corn
- B. stem
- C. root
- D. tuber

111

38. In some flowers the petals are exactly alike and are equally spaced. Such flowers are said to be:
 A. Zygomorphic
 B. Gamopetalous
 C. Polypetalous
 D. Actinomorphic
 E. Petaloid

39. PCE/96/3
 The condition in which male and female parts of a flower mature a different times is
 A. homogamy
 B. autogamy
 C. dichogamy
 D. cleistogamy

40. UME 87/30
 A flower showing radial symmetry is said to be
 A. pentamerous
 B. protandrous
 C. protogynous
 D. actinomorphic

41. PCE 94/25
 An example of a dioecious plant is
 A. pawpaw
 B. maize
 C. oil palm
 D. castor oil

12.3.2. POLLINATION

42. There are many ways in which plants prevent self-pollination. One such way is that the pistil of flowers ripen first. This is known as:
 A. Protogyny
 B. Heterostyly
 C. Dichogamy
 D. Protandry
 E. None of the above

43. UME 85/31
 If a flower is protandrous then it
 A. must be unisexual
 B. has an undeveloped androecium
 C. has no androecium
 D. must be insect-pollinated
 E. can prevent self-pollination

44. UME 89/28
 Wind-pollinated flowers usually have
 A. rough pollen grains
 B. sticky stigmas
 C. small and short stigmas
 D. long styles

45. Which of these is NOT true of wind pollinated flowers?
 A. The flowers are inconspicuous
 B. Vast quantities of pollen are released
 C. The stigma of the flowers are sticky
 D. The anthers are pendulously attached in the flower head
 E. The pollen grains are smooth

46. PCE 2004/25 Type 9
 The offspring resulting from cross pollination usually exhibit
 A. variable characteristics
 B. identical characteristics
 C. high resistance
 D. low resistance

47. UME 99/25
 Insects visit flowers in order to
 A. feed on the nectar
 B. deposit pollen on the stigma
 C. pollinate the flower

D. transfer pollens from the anthers

48. UME 85/32
 For pollen to be released in *Crotalaria* the insect must depress the
 A. wing
 B. keel
 C. standard
 D. antepetalous stamen
 E. antesealous stamen

12.3.3. FERTILIZATION

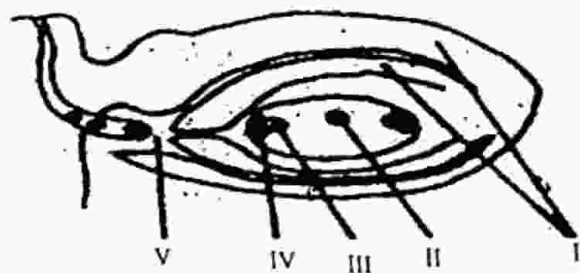
49. What is the first noticeable change in the flower after fertilization?
 A. The calyx is shed
 B. The petals fall
 C. The style imbibes water
 D. The stigma becomes unreceptive to pollen grains
 E. The ovules ripen

50. If a fruit bears above it withered sepals, stamens and the stigma, the ovary is:
 A. Epigynous
 B. Hypogynous
 C. Perigynous
 D. Semi-perigynous
 E. None of the above

51. UME 92/25
 Double fertilization in higher plants is significant because it ensures
 A. formation of a fertile embryo
 B. formation of a fertile embryo and endosperm
 C. development of the seed
 D. development of the fruit

52. UME 90/31
 Fruits which develop without fertilization are described as
 A. simple
 B. parthenocarpic
 C. aggregate
 D. compound

53. UME 84/29
 How many nuclei are found in the pollen tube during fertilization?
 A. 2
 B. 3
 C. 5
 D. 6
 E. 7



54. UME 97/26
 The function of the part labeled V is for the passage of
 A. pollen tube and pollen nucleus
 B. air, water and pollen nucleus
 C. air, antipodal cells and ovum
 D. synergids and egg cell

55. UME 97/27
 The female gamete is represented by
 A. I
 B. II
 C. III
 D. IV

23. A pistil of a flower is composed of the ovary and the style. The ovary is the part of the pistil that contains the ovules. The style is the part of the pistil that connects the ovary to the stigma.

24. Which of the following is not a fruit? A. Tomato B. Bean C. Apple D. Potato E. Strawberry

25. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

26. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

27. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

28. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

29. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

30. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

31. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

32. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

33. A fruit is a mature ovary. The ovary is the part of the flower that contains the ovules. The fruit is the part of the flower that develops from the ovary.

34. Which of the following is not a fruit? A. Tomato B. Bean C. Apple D. Potato E. Strawberry

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59. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

60. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

61. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

62. Which of the following is a fruit? A. Seed B. Pollen C. Stamen D. Pistil E. Ovary

is
marginal
ventral

Placentation shows
bicus
ide of Barbados

a fruit and not a seed

fertilization
don
variate.

er having several free
a dry dehiscent fruit
a simple fruit

several parts each
gregarious fruit
hizocarp

ne biological sense
l

gle flower having
simple fruit
dehiscent fruit

cause it



87/29
shown above is
inferior ovary
simple, regular, hermaphrodite with superior ovary
simple, regular, staminate with superior ovary
simple, regular, hermaphrodite with inferior ovary
simple, irregular, pistillate with superior ovary

8/ME 87/31
differs from a cypselid by having
pendent pericarp
pendent pericarp
pendent pericarp fused with the seed coat
pendent pericarp fused with the seed coat
pendent pericarp fused with the seed coat

8/ME 88/28
dehiscent fruit which breaks up into one-seeded parts
A. legume
B. capsule
C. follicle
D. legume

8/ME 89/29
The components of castor oil seed and the maize grain are
A. number of cotyledons
B. location of the embryo
C. number of radicles
D. numbers of plumules

7/ME 91/33
A collection of achenes formed from several carpels of a
flower is
A. multiple fruit
B. an aggregate fruit
C. schizocarp
D. a simple fruit

7/ME 95/23
The term carpel is used to describe a fruit in which the
A. testa and pericarp are separate
B. seed and endocarp are fused
C. testa and pericarp are fused
D. seed coat and fruit wall are impermeable

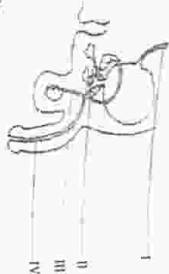
7/ME 97/29
Coconut and oil palm fruits can be grouped as
A. berry
B. legume
C. capsule
D. drupe

8/ME 98/25
A dry fruit formed from two or more carpels containing
several seeds is a
A. follicle
B. legume
C. capsule
D. schizocarp

11. Cotton seeds are dispersed by wind. What attribute or
structure assists in its dispersal?
A. Possession of flat seeds
B. Possession of "wings"
C. Possession of a floss
D. Lightness of the seeds
E. Hardness of the seeds

82. Which of these seeds are dispersed by the
explosive mechanism?
A. The para-nibbler
B. The mango
C. The oil palm
D. The coconut
E. The tung tree

12.4 REPRODUCTION IN MAMMALS



83. UME 2005/25 Type D
The structure labeled I originates from the
A. liver
B. kidney
C. small intestine
D. stomach

84. UME 2005/26 Type D
Birth control by vasectomy is achieved by severing the
structure labeled
A. I
B. II
C. III
D. IV

85. UME 2009/25
The reproductive system of a male mammal is made up of
A. clasper, prostate gland, sperm duct and vas deferens
B. testis, prostate gland, sperm duct and vas deferens
C. oviduct, urethra, testis and sperm duct
D. testis, uterus, prostate gland and sperm duct

86. UME 2010/24 Type D
In the male reproductive system of a mammal, sperm is
stored in the
A. urethra
B. epididymis
C. semitubular tubules
D. vas deferens

87. What is the function of the vesicular seminalis?
A. It is for the storage of urine
B. It is for the storage of excretory materials
C. It is for the storage of spermatozoa
D. It is for the storage of effete blood cells
E. None of the above

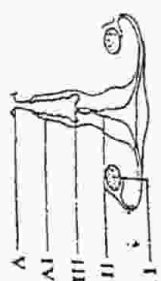
88. What is the function of the vas deferens?
A. It transports both urine and spermatozoa
B. It transports only the spermatozoa
C. It transports excretory materials from the kidney
D. It transports excretory materials from the kidney
E. None of the above

89. The duct through which urine and spermatozoa pass
out in the male mammal is:
A. the ureter
B. the penis
C. the urethra
D. the vas deferens
E. None of the above

90. What is the function of the bladder in mammals?
- A. It is for storage of urine
 - B. It is for the storage of urine and spermatozoa
 - C. It stores toxic substances
 - D. It is for the storage of water for the body
 - E. None of the above

91. UME/2003/23 Type: A
In mammalian males, the excretory and reproductive systems share the
- A. ureter
 - B. testes
 - C. vas deferens
 - D. urethra

92. PCE/2001/25 Type: 4
The spermatozoon produced in the testes is often stored in a coiled tube called
- A. spermatic cord
 - B. vas deferens
 - C. scrotum
 - D. epididymis



93. PCE/98/22
The structure labeled III is the
- A. ovary
 - B. cervix
 - C. vagina
 - D. vulva

94. PCE/98/23
The fertilized ovum is implanted in the part labeled
- A. I
 - B. II
 - C. III
 - D. IV

95. UME/2008/25
During ovulation, an egg is released from the
- A. corpus luteum
 - B. ovarian funnel
 - C. Graafian follicle
 - D. fallopian tube

96. UME/2010/30/TYPE: D
The barrier between maternal and foetal blood is the
- A. liver
 - B. umbilical cord
 - C. uterine wall
 - D. Placenta

97. UME/99/24
Fertilization in humans usually takes place in the
- A. lower part of the uterus
 - B. upper part of the uterus
 - C. lower part of the oviduct
 - D. upper part of the oviduct

98. UME/95/24
The sex of a foetus is determined during
- A. meiosis
 - B. copulation
 - C. fertilization
 - D. placentation

99. PCE/98/25/TYPE: C
Identical twins are formed when one ovum
- A. is fertilized by one sperm

- B. is fertilized by two sperms
C. divides after fertilization
D. divides before fertilization

100. PCE/06/16/TYPE: A
Ecotopic pregnancy may result from
- A. the fertilization of two ova
 - B. a focus developing in the uterus
 - C. the non-fertilization of ovum
 - D. a zygote developing in the fallopian tube

101. When the embryo has remained in the uterus for a period, it becomes a miniature adult and it is expelled from the womb. This process is known as
- A. The gestation
 - B. The afterbirth
 - C. Parturition
 - D. Foetalization
 - E. None of the above

102. PCE/2005/13/Type: P
Which of the following provides the mammalian embryo with nutrients before the formation of the placenta?
- A. Amnion
 - B. Ovary
 - C. Uterus
 - D. Oviduct

103. PCE/2002/43/Type: 5
In a developing mammalian embryo, external blood absorbed by the
- A. yolk sac
 - B. placenta
 - C. umbilical cord
 - D. amniotic fluid

104. PCE/2002/43/Type: 5
In a mammalian development, the function of the yolk sac
- A. supply nutrient to the embryo
 - B. act as shock absorber to the foetus
 - C. supply air to the embryo
 - D. facilitate the process of excretion in the foetus

105. PCE/93/27
The hormones contained in birth control pills for females are
- A. progesterone and testosterone
 - B. oestrogen and progesterone
 - C. oestrogen and testosterone
 - D. progesterone & follicle stimulating hormone

106. PCE/2000/24 Type: A
Within the mammalian female reproductive system, the of muscles which closes the lower end of the uterus is called
- A. cervix
 - B. vulva
 - C. vagina
 - D. oviduct

107. UME/81/33
Which of the following statements is NOT true of menstruation?
- A. It occurs monthly
 - B. The discharge contains amniotic fluid
 - C. It signifies no conception
 - D. It involves the lining of the uterus
 - E. It may cause pains

108. UME/08/24/TYPE: A
During gestation, the foetus is nourished by
- A. placenta
 - B. uterus
 - C. amnion
 - D. yolk sac

109. UME/08/24/TYPE: A
In a mammal, the placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

110. UME/08/24/TYPE: A
During gestation, the foetus is nourished by
- A. placenta
 - B. uterus
 - C. amnion
 - D. yolk sac

111. UME/08/24/TYPE: A
The outermost layer of the embryo is the
- A. amnion
 - B. chorion
 - C. allantois
 - D. yolk sac

112. A
The placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

113. A
The placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

114. A
The placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

115. A
The placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

116. A
The placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

117. A
The placenta is attached to the
- A. uterus
 - B. ovaries
 - C. fallopian tube
 - D. vagina

108. UME 86/37

- Gestation in mammals is the period
A. required for growth after birth
B. between the formation of foetus and birth
C. of development from zygote to birth
D. before the formation of the zygote

109. UME 88/32

- In a mammal, the placenta performs functions similar to those of the
A. lungs, kidneys and digestive system
B. lungs, heart and nervous system
C. liver, intestines and reproductive system
D. intestines, heart and digestive system

110. UME 90/29

- During the mammalian embryo development, large amounts of oestrogen and progesterone are produced in the
A. umbilical cord B. amnion
C. placenta D. amniotic fluid

111. UME 98/26

- The outermost embryonic membrane in the mammal is the
A. amnion B. chorion
C. allantois D. yolk sac

MISCELLANEOUS QUESTIONS (12)

112. An example of a product of sexual reproduction is
A. yam tuber B. cassava tuber C. ginger
D. groundnut (PCE/97/21/Type: E)

113. When a peacock displays its colourful feathers, it is
A. ready for a fight
B. protecting itself from predators
C. protecting its mate from predators
D. courting a female (UME/2005/47/Type: D)

114. In which of the following groups of plants is the pericarp inseparable from the seed coat
A. Nut
B. Follicle
C. Cypsela
D. Caryopsis (UME/2002/38/Type: Y)

115. Tomatoes, guava and pepper belong to a group of fruits called
A. berry B. drupe
C. schizocarp D. caryopsis (PCE/92/27)

116. Placentation in fruit describes the
A. arrangement of fruits on the branches
B. development of the ovary wall to form the pericarp
C. arrangement of ovules in the ovary
D. classification of fruits into different types (PCE/93/26)

117. The crown of leaves on a pineapple fruit is known as
A. offset B. suckers
C. runners D. bulbils (PCE/95/24)

118. Maturation of the stigma of a flower before the anther is described as

- A. protandry B. hypogyny
C. perigyny D. protogyny (PCE/98/24/TYPE: C)

119. What makes yam an underground stem?
A. The fact that it store starch
B. The fact that it has no roots
C. The fact that it bears dormant buds known as "eyes"
D. The fact that it is swollen
E. None of the above

120. That part of the leaf that secures it to the stem is:
A. The peduncle B. The pedicel
C. The petiole D. The patula
E. The perianth

121. The leaves of the onion are modified into a storage organ known as:
A. The rhizome B. The corm
C. The rubber D. The stem
E. None of the above

122. The food stored by the onion is:
A. Starch B. Sugar
C. Protein D. Oils E. Fats

123. The plant *Bryophyllum pinnatum* can be propagated easily because
A. The leaf stores food B. It does not die
C. The stem is stout D. It is immortal
E. None of the above

124. What are monoecious plants?
A. These are plants having only the male reproductive parts present in one individual
B. These are plants with both the male and female reproductive parts present in a single plant
C. These are plants with only pistillate flowers
D. These are monocarpellary fruits
E. These are plants with many flowers

125. What is binary fusion?
A. The splitting up of an individual into two identical parts
B. The splitting up of an animal into minute forms
C. The loss of a part of an animal due to injury
D. The fusion of the gametes
E. None of the above

126. The structure from which gametes arise is:
A. The scrotum B. The ovary
C. The ovule D. The gonad E. None of the above

127. Which of these animals do not show parental care over their young?
A. the mammal B. the fowl
C. the lizard D. the earwigs E. the boa constrictor

128. Which of these is true of seeds dispersed by water?
A. The plants bearing them grow near water
B. They are light
C. They are buoyant
D. They contain air space in the internal cavity of the seed
E. They are hollowed

CHAPTER 12

1. B	2. A	3. A	4. B	5. NH
6. A	7. A	8. D	9. B	10. E
11. A	12. D	13. B	14. D	15. E
16. D	17. E	18. D	19. A	20. E
21. D	22. D	23. B	24. D	25. B
26. C	27. A	28. A	29. D	30. D
31. E	32. C	33. A	34. D	35. A
36. B	37. B	38. D	39. C	40. D
41. A	42. A	43. E	44. D	45. C
46. A	47. A	48. B	49. B	50. A
51. B	52. B	53. B	54. A	55. C
56. A	57. D	58. A	59. C	60. C
61. D	62. B	63. D	64. B	65. D
66. B	67. E	68. D	69. B	70. C
71. B	72. C	73. C	74. A	75. A
76. A	77. B	78. C	79. D	80. C
81. C	82. A	83. B	84. C	85. B
86. B	87. E	88. B	89. C	90. E
91. D	92. D	93. B	94. B	95. C
96. D	97. D	98. C	99. A	100. D
101. C	102. C	103. D	104. A	105. B
106. A	107. B	108. C	109. A	110. C
111. B	112. D	113. D	114. D	115. A
116. C	117. A	118. D	119. E	120. C
121. E	122. B	123. E	124. B	125. E
126. D	127. C	128. D		

GROWTH

Understand the following after reading this chapter:

Meaning of growth
Growth in plants and animals
Germination of seeds

GROWTH

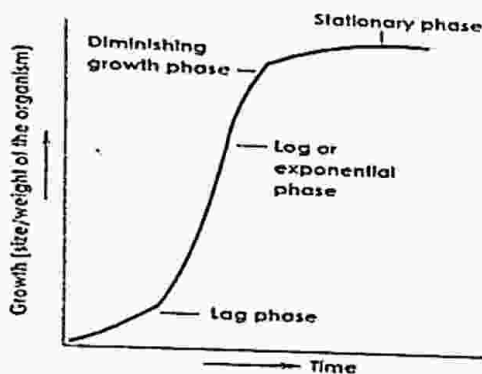
Growth is the permanent increase in volume and weight of the cells protoplasm. It occurs through food assimilation, cell expansion, cell division and cell differentiation.

Factors Affecting Growth are: nutrition/food, environment, hormones, heredity and diseases

Growth Curve

The curve can be shown appearing slowly along the line and stabilizing.

During the initial stage, i.e., during the lag phase, the rate of plant growth is slow. Rate of growth then increases rapidly during the exponential phase. After some time the growth rate slowly decreases due to limitation of nutrients. This phase constitutes the stationary phase.



The curve obtained by plotting growth and time is called a growth curve. It is a typical sigmoid or S-shaped curve.

Patterns of Growth

Growth patterns include:

Isometric growth: when parts of an organism grow at the same rate as the whole organism e.g. plant leaves.

Allometric growth: When parts or organs of an organism grow at different rates from the whole organisms e.g. the head and brain of man during the early stage of life.

Intermittent growth: This is a step-like growth; it occurs in arthropods due to moulting.

Germination

Germination is the development and emergence of the embryo from the seed coat into a seedling.

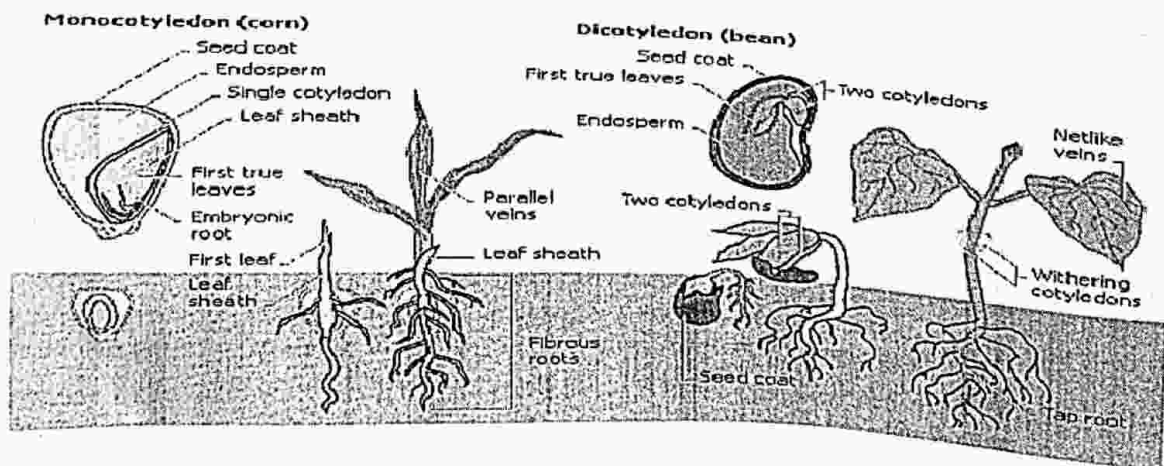
When conditions are right, the seed will take up water through the micropyle by imbibition. This triggers the beginning of the growth of the seed. The cell swells and the testa splits. With the addition of water, large molecules of carbohydrate, protein and fat can be hydrolysed (broken down) to produce substances for respiration. The water activates such enzymes as α -amylase to catalyse this digestion. The growing embryo releases a hormone called **gibberellin** and some enzymes are produced and released in response to this. The soluble products of digestion are delivered to the cotyledons, root and shoot. They respire aerobically and grow in size. By the time the food store has been used up, the shoot has grown enough to push the first leaves into the sunlight. Photosynthesis can then start.

Conditions necessary for germination: Oxygen, warmth, moisture and viable seeds

Types of germination

Epigeal germination: The hypocotyls rapidly elongates thereby pulling the cotyledons and the plumule above the ground e.g. cowpea and most dicot.

Hypogeal germination: The epicotyls rapidly pushing up the plumule above the ground and leaving the cotyledon underground e.g. Sorghum and most monocot.



Germination in monocot and dicot seed

EXPERIMENTS ON GROWTH

Experiment:1 To demonstrate the conditions necessary for germination.
The conditions necessary for germination are warmth, oxygen, water, and viable seeds.

Materials required. Four test tubes, labels, dry cotton wool and dry seeds.

Method. Four clean test tubes are collected and dry cotton wool is placed at the bottom of each. Some dry seeds are put in the tubes and the tubes are labelled 1, 2, 3, and 4. They are treated as follows:

Tube 1: This tube is placed where there is air and warmth. Water is not added so that the seeds receive warmth and air only.

Tube 2: Water that is previously boiled and cooled is poured into the tube. The boiled water lacks air. A few drops of oil are added to the water surface to prevent air from dissolving in it. The tube is then placed where there is warmth. In this tube, warmth and water are present but no air.

Tube 3: Some water is added to moisten the cotton wool. The tube is kept where there is enough air and warmth. This tube is regularly checked and water is added to keep the cotton moist. In this tube, air, water, and warmth are present for the seeds. No condition is missing.

Tube 4: Here, the seeds are treated as in tube 3. However, this is kept in a cold place such as a refrigerator. In this tube, air and water are present for the seeds but no warmth. The four tubes are then left for a few days.

Observations. Only seeds in tube 3 show normal germination.

Conclusion. Seeds in tube 3 which show normal germination were supplied with air, warmth and water. Seeds in other tubes lacked one of the conditions and hence normal germination could not take place. This shows that air, warmth and water are necessary for seed germination.

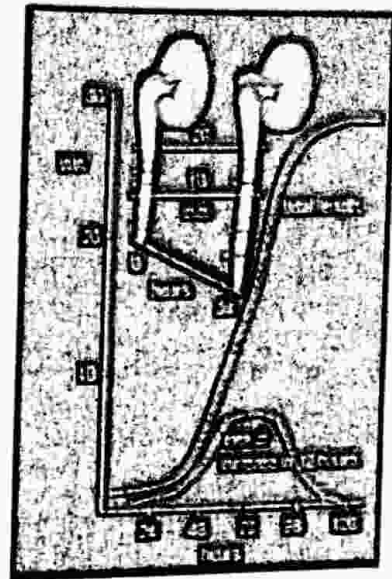
Experiment:2 To determine the area of fastest growth in a root.

Materials required. Bean seedling, blotting paper, a gas jar, water proof Indian ink.

Method. A bean seedling whose root (or radicle) is straight is selected. The selected root is just an inch long. From the tip upwards, the root is marked across with lines. The lines are marked with a water proof Indian ink. The lines are equidistant from each other, 1mm apart. Blotting papers are moistened and are used to line a gas jar. The seedling is then placed between the glass and the moistened paper. The setup is then kept in a dark place for two days. After this period, the plant root was carefully inspected.

Observations. The root increased in length. The lines at the upper portion (i.e. lines far from the root tip, the lines moved apart widely from each other.

Conclusion. The sharp increase of the spaces between lines 2, 3, 4 and 5 shows that the region of fastest growth lies behind the root tip, a few centimetre behind



Growth rate in germinating seed

Experiment:3. To demonstrate root pressure using a manometer.

Materials required. A young vigorously growing plant, a sharp pen knife, a manometer.

Method. A young vigorously growing plant is cut at the base, say 12cm from the ground level, with a sharp pen knife. Next, a manometer (an instrument for measuring root pressure) is carefully tied to the cut surface of the stem. The manometer is held in a place by means of rubber tubing. The setup is then observed for some time.

Observations. The mercury in the manometer is seen to move up gradually.

Conclusion. The upward movement of the mercury is due to the pressure generated by the flow of sap.

Revision Questions

13.1. MEANING OF GROWTH

1. UME/89/32

The irreversible life process by which new protoplasm is added to increase the size and weight of an organism can be termed.

- A. anabolism
C. growth

- B. catabolism
D. development

2. UME/99/25

Biological growth refers strictly to an increase in the

- A. protoplasm of an organism
C. number of organisms

- B. size of an organism
D. development of form

3. UME/98/24

The most reliable estimate of growth is by measuring changes in

- A. length
C. surface area

- B. volume
D. dry weight

4. UME/2000/20/TYPE: M
For growth to occur in organisms, the rate of
A. food storage must be low
B. anabolism must exceed that of catabolism
C. catabolism must exceed that of anabolism
D. food storage must be high

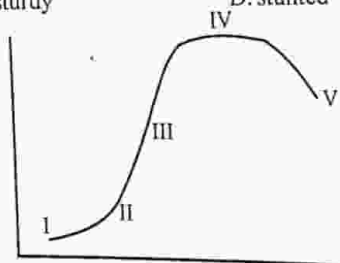
5. PCE/2000/25/TYPE: A
A unicellular organism is said to have attained his maximum growth when it
A. has reproduced itself several times
B. has used up all the nutrients in the medium
C. can no longer reproduce itself
D. can reproduce itself

6. PCE/07/1
Growth in living things is brought about by the
A. deposition of new material from the surroundings
B. synthesis of new material from within the body
C. absorption of water and minerals
D. elasticity of the cell

13.2. GROWTH IN PLANTS AND ANIMALS

7. UME/88/33
The radicle of a bean seedling grows most rapidly in the region
A. of the root tip
B. just above the root tip
C. just around the root tip
D. just below the root tip

8. UME/2001/17/TYPE: T
A seedling grown in the dark is likely to be
A. etiolated
B. dormant
C. sturdy
D. stunted



9. PCE/95/36
The area with the highest growth rate is labeled
A. V
B. IV
C. III
D. I

10. PCE/95/37
Environmental resistance is greatest at
A. II
B. III
C. IV
D. V

11. PCE/99/23
When the tip of coleoptiles is illuminated from one side, it comes towards that side because the growth hormone is
A. concentrated on that side at the tip of the coleoptiles
B. concentrated on the dark side at the tip of the coleoptiles
C. destroyed by the light rays at the tip of the coleoptiles
D. concentrated below the region of elongation

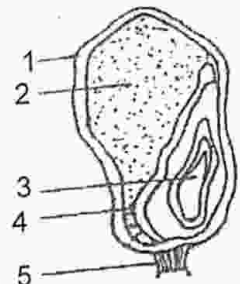
12. PCE/2002/6/TYPE: 5
Growth in animals differs from that in plants because it occurs
A. in the reproductive cells only

- B. equally in all parts of the body
C. in restricted parts of the body
D. at the apex only

13. PCE/07/22
Which of the following is the correct sequence of growth in a multicellular organism?

- A. cell enlargement → cell division → cell differentiation
B. cell differentiation → cell division → cell enlargement
C. cell division → cell differentiation → cell enlargement
D. cell division → cell enlargement → cell differentiation

13.3. GERMINATION OF SEEDS



14. UME/86/34
Which of the labeled parts will develop into a new plant?
A. 2
B. 3
C. 4
D. 5

15. UME/86/35
The structure labeled 1 is the
A. plumule
B. radicle
C. cell membrane
D. seed coat

16. UME/86/36
The main function of the structure labelled 2 is to
A. protect the inner parts of the seed
B. keep the inner parts moist
C. nourish the embryo and the growing parts
D. maintain the shape of the seed

17. PCE/96/23
Dormancy in seeds may be caused by
A. the rapid diffusion of oxygen into the seed
B. delayed development of the embryo
C. the rapid absorption of water
D. insufficient stored food for the embryo

13.3.1. TYPES OF GERMINATION

18. UME/86/47
In an epigeal germination, it is the
A. epicotyls that elongates fast
B. hypocotyls that elongates fast
C. hypogeal that elongates fast
D. plumule that elongates fast
E. roots that elongates fast

19. UME/85/37
Germination which results in the cotyledons being above ground is called
A. hypocotyls
B. epicotyls
C. epigeal
D. hypogeal
E. plumule

20. UME/92/27
Hypogeal germination is characterized by the

- A. emergence of the plumule out of the ground
- B. elongation of the hypocotyls
- C. provision of nourishment by the endosperm
- D. elongation of the epicotyls

UME/2004/18/TYPE: 4

21. Epigeal germination of a seed is characterized by
- A. lack of growth of the hypocotyls
 - B. more rapid elongation of the hypocotyls
 - C. more rapid elongation of the epicotyls than the hypocotyl
 - D. equal growth rate of both the hypocotyls and epicotyls

22. UME 2009/26

In a bean seed, absorption of water at the beginning of germination is through the

- A. hilum
- B. micropyle
- C. testa
- D. plumule

13.3.2. CONDITIONS NECESSARY FOR GERMINATION

23. UME 2006/13

Water is necessary for a germinating seed because it

- A. promotes aerobic respiration
- B. activates the enzymes
- C. wets the soil for proper germination
- D. protects the seeds from desiccation

24. PCE/2006/13/TYPE: A

The two most critical factors required for the growth of plant are the

- A. nutrients and water content of the soil
- B. light intensity and topography
- C. types of plant hormones and enzymes
- D. temperature and pH of the surroundings

25. PCE/95/28

The most important conditions necessary for seed germination are

- A. water, temperature and oxygen
- B. water, temperature and carbondioxide
- C. oxygen, water and carbondioxide
- D. oxygen, temperature and carbondioxide

26. UME/78/46

A germinating seed requires oxygen which is essential for

- A. converting carbohydrates into glucose
- B. transporting energy from one part of the plant to another
- C. the production of energy by oxidizing essential carbohydrates
- D. hydrolysis of proteins
- E. the formation of water molecules within the germinating seed

27. UME/80/12

Water is required for seed germination to take place because it

- A. activates the enzymes
- B. liberates energy for growth
- C. softens the testa
- D. permits radicle growth
- E. allows oxygen to diffuse into the seed

28. UME/79/23

Alkaline pyrogallol was used in an experiment.

The experiment must have been connected with

- A. excretion
- B. germination
- C. transportation
- D. digestion
- E. photosynthesis

MISCELLANEOUS QUESTIONS (13)

29. Growth can best be determined in a population of spirogyra by measuring the

- A. total lengths of the filaments
- B. total widths of the filaments
- C. rate of photosynthesis in the population
- D. dry weight of the organism (UME/87/33)

30. Water is required for seed germination to take place because it

- A. activates the enzymes
- B. liberates energy for growth
- C. softens the testa
- D. permits radical growth
- E. allows oxygen to diffuse into the seed (UME/80/12)

31. Seed whose cotyledons are swollen as a result of stored food are described as

- A. non-cotyledonous
- B. endosperms
- C. non-endospermous
- D. albuminous (PCE/97/24)

32. Epigeal germination can be found in

- A. sorghum
- B. maize
- C. millet
- D. groundnut (UME/91/35)

33. Which of the following chemicals is a plant growth substance?

- A. oxaloacetic acid
- B. pyruvic acid
- C. indoleacetic acid
- D. oxalosuccinic acid (PCE/92/28)

34. The storage tissue in a maize grain is the

- A. radicle
- B. plumbe
- C. endosperm
- D. embryo (PCE/03/37/TYPE: N)

35. The best method of measuring the growth rate of an animal is by taking it

- A. wet weight
- B. dry weight
- C. total volume
- D. surface area (PCE/04/27/TYPE: 9)

36. Some germinating seeds are put into a thermos flask with a cork through which passes a thermometer. This is a noticeable increase in the reading of the thermometer. The increase in temperature is due to:

- A. The external temperature of the room in which the flask is
- B. The heat given off by the decomposing seeds
- C. The dissipated heat from the respiring seeds
- D. The heat of action of the seeds
- E. The rise of temperature of the air in the flask

37. In a germinating seed, the first embryonic structure to emerge is:

- A. The plumule
- B. The tap root
- C. The coleoptile
- D. The coleorhiza
- E. The radicle

38. In the groundnuts, the seed leaves are usually brought above the soil level. This is:

- A. Hypogeal germination
- B. Hypocotyl germination
- C. Epigeal germination
- D. Epicotyl germination
- E. None of the above

39. In this type of germination the cotyledons are left in the soil, e.g. the maize:

- A. Hypogeal germination
- B. Epigeal germination
- C. Hypocotyl germination
- D. Epicotyl germination
- E. None of the above

40. In the germinating bean, the zone between the soil and the cotyledons (seed-leaves) is:

- A. The epicotyl region
- B. The internode
- C. The hypocotyls region
- D. The plumular region
- E. None of the above

41. The germinating bean plant before establishing itself, depends for its food on:

- A. its roots
- B. its first foliage leaves
- C. The hypocotyls
- D. Its radicle
- E. Its seed leaves

CHAPTER 13

1. C	2. A	3. D	4. B	5. D
6. B	7. B	8. A	9. C	10. C
11. B	12. B	13. D	14. B	15. D
16. C	17. B	18. B	19. C	20. D
21. B	22. B	23. B	24. C	25. A
26. C	27. A	28. B	29. D	30. A
31. B	32. D	33. C	34. C	35. B
36. C	37. E	38. C	39. A	40. E
41. E				

CHAPTER 14

Understand the following after reading this chapter:

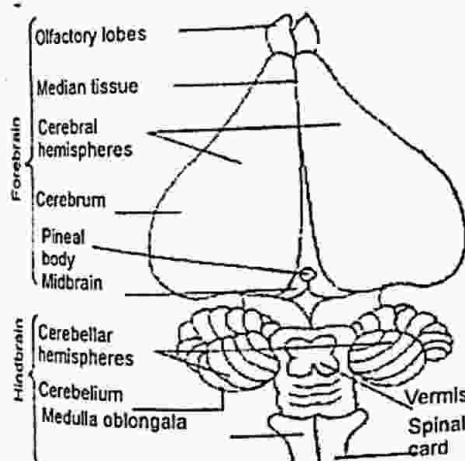
Nervous coordination
Sense organs
Hormones
Homeostasis

THE NERVOUS SYSTEM

The nervous system is made up of the Central nervous system (CNS), which consist of the brain and spinal cord, and the Peripheral nervous system (PNS), which comprises the somatic and autonomic system

THE BRAIN AND ITS FUNCTION

The brain is composed of numerous brain cells or neurons. It is confined and protected by the cranium of the skull. It is divided into three main sections, namely: the forebrain, midbrain and hind brain.



The brain of the rabbit (dorsal view)

Parts of mammalian brain	Function(s)
1. Cerebral hemisphere (Cerebrum)	Intelligence, memory, voluntary actions, sensations.
2. Hypothalamus	Regulation of body temp. and osmotic pressure in blood; appetite and emotions.
3. Pituitary gland	Secretes a number of hormones.
4. Optic lobes	Concerned with sight and movement of eyeball.
5. Cerebellum	Muscular coordination and bodily balance.
6. Medulla oblongata	Involuntary actions, e.g. heartbeat, respiratory movements, peristalsis.

THE SPINAL CORD AND ITS FUNCTION

The spinal cord is also composed of neurons. Spinal nerves emerge at intervals from the vertebral canals and extend to all parts of the body. It is responsible for some local reflex actions

THE PERIPHERAL NERVOUS SYSTEM

The peripheral nervous system consists of all the nerves that link the CNS to all parts of the body. It is divided into two parts.

Sensory somatic system, which receives information from sensory receptor and conduct same to the CNS.

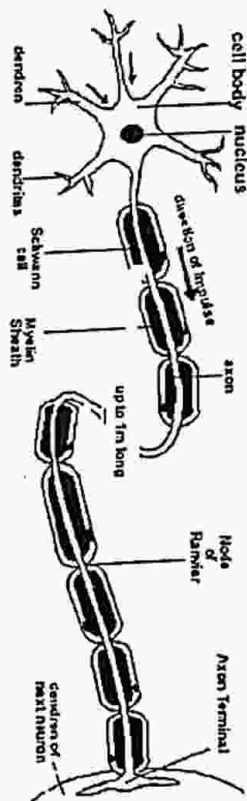
Autonomic nervous system, consists of the sympathetic and parasympathetic system.

Neurons are specialized cells which receive information (stimuli) and transmit them in form of electrical impulse to muscles and glands. A neuron consists of the cell body which contains the nucleus and dendrites, and a single long process called axon. There are three types of neurons:

Afferent or sensory Neurons: These conduct impulses from receptors or sensory organs to the CNS.

Efferent or motor Neurons: They relay impulses from the CNS to effector organs i.e muscles and glands

Intermediate or Association Neurons: They transmit impulses within the CNS. They connect the afferent to efferent neurons.



TRANSMISSION OF NEURAL IMPULSES

The transmission of neural impulses is electrochemical in nature. At resting potential, there is polarization (potential difference) on the nerve fiber (axon) membrane. The concentration of K^+ inside is greater than on the membrane and the concentration of Na^+ is greater on the membrane than inside. When the membrane is stimulated, it becomes highly permeable to Na^+ which then rush in momentarily (depolarization) and reverse the polarity of the membrane. The flow of the Na^+ is sustained down the axon producing the neural impulse or action potential.

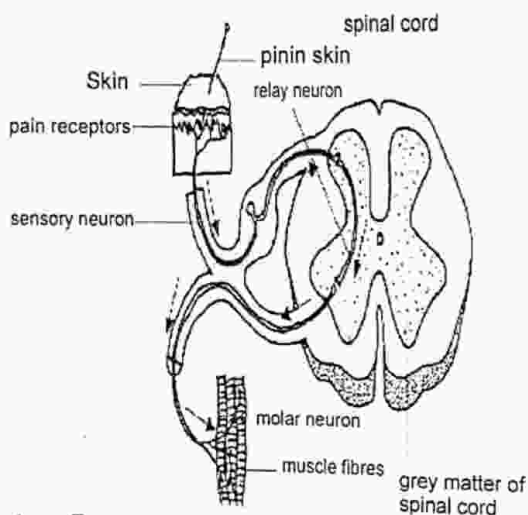
Reflex Actions

A reflex or involuntary action is a quick and automatic response by an organ or system of organs of the body to a stimulus, e.g sneezing, blinking of the eye, laughing when tickled, knee jerk etc

Reflex Action	Voluntary Action
Response is stereotyped	Response is not stereotyped it varies with circumstances
It is done unconsciously i.e. it does not involve the higher centre of the brain	It is a conscious action i.e. it involves the higher centre of the brain
It is rapid/fast	It is less rapid/slow
It involves a small number of neurons	It involves a large number of neurons

The Reflex Arc

A reflex arc is the nervous pathway taken by a reflex action. It consists of receptors which receive stimulus and transmit it through the afferent neuron to the spinal cord.



A reflex arc

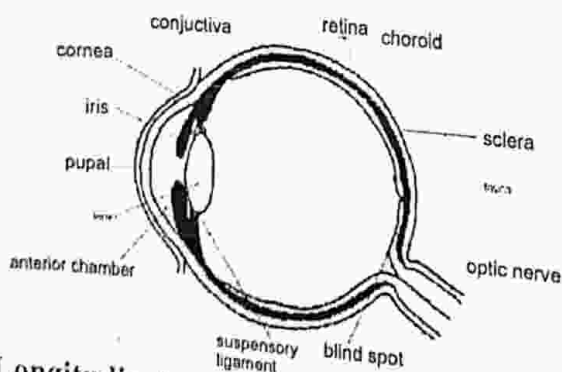
Voluntary Actions

A voluntary action is an action consciously taken in response to a stimulus.

SENSE ORGAN

There are five principal sense organs viz: eye, ear, nose, tongue and skin.

THE EYE



Longitudinal section of the mammalian eye

Parts of the eye	Function(s)
1). Aqueous humour	Refracts light; keeps eyeball firm.
2). Vitreous humour	Refracts light; keeps eyeball firm.
3). Cornea	Refracts light rays into pupil.
4). Suspensory ligament	Attaches lens to ciliary body.
5). Iris	Controls amount of light entering the eye.
6). Pupil	Allows light to enter the eye.
7). Lens	Focuses light rays into retina.
8). Ciliary muscle	Controls curvature or thickness of the lens.
9). Rectus muscle	Movement of eyeball.
10). Sclera	Protection against mechanical injury.
11). Choroid	Pigmented black to prevent internal reflection of light.
12). Yellow spot (fovea)	Region of acute vision
13). Optic nerve	Transmits impulses from eye to brain.
14). Blind spot	No photoreceptor cells; no vision when image falls on it.
15). Retina	Light sensitive layer, contains: (i). Cones concerned with colour vision in bright light. (ii). Rods concerned with vision in dim light.

Image Formation

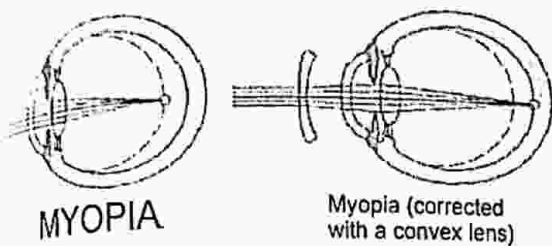
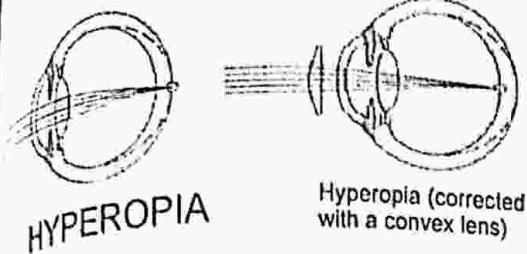
The eye sees by reflection of light from objects. Light entering the eye are refracted by the cornea, aqueous humour, lens and vitreous humour and brought to focus on the retina to form an image. The image formed by the eye is real, inverted and smaller than the object.

Accommodation: This is the ability of the eye to focus both far and near objects due to the adjustment done on the lens by the ciliary muscle.

Eye Defects

Short sightedness or myopia: This is the ability of the eye to see only near objects clearly but not distant objects. Caused by large or elongated eye balls. Image of distant object is formed before the retina. It can be corrected by wearing concave lenses.

Long sightedness or Hypermetropia: This is the ability of the eye to see distant objects clearly but not near objects. Caused by small or short eyeball. Image of near object is formed behind the retina. It can be corrected by the use of convex lenses.



Correction of eye defects

astigmatism: This is caused by unequal curvature of the cornea. It can be corrected by the use of cylindrical lenses.

Cataract: This is the opacity of the lens which may be caused by bacterial infection.

Conjunctivitis: This is the inflammation of the conjunctiva.

Presbyopia: This is the recession of the near point with age, beyond 40cm, due to increased plasticity of the lens. It can be corrected by the use of a bi-focal lens.

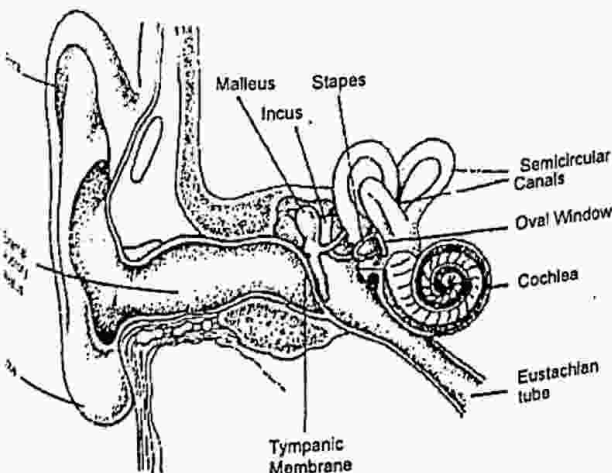
THE EAR

The ear is the organ responsible for hearing and balancing. It is made up of three parts, namely, the outer ear, middle ear and inner ear.

The outer ear consists of the pinna and the external auditory meatus. The eardrum demarcates the external ear from the middle ear.

The middle ear is oval and filled with air. It contains the malleus, incus and stapes collectively known as ossicles.

The inner ear is filled with a fluid called perilymph. It



Section of the mammalian ear

contains the semicircular canals, utricle and saccule. The cochlea is made up of spirally coiled tube containing a fluid called endolymph and the organ of Corti connected to the brain by the auditory nerves.

Mechanism of Hearing

Sound waves from the air enter the ear through the auditory meatus and hit the eardrum which is set into vibration. The vibration is picked up by the bone ossicles and amplified before entering the inner ear. The oscillation of the stapes set the endolymph into vibration which in turn stimulates the auditory nerve fibres to transmit impulses to the brain. The brain then interprets the pitch, quality and loudness of the sound and relays back and the sound is heard.

Balance

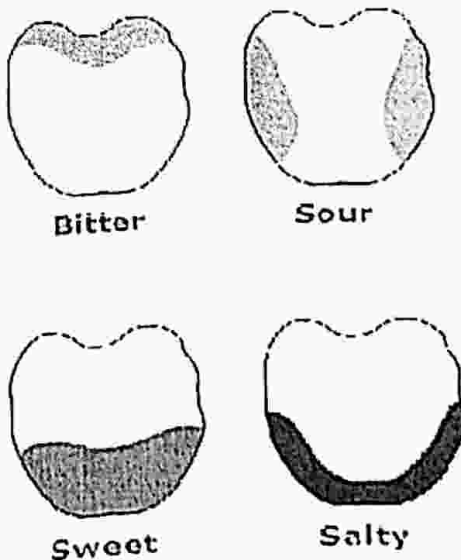
The semi circular canals are sensitive to the turning movement of the head. The utricle and saccule contain receptors sensitive to any movement of the body. These sensations are transmitted as impulses to the brain which interprets the position of the animal.

THE NOSE

The nose is the organ of smell. It has ciliated tips of olfactory cells on the surface of the nasal cavities which respond to various chemicals in the air.

THE TONGUE

The tongue is the organ of taste. It has buds which are sensitive to four primary tastes. The back of the tongue is sensitive to bitter stimuli, the sides to sour stimuli and the tip to sweet and salty stimuli.

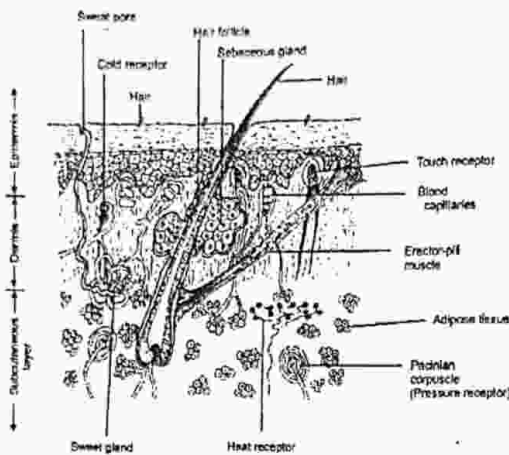


Different parts of sensitivity on the tongue for the four tastes

THE SKIN

The skin is a waterproof structure; One of the organs of the sense of touch. The skin has three parts:

1. Epidermis:
 - i. Cornified layer;
 - ii. Granular layer;
 - iii. Malpighian layer.
2. Dermis.
3. Hypodermis.



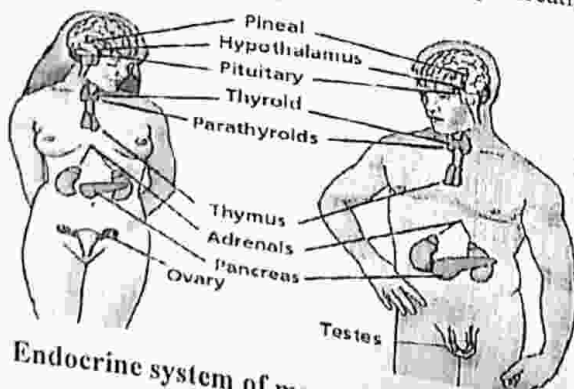
Function of the Skin

1. It protects the underlying tissues from damage and mechanical injury.
2. It protects the body against bacteria and other infective microbes.
3. It protects the body against ultraviolet rays
4. It excretes sweat
5. The nerves cells respond to change in the environment.
6. The sebaceous glands secrete sebum which prevents the skin from cracking or drying out
7. It synthesizes vitamin D.
8. It regulates body temperature through heat gain and heat loss.

Heat gain – Vaso constriction, shivering, contraction of hair erector muscles etc
Heat loss – Vasodilation and sweating

THE ENDOCRINE SYSTEM

The endocrine system is made up of the ductless glands situated in various parts of the body. Secretions of the ductless glands are called hormones. The endocrine system consists of the pituitary, thyroid, pancreatic, adrenal and gonadal glands.



Endocrine system of man

Nervous System

Transmission is through nerve fibre

It is rapid

Response is immediate, short-lived and very precise

Message passed as electrical impulses along nerve

Effect are usually localized and temporary

It occurs in animal only

Endocrine System

Transmission is through the circulatory system

It is slow

Response is slow, long-lasting, diffuse

Message or information passed as chemical substances in the blood stream

Effect may be widespread and permanent

It occurs in plants and animals

1. Pituitary gland: The pituitary gland has two divisions namely: anterior pituitary gland and posterior pituitary gland.

Anterior pituitary gland secrete the following hormones: Thyroid stimulating hormone (TSH), Follicle stimulating hormone (FSH), Luteinizing hormone (LH), Growth hormone (GH), Adenocorticotrophic hormone (ACTH), Prolactin.

Posterior pituitary gland releases two hormones: Antidiuretic hormone (ADH) and Oxytocin.

Over secretion of GH causes gigantism in infants and acromegaly in adult. Under secretion results in dwarfism in children.

2. Thyroid gland is located in the neck region. It secretes thyroxine, which regulates rate of metabolism. It stimulates growth in young mammals and metamorphosis in tadpole. Over secretion causes hyperthyroidism, goiter and mental instability. Under secretion result in cretinism and myxoedema.

3. Parathyroid glands are four bodies behind the thyroid gland. They secrete parathormone which regulates (increase) blood calcium.

4. Pancreatic gland is located between the stomach and small intestine. It helps the production of insulin and glucagon which regulates blood glucose. It also secretes digestive juice. Over secretion can cause insulinoma. Under secretion results to *Diabetes mellitus* (excess sugar).

5. Adrenal glands are located on top of each kidney. They help the production of adrenaline or emergency hormone which prepares the body for emergencies. It also controls certain behavioural patterns such as bravery and fear.

6. Gonadal or reproductive glands: These are the ovaries in females and testes in males.

Ovaries are located in the lower abdomen. They help the production of sex hormones such as oestrogen, progesterone, prolactin, relaxin. Also produces egg cells. **Testes** are located in the groin enveloped by the scrotum. They help the production of sex hormone called testosterone or androgen, which maintain secondary sexual characters and helps the production of sperm.

PLANT

Common

1. Auxin

i. Facilitate

ii. Promote

iii. Regulate

iv. Control

v. Stimulate

cutting

vi. Retard

vii. Promote

Note: A.S.

2. Gibberellins

i. Breaks

germination

ii. Enhance

presence of

iii. Induces

v. Induce

vi. Retard

vii. Promote

3. Kinins/

developing

i. Induce

ii. Promote

iii. Break

iv. Regulate

4. Abscission

gibberellins

i. Induces

ii. Inhibits

iii. Retards

iv. Accelerate

5. Ethylene:

i. Induces

ii. Promotes

14.1. NER

14.1.1 THE FUNCTIONS

1. Which of the

of the central ne

A. The axon and

B. The brain and

C. The spinal co

D. The spinal co

E. The spinal co

2. What is the fun

A. It dictates the

B. It directs the fu

C. It controls and

animals

- Common plant hormones include Auxins, gibberellins, kinins and florigens.
- Auxins:** They are produced at the shoot and root apices.
- Facilitate cell expansion in the zone of elongation
 - Promote cell division and stem elongation
 - Regulate growth in plant's apical region.
 - Control tropism
 - Stimulation of adventitious roots, hence used in stem cutting
 - Retard abscission.
 - Promote formation of pistillate (female) flowers.
- Note:** A synthetic form of Auxin is Indole Acetic Acid
- Gibberellins:** They are found in embryos of germinating seed and in roots
- Breaks dormancy in buds and seeds and promote seed germination.
 - Enhances elongation of stems at internodes in the presence of auxins.
 - Induces synthesis of plant enzymes e.g. amylase
 - Induce flowering, fruit and seed development
 - Retard leaf abscission.
 - Promote formation of staminate (male) flowers.

- 3. Kinins/Cytokinins:** They are found in the root apex, developing fruits and germinating seeds
- Induce rapid cell division
 - Promote growth of lateral buds and fruits
 - Break seed dormancy
 - Regulate shoot growth

- 4. Abscissic Acid:** Functions are opposite to those of gibberellins
- Induces dormancy in buds
 - Inhibits seed germination
 - Retards plants growth especially stem elongation
 - Accelerates abscission of leaves, fruits and flowers.

- 5. Ethylene:**
- Induces ripening of fruits
 - Promotes abscission of leaves

Revision Questions

14.1. NERVOUS COORDINATION 14.1.1 THE COMPONENTS, STRUCTURE AND FUNCTIONS OF THE C.N.S.

- Which of the following represent the essential structures of the central nervous system?
 - The axon and proprioceptors
 - The brain and axon
 - The spinal cord and the axon
 - The spinal cord and peripheral nerves
 - The spinal cord and the brain
- What is the function of the central nervous system?
 - It dictates the working condition of the body
 - It directs the functions of certain parts of the body
 - It controls and co-ordinates the activities of individual animals

- It relays impulses from one part of the body to the other
- None of the above

3. PCE/99/20

The central nervous system is connected with the receptor and effectors organs by

- motor and sensory neurons
- motor neurons and cranial nerves
- cranial nerves and spinal nerves
- spinal nerves and sensory neurons

4. UME/84/31

If an animal is very active and has good muscular control, it is likely to have well-developed

- olfactory lobes
- cerebral hemispheres
- optic lobes
- cerebellum
- spinal cord

5. UME/87/38

The centre for controlling body temperature in the brain is the

- cerebrum
- cerebellum
- medulla
- hypothalamus

6. UME/88/35

What part of the central nervous system is concerned with answering an examination question?

- cerebrum
- cerebellum
- medulla oblongata
- spinal cord

7. UME/97/31

The part of brain that regulates most biological cycles in humans is

- olfactory lobe
- optic lobe
- medulla oblongata
- pineal body

8. PCE/93/30

The part of the brain that coordinates reproductive events is the

- olfactory lobe
- medulla oblongata
- cerebellum
- cerebrum

9. PCE/94/28

If an animal is very active and has good muscular control, it is likely to have well-developed

- olfactory lobes
- spinal cord
- cerebellum
- cerebrum

10. PCE/94/29

A neuron is made up of a

- cell body, dendrites and an axon
- cell body, muscle fibre and dendrites
- sheath, an axon and nucleus
- synaptic knobs, cell body and dendron

11. PCE/98/30

The part of the brain which controls intellectual functions is called

- medulla oblongata
- cerebrum
- cerebellum
- olfactory lobe

12. PCE/98/27

Changes in the concentration of the blood is detected by

- A. medulla oblongata
- C. hypothalamus

- B. cerebellum
- D. cerebrum

13. The sensory cells situated on the surface of the body and which receive external stimuli are:

- A. The proprioceptors
 - C. The exteroceptors
 - E. The exteroceptors
- B. The external receptors
 - D. The enteroceptors

14. Which of these functions and activities is not undertaken by the cerebral hemispheres of man?

- A. Argument
 - C. Recitation
 - E. None of the above
- B. Reasoning
 - D. Blinking of the eyes

15. Samuel was riding down a hill when he had an accident. After treatment it was discovered that he was partially paralyzed and further that he lacked a sense of feeling. The part of Samuel's body much affected by the accident was:

- A. The brain
 - C. The cerebellum
 - E. The peripheral nerves
- B. The cerebrum
 - D. The spinal cord

16. A tube of stored ammonia is passed round an audience of market women. It appeared that some of the women were indifferent to the smell of the ammonia. Which part of their brain or body was defective

- A. The bones of the nasal cavity
 - B. The olfactory lobes
 - D. The olfactory nerves
- C. The olfactory organs
 - E. None of the above

17. Kalu is a child who always has a high body temperature. Hospital investigations showed that his illness was due to a defective portion of his brain. Which part of his brain was severely hit?

- A. The cerebral hemisphere
 - C. The cerebellum
 - E. None of the above
- B. The ventricles
 - D. The corpora striata

18. The spinal cord arises from:

- A. The spinal stump
 - C. The medulla oblongata
 - E. The brain
- B. The cerebral body
 - D. The cerebellum

19. One contestant in a boxing bout after a knockout cannot lift up his right arm. Which part of his brain was severely hit?

- A. The right cerebellum
- B. The medulla oblongata
- C. The left cerebral hemisphere
- D. The right cerebral hemisphere
- E. The central lobes of the cerebellum

20. The thin skin that immediately covers the spinal cord is:

- A. The dura mater
 - C. The pia mater
 - E. None of the above
- B. The arachnoid skin
 - D. The fibrous capsule

21. What is the characteristic of the ventral fissure of the

spinal cord?

- A. It is elongated
- B. It is usually wide
- C. It contains blood vessels and nerves
- D. It gives rise to the ventral nerve roots
- E. None of the above

22. Where in the spinal cord are the cell bodies found

- A. In the ventral and dorsal roots of the white matter
- B. In the dorsal root of the grey matter
- C. In the grey matter
- D. In the white matter
- E. In the dorsal root ganglia

23. The white matter is noted for:

- A. Having nerve fibres and myelin sheaths
- B. The relay of nerve impulses
- C. Possessing only the fatty sheaths
- D. The storage of nerve forming cells
- E. Possessing the intermediate or relaying neurons

14.1.2 THE COMPONENTS AND STRUCTURE OF THE P.N.S

24. What is the function of the peripheral nervous system?

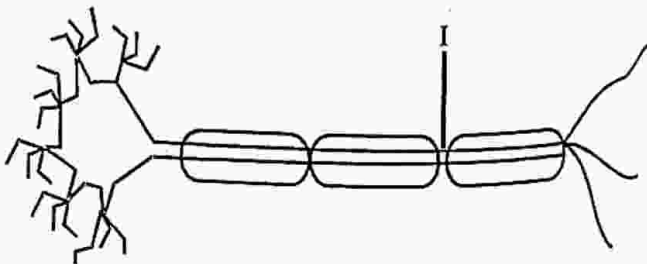
- A. It control the activities of the body
- B. It is a relay system
- C. It co-ordinates the receptive functions of the skin
- D. It serves as a link between the organs and the tissues of the body and the central nervous system
- E. None of the above reasons

25. UME/93/28

In the mammal, the autonomic nervous system consists

- A. sympathetic and parasympathetic systems
- B. brain and cranial nerves
- C. brain and spinal nerves
- D. spinal cord and spinal nerves

26. UME/2003/26/TYPE:A



The structure can be found in the

- A. peripheral and central nervous system
- B. peripheral nervous system only
- C. sympathetic and parasympathetic nervous system
- D. central nervous system only

27. UME/2003/27/TYPE:A

The point marked I is referred to as

- A. myelin sheath
- C. node of ranvier
- B. dendrites
- D. axon

14.1.3 MECHANISM OF TRANSMISSION OF NERVE IMPULSES

28. UME/2004/11/TYPE: 4

The two key cations involved in the action potentials of nervous transmission are

- A. Mg^{2+} and K^{+}
- B. Na^{+} and Fe^{2+}
- C. Fe^{2+} and Mg^{2+}
- D. Na^{+} and K^{+}

29. PCE/91/32

Which of the following plays a direct role in the transmission of impulses across a synapse?

- A. acetylcholine
- B. cholinesterase
- C. sodium ion
- D. potassium ion

30. UME/86/40

In a mammal, stimulus is transferred from the receptor muscle to the central nervous system through the

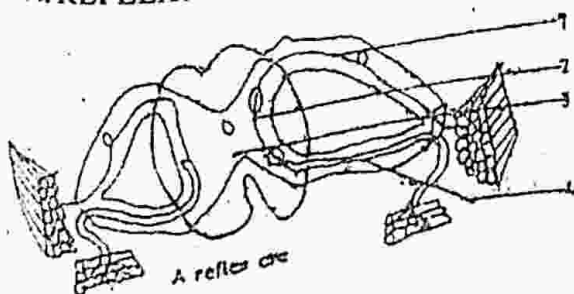
- A. motor neurones
- B. effector muscles
- C. dendrites
- D. sensory neurones
- E. synapses

31. UME/98/27

The small masses of nervous tissue in which many neurons have their nuclei are called

- A. dorsal roots
- B. ventral roots
- C. ganglia
- D. synapses

14.1.4. REFLEX ACTION



32. UME/90/33

All the cell bodies in the spinal cord are found in

- A. 1
- B. 2
- C. 3
- D. 4

33. UME/90/34

In a reflex action, impulse flows from

- A. 1 to 2
- B. 2 to 1
- C. 4 to 1
- D. 4 to 2

34. UME/80/34

The following are connected with the movement of a reflex action

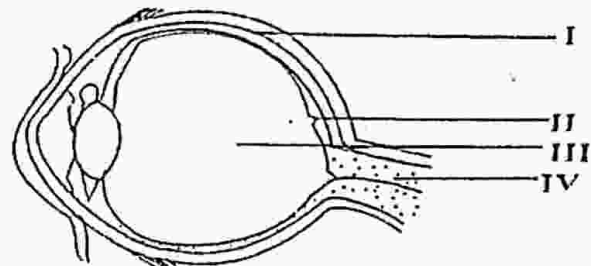
- (1) Central nervous system (2) Muscle (3) Skin (4) Sensory nerve (5) Motor nerve.

Which of the following sequences indicates a correct path?

- A. 1-2-3-4-5
- B. 2-1-4-5-3
- C. 4-1-5-2-3
- D. 4-1-5-2-3
- E. 3-4-1-5-2

14.2. SENSORY ORGANS

14.2.1. EYE



35. PCE/2004/30/TYPE: 4

The part that provides nourishment for the eye is labelled

- A. I
- B. II
- C. III
- D. IV

36. PCE/2004/31/TYPE: 4

The part labelled II is the

- A. blind spot
- B. retina
- C. yellow spot
- D. conjunctiva

37. PCE/2003/34/TYPE: N

The three layers that form the wall of the eyeball are

- A. sclera, choroid and epidermis
- B. choroid, cornea and epidermis
- C. retina, choroid and dermis
- D. sclera, choroid and retina

38. PCE/96/24

Which of the following is found in the retina?

- A. ciliated cell
- B. cone-shaped cell
- C. epithelial cell
- D. cubical cell

39. UME/82/25

The parts of the mammalian eye that strongly bend light rays are the

- A. cornea and the lens
- B. cornea and aqueous humour
- C. cornea and vitreous humour
- D. lens and aqueous humour
- E. lens and vitreous humour

40. UME/79/17

When the ciliary muscle of the eye contracts, the eye lens

- A. bulges
- B. contracts
- C. rotates
- D. flattens
- E. is rounded

41. UME/97/32

The ability of the eye to focus on both near and distant objects is termed

- A. image formation
- B. refraction
- C. hypermetropia
- D. accommodation

42. UME/85/39

An old man is likely to be long-sighted because age affects the

- A. optic nerves
- B. retina

- C. ciliary muscles
E. aqueous humour

D. cornea

44. UME/94/27

When a short-sighted person views a distant object without spectacles, the image is formed

- A. on the retina
B. in front of the retina
C. behind the retina
D. on the blind spot

45. PCE/92/29

Human beings have relatively poorer vision at night than nocturnal animals because the human eyes

- A. are meant for day vision
B. contain fewer rods
C. contain more rods
D. contain fewer cones

46. The function of concave lens in correcting myopia is to

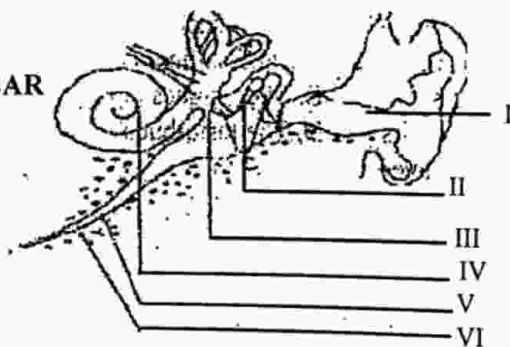
- A. diverge incoming rays
B. converge incoming rays
C. reflect incoming rays
D. screen incoming rays

47. PCE/94/30

A person is suffering from an eye defect in which he can see only distant objects clearly. This eye defect may be corrected by using

- A. convex lens
B. biconvex lens
C. concave lens
D. biconcave lens

14.2.2. EAR



48. UME/2001/22/TYPE: T

The parts which function together to bring about hearing are labelled

- A. IV, V and VI
B. I, II, IV and VI
C. I, II, III and IV
D. I, II and IV

49. UME/2001/23/TYPE: T

The part labelled II is the

- A. fenestra ovalis
B. middle ear canal
C. internal auditory meatus
D. ear ossicles

50. UME/98/32

Which path does sound entering the human ear follow?

- A. oval window, ossicles, eardrum
B. ear drum, oval window, ossicles
C. eardrum, ossicles, oval window
D. ossicles, ear drum, oval window

51. UME/2000/19/TYPE: M

The inner ear contains two main organs, namely, the

- A. eardrum and eustachian tube
B. cochlea and semi-circular canals
C. oval window and ossicles
D. pinna and cochlea

52. UME/83/39

The function of the ossicles (malleus, incus and stapes) in the mammalian ear is the

- A. transmission of vibrations
B. regulation of pressures
C. support of the inner ear
D. maintenance of balance during motion
E. secretion of oil

14.2.3. TONGUE

53. PCE/2001/27/TYPE: 4

In the tongue, the buds responsible for sweet taste are

- A. sides
B. back
C. centre
D. tip

54. The structures that respond to such sensations as sweetness, bitterness, sourness and saltiness are:

- A. The taste-hairs
B. The tongues
C. The gustatory hairs
D. The taste-buds
E. None of the above

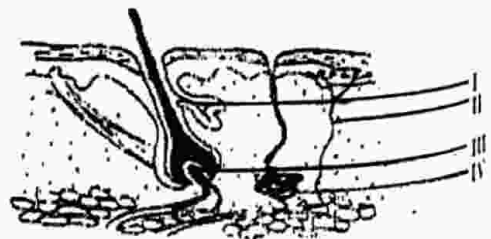
14.2.4. NOSE (OLFACTORY)

55. UME/99/34/TYPE: D

The correct sequence for the perception of smell in mammals is

- A. chemicals → olfactory nerve endings → brain
B. dissolved chemicals → nasal sensory cells → brain
C. chemicals → mucous membrane → sensory cells → brain
D. dissolved chemicals → sensory cells → olfactory nerve → brain

14.2.5. SKIN



56. UME/2004/15/TYPE: 4

The function of the part labelled III is to

- A. produce oil for the skin
B. carry blood and nitrogenous waste
C. contract to pull the hair erect
D. conduct nervous impulses

57. UME/2004/16/TYPE: 4

The sweat gland is the structure labelled

- A. IV
B. III
C. II
D. I

14.3. HORMONES

14.3.1. ANIMAL HORMONES

58. UME/85/38

The mammalian endocrine system is responsible for

- A. transmitting impulses
B. regulatory body temperature
C. regulating osmotic pressure of blood
D. chemical co-ordination

E. the manufacture of blood

59. UME/89/34

A fundamental similarity between nervous and hormonal system is that both

- A. involve chemical transmission
- B. have widespread effects
- C. shed chemical into the blood stream
- D. evoke rapid responses

60. UME/2000/9/TYPE: M

In which part of the human body does the secretion of the growth hormone occur?

- A. head region
- B. waist region
- C. neck region
- D. gonads

61. PCE/94/32

The endocrine glands responsible for the production of antidiuretic hormone is the

- A. thyroid gland
- B. pancreas
- C. adrenal gland
- D. pituitary gland

62. PCE/99/22/TYPE: E

The hormone whose deficiency in a child would cause reduced mental and physical growth is

- A. adrenalin
- B. thyroxine
- C. insulin
- D. secretin

63. UME/95/27

The gland directly affecting metabolic rate, growth and development is known as

- A. adrenal gland
- B. thyroid gland
- C. mammary gland
- D. parathyroid gland

64. UME/80/33

Which of the following hormones is produced during fright or when agitated?

- A. Insulin
- B. adrenalin
- C. thyroxine
- D. pituitrin
- E. progesterone

65. PCE/96/25

Insulin and glucagon are hormones produced by the

- A. thyroids
- B. adrenal gland
- C. pancreas
- D. pituitary gland

66. UME/86/40

The hormone which regulates the amount of glucose in the blood is called

- A. adrenalin
- B. auxin
- C. insulin
- D. thyroxine

adrenal

C. pancreatic

B. thyroid

D. pituitary

68. UME/2002/43/TYPE: Y

The hormone secreted at II serves to

- A. facilitate the development of facial hours
- B. raise the level of calcium ions in the blood
- C. lower blood glucose level
- D. make the body react to emergencies

69. PCE/92/30

Overproduction of somatotrophic hormones in the human being could result in

- A. dwarfism
- B. gigantism
- C. cretinism
- D. obesity

70. PCE/94/31

The cells responsible for the secretion of insulin in human body are called

- A. Graafian follicles
- B. islets of langerhans
- C. crypt of lieberkhun
- D. corpus luteum

71. PCE/06/5/TYPE: A

The antidiuretic hormone is secreted by the

- A. posterior pituitary gland
- B. anterior pituitary gland
- C. Gonads
- D. pancreas

72. PCE/07/23

Which of the following produces secretions that help on controlling the level of calcium in the blood?

- A. gonad
- B. parathyroid
- C. pancreas
- D. pituitary gland

14.3.2. PLANT HORMONES (PHYTOHORMONES)

73. UME/87/36

The growth of a coleoptile towards unilateral light source is due to

- A. rapid rate of photosynthesis
- B. unequal distribution of auxin
- C. the effect of geotropism
- D. the effect of photolysis

74. UME/89/33

Fruit enlargement can be induced by spraying young ovary with

- A. cytokinin, abscisic acid and ethylene
- B. gibberellins, ethylene and abscisic acid
- C. auxin, abscisic acid and ethylene
- D. auxin, cytokinin and gibberellins

75. UME/93/30

The substance that is responsible for apical dominance in plants is known as

- A. gibberellins
- B. tannin
- C. auxin
- D. kinin

76. UME/87/39

Unlike auxin, gibberellins



2019-08-

43/TYPE: Y
position labelled is the A

- A. induce the formation of adventitious roots
- B. do not affect leaf and fruit abscission
- C. cannot stimulate stem elongation
- D. are quite effective as herbicides

77. UME/91/36

A dwarf plant can be stimulated to grow to normal height by the application of

- A. Thyroxin
- B. gibberellins
- C. Insulin
- D. Kinin

78. UME/2000/33/TYPE: M

Which of the following growth activities in plants is brought about by gibberellins?

- A. Rapid cell division
- B. tropic response
- C. cell elongation
- D. main stem elongation

79. UME/2003/21/TYPE: A

The most important hormone that induces the ripening of fruit

- A. Cytokinin
- B. indole acetic acid
- C. Ethylene
- D. gibberellins

80. PCE/06/9/TYPE: A

Weeds can be effectively controlled by the use of

- A. abscisic acids
- B. gibberellins
- C. auxins
- D. cytokinins

81. One of the following pairs of plant hormones helps in seed germination

- A. Cytokinin and Auxin
- B. Gibberellin and Abscissic acid
- C. Auxin and Gibberellin
- D. Cytokinin and Gibberellin
- E. Abscissic acid and Cytokinin

82. One of the following pairs of hormones is responsible for both cell elongation and apical dominance

- A. Cytokinin and Auxin
- B. Gibberellin and Cytokinin
- C. Florigens and Abscissic acid
- D. Auxin and Gibberellin
- E. Gibberellin and Abscissic acid

14.4. HOMEOSTASIS

83. UME/82/7

An homeothermic animal kept in a room where the temperature is lower than the body temperature may lose heat by four physical processes. Which of the following processes is NOT connected with body temperature regulation?

- A. Radiation
- B. sweat
- C. Evaporation
- D. conduction
- E. convection

84. UME/83/17

The kidneys of all vertebrates act as osmo-regulators. This means that they

- A. keep the composition of the plasma constant
- B. regulate osmotic processes

- C. control the volume of blood entering the kidneys
- D. decrease the osmotic pressure of blood
- E. increase the osmotic pressure of blood

85. UME/93/49

Cold blooded animals are referred to as

- A. Poikilothermic
- B. homeothermic
- C. Polythermic
- D. homeostatic

86. PCE/98/10/TYPE: C

The group of animals that is poikilothermic is

- A. birds, fishes and amphibians
- B. reptiles, mammals and birds
- C. mammals, amphibians and fishes
- D. amphibians, fishes and reptiles

87. PCE/2001/30/TYPE: 4

Homeostasis is a feedback reaction which is largely

- A. hormonal
- B. voluntary
- C. reflex
- D. nervous

88. PCE/2003/39/TYPE: N

High body temperature in mammals can be regulated

- A. vasodilation and sweating
- B. sweating and shivering
- C. vasdilation and vasoconstriction
- D. vasoconstriction and sweating

89. PCE/92/47

The temperature of a homoeothermic animal depends

- A. temperature of the environment
- B. heat obtained from the sun
- C. heat it receives from the environment
- D. in-built mechanisms that maintain a constant temperature

90. PCE/00/26/TYPE: A

When the osmotic pressure of blood rises, the hormone released to the kidney causes the nephron to

- A. decrease the reabsorption of water
- B. increase the reabsorption of water
- C. decrease the reabsorption of salt
- D. increase the reabsorption of salt

MISCELLANEOUS QUESTIONS (14)

91. Which of the following is a homeostatic response in humans?

- A. withdrawing the hand from a hot object
- B. the mouth getting watery when food is sighted
- C. yawning owing to tiredness
- D. shivering in a cold environment

(UME/2005/31/TYPE: D)

92. The part of the brain that control heart breathing is the

- A. olfactory lobe
- B. cerebellum
- C. cerebral hemisphere
- D. medulla oblongata

(UME/94/29)

93. The centre for controlling body temperature in the brain is the
A. cerebrum
C. medulla (UME/87/38)

- B. cerebellum
D. hypothalamus

94. Movement and positions of the head in man are moved by the
A. achlea
C. tricus (UME/85/28)

- B. malleus
D. semicircular

95. The main function of the choroid is
A. protection of the eyeball
B. transmission of light
C. supply of nutrients to tissues of the eye
D. converging light (UME/88/34)

96. What part of the central nervous system is concerned with answering an examination question?
A. Cerebrum
C. medulla oblongata
B. cerebellum
D. spinal cord (UME/88/35)

97. In the mammalian skin, mitosis occurs in the
A. cornified layer
C. malpighian layer
B. granular layer
D. dermal layer (CE/99/26/TYPE: E)

98. The skin of a mammal will be rendered colourless by
A. a defective sebaceous gland
B. a defective malpighian layer
C. dead nerve endings
D. blocked sweat pores (PCE/2000/28/TYPE: E)

99. What would happen to a man whose pancreas has been surgically removed?
A. the level of blood sugar would increase
B. the glycogen content of the liver would increase
C. his blood pressure would decrease
D. his weight would increase appreciably

100. Which of the following growth activities in plants brought about by gibberellins?
A. rapid cell division
C. cell elongation
B. tropic response
D. main stem elongation (UME/2000/33/TYPE: M)

101. One basic similarity between nervous and endocrine system is that they both
A. produce widespread effects
B. transmit very fast impulses
C. involve the use of chemical substances
D. electrostatic changes (UME 2006/6)

102. The unit of the nervous system is called:
A. The axons
C. The neurons
B. The myelin sheath
D. The cell body
E. None of the above

103. The small gaps occurring between the nerve fibres are:
A. The juxtapositions
C. The synapses
B. The bridges
D. The nodes of Ranvier

104. The function of the fatty myelin sheath is:
A. To insulate the axon and prevent the loss of impulses
B. To lubricate the axon and increase its conductivity
C. To replace damaged axons
D. To transmit impulses from the receptor to the effectors
E. None of the above

105. The nucleated sheath that covers the myelin sheath is:
A. The sheath nuclei
C. The cytoplasm
E. None of the above
B. The neurilemma
D. The neurolemma

106. The nodes of Ranvier are:
A. Areas in which the axons swell up
B. Areas where the neurilemma touches the axon
C. The points in which the axon is exposed
D. The contact points found over the non-myelinated nerve fibres
E. None of the above

107. Which of these actions is not reflex actions?
A. Jumping up when on a hot iron
B. The beating of the heart
C. The secretion of pepsin into the stomach
D. Ultra filtration of urea in the kidney
E. None of the above

108. What is the main difference between a reflex and a voluntary action?
A. A voluntary action is premeditated
B. A reflex action is initiated by the spinal cord while a voluntary action is the result of activity of the cerebral hemisphere of the brain
C. A reflex action has a longer period of execution
D. None of the above

109. What are the cranial nerves?
A. These are the nerves of the cranium
B. These are the nerves that control the brain
C. These are the shortest nerves of the body
D. These are the main nerves of the body
E. These are the nerves of the brain

110. While the ears and the eyes respond to the stimulus of sound and light respectively the taste buds and olfactory organs respectively responds to:
A. The stimulus of smell
B. The stimulus of bitter taste
C. The stimulus of chemicals
D. Bitterness, sourness and sweetness of substances
E. None of the above

111. The structures that respond to such sensations as sweetness, bitterness, sourness and saltiness are:
A. The taste-hairs
C. The gustatory hairs
E. None of the above
B. The tongues
D. The taste-buds

112. The vitreous humour differs from the aqueous humour in:
A. Being much more watery
B. Being more translucent

- C. Having a less refractive index
- D. Being more viscous
- E. None of the above

113. Why is an area of the eye known as the blind spot?

- A. This spot is dead
- B. The spot is outside the receptive surfaces of the retina
- C. The spot is covered by nerves from the retina
- D. This spot being without visual cells is non-receptive to light stimulus
- E. None of the above

114. The function of the iris is:

- A. To stir the aqueous humour and keep it always clean
- B. To protect the lens
- C. To regulate the size of the pupil
- D. To direct the rays of light to the retina
- E. To assist the ciliary muscles in varying the focal length of the lens

115. Which of these structures has practically no effect on the rays of light that enter the eye?

- A. The cornea
- B. The lens
- C. The aqueous humour
- D. The vitreous humour
- E. None of the above

116. A man discovers that he cannot clearly see a far distant object. What is the defect of his eye?

- A. Myopia
- B. Astigmatism
- C. Long-sightedness
- D. Hypermetropia
- E. A cataract of the eye

117. The watery fluid in the inner ear of mammals is known as

- A. The endolymph
- B. The perilymph
- C. The labyrinth
- D. None of the above

118. The most important structure of the utricle of the inner ear responsible for balance is:

- A. The cochlea
- B. The ampulla
- C. The semicircular canal
- D. The sacculus
- E. None of the above

119. Which of these is not an ossicle of the middle ear?

- A. Malleus
- B. Tympanic bone
- C. Incus
- D. Stapes
- E. None of the above

120. If a student closes tightly his nostrils and mouth and blows from within him, he feels some unpleasantness on his tympanic membranes. Through which of these structures did air get into his ear?

- A. The auditory meatus
- B. The pharynx
- C. The tympanic chamber
- D. The lungs
- E. The eustachian tube

121. The function of the pinna of the ear is:

- A. To locate the direction of sound
- B. To distinguish between different types of sound reaching the ear
- C. To pick up the sound reaching the ear
- D. To amplify sound reaching the ear
- E. None of the above

122. Which of these is a function of the cochlea?

- A. Hearing
- B. To air the ear ossicles or bones
- C. Detection of change in the angular disposition of animal
- D. Orientation or maintaining body posture in space
- E. None of the above

123. Diabetes is an illness resulting from under-secretion of the hormone

- A. Corticin
- B. Thyroxin
- C. Secretin
- D. Insulin
- E. Adrenalin

124. Which of these hormones is responsible for growth and metamorphosis of tadpoles?

- A. Adrenalin
- B. Insulin
- C. Pituitrine
- D. Thyroxin
- E. None of the above

125. A truant of a pupil escaping into a bush is caught by a snake. Which of these hormones will be released into his blood-stream?

- A. Insulin
- B. Parathymine
- C. Adrenalin
- D. Thyroxin
- E. Gastrin

CHAPTER 14

1. E	2. C	3. A	4. D	5. D
6. A	7. C	8. B	9. C	10. A
11. B	12. C	13. C	14. D	15. E
16. B	17. E	18. C	19. Nil	20. B
21. B	22. E	23. A	24. D	25. A
26. D	27. C	28. D	29. A	30. B
31. D	32. C	33. C	34. A	35. E
36. A	37. C	38. D	39. B	40. A
41. A	42. D	43. C	44. B	45. B
46. A	47. A	48. C	49. D	50. C
51. B	52. A	53. D	54. D	55. D
56. B	57. A	58. D	59. A	60. A
61. D	62. B	63. B	64. B	65. C
66. C	67. D	68. B	69. B	70. B
71. A	72. B	73. B	74. D	75. C
76. A	77. B	78. C	79. C	80. C
81. D	82. D	83. B/C	84. A	85. A
86. D	87. C	88. A	89. D	90. B
91. D	92. D	93. D	94. D	95. C
96. A	97. C	98. B	99. A	100. C
101. C	102. C	103. C	104. A	105. B
106. C	107. E	108. B	109. E	110. C
111. D	112. D	113. D	114. C	115. E
116. A	117. B	118. C	119. B	120. E
121. C	122. A	123. D	124. D	125. C

CHAPTER 15

Understand the following after reading this chapter:

Factors affecting the distribution of organisms
Methods of measuring abiotic and biotic factor

Ecology is the study of living things in relation to their environment.

SOME BASIC ECOLOGICAL CONCEPTS

1. **Autecology:** The study of an individual organism and its environment.
2. **Synecology:** The study of two or more organism and their environments.
3. **Environment:** The totality of living and non living external factors which influence the life of an organism.
4. **Habitat:** The natural home of an organism.
There are two major types of habitats- **Terrestrial habitat** (living on land) and **Aquatic habitat** (living in water).
There are three types of aquatic habitats:
 - i. **Fresh water:** A water body that has little or no salt concentration e.g. ponds, streams, rivers etc
 - ii. **Marine or salt water:** A water body with high salinity e.g. seas and oceans.
 - iii. **Brackish or estuarine:** A mix-up of fresh and marine waters with seasonally fluctuating salinity e.g. Deltas and Lagoons.
5. **Population:** Total number of individuals of the same species living in a particular area.
6. **Community:** A naturally occurring group of plants and animals living in a given area.
7. **Niche:** The smallest area within an habitat where an organism is found. It also means the role performed by an organism in the habitat.
8. **Biome:** A large land community with peculiar plants and animals.
9. **Atmosphere:** The open space above the earth surface which contains a mixture of gases and water vapour in different proportions.
10. **Biosphere:** The part of the earth that support the existence of living things.
11. **Hydrosphere:** Biosphere is made up of water e.g. river, stream etc.
12. **Lithosphere:** The solid part of the earth's crust e.g. rock, soil etc.

ECOLOGICAL OR ENVIRONMENTAL FACTORS

Ecological factors are broadly divided into two:

1. **Abiotic or non-living factors** i.e.
 - a. Climatic factors e.g. rainfall, light, wind, temperature, etc.
 - b. Topographic factors i.e. nature of soil surface e.g. slopes, valleys elevation, exposures etc.
 - c. Edaphic factors: soil types and properties i.e. moisture content, pH, organic matter content etc.
2. **Biotic or living factors:** activities of living things e.g.

feeding, shading from light, competition dispersal, predator etc.

Ecological Factors in Aquatic Habitat

1. Salinity
2. Turbidity
3. Tides
4. Current

Ecological Instruments and their uses

Instrument	Uses
Anemometer-	For measuring the speed of wind
Barometer -	For measuring atmospheric pressure
Hygrometer -	For measuring relative humidity
Sweep net/ insect net-	For collecting flying insects
Min.Max. thermometer-	For measuring day temperature
Wind vane-	For determining the direction of wind
Rain gauge-	For measuring amount of rain fall
Quadrant-	For determining the population of organism
Pooter-	For collecting small insects
Photometer/Light meter-	For measuring light intensity
Secchi disc-	For measuring turbidity of water
pH indicator-alkalinity	For measuring water or soil acidity or alkalinity

Revision Questions

15.1. BASIC ECOLOGICAL CONCEPTS

1. UME/78/18

The study of the organism and the environment of an abandoned farmland is the ecology of

- | | |
|-----------------|-----------------|
| A. a community | B. a population |
| C. a species | D. a habitat |
| E. an ecosystem | |

2. UME/92/33

A population is defined as a collection of similar organisms that

- | | |
|----------------------------------|----------------------|
| A. behave in the same way | B. interbreed freely |
| C. are found in the same habitat | D. eat the same food |

3. UME/93/35

What is the term used to describe the sum total of biotic and abiotic factors in the environment of the organism?

- | | |
|--------------|---------------------|
| A. habitat | B. biome |
| C. ecosystem | D. ecological niche |

Use the list of ecological constituents below to answer questions 4 and 5

- | | | | |
|----------------|----------------|---------|----------|
| 1. mango | 2. Spear grass | 3. Goat | 4. Sheep |
| 5. Temperature | 6. Beans | 7. Rock | 8. Water |

4. UME/94/30

Item 1-4 can be regarded as

A. a population
C. an ecosystem

B. a community
D. a niche

5. UME/94/31

The physical factors are represented by

- A. 1, 5, 6 B. 4, 5, 7
C. 5, 7, 8 D. 6, 7, 8

6. UME/2001/28/TYPE: T

The number of plant species obtained from a population study of a garden is as follows: Guinea grass (15), pornea spp (5), sida spp (7) and imperata spp (23). What is the percentage of occurrence of imperata spp.?

- A. 35% B. 16%
C. 46% D. 13%

7. UME/2004/41/TYPE: 4

A caterpillar and an aphid living in different of the same habitat can be said to

- A. be in similar microhabitats
B. occupy different ecological niches
C. occupy the same ecological niches
D. be in different habitats

8. PCE/96/26

The correct order of the ecological sequence in nature is

- A. species → population → community → ecosystem
B. species → population → ecosystem → community
C. species → ecosystem → population → community
D. species → community → ecosystem → population

9. UME 2007/7

The highest level of ecological organization is the

- A. ecosystem B. population
C. biosphere D. Niche

10. PCE/93/34

The functional position of an organism on the ecosystem is referred to as its

- A. habitat B. niche
C. community D. micro-habitat

11. PCE/98/30/TYPE: C

A group of organisms of the same species living together in the same habitat is referred to as

- A. community B. a population
C. an ecosystem D. an association

12. PCE/06/40/TYPE: A

A group of closely related organisms capable of interbreeding are termed

- A. species B. phylum
C. population D. community

15.2. FACTORS AFFECTING THE DISTRIBUTION OF ORGANISMS

13. UME/84/44

Which of the following groups of factors is completely abiotic?

- A. salinity, tide, plankton, turbidity
B. temperature, pH, soil, insect
C. wind, altitude, humidity, light

D. conifers, wind, pH, rainfall
E. soil, water, bacteria, salinity

14. UME/86/43

In an agricultural ecosystem, the biotic component consists of

- A. crops, pest and beneficial insects
B. crops, temperature and humidity
C. pests, beneficial insects and water
D. crops, water and soil

15. UME/86/45

Which of the following ecological factors are common to both terrestrial and aquatic habitats?

- A. rainfall, temperature, light and wind
B. salinity, rainfall, temperature and light
C. tides, wind, rainfall and altitude
D. pH, salinity, rainfall and humidity

16. UME/87/45

Which of the following relates to edaphic factors?

- A. the structure of the earth's surface
B. the influence of living organisms on each other
C. Temperature, rainfall and humidity
D. the influence of soils on plants and animals

17. UME/88/42

Which of the following has the greatest influence on the distribution of animals in marine and fresh water habitats?

- A. PH B. salinity
C. water current D. turbidity

18. UME/91/41

The most important factors which influence an organism's way of life in its habitat are

- A. the physical and biotic environment
B. food and water availability
C. temperature, water, light and predator-prey relationships
D. competition for food and space

19. UME/93/37

The most important physical factor which affects an organism living in the intertidal zone of the seashore is

- A. pH B. salinity
C. wave action D. temperature

20. UME/2003/33/TYPE: A

A density-dependent factor that regulates the population size of organisms is

- A. sudden flood B. disease
C. fire outbreak D. drought

21. UME 2007/9

A biotic factor which affects the distribution and abundance of organisms in a terrestrial habitat is

- A. temperature B. competition
C. pH D. light

22. UME 2008/34

An ecological factor that will have the most limiting effect on the abundance of phytoplankton in a turbid pond is